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Intelligence artificielle neuro-symbolique pour la conception accélérée et la découverte de connaissances en impression 4D

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بِالعِلْمِ يُبْنَى العُمْرَانُ، وَبِالفِكْرِ تَنْهَضُ الأُمَم.

Through knowledge, civilization is built; through thought, nations rise.

Dedication

To my beloved parents,

Thank you for your unconditional love, your sacrifices, and your prayers throughout my life. You have always given me everything you could, with patience, generosity, and faith. Your support has been one of the greatest sources of strength during this journey.

You taught me the value of hard work, humility, perseverance, and trust in Allah. Every step I have taken and every achievement I reach carries a part of your efforts and your sacrifices.

After Allah's grace, this thesis is also the result of your love, your trust, and everything you have done for me. With all my love and gratitude, I dedicate this work to you.

إلى والديّ العزيزين، شكراً لحبّكما ودعائكما وتضحياتكما. هذا العمل ثمرة دعمكما وثقتكما بي.

To my friends,

Thank you for your encouragement, your presence, and the moments of joy that helped me move forward during difficult times. Your support, advice, and sincere companionship made this journey lighter and more meaningful.

To my future children,

May this work remind you that every dream requires patience, discipline, faith, and effort. I hope that one day, when you read these words, you will understand that success is not only measured by titles or achievements, but also by the values we carry, the people we honor, and the good we try to leave behind.

To my city, Safi,

To the city where my roots began, where the Atlantic meets history, patience, and dignity. Safi is not only a place on the map; it is a memory, an identity, and a source of belonging. Known for its ancient walls, its port, its pottery, its ocean, and the strength of its people, Safi has always carried a deep cultural and historical richness. It has also been an important coastal and economic gateway in Morocco, shaped by trade, craftsmanship, fishing, ceramics, and industry. Its region is connected to Jebel Irhoud, where some of the oldest traces of Homo sapiens were discovered, reminding us that this land is linked to the deep history of humanity.

Despite the challenges it faces today, I still believe in its potential, in its youth, and in its future. I hope to see Safi grow, flourish, and receive the attention and development

it deserves. May this city, which has given so much to its people, one day become a place where dreams are not only born, but also fulfilled.

كُنْ شُجَاعًا كَالقِرْشِ فِي بَحْرِ آسْفِي، لا تَهَابُ المَوْجَ، وَاجْعَلْ مِنْ كُلِّ عَاصِفَةٍ طَرِيقًا نَحْوَ النُّور.

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Abstract

Four-dimensional (4D) printing combines additive manufacturing with stimuli-responsive materials to produce structures whose shape, properties, or functionality can evolve after fabrication. However, designing such structures requires the concurrent selection of materials, spatial material distributions, manufacturing processes, geometries, activation conditions, and boundary conditions. The interactions between these variables generate large and highly coupled design spaces. Their exploration through finite element (FE) simulation and physical experimentation is computationally and materially expensive, while the scientific knowledge required to guide design decisions remains distributed across heterogeneous publications, figures, databases, and numerical results. This thesis investigates how scientific knowledge and artificial intelligence methods can support the design of smart materials and transformable structures according to prescribed property, behavior, and performance objectives. The proposed approach combines neuro-symbolic artificial intelligence (NSAI), ontology engineering, retrieval-augmented multimodal models, graph neural networks, graph-based inverse design, and multi-agent orchestration. A conceptual study analyzes the status, challenges, and perspectives of NSAI for accelerated 4D-printing design. The principal NSAI architectures are subsequently formalized, extended, and evaluated using literature-grounded criteria covering generalization, scalability, data efficiency, reasoning, robustness, transferability, and interpretability. Within this qualitative evaluation, the Neuro $\rightarrow$ Symbolic $\leftarrow$ Neuro fibring architecture exhibits the most balanced profile across the selected criteria, particularly for coordinating specialized neural components through a shared symbolic layer. For scientific knowledge structuring, a retrieval-augmented multimodal framework extracts and validates ontology-aligned relations from textual, visual, and structured sources. The resulting workflow expands an initial 4D-printing ontology from 170 classes and a limited number of instances to more than 5,800 classes and over 12.5 million instances. An automated numerical workflow generates voxel-level material configurations, performs FE simulations, and converts the resulting data into graph representations. After training, the retained GraphSAGE surrogate predicts nodal thermo-mechanical displacements in approximately 0.0269 s per structure, compared with 3,600–10,800 s for the complete FE workflow. This corresponds to a per-structure speedup ranging from approximately 1.34×10^5 to 4.01×10^5 and enables the screening of 800,000 structures

under four boundary condition scenarios. Physics-based deformation descriptors are then used to identify candidates exhibiting potentially novel coupled shape-morphing responses. The predictive model is subsequently incorporated into a multidimensional inverse design framework based on deformation-aware graph representations and latent-space exploration. Seven search and optimization strategies are evaluated on 153 target shapes. The combination of top-k candidate search and local material mutation achieves the lowest final target-shape error for 112 targets. Finally, four specialized agents for scientific knowledge, material inverse design, thermo-mechanical simulation, and behavior discovery are integrated through a central orchestrator. Five representative workflows evaluate their coordination through explicit capability specifications, execution plans, and typed intermediate artifacts. Overall, the thesis establishes a computational framework connecting structured scientific knowledge, accelerated physical prediction, inverse design, and coordinated scientific workflows for property-, behavior-, and performance-oriented 4D-printing design.

Keywords: 4D printing, smart materials, neuro-symbolic artificial intelligence, ontology enrichment, knowledge graph, retrieval-augmented generation, graph neural networks, thermo-mechanical simulation, inverse design, behavior discovery, multi-agent systems.

Résumé

L'impression 4D constitue un paradigme de fabrication qui associe la fabrication additive à des matériaux capables de répondre à des sollicitations extérieures. Contrairement aux objets issus de la fabrication additive conventionnelle, dont la géométrie reste généralement fixe après leur production, les structures imprimées en 4D peuvent modifier leur forme, leurs propriétés ou leur fonctionnalité au cours du temps lorsqu'elles sont exposées à une variation de température, à la lumière, à l'humidité, à un champ magnétique ou électrique, à un environnement chimique ou à une variation de pH [1, 2]. Ces transformations peuvent prendre la forme d'une expansion, d'une contraction, d'un pliage, d'une flexion, d'une torsion ou d'une évolution plus complexe vers une géométrie prescrite. Cette capacité ouvre des perspectives dans les domaines des dispositifs biomédicaux, de la robotique souple, des structures adaptatives, du textile, de l'automobile, de la défense et de l'espace.

La réalisation de ces fonctions repose sur l'intégration de matériaux sensibles aux stimuli, notamment les polymères à mémoire de forme, les élastomères à cristaux liquides, les alliages à mémoire de forme, les polymères électroactifs et les hydrogels. Ces matériaux peuvent être mis en forme par différentes techniques de fabrication additive, telles que le dépôt de fil fondu, l'écriture directe à l'encre, le traitement numérique de la lumière ou la fusion sur lit de poudre. La réponse finale d'une structure ne dépend cependant pas uniquement de la nature du matériau actif. Elle résulte des interactions entre la formulation du matériau, son orientation locale, sa distribution spatiale, la géométrie de la structure, les paramètres de fabrication, les conditions aux limites, l'intensité du stimulus et la cinétique d'activation. La conception d'une structure transformable constitue ainsi un problème multidimensionnel dans lequel plusieurs paramètres physiques, géométriques et technologiques doivent être considérés simultanément.

Les connaissances nécessaires à cette conception sont dispersées dans des articles scientifiques, des bases de données, des figures, des tableaux, des modèles numériques et des résultats expérimentaux. Les terminologies employées diffèrent selon les communautés scientifiques, et les relations entre matériaux, procédés, stimuli, propriétés, comportements et applications sont rarement représentées dans un format directement exploitable par une machine. La sélection d'une solution repose par conséquent sur une recherche documentaire importante et sur l'expérience du concepteur. Cette fragmentation limite la réutilisation des résultats existants et complique la formulation de recommandations

scientifiques adaptées à un objectif de conception précis.

La prédiction du comportement physique représente une autre difficulté majeure. Les simulations par éléments finis permettent d'évaluer la déformation thermo-mécanique d'une structure, mais leur coût devient prohibitif lorsqu'un grand nombre de configurations matérielles doit être étudié. Chaque nouvelle distribution de matériaux, orientation locale ou condition aux limites peut nécessiter une nouvelle simulation. Cette contrainte est particulièrement importante pour la conception inverse, où plusieurs milliers, voire plusieurs millions, de candidats peuvent être évalués avant d'identifier une solution compatible avec une déformation cible. Le passage du comportement souhaité vers la configuration matérielle correspondante est également non unique : plusieurs distributions de matériaux peuvent conduire à des réponses géométriques proches, tandis que des modifications locales limitées peuvent produire des transformations globales sensiblement différentes.

L'intelligence artificielle offre des moyens de traiter ces difficultés, mais les approches purement neuronales et purement symboliques présentent des limites complémentaires. Les modèles neuronaux apprennent des relations complexes à partir de données et peuvent approximer rapidement des phénomènes physiques coûteux à simuler. Leur fonctionnement dépend toutefois de la représentativité des données d'apprentissage, et leurs prédictions peuvent devenir peu fiables lorsqu'ils sont appliqués à des configurations éloignées de ce domaine. Les approches symboliques permettent de représenter explicitement des concepts, des relations, des contraintes et des règles scientifiques. Elles sont adaptées à l'organisation des connaissances et à la vérification de relations, mais leur construction manuelle est coûteuse et leur capacité à apprendre directement à partir de données multimodales ou à approximer des champs physiques complexes demeure limitée.

L'intelligence artificielle neuro-symbolique étudie l'association de ces capacités complémentaires. Elle vise à combiner l'apprentissage à partir de données avec des représentations structurées et des opérations symboliques. Dans le contexte de l'impression 4D, cette association peut permettre d'extraire des connaissances à partir de la littérature, de les organiser dans une ontologie, de prédire le comportement physique de structures candidates, d'explorer un espace de conception sous contraintes et de coordonner plusieurs outils scientifiques. Elle ne désigne cependant pas une architecture unique. Les interactions entre composants neuronaux et symboliques peuvent être organisées selon différentes directions, différents niveaux de couplage et différents mécanismes d'échange d'information.

Cette thèse s'articule autour de la question de recherche suivante : comment recommander des connaissances pour concevoir des matériaux intelligents et des structures transformables par impression 4D afin d'atteindre une propriété, un comportement et une performance souhaités ? Trois objectifs scientifiques en découlent. Le premier consiste à acquérir et structurer les connaissances scientifiques existantes relatives au domaine

interdisciplinaire de l'impression 4D. Le deuxième vise à accélérer la découverte de nouvelles connaissances concernant les distributions matérielles et les comportements de transformation qui en résultent. Le troisième consiste à fournir des recommandations traçables pour la conception en impression 4D. Dans cette thèse, la recommandation est considérée comme un processus scientifique reliant un objectif exprimé par le concepteur à des connaissances pertinentes, à une configuration matérielle candidate, à une prédiction physique et à une évaluation quantitative. Elle ne se limite donc pas au classement de matériaux ou de solutions similaires. Elle vise à établir une continuité entre la connaissance scientifique, la représentation du problème, la prédiction du comportement et la construction d'une solution de conception.

La démarche développée s'organise en deux niveaux complémentaires. Les deux premiers chapitres établissent les fondations conceptuelles et architecturales de l'intelligence artificielle neuro-symbolique pour la conception en impression 4D. Les quatre chapitres contributifs portent ensuite sur la structuration des connaissances scientifiques, l'accélération de la simulation thermo-mécanique et la découverte de comportements, la conception inverse de structures multi-matériaux, puis l'intégration de ces capacités au moyen d'agents spécialisés coordonnés par un orchestrateur.

Le premier chapitre établit un état critique de l'intelligence artificielle neuro-symbolique appliquée à la conception accélérée pour l'impression 4D. Il analyse les capacités attendues d'un système destiné à ce domaine, notamment la représentation de connaissances hétérogènes, l'apprentissage à partir de données limitées, la prédiction physique, la gestion de contraintes, le raisonnement sur des relations matériau–procédé–structure–comportement et le transfert vers de nouvelles configurations. Cette étude met également en évidence plusieurs obstacles : l'absence de jeux de données normalisés, l'hétérogénéité des représentations, le coût de la génération de données physiques, la difficulté à intégrer des connaissances multi-échelles et le manque de protocoles communs pour évaluer les architectures neuro-symboliques dans un contexte d'ingénierie.

Le deuxième chapitre formalise et étend plusieurs architectures neuro-symboliques, puis les compare selon des critères issus de la littérature. L'évaluation considère la généralisation, le passage à l'échelle, l'efficacité en données, le raisonnement, la robustesse, la transférabilité et l'interprétabilité. Elle clarifie notamment la direction des échanges entre composants et distingue les situations où le module symbolique fournit des contraintes ou une structure au modèle neuronal, celles où un module neuronal alimente un raisonnement symbolique, et celles où plusieurs composants spécialisés communiquent par l'intermédiaire d'une représentation structurée. Dans les conditions de cette évaluation qualitative, l'architecture fibrée représentée par Neuro $\rightarrow$ Symbolic $\leftarrow$ Neuro présente un compromis particulièrement équilibré entre les critères étudiés. Ce résultat ne constitue pas un classement empirique universel. Il indique qu'une organisation faisant intervenir plusieurs modules neuronaux spécialisés autour d'une représentation ou d'un opérateur

symbolique commun peut être adaptée aux problèmes hétérogènes qui nécessitent simultanément perception, prédiction, structuration et coordination.

La première contribution traite de l'acquisition et de la structuration des connaissances scientifiques relatives aux matériaux intelligents et à l'impression 4D. Une ontologie initiale représentait les principaux concepts du domaine, mais son contenu restait insuffisant pour couvrir la diversité des matériaux, des stimuli, des procédés, des comportements et des applications décrits dans la littérature. Son enrichissement manuel aurait nécessité un effort important et aurait été difficile à maintenir au rythme de publication de nouveaux résultats. Un cadre d'enrichissement fondé sur la génération augmentée par récupération et sur des modèles multimodaux a donc été développé afin d'automatiser une partie de ce processus.

Le cadre commence par récupérer des informations pertinentes dans des sources scientifiques textuelles, visuelles et structurées. Les documents sont segmentés et représentés dans un espace permettant une recherche sémantique. Les passages, figures ou données les plus proches de la requête sont ensuite transmis aux modèles chargés d'identifier des entités et des relations candidates. Cette récupération préalable réduit la dépendance à la mémoire paramétrique des modèles et fournit un contexte scientifique directement associé aux informations à extraire. Les sorties ne sont pas intégrées immédiatement dans l'ontologie. Elles sont alignées avec les concepts existants, comparées au schéma ontologique et soumises à des opérations de validation destinées à écarter les relations incompatibles, insuffisamment étayées ou redondantes.

L'enrichissement porte sur des entités telles que les matériaux actifs et passifs, les formulations, les procédés de fabrication, les paramètres d'impression, les types de stimulus, les propriétés thermiques et mécaniques, les modes de déformation et les applications. Il permet également de représenter des relations entre ces entités, par exemple le fait qu'un matériau répond à un stimulus donné, qu'un procédé permet de mettre en forme une famille de matériaux ou qu'une architecture produit un comportement particulier. La combinaison de la récupération sémantique, de l'extraction multimodale et de la validation symbolique constitue le mécanisme neuro-symbolique le plus explicite de la thèse : les modèles neuronaux traitent des contenus hétérogènes, tandis que l'ontologie fournit une structure conceptuelle et des contraintes d'intégration.

Le processus a étendu l'ontologie initiale d'environ 170 classes et d'un nombre limité d'instances à plus de 5 800 classes et plus de 12,5 millions d'instances. Cette augmentation ne constitue pas uniquement un accroissement quantitatif. Elle fournit une représentation beaucoup plus large des relations nécessaires à la recherche scientifique et à la sélection de matériaux, de stimuli ou de procédés. La base de connaissances obtenue peut être interrogée pour identifier des solutions compatibles avec une exigence fonctionnelle, retrouver des relations entre propriétés et comportements ou fournir un contexte structuré aux autres composants du système. Elle constitue ainsi la mémoire sémantique de la

démarche proposée.

Cette contribution reste dépendante de la qualité des documents récupérés, de la couverture de la littérature et de la capacité des modèles à distinguer des relations scientifiques proches. Le volume obtenu rend impossible une validation manuelle exhaustive. La validation symbolique réduit le nombre d'incohérences, mais elle ne garantit pas que chaque relation soit complète, correcte et applicable à toutes les conditions expérimentales. L'ontologie doit donc être considérée comme une ressource évolutive dont les relations importantes nécessitent une traçabilité vers leurs sources, une estimation de confiance et, lorsque l'enjeu scientifique l'exige, une vérification par un expert.

La deuxième contribution porte sur l'accélération de la prédiction thermo-mécanique et son exploitation pour l'exploration de grands espaces de conception et la découverte de comportements. Un processus automatisé a été développé pour générer des configurations voxelisées, leur attribuer des matériaux, produire les modèles par éléments finis correspondants et convertir les résultats en graphes adaptés à l'apprentissage. Le système étudié combine des domaines actifs en élastomère à cristaux liquides, différentes orientations locales, des régions passives et des zones représentant le vide. Chaque structure est représentée par un graphe décrivant sa géométrie, ses propriétés matérielles et ses relations d'adjacence, afin de préserver l'organisation spatiale nécessaire à la prédiction du champ de déplacement.

Plusieurs architectures d'apprentissage, notamment des modèles multicouches, convolutionnels, attentionnels et transformeurs sur graphes, ont été comparées selon un protocole commun. GraphSAGE [3] a fourni la meilleure performance prédictive globale parmi les modèles évalués dans la configuration de référence, avec une erreur absolue moyenne globale de 0,01237 et une racine de l'erreur quadratique moyenne de 0,02485. La stabilité des performances a également été vérifiée sur un ensemble de test indépendant par rééchantillonnage bootstrap. Le modèle retenu fournit ainsi un substitut rapide aux évaluations répétées par éléments finis pour l'exploration de grands espaces de conception.

Le principal bénéfice du modèle est la réduction du temps d'évaluation. Une inférence GraphSAGE nécessite environ 0,0269 s par structure, alors que le processus complet de préparation et de résolution par éléments finis nécessite approximativement 3 600–10 800 s, selon la configuration et les conditions de convergence. Sur cette base, le facteur d'accélération est compris approximativement entre $1{,}34 \times 10^5$ et $4{,}01 \times 10^5$. Cette différence rend possible l'exploration d'un nombre de configurations qui ne pourrait pas être traité par des simulations séquentielles à haute fidélité dans un temps compatible avec une étude de conception.

La capacité de prédiction a été utilisée pour examiner 800 000 structures candidates dans quatre scénarios de conditions aux limites. Des descripteurs physiquement interprétables ont été appliqués aux champs de déplacement afin de caractériser la flexion, la torsion, l'asymétrie et les réponses couplées. Le classement selon ces descripteurs a permis

d'identifier des candidats présentant des combinaisons de comportements différentes des réponses de référence. Cette démarche transforme le modèle de substitution en outil d'exploration : il ne sert pas uniquement à reproduire une simulation connue, mais également à rechercher des régions moins explorées de l'espace des comportements.

Les comportements identifiés demeurent des candidats numériques. Une réponse prédite par le modèle ne constitue pas une démonstration expérimentale. Les configurations les plus intéressantes doivent être vérifiées par une simulation à haute fidélité, puis fabriquées et caractérisées. Les descripteurs employés influencent également le type de nouveauté recherché. Un comportement qui n'est pas représenté par les critères de flexion, de torsion ou d'asymétrie peut ne pas être détecté. La découverte computationnelle doit donc être considérée comme une étape de présélection destinée à orienter des validations plus coûteuses.

La troisième contribution développe une méthode de conception inverse multidimensionnelle appliquée à des domaines de conception unidimensionnels, bidimensionnels et tridimensionnels. Cette formulation permet d'évaluer le cadre sur des niveaux croissants de complexité géométrique et vise à soutenir son applicabilité à une gamme plus large de structures et d'applications en impression 4D. La méthode part d'une déformation cible et recherche des distributions de matériaux dont la réponse prédite se rapproche de cette cible. La méthode associe des représentations de graphes, un espace latent tenant compte de la déformation, la récupération de candidats et des opérations locales de modification de la distribution matérielle. L'utilisation d'un espace latent permet de représenter et de comparer des structures selon leurs caractéristiques géométriques et leurs transformations, sans effectuer une recherche exhaustive sur toutes les affectations possibles de matériaux.

Sept stratégies de recherche et d'optimisation ont été comparées sur 153 formes cibles. Elles associent de différentes manières l'exploration de l'espace latent, la récupération de candidats proches et la modification locale des matériaux. La qualité d'une proposition est évaluée au moyen d'une mesure géométrique comparant la forme déformée prédite à la cible prescrite. La stratégie combinant la récupération des candidats top-k avec une mutation locale de la distribution matérielle a obtenu la plus faible erreur finale pour 112 des 153 cibles. Ce résultat indique que la récupération d'une solution initiale pertinente et son raffinement local sont plus efficaces, dans le domaine étudié, qu'une exploration entièrement globale ou qu'une génération non guidée.

La conception inverse traite ainsi la non-unicité du problème en produisant et en classant plusieurs candidats plutôt qu'en supposant l'existence d'une solution unique. La prédiction rapide du modèle GraphSAGE permet d'évaluer ces candidats à chaque étape de l'optimisation. Les opérateurs explicites de modification matérielle maintiennent la recherche dans l'espace des configurations admises. La méthode fournit finalement une distribution voxelisée, une forme déformée prédite et une mesure d'accord avec la cible. Les exemples étudiés montrent que le cadre peut être appliqué à plusieurs familles de

formes et à des assemblages destinés à des fonctions différentes.

Les résultats de conception inverse sont néanmoins conditionnés par la précision du prédicteur utilisé, la résolution voxelisée et la représentation de la cible. Une faible erreur géométrique ne garantit pas qu'une structure soit imprimable, stable, réversible ou durable. La formulation actuelle ne représente pas encore toutes les contraintes liées à la compatibilité des matériaux, à la fabrication multi-matériaux, à la fatigue, à l'énergie d'activation ou au coût. Les solutions proposées doivent donc être considérées comme des candidats numériques devant être filtrés par des critères de fabrication et validés physiquement.

La quatrième contribution intègre les capacités développées dans la thèse sous la forme de quatre agents spécialisés. L'agent de connaissances scientifiques fournit l'interrogation de l'ontologie, la récupération sémantique, l'extraction multimodale et la validation de relations. L'agent de conception inverse recherche des distributions matérielles correspondant à une déformation cible. L'agent de simulation thermo-mécanique prédit les champs de déplacement à partir d'un graphe décrivant la structure et les matériaux. L'agent de découverte analyse les réponses prédites, calcule des descripteurs et classe les comportements candidats.

Un orchestrateur constitue l'opérateur central de ce système. Il interprète une requête scientifique, construit un plan d'exécution, sélectionne les agents et les outils nécessaires, ordonne leurs appels et transmet les résultats intermédiaires entre les opérations. Le modèle de langage utilisé pour l'orchestration ne génère pas directement les prédictions physiques ou les distributions optimisées. Ces calculs restent délégués aux modèles scientifiques correspondants. Cette séparation réduit le risque de substituer une génération textuelle à un calcul numérique et permet de conserver des sorties structurées à chaque étape.

L'intégration a été examinée au moyen de scénarios portant sur la sélection de matériaux à partir de la littérature, la conception inverse, la simulation d'une structure candidate, la vérification d'une hypothèse et la découverte de comportements. Ces démonstrations montrent qu'une demande peut être transformée en une suite d'opérations utilisant des ressources hétérogènes. Elles ne constituent cependant pas encore une évaluation complète de l'autonomie du système. La fiabilité de la planification, la propagation des erreurs, la reprise après échec et la communication de l'incertitude doivent être étudiées sur des jeux de tâches plus larges.

Les quatre contributions computationnelles mettent en œuvre l'intelligence artificielle neuro-symbolique à l'échelle du système, sans reproduire une architecture identique dans chaque contribution. Dans l'enrichissement de l'ontologie, les modèles multimodaux neuronaux extraient et alignent l'information, tandis que l'ontologie, le graphe de connaissances et les règles de validation fournissent la représentation symbolique. Dans la prédiction thermo-mécanique, GraphSAGE constitue le composant neuronal ; le graphe

typé, les relations d'adjacence, les conditions aux limites et les descripteurs physiques fournissent une structuration explicite du problème. Cette interface structurée–neuronale ne doit pas être assimilée automatiquement à un raisonnement symbolique, car un graphe utilisé comme entrée d'un réseau n'est pas suffisant pour constituer une architecture neuro-symbolique complète. Dans la conception inverse, les représentations apprises et les prédictions neuronales sont associées à des mesures géométriques explicites, à des contraintes matérielles et à des opérateurs de recherche définis. Dans le système d'agents, l'orchestrateur coordonne les modèles neuronaux, les ressources symboliques et les outils numériques en respectant un plan et des dépendances explicites.

La réponse apportée à la question de recherche résulte de cette complémentarité. La recommandation de connaissances pour l'impression 4D peut être soutenue par une représentation structurée des relations scientifiques, une prédiction rapide du comportement physique, une exploration inverse guidée par des critères explicites et une orchestration des ressources spécialisées. La couche de connaissances identifie les matériaux, stimuli, procédés et relations compatibles avec l'objectif. Le modèle prédictif estime la réponse des configurations candidates. La conception inverse traduit le comportement demandé en distributions matérielles possibles. L'orchestrateur assure la continuité entre ces opérations et organise les informations nécessaires à leur exécution.

La contribution globale de la thèse ne correspond donc pas à la juxtaposition de plusieurs modèles indépendants. Elle réside dans la construction d'une chaîne reliant la connaissance scientifique à la proposition et à l'évaluation d'une structure. Cette chaîne associe des informations documentaires, des représentations sémantiques, des données de simulation, des modèles neuronaux et des critères de conception. Dans le domaine étudié, elle permet de passer d'une demande formulée en termes de propriété, de comportement ou de performance à un ensemble de candidats décrits par leurs matériaux, leur configuration spatiale, leur réponse prédite et leur erreur par rapport à l'objectif.

Plusieurs limites encadrent ces résultats. L'analyse des architectures neuro-symboliques repose sur une évaluation qualitative et sur les résultats rapportés dans la littérature. Les critères permettent une comparaison systématique, mais les architectures n'ont pas toutes été réimplémentées sur un même jeu de données et avec les mêmes ressources. Le résultat concernant l'architecture fibrée doit donc être interprété comme l'identification d'un équilibre conceptuel entre plusieurs capacités, et non comme la preuve d'une supériorité générale. Un protocole empirique commun serait nécessaire pour comparer quantitativement les architectures dans des tâches d'ingénierie représentatives.

La base de connaissances dépend de la couverture et de la qualité des sources utilisées. Les articles peuvent employer des terminologies incompatibles, omettre certaines conditions expérimentales ou rapporter des résultats contradictoires. Les modèles d'extraction peuvent également produire des relations ambiguës ou incomplètes. La validation symbo-

lique améliore la cohérence avec l'ontologie, mais une relation cohérente sur le plan formel n'est pas nécessairement correcte sur le plan scientifique. Les futures versions devront renforcer la provenance, les scores de confiance, la gestion des contradictions et la validation ciblée par des experts.

Les modèles thermo-mécaniques sont limités par les hypothèses constitutives, les propriétés matérielles, la discrétisation voxelisée et les conditions d'activation étudiées. Le modèle GraphSAGE apprend à partir des données générées dans ce cadre et ne doit pas être appliqué sans vérification à des matériaux, géométries ou stimuli éloignés de son domaine d'apprentissage. La conception inverse et la découverte de comportements héritent de cette limitation, car elles utilisent le prédicteur comme évaluateur. Une erreur locale du modèle peut modifier le classement des candidats ou orienter l'optimisation vers une configuration qui ne reproduira pas la réponse attendue lors d'une simulation à haute fidélité ou d'une expérience.

Le cadre a été principalement validé de manière computationnelle. La fabrication et la caractérisation expérimentale d'un ensemble représentatif de structures sont nécessaires pour mesurer l'écart entre la réponse réelle, la simulation par éléments finis et la prédiction du modèle. Les défauts de fabrication, l'adhésion entre matériaux, les dispersions de propriétés et les variations des conditions d'activation ne sont pas entièrement décrits dans les données numériques. Les agents spécialisés ont en outre été démontrés sur des scénarios ciblés. Une évaluation plus large devra mesurer le taux de réussite des tâches, le temps d'exécution, la robustesse aux erreurs intermédiaires et la reproductibilité des plans proposés par l'orchestrateur.

La première perspective consiste par conséquent à fermer progressivement l'écart entre prédiction numérique et validation physique. Des structures issues de la conception inverse et de la découverte de comportements devront être sélectionnées, fabriquées et activées dans des conditions contrôlées. Leurs champs de déplacement devront être mesurés et comparés aux prédictions par éléments finis et GraphSAGE. Les écarts observés permettront de recalibrer les modèles de matériaux, d'identifier les sources d'erreur et d'enrichir les données d'apprentissage avec des observations expérimentales.

L'espace de conception devra être étendu à d'autres matériaux actifs, stimuli, géométries et procédés de fabrication. L'augmentation de la résolution permettra de mieux représenter les interfaces et les variations locales, mais elle augmentera également le coût de génération des données et de l'apprentissage. Une stratégie multi-fidélité pourra combiner des prédictions rapides, des simulations plus précises et des mesures expérimentales. L'incertitude du modèle et la détection des entrées hors distribution devront être intégrées afin de distinguer les candidats pouvant être évalués par le modèle de substitution de ceux qui nécessitent une simulation ou une expérience.

La conception inverse devra évoluer vers une formulation multi-objective intégrant la précision géométrique, la fabricabilité, la stabilité, la réversibilité, la fatigue, la

consommation énergétique et le coût. Les contraintes ne devront pas être appliquées uniquement après l'optimisation, mais incorporées dans la génération, le raffinement et le classement des candidats. L'ontologie pourra fournir certaines relations de compatibilité et certains domaines de validité, tandis que les données expérimentales permettront d'estimer les contraintes qui ne peuvent pas être décrites par des règles déterministes.

La perspective principale à plus long terme est le développement d'un laboratoire autonome pour les matériaux intelligents et l'impression 4D, organisé autour d'un modèle du monde scientifique [4, 5]. Ce modèle représenterait l'état du système à travers la formulation et la distribution des matériaux, la géométrie, les conditions de fabrication et d'activation, les réponses mesurées et l'incertitude associée. Il combinerait plusieurs niveaux de représentation et de fidélité : l'ontologie et le graphe de connaissances pour la mémoire sémantique, GraphSAGE pour la prédiction rapide, les éléments finis pour la vérification à haute fidélité, la conception inverse pour la proposition de candidats et les observations expérimentales pour la mise à jour du système.

À partir d'un objectif scientifique, le système pourrait identifier les connaissances pertinentes, proposer plusieurs configurations matérielles, prédire leurs réponses et sélectionner les candidats les plus utiles pour une simulation ou une expérience. Les structures retenues seraient ensuite fabriquées, activées et caractérisées. Les écarts entre réponses prédites et mesurées seraient convertis en nouvelles observations afin de mettre à jour les données, les modèles prédictifs et les relations de connaissances. Cette boucle fermerait progressivement le cycle entre définition de l'objectif, récupération de connaissances, conception, évaluation virtuelle, expérimentation et apprentissage.

Cette perspective s'inscrit dans la continuité des laboratoires autonomes déjà développés pour l'optimisation et la découverte de matériaux [6–8]. Son extension à l'impression 4D nécessitera toutefois de gérer simultanément la fabrication multi-matériaux, l'activation contrôlée, la mesure des déformations, l'incertitude et la variabilité expérimentale. Le système devra donc être conçu selon une approche multi-fidélité et neuro-symbolique, tout en conservant une supervision humaine pour les décisions critiques, l'introduction de nouveaux matériaux et l'interprétation de phénomènes inattendus.

En conclusion, cette thèse établit un cadre conceptuel et computationnel pour la recommandation de connaissances en conception pour l'impression 4D. Elle structure des informations scientifiques hétérogènes, réduit le coût de la prédiction thermo-mécanique, permet l'exploration de grands espaces de configurations, produit des distributions matérielles candidates pour des déformations prescrites et coordonne ces capacités au moyen d'agents spécialisés. Les résultats restent limités aux matériaux, modèles, données et conditions étudiés. Leur intégration future dans un laboratoire autonome fondé sur un modèle du monde constitue la principale voie permettant de relier la connaissance scientifique, la conception numérique et l'apprentissage à partir d'expériences physiques.

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List of Abbreviations

3D	Three-Dimensional
4D	Four-Dimensional
AI	Artificial Intelligence
AM	Additive Manufacturing
ASP	Answer Set Programming
BC	Boundary Condition
BERT	Bidirectional Encoder Representations from Transformers
BESO	Bidirectional Evolutionary Structural Optimization
BLEU	Bilingual Evaluation Understudy
CAD	Computer-Aided Design
CLIP	Contrastive Language-Image Pre-training
CMA-ES	Covariance Matrix Adaptation Evolution Strategy
CNN	Convolutional Neural Network
CNRS	Centre National de la Recherche Scientifique
CoT	Chain-of-Thought
DA	Discovery Approach
DIW	Direct Ink Writing
DL	Deep Learning (also Description Logic, in the ontology chapter)
DLP	Digital Light Processing
EA	Evolutionary Algorithm
EAP	Electroactive Polymer
EMMO	European Materials Modelling Ontology
FE	Finite Element
FEM	Finite Element Method
FFF	Fused Filament Fabrication

GAN	Generative Adversarial Network
GAT	Graph Attention Network
GCN	Graph Convolutional Network
GELU	Gaussian Error Linear Unit
GNN	Graph Neural Network
GPT	Generative Pre-trained Transformer
ICB	Laboratoire Interdisciplinaire Carnot de Bourgogne
ICL	In-Context Learning
ILP	Inductive Logic Programming
IOF	Industry Ontology Foundry
IUF	Institut Universitaire de France
KG	Knowledge Graph
LCE	Liquid Crystal Elastomer
LLM	Large Language Model
LNN	Logical Neural Network
LoRA	Low-Rank Adaptation
LTN	Logic Tensor Network
MAE	Mean Absolute Error
MatKG	Materials Knowledge Graph
MD	Molecular Dynamics
MDO	Materials Design Ontology
ML	Machine Learning
MLLM	Multimodal Large Language Model
MLP	Multilayer Perceptron
MoE	Mixture of Experts
MSE	Mean Squared Error
NER	Named Entity Recognition
NIR	Near-Infrared
NLP	Natural Language Processing
NN	Neural Network
NSAI	Neuro-Symbolic Artificial Intelligence
OCR	Optical Character Recognition

OOD	Out-of-Distribution
OWL	Web Ontology Language
PBF	Powder Bed Fusion
PDMS	Polydimethylsiloxane
PEFT	Parameter-Efficient Fine-Tuning
PINN	Physics-Informed Neural Network
RAG	Retrieval-Augmented Generation
RDF	Resource Description Framework
RDFS	RDF Schema
RL	Reinforcement Learning
RLHF	Reinforcement Learning with Human Feedback
RMSE	Root Mean Squared Error
RNN	Recurrent Neural Network
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
Seq2Seq	Sequence-to-Sequence
SIMP	Solid Isotropic Material with Penalization
SKOS	Simple Knowledge Organization System
SM	Smart Material
SMA	Shape Memory Alloy
SMP	Shape Memory Polymer
TPR	Tensor Product Representation
VAE	Variational Autoencoder
XAI	Explainable Artificial Intelligence
UTBM	Université de Technologie de Belfort-Montbéliard

General Introduction

4D printing is a manufacturing paradigm that combines additive manufacturing (AM) with smart materials (SMs). This technology allows printed objects to change their shape, properties, or functionality over time when exposed to external stimuli, including temperature, light, moisture, magnetic or electric fields, chemical environments, or pH [1, 2]. Through these programmed transformations, objects can autonomously expand or contract, fold, bend, twist, or morph without direct mechanical intervention (Figure 1). Since its emergence, the field has advanced rapidly through converging developments in materials science, additive manufacturing, chemistry, physics, computational mechanics, and design science, thereby opening opportunities in biomedicine, soft robotics, smart textiles, automotive systems, defense, and space applications [9, 10].

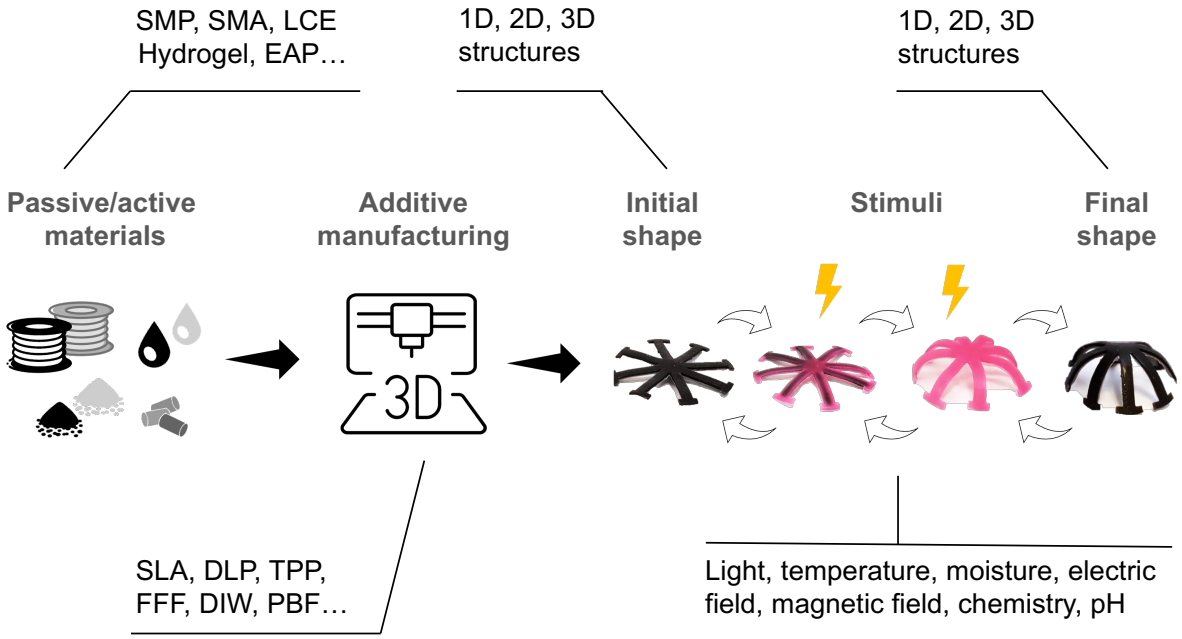

Figure 1: The 4D-printing paradigm illustrated through a structure whose shape or properties evolve over time in response to an external stimulus, reproduced from [11].

At the core of 4D printing lies the integration of stimuli-responsive materials such as shape-memory polymers (SMPs), liquid-crystal elastomers (LCEs), shape-memory alloys (SMAs), electroactive polymers (EAPs), and hydrogels. These materials can be processed using AM techniques such as fused-filament fabrication (FFF), direct-ink

writing (DIW), digital-light processing (DLP), and powder-bed fusion (PBF) to fabricate geometrically complex and functionally adaptive structures [9, 10]. SMPs, for example, can be programmed into a temporary configuration and recover a memorized shape when activated, commonly by heating above a transition temperature [12]. Hydrogels can swell or shrink in response to changes in water content, temperature, ionic concentration, or pH, whereas LCEs can produce large, reversible, and direction-dependent deformations through changes in molecular order. The suitability of a smart material therefore depends not only on the magnitude of its response, but also on its printability, activation conditions, reversibility, mechanical integrity, response time, durability, and compatibility with the intended application.

The design of 4D-printed structures involves strongly coupled material, geometric, manufacturing, and activation variables, including the selection and spatial orientation of active and passive materials, the manufacturing process, the applied stimulus, the boundary conditions, and the temporal evolution of the material response. A material that performs well in an isolated test may behave differently once printed within a heterogeneous structure. Likewise, a geometry that appears suitable in its initial configuration may fail to reach the target shape because the local actuation directions, interfaces, or mechanical constraints have been inadequately specified. The designer must consequently reason across the entire material–process–structure–stimulus–behavior chain rather than optimize each element independently [13, 14].

This coupling makes the design of transformable 4D-printed structures a high-dimensional inverse problem, in which a prescribed property, behavior, or performance objective must be associated with a suitable combination of materials, spatial material distribution, geometry, printing conditions, and activation strategy [10, 13, 15]. The inverse mapping is generally non-unique, since different material configurations may produce similar macroscopic responses, while local variations in material assignment or orientation can lead to substantially different global deformations [16, 17]. This complexity increases further when active and passive materials are distributed and oriented at the voxel scale, where the number of admissible designs grows rapidly with spatial resolution and the number of available material states [13, 18].

Finite element (FE) simulation provides a physics-based means of evaluating candidate designs prior to fabrication and remains essential for resolving coupled multiphysical responses, including thermo-mechanical, magneto-mechanical, electro-mechanical, and swelling-induced hydro-mechanical interactions in 4D-printed structures [13, 19–22]. However, repeated high-fidelity simulations become computationally prohibitive when inverse design, optimization, or behavior discovery requires the evaluation of thousands or millions of candidates [15, 17, 23]. Physical experiments are therefore indispensable for validating numerical predictions and assessing the actual response of fabricated structures, but their time and resource requirements prevent them from serving as the sole mechanism

for exploring large design spaces [10, 22].

Artificial intelligence (AI) provides a data-driven approach for addressing part of the computational complexity associated with 4D-printing design. In particular, machine-learning models can learn nonlinear relationships between material, geometric, process, and response variables from experimental or numerical data, including relationships that are difficult to formulate explicitly through analytical models [23, 24]. When trained on high-fidelity numerical data, such models can also serve as surrogate approximations of computationally expensive physical models, thereby reducing the cost of repeated evaluations during prediction, optimization, inverse design, and large-scale exploration [15, 16, 25]. This capability is particularly relevant to 4D printing, where the exploration of large combinatorial design spaces would otherwise require repeated numerical simulations or physical experiments.

However, the predictive capability of a learned model is intrinsically linked to the domain represented by its training data. Its reliability can therefore decrease when material properties, geometries, loading conditions, stimuli, or boundary conditions depart substantially from those encountered during training. Limited or heterogeneous datasets further restrict the extent to which the learned relationships can be generalized [23, 26]. Moreover, purely data-driven models do not necessarily encode the physical laws, material-compatibility conditions, manufacturing constraints, or other domain knowledge that determine whether a predicted solution is scientifically and technologically admissible. Their use in scientific design consequently requires both validation against numerical or experimental reference data and, when appropriate, the explicit incorporation of physical or domain knowledge into the learning and decision process [25, 27]. These limitations motivate approaches that complement statistical learning with explicit representations of scientific knowledge and constraints.

A complementary approach to data-driven learning is provided by symbolic knowledge representation. Symbolic AI represents concepts and their relationships explicitly through ontologies, knowledge graphs, rules, and logical constraints. An ontology provides a formal vocabulary through which materials, stimuli, processes, geometries, transformations, functions, and applications can be described consistently [28, 29]. In 4D printing, this formalization makes heterogeneous domain knowledge machine-interpretable and supports semantic querying, consistency checking, and knowledge reuse across disciplines [14]. Symbolic systems also make reasoning structures more explicit by representing the relations and constraints on which inferences are based. However, their reliability depends on the completeness and quality of the encoded knowledge. Constructing and maintaining a large domain ontology manually is labor-intensive, while predefined rules are not well suited to capturing all continuous, uncertain, and multimodal relationships contained in scientific text, images, experimental measurements, or simulated deformation fields.

The complementary limitations of neural and symbolic AI have progressively brought

these two traditions together. Neuro-symbolic artificial intelligence (NSAI) combines data-driven learning and perception with explicit knowledge representation and reasoning [30, 31]. Neural components can extract information from heterogeneous data and approximate complex physical responses, whereas symbolic components can organize domain knowledge, represent semantic relations, and impose explicit rules or consistency constraints. Their combination therefore provides a framework for coupling learning, knowledge representation, and reasoning within scientific decision-making processes. The emergence of NSAI in Gartner's 2025 artificial intelligence landscape reflects the growing interest in hybrid AI systems that combine machine-learning methods with symbolic mechanisms to improve robustness, trustworthiness, and reasoning capabilities [32].

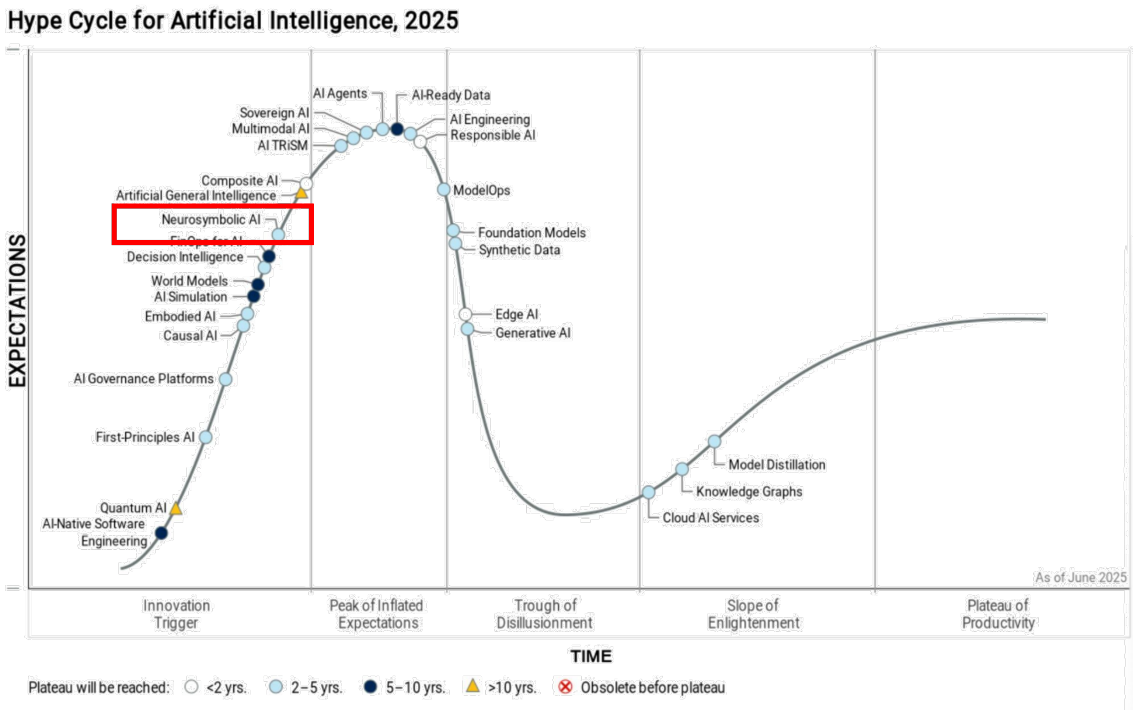

Figure 2: Positioning of neuro-symbolic artificial intelligence (AI) in the Gartner Hype Cycle for Artificial Intelligence, 2025, illustrating its emergence within a broader landscape of AI agents, knowledge graphs, foundation models, and responsible AI. Adapted from [32].

Despite substantial advances in knowledge representation, data-driven physical prediction, and inverse design, important scientific gaps remain in the range of design functions covered by these approaches and in the way domain knowledge is incorporated into computational models. Ontological approaches provide formal representations of materials, stimuli, processes, properties, and behaviors, and support the retrieval and semantic interpretation of scientific knowledge [10, 14]. However, such representations do not directly establish the quantitative relationship between a material configuration and its resulting physical response. Conversely, data-driven surrogate models can approximate computationally expensive simulations and have been increasingly used for forward prediction

and optimization in 4D-printing design [15, 17, 23]. Their potential as computational instruments for systematically exploring very large design spaces and identifying previously uncharacterized deformation behaviors remains comparatively less developed. In addition, the physical, manufacturing, and semantic knowledge used to constrain design is often treated externally to the learned predictor rather than represented explicitly within, or tightly coupled to, the predictive process. Rules, material-compatibility relations, admissible design conditions, and other structured domain knowledge therefore remain only partially connected to data-driven surrogate modeling [25, 27, 30]. Inverse-design approaches address another part of the problem by searching for configurations that satisfy prescribed responses, but they generally operate within predefined representations, objectives, and admissible design spaces [16, 17].

Taken together, these limitations reveal a broader scientific gap in how knowledge, prediction, exploration, and inverse reasoning are supported across the 4D-printing design process. The challenge is not only to accelerate the prediction of a known physical response, but also to exploit such predictive capabilities to explore large combinatorial spaces and identify relevant or previously uncharacterized behaviors. It is not only to represent scientific knowledge independently of numerical models, but also to determine how explicit domain knowledge, constraints, and symbolic representations can inform the construction, evaluation, and interpretation of data-driven predictions. Likewise, solving an inverse problem requires more than retrieving a configuration that approaches a prescribed target: the resulting candidate must remain related to its material assumptions, activation conditions, physical response, and admissibility within the considered design domain.

A further limitation arises when these functions are considered together. Knowledge acquisition, forward prediction, behavior exploration, and inverse design are frequently implemented as separate computational stages, which makes it difficult to preserve the dependencies between the scientific evidence, model assumptions, constraints, predictions, and design decisions involved in a recommendation. A scientifically grounded design process therefore requires not only specialized methods for each of these functions, but also explicit mechanisms for exchanging their intermediate results and preserving their traceability. Coordination is consequently one component of the research problem, rather than the research problem itself.

In this thesis, a *recommendation* is therefore understood as more than the retrieval of a material, process, or geometry. It denotes a scientifically grounded design proposition in which relevant knowledge can be identified and structured, candidate physical behaviors can be predicted and explored, prescribed behaviors can be translated into candidate configurations, and the evidence, assumptions, constraints, and computational results supporting the proposition remain traceable.

This problem leads to the following central research question:

How can knowledge be recommended when designing smart materials and transformable structures via 4D printing to achieve a desired property, behavior, and performance?

To address the central research question, the thesis pursues three complementary scientific objectives:

O1. Capture and structure existing scientific knowledge related to the interdisciplinary field of 4D printing.

O2. Accelerate the discovery of new knowledge in terms of material distributions and resulting shape-morphing behaviors.

O3. Provide traceable recommendations in design for 4D printing.

Together, these objectives define the scientific scope of the thesis. The manuscript is structured into two complementary levels, as summarized in Figure 3. Chapters 1 and 2 establish the state-of-the-art and conceptual foundations required to position neuro-symbolic artificial intelligence within 4D-printing design. The contributive part of the thesis is then organized according to the three scientific objectives: Chapter 3 addresses Objective 1, Chapter 4 addresses Objective 2, while Chapters 5 and 6 jointly address Objective 3.

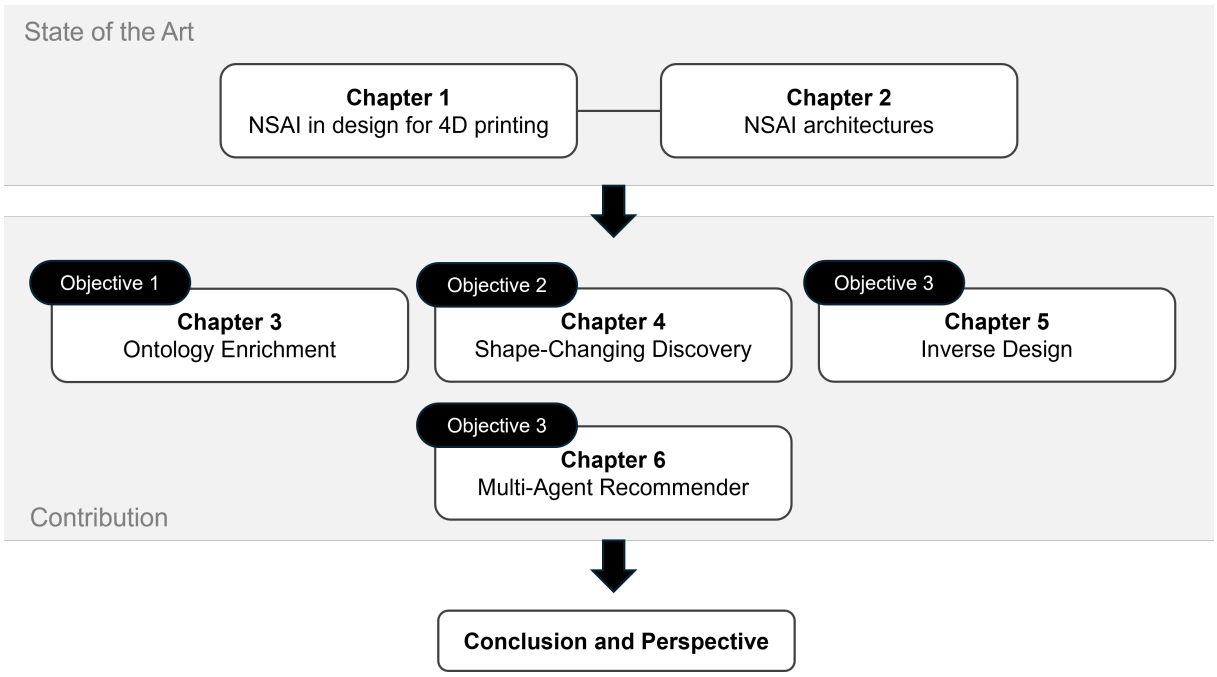

Figure 3: Synoptic representation of the thesis organization.

Chapter 1, entitled *Neuro-Symbolic Artificial Intelligence in Accelerated Design for 4D Printing: Status, Challenges, and Perspectives*, establishes the conceptual

foundation of the thesis. It examines the status of neural and symbolic methods across the 4D-printing design cycle, identifies the scientific and engineering conditions that motivate their integration, and formulates research perspectives for knowledge-informed material discovery, process selection, inverse design, simulation, and semantic knowledge integration. This chapter positions NSAI as a framework for connecting learning and reasoning across a multidisciplinary design domain, rather than as a single algorithmic approach.

Building on this foundation, Chapter 2, *Neuro-Symbolic Architectures: Formalizing, Extending, Evaluating, and Empowering Generative AI*, formalizes and extends the principal NSAI architectures and compares their capabilities using literature-grounded criteria related to generalization, scalability, data efficiency, reasoning, robustness, transferability, and interpretability. Within this qualitative evaluation, the fibring configuration, expressed as Neuro $\rightarrow$ Symbolic $\leftarrow$ Neuro, receives the broadest support across the selected criteria. Its balanced profile arises from the capacity of a shared symbolic layer to coordinate multiple specialized neural components, enforce explicit constraints, and resolve inconsistencies while preserving distributed learning capabilities.

Objective 1 focuses on the capture and structuring of existing scientific knowledge and is addressed in Chapter 3, *Structuring Scientific Knowledge for Smart Materials and 4D Printing: An Ontology Enrichment Framework Using Retrieval-Augmented Multimodal Large Language Models*. The chapter combines multimodal extraction, semantic retrieval, retrieval-augmented generation, and symbolic validation to transform heterogeneous scientific sources into ontology-aligned relations. The resulting workflow expands an initial ontology of 170 classes and a limited number of instances to more than 5 800 classes and over 12.5 million instances. Beyond increasing scale, the contribution links extracted knowledge to evidence and preserves a formal semantic structure that can support querying, consistency assessment, and knowledge-grounded recommendation.

Objective 2 addresses the accelerated discovery of new knowledge through the exploration of material distributions and their resulting shape-morphing behaviors. It is developed in Chapter 4, *Accelerating Thermo-Mechanical Simulation and Behavior Discovery in Voxelized 4D-Printed Active Structures through Automated Dataset Generation and Graph Neural Networks*. The chapter presents an automated workflow for generating voxel-level material configurations, performing FE simulations, and converting the resulting numerical data into graph representations. After training, the GraphSAGE surrogate predicts nodal thermo-mechanical displacements in approximately 0.0269 s per structure. By comparison, the complete FE workflow, including model generation and numerical solution, requires approximately 3,600–10,800 s per structure. This corresponds to a per-structure speedup ranging from approximately 1.34×10^5 to 4.01×10^5. The resulting reduction in computational cost enables the screening of 800,000 candidate structures under four boundary-condition scenarios. The chapter therefore contributes an accelerated

forward-prediction workflow and a discovery method based on physics-based deformation descriptors for identifying potentially novel coupled shape-morphing responses.

Objective 3, which concerns the provision of traceable recommendations for 4D-printing design, is addressed through two complementary contributions developed in Chapters 5 and 6. The first contribution addresses target-driven design recommendation through inverse design. Chapter 5, *Multidimensional Inverse Design of Multi-Material 4D-Printed Structures via Graph-Based Learning*, reverses the predictive problem. It develops a deformation-aware graph representation and a variational latent space for retrieving and refining candidate structures from prescribed target shapes across one-, two-, and three-dimensional design domains, with the aim of evaluating the framework across increasing dimensional complexity and supporting its applicability to a broader range of 4D-printing applications. Seven search and optimization strategies are compared under a common shape-based objective. The combination of top-k candidate search and local material mutation provides the strongest overall compromise, obtaining the lowest final target-shape error for 112 of the 153 benchmark targets. This contribution connects a prescribed deformation to a candidate material distribution while retaining the rapid forward evaluation required for iterative design.

The second contribution toward Objective 3 is presented in Chapter 6, *A Multi-Agent System for Scientific Reasoning, Design, and Discovery in 4D Printing*. This chapter extends the recommendation process to the system level by integrating the preceding knowledge, prediction, discovery, and inverse-design capabilities through a central orchestrator while preserving the traceability of their intermediate results. Four specialized agents provide capabilities for scientific knowledge processing, material inverse design, thermo-mechanical simulation, and behavior discovery. Rather than using a fixed sequence, the orchestrator constructs a request-specific execution plan, manages dependencies, and records how the output of one operation becomes the input of another. Five representative workflows demonstrate cooperation across literature-based material selection, deployable-structure inverse design, knowledge-guided simulation, hypothesis verification, and behavior discovery. This system-level contribution builds on the architectural principles examined earlier in the thesis by coordinating specialized computational capabilities within a unified orchestration framework.

Together, these contributions establish a progression from the capture and structuring of existing scientific knowledge (Objective 1), to the accelerated discovery of new knowledge through large-scale prediction and exploration (Objective 2), and finally to the provision of recommendations through inverse design and their integration within a traceable multi-agent framework (Objective 3). This progression provides the scientific basis for addressing the central research question. The proposed framework is not intended to replace physical theory, FE verification, experimental validation, or expert judgment. Rather, it aims to connect these resources more effectively, reduce the computational

and cognitive cost of exploring complex design spaces, and ensure that a recommended 4D-printing solution remains associated with the evidence, assumptions, predictions, and constraints that support it.

This doctoral research was conducted within the ARTEMIS project (ANR-22-PEXD-0016), coordinated by Prof. Frédéric Demoly within the PEPR DIADEM programme [33]. ARTEMIS provides the broader scientific framework in which the contributions of this thesis were developed. As illustrated in Figure 4, the project combines heterogeneous sources of scientific information, including numerical simulations, physical experiments, published literature, and online databases, with materials and design informatics and machine-learning approaches. This integration aims to support the exploration and discovery of both smart materials and 4D-printed structures, encompassing material selection and distribution, interfaces, metamaterials, and functional behaviors. Within this framework, the present thesis particularly contributes to the computational-intelligence dimension by developing neuro-symbolic approaches for knowledge engineering, accelerated simulation, inverse design, reasoning, and automated discovery.

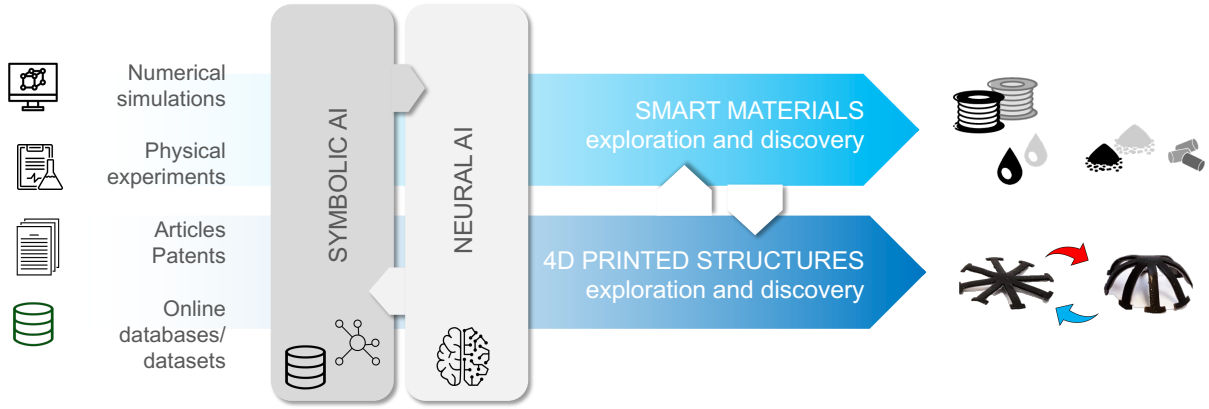

Figure 4: Scientific framework of the ARTEMIS project, integrating heterogeneous scientific data, materials and design informatics, and machine learning to support the exploration and discovery of smart materials and 4D-printed structures.

Chapter 1

Neuro-Symbolic Artificial Intelligence in Accelerated Design for 4D Printing: Status, Challenges, and Perspectives

1.1 Introduction

Current research in 4D printing focuses on enhancing the printability, stimuli sensitivity, functional fatigue resistance, and mechanical performance of SMs [34]. However, despite these advancements, significant challenges remain in predicting and controlling the behavior of stimuli-responsive materials, especially when targeting complex properties and functionalities [9, 22]. Achieving precise control over material transformations is essential to ensure that 4D-printed structures perform as intended in real-world applications. Another critical challenge lies in the development of multi-material structures, where materials with diverse mechanical, thermal, and chemical properties must bond effectively and work cohesively under stimulation [10]. To address these challenges, it is necessary to leverage computational intelligence that account for the dynamic, time-dependent interactions between active and passive materials. Indeed, the development of computational intelligence methods can accelerate the exploration and discovery at both the material and structure design levels, while improving their printability and functional performance over time.

Advanced methodologies such as inverse design have emerged, allowing mechanical engineers and designers to work backward from target material properties, shapes, or structural behaviors in order to determine the optimal design architecture and material composition [13, 15, 16]. This enables the creation of tailored materials and structures that

can meet specific performance criteria. Machine learning (ML) has become indispensable in inverse design as it offers the capacity to model complex datasets and predict material behavior, thus accelerating the discovery and optimization cycles [24]. By transforming raw data into actionable insights, ML aids in decision-making for material selection and design. However, traditional ML techniques often required manual feature extraction, which is both complicated and time-consuming. As such, the rise of deep learning (DL) has automated feature extraction using neural networks (NNs), leading to major breakthroughs in various fields such as computer vision, natural language processing (NLP), and autonomous driving, where large datasets need to be processed efficiently [35].

Deep Learning (DL) fundamentally relies on NNs, which consist of interconnected layers of artificial neurons that process and transmit information. As inputs are propagated through the layers, the network learns complex patterns and improve its predictive accuracy over time. However, while NNs are highly effective at pattern recognition, they encounter challenges in tasks that require logical reasoning, explainability, and interpretability, which are essential for high-stakes design and engineering applications [26]. To address these limitations, symbolic AI presents a complementary approach via a structured, rule-based reasoning process. By combining these two AI methods, neuro-symbolic AI (NSAI) has emerged as a promising paradigm which considers the dual-process theories of human cognition [36–38]. In these theories, the fast and intuitive processing is represented by sub-symbolic NNs, while deliberate, logical thinking aligns with symbolic reasoning systems. NSAI integrates these complementary modes of processing, which enables systems to learn from experience while reasoning based on structured knowledge, thus bridging the gap between intuitive learning and explicit logical reasoning [27, 30, 39–43]. By combining the learning capabilities of NNs with the logical structure of symbolic reasoning, NSAI creates a hybrid architecture that not only improves decision-making efficiency but also enhances the explainability of results. In the context of design for 4D printing, NSAI allows for the generation of explainable recommendations by leveraging symbolic reasoning to break down complex material interactions into understandable, rule-based decisions. This makes it easier to trace and understand the reasoning behind specific design decisions related to active material composition, formulation, selection, and distribution in the 2D/3D design space.

Therefore, this paper aims to explore new opportunities in design for 4D printing and guide future research focused on enhancing the properties of active materials and performance of structures to unlock real-world applications. Section 2 reminds the main challenges in design for 4D printing in terms of printable active materials and inverse design methods. Section 3 studies the potential of NSAI as a transformative tool for

addressing these challenges. It highlights the benefits of combining ML with symbolic reasoning for enhanced explainability and interpretability. Then, Section 4 outlines critical future research directions needed to accelerate the development of next-generation SMs and transformable structures as well as the discovery of latent applications.

1.2 Status and Challenges in 4D Printing Design

1.2.1 Printable and Stimuli-Responsive Materials

SMs play a pivotal role in advancing 4D printing as they allow structures to dynamically respond to stimuli. Despite their potential, these materials still face limitations that hinder their widespread application [10]. For instance, SMPs are valued for their lightweight nature, cost-effectiveness, and shape recovery capabilities. However, the mechanical training requirements and lack of reversibility make them less suitable for applications requiring repeatable actuation [44–49]. LCEs, known for their large deformation capabilities and rapid, reversible actuation, are well-suited for dynamic and repeatable applications. Nonetheless, they encounter challenges with high-resolution printing and exhibit functional fatigue over multiple actuation cycles [50–52]. Hydrogels, widely used in biomedical contexts, can effectively interact with biological fluids, yet they suffer from low mechanical strength, long response, and recovery times although reversible. Their tendency to dry out in non-aqueous environments further limits their reliability in diverse conditions [53–55]. Last, SMAs, such as nickel-titanium alloys, provide robust, reversible actuation upon heating. However, their brittleness, long response times, and nickel-related toxicity concerns complicate their use, especially for biomedical applications [56–58].

These limitations require further effort in material composition, formulation, processing, and the creation of composites that blend the strengths of different SMs to expand their capabilities and applications across diverse fields. In addition to this, researchers in materials science are increasingly focused on discovering new SM compositions that enhance their printability, sensitivity to stimuli, functional durability, and mechanical performance. Advancing these materials is crucial for improving the adaptability and efficiency of 4D-printed structures, enabling faster and more reliable responses to environmental changes. A key challenge is overcoming functional fatigue, which currently restricts the reliability of many SMs in applications requiring repeated or sustained use [9, 10, 12]. Developing the next-generation of printable SMs is then essential for ensuring the adoption of 4D printing in the industry. This effort involves engineering materials that exhibit faster actuation, greater mechanical strength, and enhanced responsiveness while also addressing critical concerns like biocompatibility and reducing toxicity risks,

especially in medical and sensitive environments. As researchers push the boundaries of materials science and AM, the potential of SMs in fields like healthcare, space, automotive, and soft robotics continues to grow, opening new doors for innovative and adaptive solutions.

However, the development of SMs is intrinsically tied to their responsiveness to stimuli. The stimuli sensitivity plays a critical role in optimizing energy efficiency and ensuring precise control over material behavior. Each stimulus interacts with SMs in unique ways, enabling transformations in shape, function, or physical properties, and making them versatile to a variety of applications. For example, temperature can be used to precisely control deformations in materials such as SMPs, LCEs, hydrogels, and SMAs, allowing them to shape change in a predictable manner. Light, particularly ultraviolet (UV) and near-infrared (NIR), can rapidly trigger localized changes in material properties, making them ideal for high-resolution applications. Similarly, water and pH levels influence hydrogels that can expand or contract, which is especially useful in the biomedical field for drug delivery or tissue engineering applications. Electric fields, on the other hand, provide fast and precise control over material transformations, making them suitable for the development of artificial muscles. Meanwhile, magnetic fields offer the advantage of non-invasive, remote control, which is particularly effective in soft robotics and medical devices where safety and precision are paramount [21, 53, 59].

Beyond formulating and printing single active materials, researchers are exploring the integration of multiple materials within a single design to leverage the complementary properties of the involved materials [60]. These advancements highlight the growing need to optimize material distribution within structures in order to achieve sophisticated shapes and actuation performance. This involves more than simply combining active and passive materials, but also requires a deep understanding of how different materials interact via stimuli and how their arrangement influences the overall behavior of the structure. By carefully tuning material placement, it is possible to fabricate objects that not only respond dynamically to stimuli but also meet specific mechanical and transformation requirements, such as strength, flexibility, and reversibility [48]. To achieve targeted shape changes, advanced computational design approaches are essential to predict the right material at the right place in the 2D/3D structure. This strategy opens up exciting possibilities for the discovery of new digital material distributions, which could expand the capabilities of SMs.

1.2.2 AI-Enabled Inverse Design

Through inverse design, material composition and distribution, printing parameters, and desired properties can be systematically determined to achieve precise target shapes and

performance. In the field of materials science through materials by design, significant progress has been made in utilizing ML for predicting and optimizing the mechanical properties of composite and heterogeneous materials. For example, convolutional neural networks (CNNs) have been used to predict the effective modulus of these composites, allowing for rapid design iterations [61, 62]. Regarding strength and toughness, Gu et al. [63] and Yu et al. [64] developed ML classification and regression models that optimize the mechanical performance of composites under shearing. Hierarchical design approaches have been developed to consider microstructures as building blocks (pixels) to predict mechanical properties [65].

Hybrid methods that integrate ML with evolutionary algorithm (EA) have proven successful in optimizing shape-change responses in 4D-printed active composite structures, offering faster and more efficient inverse design solutions [23]. For instance, finite element (FE) simulations combined with EA have been used to explore the vast design space of active composite beams with voxel-level material distributions, leading to significant advancements in shape optimization [13, 13, 66]. While traditional EA approaches are computationally expensive, ML models, such as ResNet and inverse NN, have been integrated to drastically reduce computational costs, enabling quicker and more accurate predictions of shape-change responses [15, 16, 19] (**Figure 1.1a,b**). These models have demonstrated remarkable success in handling complex, high-dimensional inverse design problems, such as buckling mesosurfaces and inflatable composite membranes [67, 68]. Athinarayanarao et al. [18] extended the use of voxel-based approaches by incorporating void as a third material in EA to optimize the use of active material in 3D active composite structures (**Figure 1.1e**). As for stress and strain fields prediction, conditional generative adversarial networks (GANs) have been employed in materials with pixel-level material distribution. For example, Yang et al. [69] demonstrated how conditional GANs could predict stress and strain fields in hierarchical composites, handling varying component shapes, boundary conditions, and material hierarchies. These models have significantly improved the generalization and robustness of designs under various conditions, so as to enable more efficient and reliable optimization of composite structures.

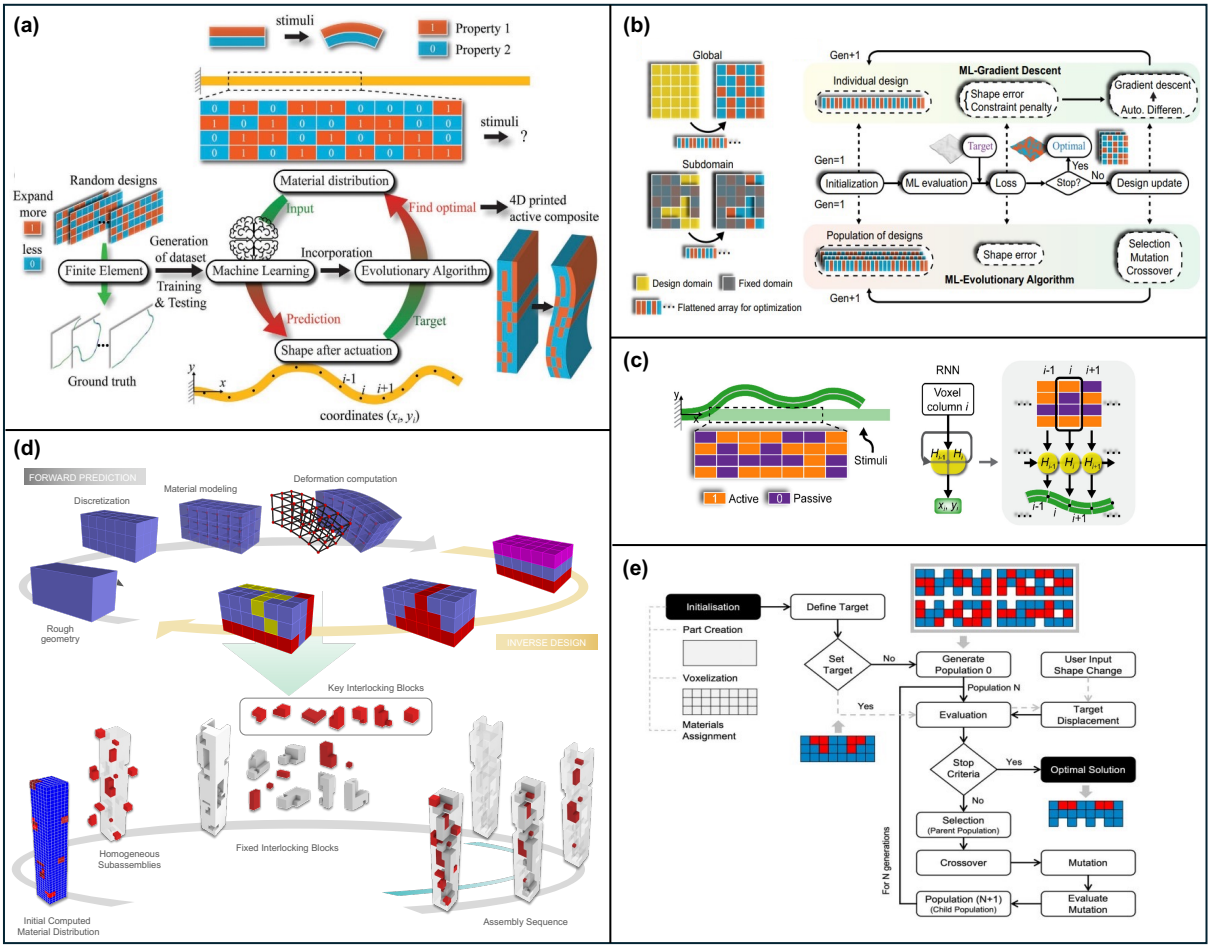

Figure 1.1: Inverse design and optimization architectures of 4D-printed multi-material structures. (a) Design of 4D-printed active composite beam using ML-based prediction and EA-based optimization [16] (Reproduced with permission from Wiley). (b) Inverse design strategies based on ML and EA [15]. (c) Active composite determination using RNNs [17] (Reproduced with permission from Elsevier). (d) Computational design for building multi-material structures using interlocking blocks and voxel-based modeling [70] (Reproduced with permission from Elsevier). (e) EA-driven topology optimization of active composite structures [18].

Furthermore, physics-informed neural networks (PINNs), introduced by Raissi et al. [25], have been applied to inverse problems like material characterization, where they enable the identification of heterogeneous material properties based on observed deformation data. This innovative approach integrates physical laws directly into the ML models in order to improve precision in material property identification. Such integration not only reduces the need for extensive labeled data but also enhances the overall efficiency and accuracy of predictions. Mechanical metamaterials have seen the application of deep generative models such as GANs and variational autoencoders (VAEs) to design architectures with tailored anisotropic elastic properties. These models allow for the rapid exploration of new microstructural designs with enhanced stress-strain responses, then facilitating the discovery of novel metamaterials with target mechanical properties

[71–77]. Despite these research efforts, the resulting material distributions may introduce printability issues, especially when dissimilar materials are employed. To overcome this barrier, Benyahia et al. [70] developed a computational design approach that converts any voxel-based distribution achieving a desired shape change into interlocking blocks suitable for manual or robotic assembly (**Figure 1.1d**). Adding to these advances, Jin et al. [78] proposed a hybrid optimization framework for ultrafast inverse design of 4D-printed active composite structures. Sun et al. [17] used recurrent neural networks (RNNs) for the design of active composite structures, enabling the precise prediction of material distribution that react dynamically to environmental stimuli (**Figure 1.1c**).

Recent efforts have also explored ML applications in soft robotics, where sustainable robotic joints with variable stiffness have been developed using reinforcement learning to predict and optimize the behavior of 4D-printed structures. This enabled the precise actuation and control of soft robotic systems [79]. ML was employed in the design of dielectric elastomer to fabricate actuators that can respond to electric field in order to enhance the potential for human-machine interaction [20]. Soft pneumatic actuators demonstrated that ML can significantly enhance the precision of shape changes through time-efficient predictions compared with numerical simulations [80]. Additionally, AI-based control systems have been integrated with 4D printing processes to address the nonlinear behavior of active materials, improving the adaptability and precision of actuators and paving the way for more autonomous and intelligent structures [22].

These initiatives highlight the growing importance of ML in advancing 4D printing by accelerating material discovery, improving design and optimization processes, and enhancing the performance of 4D-printed structures across various applications. However, to fully harness the potential of ML, especially when working with small datasets or when detailed explanations are required in quantitative research, integrating symbolic AI becomes crucial. This is particularly relevant for the emerging field of 4D printing. Symbolic AI complements data-driven approaches by structuring background knowledge and providing more transparent and interpretable results. This integration ensures that ML-driven discoveries in 4D printing are not only accurate but also easily validated, making them more applicable to real-world scenarios.

Complementary research efforts have been made on symbolic AI for 4D printing. Dimassi et al. [14] have developed an ontology for structuring and formalizing knowledge related to stimuli, SMs, product design, and AM. By leveraging this domain ontology, researchers can ensure that the captured knowledge is not only accessible but also interoperable with other domains, as advocated by the Industry Ontology Foundry (IOF) initiative [81]. Ontologies also serve as a foundation for the development of reasoning procedures, offering qualitative decision support in material and process selection, design

recommendations, and the alignment of 4D printing practices with emerging standards. By enabling real-time updates and reasoning over structured knowledge, ontologies can bridge the gap between symbolic AI and neural AI approaches, providing a more holistic and context-aware design strategy [82] [83]. Effectively integrating this structured knowledge into a broader design and material informatics frameworks will be key to the future of 4D printing.

As the complexity of 4D printing continues to increase, the need for explainable and traceable recommendations becomes ever more critical. Transparent decision-making tools are vital, especially in fields such as materials science, engineering, design, and manufacturing. By combining the data-driven capabilities of ML with the logical reasoning of symbolic AI, it is possible to create more transparent and scientifically grounded design insights. This fusion enables more efficient, reliable, and scalable 4D printing processes, fostering innovations in multiple industrial sectors.

1.3 The Rise of Neuro-Symbolic AI

NNs have shown remarkable proficiency in processing unstructured data such as images, audio, and text. Their ability to learn complex patterns and representations from vast datasets has led to significant breakthroughs in various fields, including computer vision, speech recognition, and natural language processing (NLP) [84, 85]. One of the key strengths of NNs lies in their adaptability and capacity to learn from raw data without requiring predefined rules or expert knowledge. This makes them highly scalable and effective in applications that involve large amounts of unstructured data. However, despite these strengths, NNs have several notable weaknesses. One of the most significant issues is their lack of transparency. Neural models are difficult to interpret, making it challenging to understand how they arrive at specific decisions or predictions. This opacity poses problems for critical applications where explainability is crucial, such as healthcare, finance, legal systems, or engineering. Additionally, NNs are highly data-hungry, requiring vast amounts of labeled training data to perform effectively. This reliance on large datasets makes them inefficient in scenarios where data is limited or costly to acquire. Furthermore, neural models struggle with reasoning and generalizing beyond their training data, which leads to poor performance in tasks requiring logical inference or commonsense reasoning. Specifically, tasks such as understanding causal relationships, solving problems involving sequential logic, and making decisions that rely on external world knowledge (**Table 1.1**).

Table 1.1: Comparison between symbolic AI and neural AI.

Aspect	**Symbolic AI**	**Neural AI**
Generalization	Strong at handling abstract concepts and generalizing beyond training data (out-of-distribution reasoning)	Weak in generalizing beyond training data, primarily excels within the dataset it has seen (overfitting potential)
Data handling	Struggles with unstructured data (e.g., images, raw text)	Excels at handling unstructured data (e.g., visual, auditory, textual data)
Learning speed	Slow learning, requires pre-defined rules	Fast learning from large datasets, can learn patterns without explicit rules
Interpretability	Highly interpretable (decisions can be traced to explicit rules and logic)	Lacks interpretability (black-box models, hard to understand the decision-making process)
Robustness	Weak robustness, brittle in noisy environments or with incomplete data	Highly robust, adaptable to noisy and varied data environments
Scalability and optimization	Limited scalability, struggles with large-scale, complex datasets	Highly scalable, efficient in large-scale data processing and optimization
Reasoning and explainability	Strong reasoning capabilities, excels in logical reasoning and providing explanations	Weak reasoning and explainability, struggles to provide transparent logic behind decisions
Data requirements	Requires less data, relies more on rules and knowledge bases	Requires large labeled datasets for training, data-intensive
Uncertainty and inference	Struggles with uncertainty, less flexible in probabilistic inference	Strong at learning and inference under uncertainty, better at probabilistic reasoning

Symbolic AI excels in areas where NNs fall short. Symbolic systems are based on explicit rules and structured representations, which enable them to perform complex reasoning tasks, such as mathematical proofs, planning, and expert systems. The transparency of symbolic AI is one of its greatest strengths. Since symbolic methods rely on predefined rules and logical structures, their decision-making processes are easily

interpretable and explainable. However, symbolic AI has its own limitations. One of the major drawbacks of symbolic approaches is their rigidity and difficulty in adapting to new scenarios. Symbolic systems require manually crafted rules and structured input data, making them difficult to scale in real-world environments where data is often noisy, incomplete, or unstructured. They are also vulnerable to combinatorial explosions when dealing with large datasets or complex reasoning tasks, which significantly hampers their performance at scale. Moreover, symbolic systems are not well-suited for tasks involving perception, such as image or speech recognition, as they cannot autonomously extract knowledge from unprocessed data (**Table 1.1**).

1.3.1 Advantages of NSAI

While traditional NNs are great at recognizing patterns in data, they often struggle when faced with situations they have never seen before. On the other hand, symbolic reasoning provides a logical framework for decision-making but is constrained in its ability to acquire knowledge from new data and adapt dynamically. By bringing these two approaches together, NSAI manages to fill the gaps and create a more adaptable, interpretable, and efficient AI system (**Figure 1.2**).

One of the standout features of NSAI is its ability to generalize beyond its training data. Traditional AI models often falter when encountering something new, but NSAI, with its blend of learning and logic, can handle these situations better. This quality is essential in real-world applications such as autonomous vehicles or healthcare, where systems must reliably function even in unpredictable environments. In addition, NSAI can offer better interpretability of the decision-making process in designing for 4D printing by providing insights into how and why specific design decisions are made, making this approach both more trustworthy and valuable. Another significant benefit of NSAI is its reduced need for large datasets. Traditional AI-based systems often require massive amounts of data to perform well, which can be costly in terms of both time and resources. However, NSAI can achieve high performance with far less data, thanks to its ability to reason symbolically. This makes it a more sustainable and practical option, especially for smaller organizations or emerging research field working with limited resources. In addition to this data efficiency, NSAI models also show remarkable transferability, meaning they can take what they have learned in one area and apply it to another with minimal retraining. This is incredibly useful in situations where there is not much data available for a new

task.

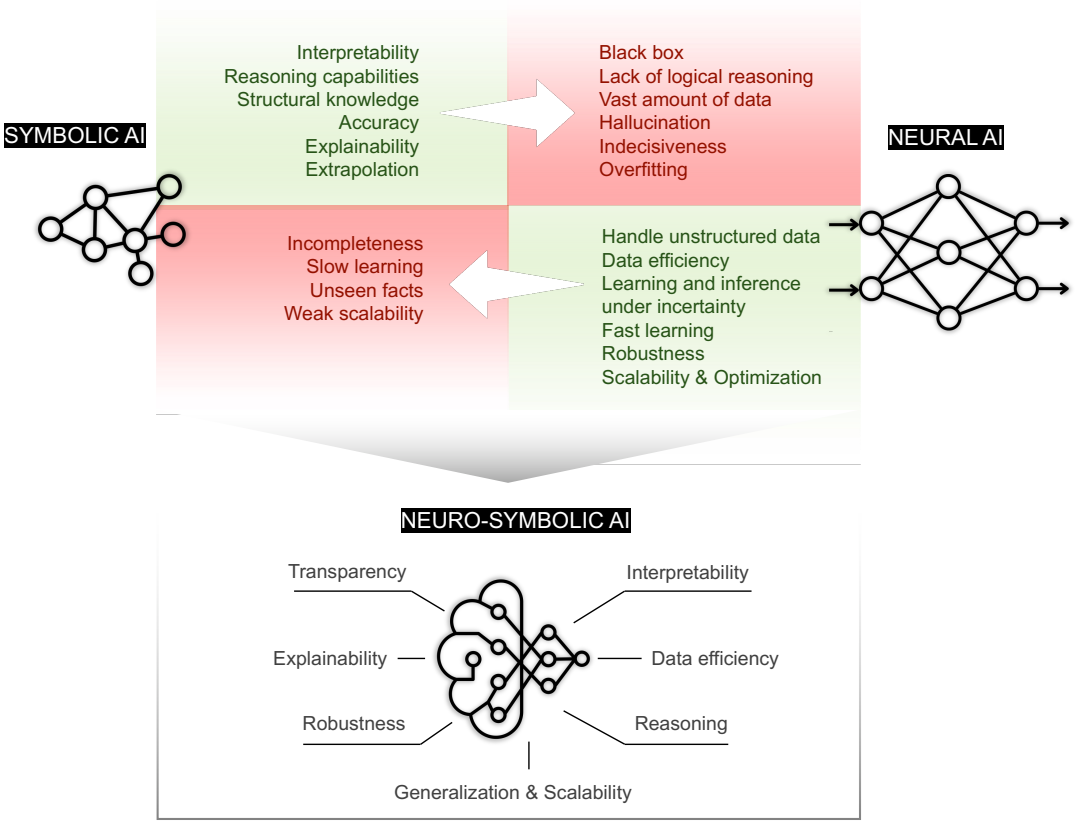

Figure 1.2: Neuro-symbolic AI as a combination of symbolic AI and neural AI strengths (illustrated in green parts), reproduced from [11].

What really sets NSAI apart is its enhanced reasoning capabilities. While NNs are proficient in detecting correlations and underlying structures within datasets, they often fall short in tasks requiring logical reasoning or problem-solving. NSAI bridges this gap by handling more complex challenges, whether it is diagnosing a medical condition, analyzing legal cases, or making scientific discoveries. Additionally, NSAI is highly scalable and adaptable, making it capable of tackling increasingly complex problems across various fields. Its ability to combine unstructured data processing with structured logic means it can continue to grow and evolve as AI technologies advance, positioning NSAI as a key player in the future of AI-based systems.

1.3.2 Taxonomy of NSAI

The taxonomy of NSAI presents the main elements that combine NNs and symbolic reasoning to build a hybrid approach. It shows how neural components are designed to learn from large datasets, while symbolic components offer logical frameworks that support decision-making and explainability. By categorizing this AI paradigm, researchers

and practitioners can understand how these two pillar systems work together in a way to address real-world applications.

Neural Component The neural component serves as the foundation for data-driven learning that is crucial for tasks like vision, language processing, and reasoning. Numerous neural architectures have been investigated and developed to advance the capabilities of NSAI, as illustrated in Figure 1.3. NNs, such as CNNs, are important in handling grid-like data, particularly images, by learning local patterns through convolutional layers. These CNNs have been fundamental in revolutionizing fields like computer vision, enabling tasks such as image recognition and object detection with their hierarchical feature extraction.

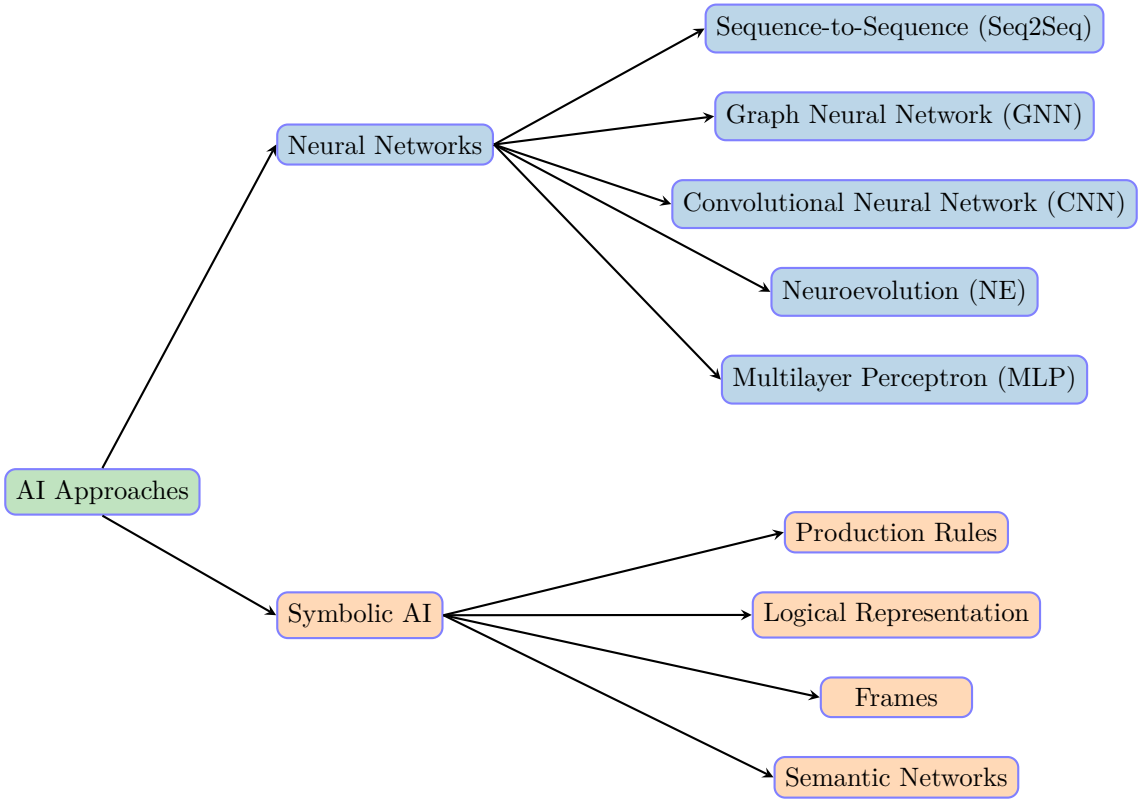

Figure 1.3: Existing neural and symbolic AI architectures, reproduced from [11].

Another important architecture is the multilayer perceptron (MLP), one of the earliest and simplest NNs. MLPs are built with fully connected layers and are effective for classification and regression tasks, especially when the relationships between input and output are straightforward. Although more complex models have emerged, MLPs remain valuable in simpler tasks due to their efficiency and ease of use. For graph-structured data, graph neural networks (GNNs) excel by processing relationships between nodes, allowing for advanced reasoning in areas such as social networks, molecular chemistry, and recommendation systems. GNNs leverage node connections to perform neighborhood aggregation in order to help the network learning meaningful patterns from graph-based

inputs. In tasks where sequences are involved, Sequence-to-Sequence (Seq2Seq) models have made significant strides [86]. These models manage both input and output sequences using an encoder-decoder structure in tasks such as language translation and time-series analysis. Many improvements have been made to enhance their accuracy such as attention mechanisms [85]. Transformers, a category of seq2seq models, have revolutionized NLP by using a self-attention mechanism that allows for parallel data processing. Based on the transformer architecture, models, such as BERT [84], RoBERTa [87], and T5 [88], have achieved promising results in question answering, text generation, and machine translation. Larger versions of these models, such as ChatGPT [89] and PaLM [90], have expanded the possibilities of AI, therefore showing potential for applications like education and personalized recommendations, and hinting at their future role in artificial general intelligence.

More recently, while large language models (LLMs) have demonstrated impressive performance in various tasks. Their ability to generate reliable factual knowledge remains a concern. They tend to memorize facts from the training data but struggle to recall them accurately, which often lead to hallucinations with incorrect statements [91–93]. For example, an LLM might claim that "Einstein discovered gravity in 1687", which is not true as Isaac Newton formulated gravitational theory. This undermines the trustworthiness of LLMs. Moreover, seen as black box models, LLMs are difficult to interpret due to their implicit representation of knowledge within their parameters [94]. This lack of interpretability, combined with their probabilistic reasoning processes, makes their predictions and decisions difficult to validate or explain [95]. Even when LLMs provide explanations using techniques such as chain-of-thought reasoning [96], these explanations can still suffer from hallucination issues [97]. This severely limits the application of LLMs in high-stakes engineering applications like 4D printing, where inaccurate reasoning could have significant consequences on the multiple inherent aspects of this technology. Additionally, training LLMs on broad datasets may face difficulties in effectively generalizing to specialized fields or integrating new knowledge, as they often lack domain-specific information [89]. LLMs are based on the transformer architecture, which utilizes an attention mechanism. The latter enables the model to focus on the most important parts of the text during processing. Transformers consist of two main components: the encoder and the decoder. In the context of language models, the decoder often plays a central role.

In an encoder, the input text – such as a question or sentence – is first tokenized. The tokenization breaks the text into smaller units called "tokens", which can be words or sub-words. Each token is then converted into a numerical representation, called an embedding, using an embedding model. These embeddings are vectors in a

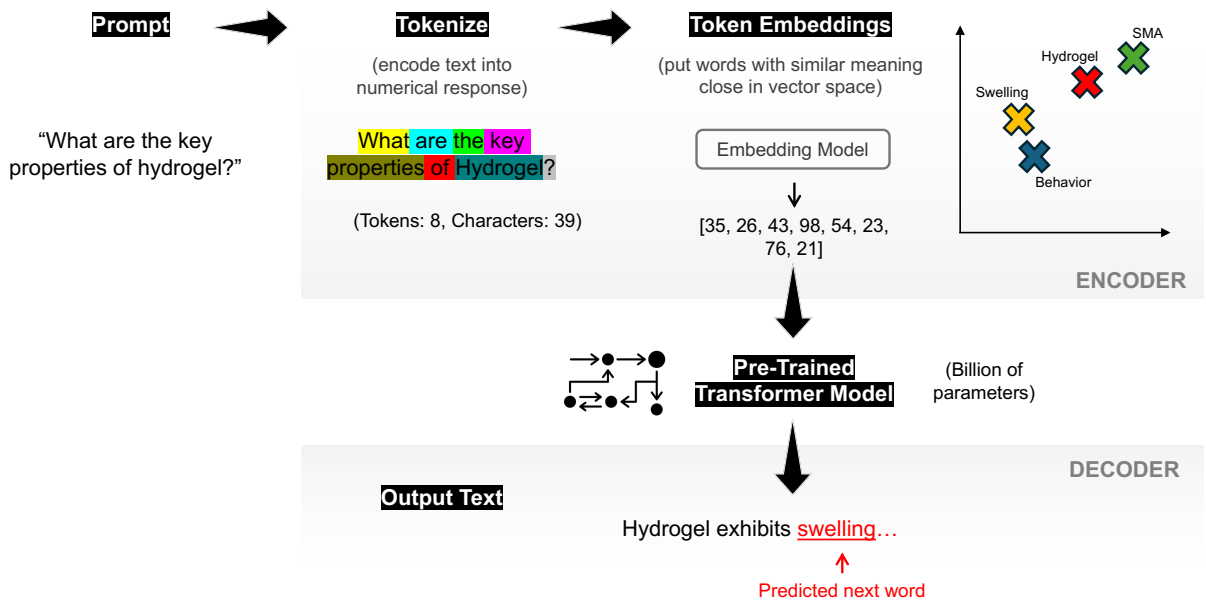

Figure 1.4: Process flow of a pre-trained transformer model for text generation in 4D printing, reproduced from [11].

multidimensional space, where tokens with similar meanings are positioned close to each other, capturing the semantic relationships between them. For instance, the prompt "What are the key properties of hydrogel?" can be converted into a sequence of numerical tokens. These tokens are then represented in a vector space where words like "hydrogel" and "swelling" are placed near each other due to their contextual relationship. Once the embeddings are generated, they are fed into the pre-trained transformer model. Using its attention mechanism, the model focuses on the relationships between different tokens while considering the overall context of the sentence. This allows it to understand not only the individual meanings of the words but also their interactions within the sentence. One of the strengths of the transformer architecture is its ability to capture long-range dependencies in text. This means the model can grasp how words that are far apart in a sentence influence each other, contributing to a deeper understanding of the sentence's meaning. In the decoding phase, the decoder generates the output text one token at a time. Based on the context provided by the encoder and previous predictions, the decoder predicts the next word in the sequence. For instance, after "Hydrogel exhibits", the model might predict the word "swelling", demonstrating its understanding of the contextual relationship between "Hydrogel" and "swelling" (see Figure 1.4).

Recent years have witnessed significant advancements in the field of multimodal large language models (MLLMs), driven by the complementary strengths of LLMs and large vision models (LVMs). MLLMs bridge the gap between text-based reasoning and visual understanding, enabling models to process, reason, and generate content across multiple media, including text, images, videos, and audio [98, 99]. These models leverage the

vast pre-training of LLMs, incorporating visual information to enhance their reasoning capabilities, thus overcoming the inherent limitations of conventional LLM models. MLLMs exhibit two distinct traits that set them apart from traditional models: (i) they are built on LLMs with billions of parameters, which offer unprecedented generalization and scalability [100], and (ii) they employ novel training paradigms such as multimodal instruction tuning [98, 99]. This combination empowers MLLMs with unique capabilities, such as generating website code from images [100], interpreting complex memes [101], and performing optical character recognition (OCR)-free mathematical reasoning [102]. The launch of GPT-4 marked a pivotal moment, which showcased the potential of MLLMs and increased research and development efforts from both academia and industry [103].

The progress of MLLMs has been rapid and multifaceted towards applications across various domains. These include enhanced support for fine-grained control in user prompts [104], better handling of diverse input and output modalities [105, 106], and extension to multilingual contexts [107, 108]. Additionally, MLLMs are being adapted for specific applications, such as medical image analysis [109, 110] and document parsing [111, 112]. As research in this field continues to evolve, MLLMs are poised to play a crucial role in the future of AI, particularly in scenarios requiring complex multimodal reasoning and interaction. Building upon MLLMs, the integration of retrieval-augmented generation (RAG) has emerged as a crucial approach to address several inherent limitations of LLMs, such as hallucinations and knowledge obsolescence. This particularly relevant in the rapidly evolving field of 4D printing, where the volume of new research and published knowledge is increasing at an unprecedented pace. RAG enhances LLMs by incorporating relevant external knowledge, retrieved based on semantic similarity, into the generation process. This significantly improves the factual accuracy and reliability of the outputs [113], especially in knowledge-intensive tasks like designing for 4D printing. RAG operates by retrieving pertinent sections of documents from an external knowledge base and incorporating them into the LLM's response generation. This retrieval process mitigates the risk of generating factually incorrect or outdated content [114]. The development of RAG has led to three main paradigms, namely naive RAG, advanced RAG, and modular RAG, each offering progressive improvements in retrieval and generation capabilities. Naive RAG relies on a basic retrieve-read framework, while advanced RAG incorporates sophisticated pre- and post-retrieval strategies to enhance context relevance. Modular RAG, the most flexible approach, introduces additional specific components that further refine retrieval accuracy and content generation [112]. The integration of RAG into MLLMs addresses the "blindness" of LLMs to non-textual data by incorporating multimodal retrieval capabilities, thus enabling more informed and contextually accurate responses. This synergy between MLLMs and RAG opens up new avenues for high-stakes applications, like those involved in design and 4D printing, where accuracy and factual

correctness are paramount [115].

Moreover, neuroevolution is another neural component that represents a pivotal approach in optimizing NNs by leveraging EAs rather than traditional gradient-based methods like backpropagation. This approach relies on genetic algorithms to evolve the topologies, weights, or ensembles of NNs across multiple generations. Neuroevolution is particularly effective in dynamic environments, such as those encountered in reinforcement learning, where predefined architectures may struggle to adapt effectively. By evolving neural structures over time, neuroevolutionary techniques can discover and refine architectures better suited to specific tasks, making them especially valuable in fields like robotics and autonomous systems such as those enabled by 4D printing [116].

Symbolic Component In NSAI, the symbolic component is critical for reasoning and decision-making. It complements NNs by structuring knowledge in a way that allows logical inference and interpretability. This symbolic reasoning can be categorized into four key knowledge representation frameworks: production rules, logical representation, frames, and semantic networks. Production rules are fundamental in rule-based systems, where the knowledge is represented as "If-Then" statements that guide decision-making. Each rule consists of an antecedent (the conditions) and a consequent (the actions to be taken if the conditions are met) [117]. In addition, logical representation provides a framework for reasoning using formal logic systems. It involves defining syntax (symbols and sentence structure), semantics (meaning and truth values), and an inference procedure (rules for deriving new truths). In 4D printing, logical representations can be used to model complex relationships between stimuli and material responses. For instance, we can suppose the following predicates in Table 2.

Table 1.2: Predicates used to describe logical rules in 4D printing.

Predicate	**Description**
$A(x)$	"x is an Active Material"
$P(x)$	"x is a Passive Material"
$S(s)$	"s is a Stimulus"
$App(s, x)$	"Stimulus s is applied to Material x"
$I(x, y)$	"Material x interacts with Material y"
$D(x)$	"Material x deforms"
$Time(t)$	"Time t"
$C(x)$	"Material x contracts"
$E(x)$	"Material x expands"
$FinalState(x, t)$	"Material x reaches its final state at time t"

From this predicate logic, several rules can be defined to describe material response to stimuli (Equation 1), active and passive material interaction (Equation 2), and time-dependent behavior (Equation 3) for instance.

$$\forall x, \exists s \; A(x) \wedge S(s) \rightarrow \; C(x) \tag{1.1}$$

$$\forall x, y, \exists s \; A(x) \wedge P(y) \wedge I(x, y) \wedge S(s) \wedge App(s, x) \wedge App(s, y) \rightarrow D(x) \wedge D(y) \tag{1.2}$$

$$\forall x, \exists s, \exists t'' > t' > t \;\; A(x) \wedge S(s) \wedge T(t) \rightarrow C(x) \vee E(x) \wedge T(t') \rightarrow FinalState(x) \wedge T(t'') \tag{1.3}$$

Through these representations, Rule 1 describes that for an active material x under the effect of a stimulus (temperature for instance), the material will contract over time t. Rule 2 makes explicit that if the active material x interacts physically with passive material y under a stimulus, a deformation of both materials occurs. As for Rule 3, it is stated that the active material x may exhibit a time-dependent transformation (contraction or expansion) to reach its final state after the stimulus is applied. These simple rules are capable of capturing various scenarios involving active/passive materials, stimuli, and their interaction over time. They illustrate the flexibility of logical representation to model existing behaviors in 4D printing [117, 118].

In addition, frames are objects that encapsulate entities, along with their properties and methods [117]. One type of frame is an ontology, a term that is used in different contexts with various meanings. In the realm of computer science, an ontology refers to a structured representation of concepts within a specific domain, along with the relationships that connect those concepts [28]. As such, an ontology can be considered as a formal specification of a share conceptualization [119]. The notion of shared conceptualization underscores the importance of consensus among stakeholders, ensuring that the ontology reflects a collective understanding rather than an individual perspective. In practical terms, computational ontologies serve as a means to formally model the structure of systems, encompassing relevant entities and their interrelationships. These models enable various applications, from knowledge representation to automated reasoning, by providing a consistent and machine-interpretable framework for understanding complex domains. Ontologies are often written using the Web Ontology Language (OWL) [29], a highly expressive standard of the Semantic Web that enables detailed descriptions of data structures and advanced reasoning capabilities. Furthermore, standards such as RDF [120], RDFS [121], and SKOS [122] serve as foundational components of the Se-

mantic Web, facilitating data consistency and enhancing interoperability across diverse graphs.

As such, ontologies are not static artifacts but are dynamic constructs that evolve alongside scientific knowledge and the needs of their users. These representations are designed to be both accurate and computable, enabling a wide range of applications from data integration to advanced reasoning. However, the construction and maintenance of ontologies are resource-intensive tasks that require substantial collaborative efforts from domain experts, curators, and ontology specialists. These efforts are essential to ensure that ontologies remain active and reflective of the latest scientific achievements.

In life sciences, for example, ontologies such as the Gene Ontology (GO) [123], Mondo [124], Uberon [125], and FoodON [126] are critical for tasks like gene function annotation, disease classification, and food dataset curation. These ontologies also form the backbone of major data generation projects, including The Encyclopedia of DNA Elements (ENCODE) [127] and the Human Cell Atlas [128]. In 4D printing, ontologies can be incorporated into the frame structure to organize knowledge about SMs, AM techniques, and stimuli. For instance, the HERMES ontology has been developed to formalize the 4D printing knowledge involved in materials science, AM, design, mechanics, and physics [14]. **Figure 1.5** presents a portion of the 4D printing ontology, emphasizing the design and engineering view as well as the transformation process view [14]. The development of these ontologies is an ongoing process, as the pace of scientific change, the rapid generation of new data, and the discovery of new concepts continuously drive the need for updates and expansions. The ontology serves as the backbone of structured knowledge, comprising four essential elements. First, classes represent overarching concepts or categories within a domain, such as Voxel in the design and engineering view (**Figure 1.5a**) and Transformation_Function, Transformation_Step, and Technical_Function in the transformation process view (**Figure 1.5b**). Second, individuals can be defined as specific instances of these classes, representing concrete entities (e.g., *1D_to_2D* as an instance of Transformation_Dimension, or *Repeatable* as an instance of Transformation_Effect from [14]). Third, object properties define relationships between individuals, establishing meaningful connections within the ontology (e.g., the relationship `isFulfilledBy` between the individuals of the classes Transformation_Function and Transformation_Process). Finally, data properties describe specific attributes or characteristics of classes (e.g., `Folding_angle` or `Bending_torque`, which are properties related to transformations like folding or bending) [14].

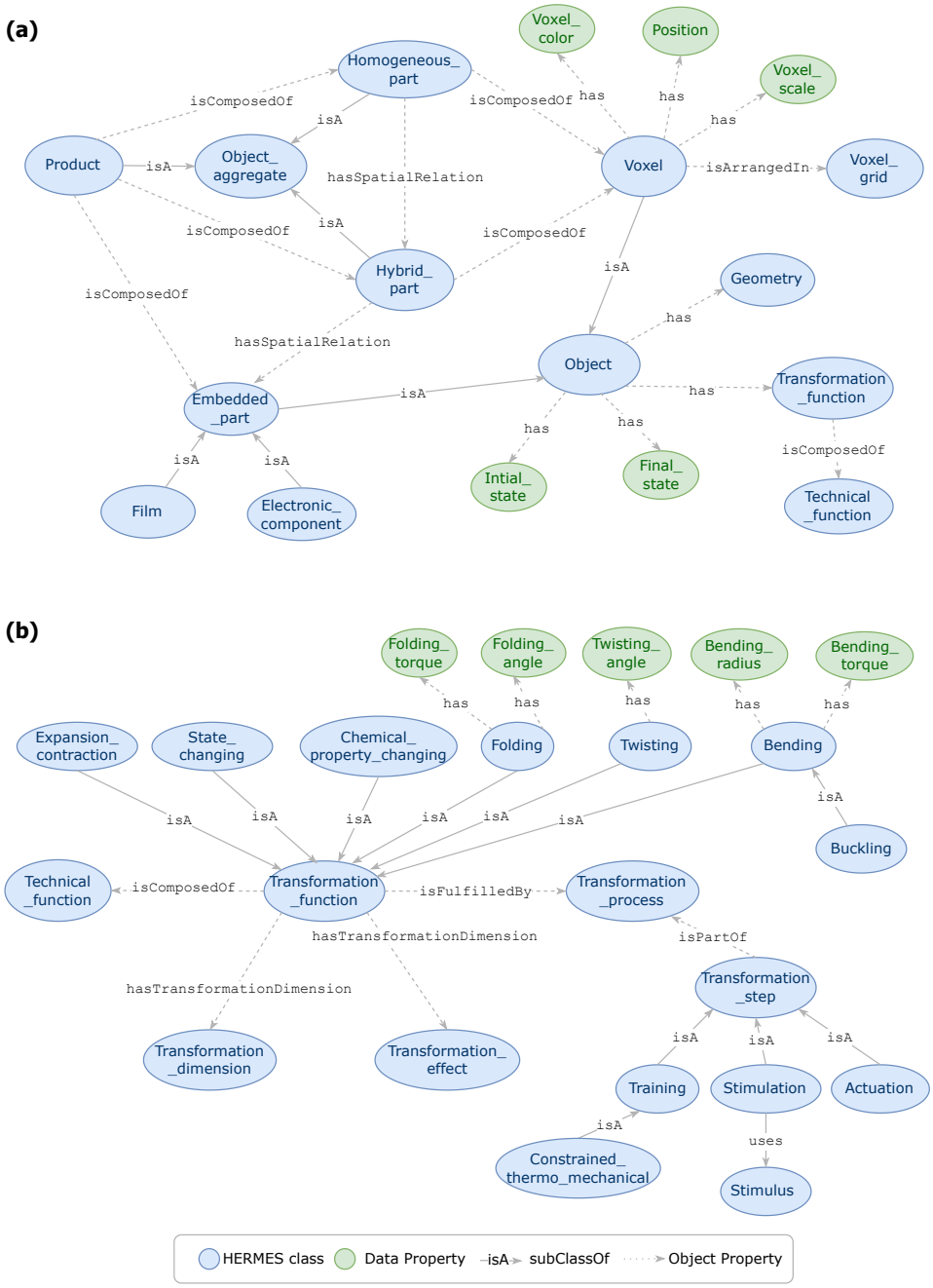

Figure 1.5: Extraction of a portion of an ontology on 4D printing [14]. (a) Design and engineering view with classes and object properties. (b) Transformation process view with classes and object properties. (Adapted with permission from Elsevier).

Given the increasing complexity and scale of data, there is a growing need for efficient methods to extend and refine existing ontologies. LLMs show significant potential in this regard. Trained on vast corpora, they are capable of generating, suggesting, and refining ontology terms based on their contextual understanding of language. This capability makes LLMs powerful tools for automating parts of the ontology expansion process, enabling more rapid adaptation to new scientific discoveries and evolving knowledge landscapes. By leveraging LLMs, ontology development can become more scalable and

responsive, reducing the manual effort required while ensuring that ontologies remain comprehensive and up-to-date.

Alongside with ontologies, semantic networks represent knowledge as a graph, where nodes represent entities and edges represent relationships between them. These networks are used to capture the interconnectedness of concepts and facts. In 4D printing, a semantic network could model the relationships between different types of stimuli (e.g., temperature, light, moisture, etc.) and their effects on SMs (e.g., shape change, property change). For example, one node could represent an active polymer and another could represent the temperature. The edge between them would describe the reaction, e.g., the polymer expands when heated up to 60 °C. This interconnected representation allows for dynamic reasoning about how different stimuli will affect the printed object over time [129]. In addition, modern implementations of semantic networks in AI often take the form of knowledge graphs (KGs), leveraging technologies such as Neo4j [130], PuppyGraph [131], AWS Neptune [132], TigerGraph [133], OrientDB [134], ArangoDB [135], and NebulaGraph [136]. These tools provide powerful frameworks for representing and reasoning over complex relationships, which can be adapted to model the entire lifecycle of a 4D-printed object, including its design, printing, and behavior when exposed to stimuli [129, 137].

Neuro-Symbolic Architectures Neuro-symbolic systems have emerged by integrating the learning capacities of NNs with the structured reasoning strengths of symbolic AI. This integration enables the execution of logical inferences alongside data-driven learning. Building on the recent taxonomy proposed by Kautz [138] and later extended by Hamilton et al. [116] in terms of learning/reasoning capabilities, the existing NSAI architectures can be categorized into seven distinct types, as illustrated in **Figure 1.6**. This figure synthesizes the relationships between neural and symbolic components through key integration strategies such as sequential, nested, ensemble, cooperative, and compiled. The classification includes the original six categories introduced by Kautz, which are expanded to include a seventh category (i.e., fibring), as proposed by Garcez and Gabby [139] and Hamilton et al. [116]. Each category represents a unique approach to integrating neural and symbolic methods, enabling the exploitation of their respective strengths:

- *symbolic Neuro symbolic* refers to systems where both input and output are represented in symbolic form, with a neural component acting as an intermediary processor. Symbolic knowledge, such as sequences of words or logical expressions, is first converted into vector representations, processed by a neural network, and then transformed back into a symbolic output. These architectures are particularly effective for tasks requiring symbolic reasoning enhanced by statistical generalization [14, 140].

- *Symbolic[Neuro]* uses a neural component as a subroutine within a predominantly symbolic system. This allows statistical pattern recognition within a symbolic problem-solving framework. Such architectures are widely applied in robotics and autonomous systems, where symbolic planning integrates sensory data interpreted by neural networks [141].

- *Neuro[Symbolic]* embeds a symbolic reasoning engine within a neural system. This architecture is particularly effective for tasks that require a neural system to generalize from unstructured data while leveraging symbolic rules or relationships at certain conditions to enhance reasoning or interpretability. Therefore, it is capable of efficient combinatorial reasoning and currently represents one of the most promising approaches for combining the strengths of logic-based and neural-based AI [142, 143].

- $Neuro|Symbolic$ processes non-symbolic input and converts it into a symbolic representation for reasoning, which in turn provides a feedback signal that is used to train the NN. For this case, the neural component – considered as a coroutine rather than a subroutine – can then learn from the errors of the symbolic component. Convergence is achieved through iterations. [144].

- $Neuro_{Symbolic}$ integrates symbolic rules as templates to guide the NN structure through weights or the loss function. The mapping of symbolic logic rules onto embeddings are compiled away during training [145–147].

- *Neuro:Symbolic* $\rightarrow$ *Neuro* incorporates symbolic rules during the training of NNs to guide their learning process. Using a structured training regime, neural models are trained to map input-output pairs without requiring step-by-step reasoning [148].

- *Fibring* operates on the principle of recursion, connecting multiple NNs through symbolic components [139].

Several notable algorithms have been developed in these categories. For instance, in the $Neuro_{Symbolic}$ category, logic tensor networks (LTNs) merge the power of first-order logic with NN learning. LTNs utilize tensors to represent logical predicates and relationships, allowing NNs to reason over data through logical inference and optimize through gradient descent. This architecture serves as a critical bridge between symbolic logic and neural learning, facilitating structured reasoning tasks [145]. Besides, tensor product representations (TPRs) encode symbolic entities and their relationships into distributed neural representations, allowing NNs to manipulate these structures within a tensor space. TPRs are critical for relational reasoning tasks, such as those involving multi-agent systems and knowledge representation [146]. In the *Neuro:Symbolic* $\rightarrow$ *Neuro* category, recursive neural knowledge networks (RNKNs) introduce symbolic knowledge

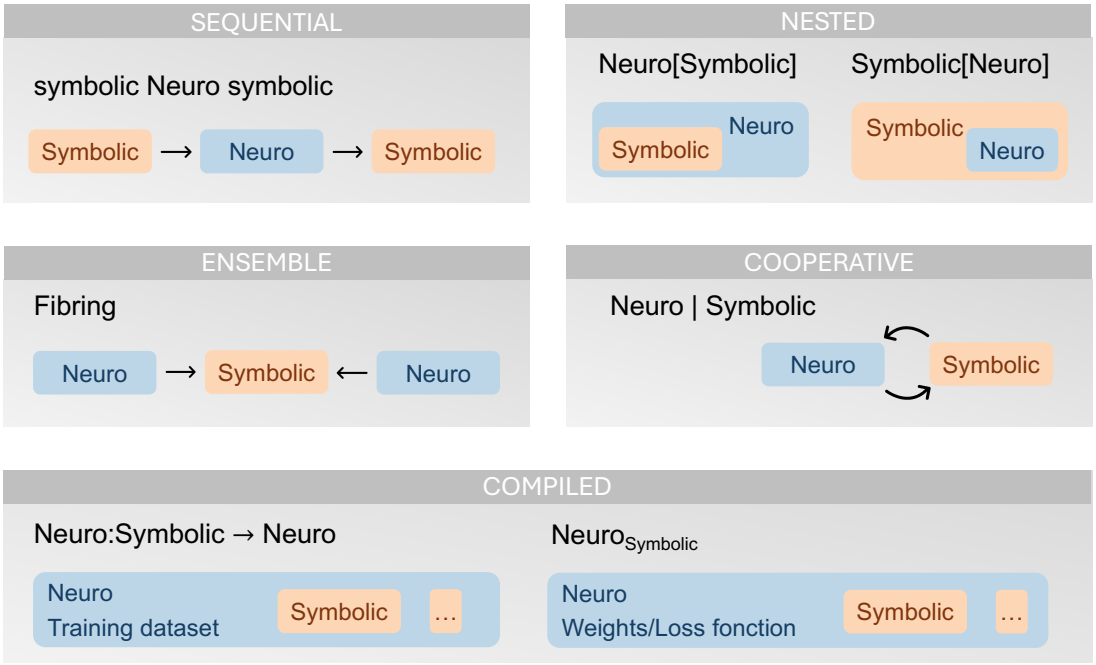

Figure 1.6: Classification of NSAI architectures according to the relationships of the neural and symbolis components and their capabilities, reproduced from [11].

into recursive NN structures. By recursively applying symbolic knowledge representations, RNKNs enable the propagation of knowledge across layers of reasoning, allowing for complex, multi-step logical inference. This is particularly beneficial for tasks that demand structured data reasoning [147]. In addition, logical neural networks (LNNs) directly incorporate symbolic reasoning within NN architectures. By embedding logical rules into the learning process, LNNs maintain logical consistency while still being trainable through gradient descent. This architecture is particularly well-suited for AI applications where adherence to predefined logical constraints is essential, such as in reasoning-based systems [148].

1.3.3 Explainability

NSAI addresses the challenges of traditional AI-based systems, by making AI models more transparent, interpretable, and trustworthy, without compromising their performance. Explainability provides insights into how inputs are processed, how decisions are made, and the reasoning behind those decisions. Symbolic reasoning contributes to this through logical and human-readable explanations of decisions. Typically in AI, the terms black box, gray box, and white box are utilized to describe the varying degrees of openness and transparency within a system. The black box problem in AI refers to the difficulty to justify and legitimize a decision. In contrast, a white box system ensures an internal

decision-making process fully explainable.

Building on the foundation of explainable AI, NSAI integrates key explainability concepts that are crucial for addressing the black box problem and enhance trustworthiness in AI-based systems. Explainability, which allows human to comprehend how a system operates, is tightly linked to interpretability and extends into other principles. For instance, interactivity can foster a dynamic engagement between users and AI, enabling them to query and better understand its decisions. Transparency ensures that the decision-making process is clear and accessible, while responsibility emphasizes ethical considerations that ensure the system to align with societal values. Fairness guarantees that the AI system makes unbiased decisions across different population groups, and stability alongside satisfaction ensures the system's behavior remains consistent, with its explanations meeting user needs. Lastly, robustness refers to the system's ability to handle changes or uncertainties in its inputs without significantly altering its performance. All these interconnected principles work together to create a more reliable AI system, therefore fostering wider adoption for real-world applications.

1.3.4 Main Achievements

NSAI has led to promising achievements across various fields. In NLP and automated theorem proving, NSAI transcends simple pattern recognition, enabling deep contextual understanding and precision in verifying complex proofs [36]. This approach has set new standards in solving intricate theorems that were previously out of reach. In healthcare, NSAI has accelerated drug discovery and diagnostics by bridging the gap between understanding complex biochemical interactions and predicting therapeutic outcomes. It offers interpretable insights, which not only streamline research but also improve treatment planning through personalized diagnostics. Similarly, in robotics and autonomous systems, NSAI empowers machines with both perceptual awareness and logical decision-making, enhancing their ability to navigate real-world environments safely and efficiently. Industries such as finance and law benefit from NSAI's capacity to automate structured information extraction from unstructured data, improving accuracy and speed in data-intensive processes. In supply chain management, NSAI enhances demand forecasting and resource allocation by integrating predictive insights with rule-based decision-making, optimizing operations in complex and variable conditions [149, 150].

Some quantitative results can highlight NSAI's significant improvements in efficiency, scalability, and accuracy. Susskind et al. (2021) [151] evaluated the neuro-symbolic concept learner (NSCL) using the CLEVR dataset [152], a benchmark designed to assess visual reasoning and compositional generalization. The objective of NSCL is

to combine visual understanding with reasoning capabilities by parsing scenes and questions into executable symbolic programs. They compared NSCL against purely neural models, specifically TbD-Net and MAC, which rely on end-to-end neural architectures for reasoning tasks. NSCL demonstrated a clear advantage, achieving 93% accuracy with only 10% of the training data, whereas TbD-Net and MAC plateaued at approximately 80% accuracy under the same conditions. This comparison highlights the superior data efficiency of NSAI, making it highly effective in scenarios with limited training data. In addition, Hersche et al. (2023) [153] introduced the neuro-vector-symbolic architecture (NVSA), whose objective is to combine NN representation learning with vector-symbolic algebra to address two key challenges: the "binding problem" in NNs, which limits their ability to disentangle and represent distinct objects within a compositional structure, and the computational inefficiency of exhaustive symbolic reasoning. NVSA leverages high-dimensional distributed representations to enable end-to-end learning that maintains both compositional transparency and computational efficiency. They applied NVSA to the I-RAVEN dataset [154], which focuses on solving complex visual analogies, and compared it with baseline CNN models. NVSA achieved 88.1% accuracy while significantly reducing computational costs compared to these neural approaches. This highlights NVSA's effectiveness in tasks requiring both efficiency and accuracy.

Moreover, Han and Srivastava (2024) [155] focused on complex event detection (CED) using multimodal sensor data streams, such as inertial measurement unit(IMU) [156] and acoustic signals. The objective was to identify temporal patterns that signify high-level events, even when the data is noisy and spans long durations. They developed a neuro-symbolic event detection framework that integrates NNs for feature extraction with symbolic reasoning for temporal analysis. The framework was compared against widely-used neural methods, including LSTM, temporal convolutional models (TCM), and transformers, which are optimized for event detection tasks. Their neuro-symbolic approach demonstrated a 41% increase in F1 score, a widely-used metric in ML that balances precision (the proportion of correctly predicted positive events among all predicted positives) and recall (the proportion of correctly predicted positive events among all actual positives). This result highlights the framework's superior ability to combine contextual reasoning with data-driven insights, making NSAI particularly well-suited for applications requiring an understanding of temporal and causal relationships. Cunnington et al. (2024) [157] introduced a novel architecture called NeSyGPT, designed to integrate a vision-language foundation model (BLIP) with symbolic reasoning. The objective of NeSyGPT is to enable efficient neuro-symbolic learning by leveraging pre-trained vision-language models to extract symbolic features from raw data, followed by symbolic reasoning using answer set programming (ASP). This approach was applied across four problem domains: MNIST arithmetic [158], follow suit, plant hitting sets,

and CLEVR-Hans, addressing tasks requiring both scalability and explainability. For example, in the follow suit domain, NeSyGPT demonstrated its ability to handle tasks with massive symbolic feature choices—scaling from 6.5 million in a 4-player card game to 5.74×10^{16} in a 10-player variant—while consistently achieving perfect accuracy. It significantly outperformed other neuro-symbolic methods, including NeurASP [159], SLASH [160], Embed2Sym [161], and BLIP+ABL [162], particularly in low-data scenarios. By integrating neural and symbolic components seamlessly, NeSyGPT demonstrated state-of-the-art scalability, accuracy, and explainability in highly demanding problem domains.

These findings underscore the major leap NSAI brings over conventional methods by uniting computational efficiency, advanced precision, and interpretative capabilities. NSAI not only addresses intricate real-world challenges more effectively, but also offers a transparent and adaptable framework—laying the groundwork for robust, next-generation AI systems. Given these achievements, it is essential to consider applying NSAI within 4D printing to harness its potential for managing the complexity of multi-material designs, stimuli-responsive structures, and real-time decision-making. NSAI could provide a transformative framework that could advance 4D printing applications across sectors that demand precision, adaptability, and reliability, setting the foundation for innovative and highly functional smart structures.

1.4 Opportunities in 4D Printing Design

The current achievements of NSAI and challenges in the design of active materials and transformable structures using 4D printing have highlighted the need for deeper exploration of material formulation and distribution, while maintaining control over AI-generated results. The following sections – illustrated in **Figure 1.7** – present key opportunities in designing for 4D printing.

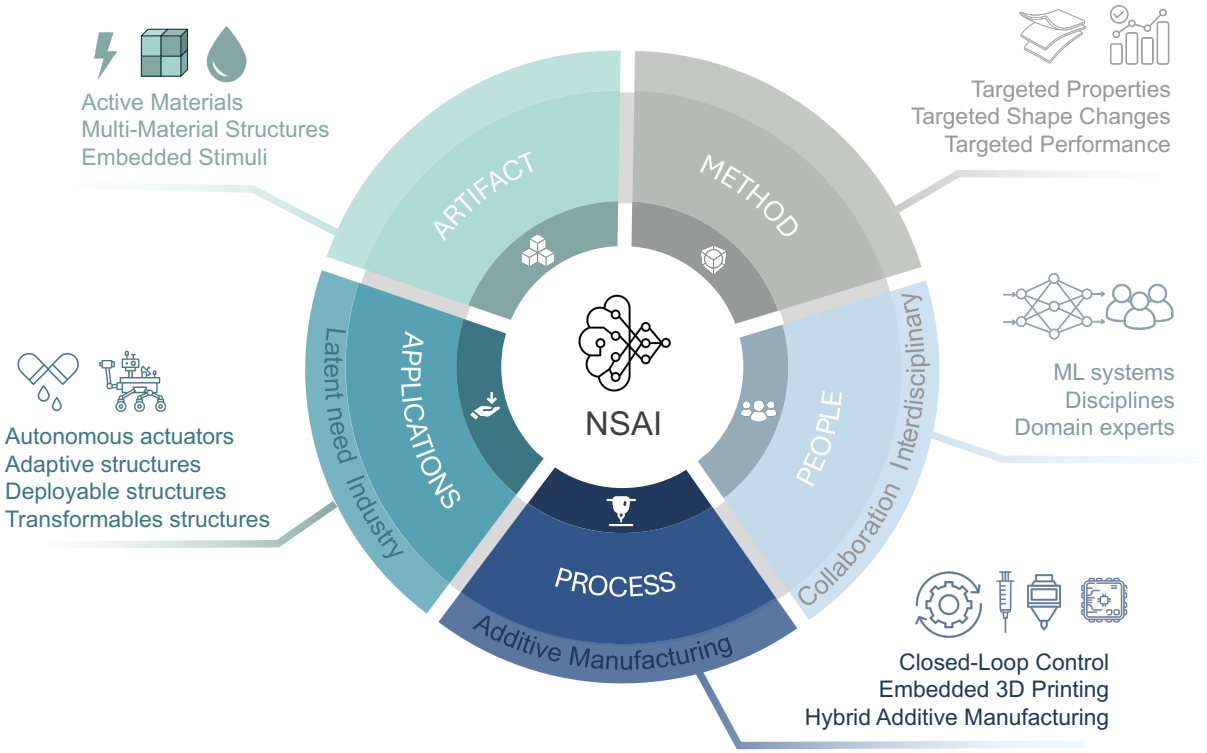

Figure 1.7: Overview of NSAI in accelerated design for 4D printing with regard to AM, people involved from various fields, artifacts, and applications, reproduced from [11].

1.4.1 Multidisciplinary Integration and Coherence

The interdisciplinary nature of 4D printing presents substantial challenges in integrating expertise from diverse fields, such as engineering, design, materials science, AM, computational mechanics, design, chemistry, physics, and computer science. Each domain brings its own data, constraints, models, methodologies, and terminologies, making it difficult to establish cohesive and coordinated design processes. Demoly et al. (2021) [10] highlighted the importance of integrating multiple actors (e.g., product architects, mechanical designers, material experts, and AM process planners), while managing complexities such as multi-material structures, various AM techniques, functionalities, physical principles, scales, and temporal dimensions. NSAI is particularly valuable in addressing these challenges, as it facilitates collaborative design by enabling experts from different fields to work within a shared and explainable framework.

In industrial contexts, the multi-disciplinary integration is further complicated by distinct departments, such as engineering, manufacturing, and production, each with its specific goals, priorities, and methods. Unlike the discipline-oriented organization of research, industry departments are often project and task-driven and focused on addressing practical challenges. This distinction adds another layer of complexity to collaboration. NSAI bridges the gap between the theoretical focus of research and the practical needs of industry by providing a unified framework that fosters communication

and collaboration across domains.

Achieving effective integration in 4D printing design requires both data interoperability and semantic alignment of knowledge from various sources. NSAI, with its dual capabilities of processing unstructured data (via NNs) and reasoning over structured rules (via symbolic reasoning), facilitates this alignment by creating a common ground for collaboration. For example, NSAI can integrate physics-based simulation results with material property datasets and user-defined design objectives into a coherent and dynamic decision-making framework. This capability not only enhances cross-disciplinary collaboration but also accelerates iterative design processes through more efficient exploration and refinement of design options. By supporting seamless communication, data integration, and informed decision-making, NSAI becomes a cornerstone for advancing collaborative and integrated design in 4D printing.

1.4.2 Fostering Collaboration Between ML and 4D Printing Experts

NSAI can can play a transformative role in fostering collaboration ML systems and the diverse set of actors involved in 4D printing. A key challenge lies in bridging the opaque and data-driven nature of ML models with the clarity and precision required by experts to make informed decisions. Explainability ensures that the decision pathways of the ML model are accessible and interpretable to domain experts, allowing them to understand, validate, and modify how the model arrives at specific conclusions. This capability is crucial for establishing trust between 4D printing experts (i.e. designers, engineers, material expert, process planner, etc.) and AI systems, as it allows these experts to assess whether the model's decisions align with their domain knowledge.

In 4D printing, experts often possess domain-specific knowledge that is difficult to encode into traditional ML models. NSAI can address this by incorporating symbolic reasoning, allowing experts to embed their qualitative insights into the AI-driven decision-making process. This integration ensures that expert knowledge, which is based on years of experience and understanding of complex physical and/or chemical phenomena, can enhance and guide the ML models toward more accurate and contextually relevant predictions by explicitly incorporating expert-defined rules and constraints, which influence the resulting decisions of the model. These rules can help limit the model's outputs to feasible and realistic scenarios, ensuring that predictions align with practical constraints known to experts.

Explainability can also allow 4D printing experts to iteratively refine the AI model based on their insights. For instance, when an expert identifies a decision that does not meet practical expectations, they can trace back through the explainable components

to understand why the decision was made, then update the symbolic reasoning layer accordingly. This iterative process helps the ML system evolve in a way that consistently incorporates new expert knowledge from new materials and structures development, making the model increasingly accurate and aligned with real-world needs.

1.4.3 Closed-Loop Additive Manufacturing

ML techniques have attracted increasing attention in the field of AM due to their potential to optimize processes and improve printing quality. However, despite their promise, these approaches face several inherent limitations due to the complexity of AM techniques – especially when combined or hybridized, and the characteristics of the available data [163] [164]. These limitations can be grouped into five main aspects: process parameter variability, dependence on large datasets, inability to detect unknown defects, interpretability and integration challenges, and the high cost of real-time monitoring. AM process, such as powder bed fusion (PBF), involve complex interactions between energy sources and materials, resulting in significant variability in parameters that affect part quality. With more than 130 parameters influencing printing, it is then difficult to accurately predict mechanical properties using ML techniques [165, 166]. Traditional ML models show limited performance in the face of this complexity, often failing to capture all the dynamic interactions between these parameters. Next, training ML models, particularly those using deep NNs, requires large and diverse datasets to ensure accurate predictions. In the context of AM, collecting such data is challenging due to the diversity of materials, techniques, and geometries [167]. The data needed for modeling complex phenomena during AM can be costly to obtain, thus limiting the effectiveness of ML approaches.

Another significant limitation is the inability of supervised models to detect unknown defects. These models rely on the availability of historical labeled data and are therefore limited in their ability to identify anomalies or defects that have not been previously observed during training [168]. This represents a major constraint, as defects in AM, such as porosity, cracking, or delamination, can vary greatly depending on printing conditions. In addition, there are specific limitations associated with CNNs and image processing techniques in AM. CNNs have been widely used for defect detection and quality control by analyzing images from various stages of 3D printing process. However, CNNs require a large amount of labeled image data for training, which is often not available in AM [166]. CNNs can also be sensitive to changes in imaging conditions, such as lighting or camera angle, which can significantly impact the model's performance [166]. Image processing also tends to be computationally intensive, requiring substantial processing power and infrastructure to analyze high-resolution images in real-time. Additionally, CNNs often face difficulties in capturing subtle, small-scale defects, which may be critical in AM parts,

potentially leading to undetected quality issues [167].

The lack of interpretability of ML models complicates the identification of defect causes and the implementation of corrective actions during the manufacturing process [169]. Integrating these models into existing AM systems presents challenges, particularly regarding the adaptation of equipment and operational practices. One can say that ML approaches applied to real-time monitoring of AM processes require sophisticated sensor systems and adapted infrastructure. These sensors must be capable of collecting in situ data with high resolution and precision, which increases the cost of implementing and maintaining these systems. Additionally, volumetric matching of validation data, such as X-ray computed tomography, remains difficult to achieve, thus limiting the effectiveness of ML models based on imaging [166].

These limitations illustrate the need to develop more robust ML approaches that can manage the variability and complexity of AM techniques, while reducing dependence on large datasets and improving model interpretability. In this context, NSAI can be employed to enhance the interpretability of ML models by integrating domain knowledge into the learning process, thereby making predictions more explainable. Furthermore, the symbolic reasoning component can help to identify and manage unknown defects by applying rule-based logic, even when sufficient labeled data is unavailable. NSAI also allows for more efficient use of available data, reducing the dependence on large datasets, and enabling more robust decision-making in real-time monitoring and optimization of AM processes. By using symbolic rules to fill in knowledge gaps, NSAI can operate effectively even with limited training data, which is particularly advantageous in 3D/4D printing scenarios where data collection is costly or time-consuming.

1.4.4 Optimizing Multi-Material and Metamaterial Structures

The development of mechanical metamaterials faces significant challenges in computational design and experimental validation, especially when coupled with ML technologies. The geometric complexity of these materials, characterized by hierarchical multi-scale structures such as lattices, plates, and shells, demands sophisticated computational models to predict their performance. Integrating complex architectural designs and printing parameters into these models is essential to meet specific constraints, such as optimal stiffness, high thermal dissipation, lightweight, and low cost [170, 171]. Numerical tools like finite element methods and topology optimization algorithms, including bidirectional evolutionary structural optimization (BESO) and the solid isotropic material with penalization (SIMP) method, enable progressive improvement of structures under complex constraints. However, these approaches are limited by vast design spaces and high com-

putational demands [172–174].

ML offers a novel approach to material design, enabling the establishment of reporting standards and the use of benchmark datasets, its application in metamaterial optimization faces obstacles related to data quality, availability, and representativeness [175]. High-throughput simulations, such as those combining ML with molecular dynamics (MD), have shown promise in translating architected materials to the atomistic scale and discovering disordered mechanical metamaterials. However, these methods require extensive computational resources and rely on accurate forcefield models, which may not always generalize across different material systems [176]. Additionally, ML models often need large, diverse, and curated datasets to avoid biases and ensure reliable predictions. The lack of standardized datasets and unified frameworks for model evaluation further complicates the integration of ML into the metamaterial design process.

Finally, the experimental validation of ML-predicted metamaterials remains a bottleneck. Ensuring that the properties predicted by ML align with real-world performance requires precise fabrication techniques and rigorous testing methodologies. For example, advanced designs leveraging programmable Poisson's ratios and microstructural geometric adjustments [177] demand meticulous validation to confirm their in-situ tunability and mechanical robustness. These challenges underscore the importance of interdisciplinary strategies that combine advanced computational tools, innovative manufacturing processes, robust ML frameworks, and tailored experimental techniques.

Effective multi-scale topology optimization integrates both continuous and discrete optimization techniques, which is essential for addressing the complexities of multi-material design. NSAI not only reduces the solution space but also ensures that the optimization aligns with the specific requirements and constraints of the design, improving both functionality and printability. Additionally, NSAI supports bio-inspired strategies such as voxel automata, L-systems, and digital morphogenesis, which are vital for creating multi-scale structures [10]. By simulating growth patterns and evolutionary processes with NNs and applying structural and functional constraints through symbolic AI, this dual approach not only streamlines the optimization process for efficiency but also ensures it conforms to the specific requirements and constraints of multi-material designs, thereby enhancing functionality and printability.

NSAI can adopt an inverse design approach where NNs rapidly identify and adapt material distribution patterns based on targeted outputs such as target shape, mechanical strength, flexibility, and responsiveness to stimuli. NNs maximize mechanical and actuation performance by exploring large design spaces and generating configurations that leverage historical data to determine optimal distributions of active and passive

materials within the 3D structure [15, 16]. Symbolic AI then verifies and refines these patterns, ensuring alignment with physical laws and enforcing expert-driven design constraints. This hybrid capability supports voxel/octree-based modeling and uses rules based on expert knowledge to validate configurations, ensuring they meet design specifications and AM constraints. In cases of extreme environmental conditions, symbolic AI may recommend targeted adjustments to material placement, enhancing the structure's adaptation to fatigue and ensuring that multi-material structures remain robust, responsive, and reliable across diverse operational conditions.

In cases where datasets are limited, the integration of physics-informed NNs (PINNs) offers an effective solution. PINNs embed physical laws directly into NN models, enabling them to achieve higher precision in predicting material properties and interactions without relying on extensive data. However, while PINNs are excellent for incorporating physical laws, they can lack the versatility needed to handle complex design requirements that extend beyond pure physics. Symbolic AI becomes essential here, as it employs logical rules that represent expert-driven insights and practical constraints, guiding the optimization process in areas that PINNs alone may not sufficiently address. By combining these elements, NSAI validates predictions and embeds expert knowledge, offering a more adaptable and comprehensive approach that maintains high performance even when data is scarce or incomplete.

Integrating NSAI into the field of multi-material 4D printing provides a strategic pathway to overcoming current limitations. By harnessing the learning and adaptability of NNs alongside the logical, rule-based capabilities of symbolic AI, NSAI enables a holistic and scalable approach to designing, optimizing, and simulating multi-material structures across diverse scales. This approach not only accelerates discovery but also enhances the precision and practicality of designs in real-world applications. The future potential of NSAI in the realm of multi-material structures, metamaterials, and metastructures rests on its capacity to integrate predictive analytics with real-time adaptive capabilities. Advancing research should prioritize the incorporation of NSAI with real-time data acquisition and feedback mechanisms, facilitating continuous learning and structural adaptation over the lifecycle of a material. This progression could enable the creation of truly intelligent structures with self-monitoring and self-adjustment capabilities, unlocking unparalleled levels of performance, resilience, and longevity in complex and variable environments.

1.4.5 Exploring and Discovery of New Material Compositions

The discovery and integration of new materials, particularly in multi-material and metamaterial designs, is one of the most promising areas in 4D printing. However, the

vast design space of material composition presents a significant challenge. Understanding how different materials interact with each other and respond to stimuli requires careful exploration. While traditional NNs are highly effective at analyzing large datasets and identifying patterns in material behavior, they often lack the ability to provide transparent explanations or justify the use of specific material combinations. In 4D printing, knowing not just what material works but understanding why it works is essential for innovation. NSAI overcomes these limitations by integrating symbolic reasoning into the material discovery process. This allows the system not only to generate promising material combinations based on learned data but also to incorporate expert hypotheses derived from scientific literature and research. By combining NNs' ability to explore vast configurations of materials with symbolic AI's capacity to encode performance constraints such as cost, durability, and stimuli responsiveness, NSAI can systematically evaluate material combinations that satisfy multiple criteria.

For instance, experts' intuitive hypotheses, drawn from state-of-the-art research, can be encoded into the symbolic system to guide the exploration of material configurations. NSAI can then prioritize combinations based on criteria defined by symbolic reasoning, such as mechanical performance, environmental stability, or biocompatibility. As the NN proposes potential material candidates, the symbolic system evaluates these combinations to ensure they align with established scientific principles, thus bridging the gap between data-driven insights and knowledge-based reasoning.

Additionally, NSAI enhances the material discovery process by enabling the verification of predictions made by NNs through a structured symbolic reasoning framework. While NNs can generate potential material candidates by analyzing large datasets and identifying patterns, they may not provide insight into the chemical feasibility or performance of the predicted materials. This is where symbolic reasoning becomes crucial. The symbolic component of NSAI evaluates the proposed material combinations by applying knowledge-based rules derived from scientific principles, ensuring that the predicted materials are chemically viable and meet specific performance criteria. This knowledge-guided exploration helps verify whether the material will function as expected under real-world conditions, such as withstanding environmental changes or complying with printability constraints. Symbolic reasoning ensures that each material combination adheres to established scientific knowledge, including chemical properties, reactivity, and mechanical behavior, thus filtering out combinations that are unlikely to succeed. This process guarantees that the materials being considered for further development are not only innovative but also scientifically valid and technically feasible.

1.4.6 Discovery of Emerging Applications

In the realm of 4D printing, identifying practical applications remains a significant challenge. This difficulty arises from the disruptive and unconventional nature of printed shape-changing solutions, which often do not align seamlessly with existing market demands or user expectations. To bridge this gap, employing NSAI offers a promising avenue for discovering latent user needs, market experiences, and tacit desires that might not be immediately apparent to researchers involved in this emerging field [178]. Indeed, NSAI can explore both structured and unstructured data across industries, mining patent databases [179–181], industrial reports, ontologies and knowledge graphs, and scientific literature to uncover patterns, trends, and opportunities for new applications in 4D printing. Recent advances, including deep learning, AI-driven agent-based simulations, and LLM-based empathic mental inference, have demonstrated the potential to explore and analyze these sources for forecasting emerging economies [182], identifying and prioritizing customer needs, and eliciting design requirements [178, 183–185]. By proactively identifying latent needs, technological gaps, and subsequenty emerging innovations, NSAI can position 4D printing as a transformative technology capable of addressing mid- and long-term real-world challenges. By capturing and understanding these intangible things, NSAI can guide 4D printing efforts towards discovering more promising and high-impact applications, ensuring that the technology addresses real-world challenges effectively.

Beyond industrial applications, NSAI can also explore other disciplines such as biology, physics, and chemistry to further develop the potential of 4D printing. For instance, by analyzing biological data and processes, NSAI can use NNs to identify hidden relationships and suggest biomimetic designs, and uncover biological principles that can be replicated with smart materials and stimuli. This cross-disciplinary capability allows for the creation of solutions that go beyond traditional cases developed in the literature, therefore applying 4D printing in areas like tissue engineering, smart implants, and responsive drug delivery systems, among others.

1.5 Conclusion

The integration of NSAI into 4D printing represents a critical opportunity to address the complexities of multi-material design, stimuli responsiveness, and interdisciplinary collaboration. As 4D printing is a booming and interdisciplinary research field, the need for precise material behavior control, scalable design frameworks, and transparent decision-making processes becomes increasingly important. NSAI not only bridges gaps between fields such as materials science, chemistry, AM, mechanics, engineering, and design optimization but also enhances the capabilities of existing computational

methods, ensuring that design decisions are grounded in logical reasoning and backed by comprehensive datasets. By synthesizing diverse data from various domains and offering explainable recommendations, NSAI allows involved experts to navigate the intricate interactions between materials and stimuli with a level of precision that traditional approaches cannot achieve. This transparency in decision-making fosters greater collaboration across disciplines, optimizing both material composition, distribution, printing, and stimuli control at multiple scales. Ultimately, NSAI unlocks new possibilities for innovation, enabling the development of 4D-printed structures that are more reliable, efficient, and tailored to meet the specific demands of diverse industries, from biomedical engineering to automotive and space. As the field moves forward, the adaptability and comprehensive decision-making capabilities of NSAI will play a pivotal role in driving the future of 4D printing.

Having established the relevance of NSAI for addressing the multidisciplinary challenges of 4D printing, the next chapter moves from this domain-level perspective to the architectural foundations of neuro-symbolic integration. It examines how neural and symbolic components can be combined, formalizes and extends existing architectural categories, and evaluates their respective characteristics, thereby providing a conceptual basis for the computational contributions developed in the subsequent chapters.

Chapter 2

Neuro-Symbolic Architectures: Formalizing, Extending, Evaluating, and Empowering Generative AI

2.1 Introduction

Neuro-symbolic artificial intelligence (NSAI) is fundamentally defined as the combination of deep learning and symbolic reasoning [30]. This hybrid approach aims to overcome the limitations of both symbolic and neural artificial intelligence (AI) systems while harnessing their respective strengths. Symbolic AI excels in reasoning and interpretability, whereas neural AI thrives in learning from vast amounts of data. By merging these paradigms, NSAI aspires to embody two fundamental aspects of intelligent cognitive behavior: the ability to learn from experience and the capacity to reason based on acquired knowledge [30, 42].

The importance of NSAI has been increasingly recognized in recent years, especially after the 2019 Montreal AI Debate between Gary Marcus and Yoshua Bengio. This debate highlighted two contrasting perspectives on the future of AI: Marcus argued that "expecting a monolithic architecture to handle abstraction and reasoning is unrealistic," emphasizing the limitations of current AI systems, while Bengio maintained that "sequential reasoning can be performed while staying in a deep learning framework" [186]. This discussion brought attention to the strengths and weaknesses of neural and symbolic approaches, catalyzing a surge of interest in hybrid solutions. Bengio's subsequent remarks at IJCAI 2021 underscored the importance of addressing out-of-distribution (OOD) generalization, stating that "we need a new learning theory" to tackle this critical challenge [187]. This aligns with the broader consensus within the AI community that combining neural and symbolic paradigms is essential to developing more robust and adaptable systems. Drawing on concepts like Daniel Kahneman's dual-process theory of

reasoning, which compares fast, intuitive thinking (System 1) to deliberate, logical thought (System 2), NSAI seeks to bridge the gap between learning from data and reasoning with structured knowledge [188]. Despite ongoing debates about the optimal architecture for integrating these two paradigms, the 2019 Montreal AI Debate has played a pivotal role in shaping the trajectory of research in this promising field [26, 27, 39–41, 43].

NSAI offers a promising avenue for addressing limitations of purely symbolic or neural systems. For instance, while neural networks (NNs) often struggle with interpretability, symbolic AI systems are rigid and require extensive domain knowledge. By combining the adaptability of neural models with the explicit reasoning capabilities of symbolic methods, NSAI systems aim to provide enhanced generalization, interpretability, and robustness. These characteristics make NSAI particularly well-suited for solving complex, real-world problems where adaptability and transparency are critical [116]. Kautz [138] identified several NSAI architectures that effectively integrate these paradigms, each architecture offers unique advantages but also poses specific challenges in terms of scalability, interpretability, and adaptability. A systematic evaluation of these architectures is imperative to understand their potential and limitations, guiding future research in this rapidly evolving field.

Generative AI has witnessed remarkable advancements, encompassing a diverse range of technologies that address various challenges in data processing, reasoning, and decision-making. These advancements can be categorized into several major branches of AI. Natural language processing (NLP) [85] includes technologies such as retrieval-augmented generation (RAG) [189], sequence-to-sequence models [190], semantic parsing [191], named entity recognition (NER) [192], and relation extraction [193], which focus on understanding and generating human language. Reinforcement learning techniques, like reinforcement learning with human feedback (RLHF) [194], enable systems to learn optimal actions through interaction with their environment. Advanced NNs include innovations such as graph neural networks (GNNs) [195] and generative adversarial networks (GANs) [196], which excel in handling structured data and generating realistic data samples, respectively. Multi-agent systems [197, 198] explore the coordination and decision-making among multiple intelligent agents. Recent advances leverage mixture of experts (MoE) architectures to enhance scalability and specialization in collaborative frameworks. In MoE-based multi-agent systems, each expert operates as an autonomous agent, specializing in distinct sub-tasks or data domains, while a dynamic gating mechanism orchestrates their contributions [199, 200]. Transfer Learning [201], including pre-training [84], fine-tuning [202], and few-shot learning [203], allows AI models to adapt knowledge from one task to another efficiently. Explainable AI (XAI) [204] focuses on making AI systems transparent and interpretable, while efficient learning techniques, such as model distillation [205], aim to optimize resource usage. Reasoning and inference methods like chain-of-thought (CoT) [206] reasoning and link prediction

enhance logical decision-making capabilities. Lastly, continuous learning [207] paradigms ensure adaptability over time. Together, these technologies form a comprehensive toolkit for tackling the increasingly complex demands of generative AI applications.

The classification of generative AI technologies within the NSAI framework is crucial for several reasons. Firstly, it provides a structured approach to understanding how these diverse technologies relate to and enhance NSAI capabilities. By mapping these techniques to specific NSAI architectures, researchers and practitioners can better grasp their potential applications and limitations. This classification also facilitates the identification of synergies between different AI approaches, potentially leading to more robust and versatile hybrid systems. Furthermore, it aids in decision-making processes when selecting appropriate technologies for specific tasks, considering factors like interpretability, reasoning capabilities, and generalization. As AI continues to evolve, this systematic categorization becomes increasingly valuable for bridging the gap between cutting-edge research and practical implementation, ultimately driving the field towards more integrated and powerful AI solutions.

Therefore, this research aims to examine how selected generative AI technologies may relate to the core categories of NSAI and what insights such a comparison can provide regarding their strengths and limitations. Building on Kautz's taxonomy, our contribution is fourfold: (i) to clarify, formalize, and extend existing NSAI architectures, (ii) to provide an illustrative and conditional mapping of selected recent generative AI methods onto NSAI paradigms, explicitly distinguishing literature-grounded instantiations from prospective integrations, (iii) to develop a qualitative and literature-grounded evaluation framework for comparing NSAI architectures across multiple criteria, and (iv) to propose and illustrate NSAI-based application scenarios for 4D printing, showing how each paradigm can be instantiated in this domain and what opportunities it may support for design workflows.

2.2 Neuro-Symbolic AI: Combining Learning and Reasoning to Overcome AI's Limitations

NNs have been exemplary in handling unstructured forms of data, e.g., images, sounds, and textual data. The capacity of these networks to acquire sophisticated patterns and representations from voluminous datasets has provided major breakthroughs in a series of disciplines, from computer vision, speech recognition, to NLP [84, 85]. One of the major benefits of NNs is that they learn and become better from raw data without requiring pre-coded rules or expert knowledge. This makes them highly scalable and efficient to utilize in applications with large raw data. However, despite these benefits, NNs also have some very well-documented disadvantages. One of the major ones of these might be that

they are not transparent. Indeed, neural models pose interpretability challenges, making it difficult to understand the process by which they arrive at specific decisions or predictions. Such opacity causes problems for critical applications where explanation is necessary, such as in healthcare, finance, legal frameworks, and engineering. Additionally, NNs have a high requirement for data, requiring substantial amounts of labeled training data in order to operate effectively. This reliance on large data makes them ineffective when applied to data-scarce or data-costly environments. Neural models also struggle with reasoning and generalizing beyond their training data, which makes their performance less impressive when it comes to tasks in logical inference or commonsense reasoning. Specifically, tasks including understanding causality, sequential problem-solving, and decision-making relying on outside world knowledge.

Symbolic AI is better at handling areas that are weaker for NNs. Symbolic systems function on explicit rules and structured representations, which enables them to achieve reasoning tasks related to complicated issues, such as mathematical proofs, planning, and expert systems. Symbolic AI is most important because it is transparent. Since symbolic methods are grounded in known rules and logical formalisms, decision-making processes are easy to interpret and explain. However, symbolic AI systems have some drawbacks. One of the biggest ones is that they are rigid and difficult to respond to new circumstances. They require rules to be manually defined and require structured input data, leading them difficult to apply to real-world situations where data might contain noise, incompleteness, or unstructured form. They are also susceptible to combinatorial explosions in handling big data or hard reasoning problems, which significantly slows down their performance at scale. Symbolic systems are also not well suited for perception tasks like image or speech recognition since they are unable to draw knowledge from raw data alone.

While traditional NNs are strong at recognizing patterns in collections of data but falter when presented with new situations, symbolic reasoning provides a rational foundation for decision-making but is limited in the manner in which it can learn knowledge from new information and adapt in a dynamic process. The combination of these two approaches in NSAI effectively minimizes these limitations, producing a more flexible, explainable, and effective AI system. Another distinguishing feature of NSAI is that it is able to generalize outside its training set. Traditional AI systems are prone to fail in novel situations; however, NSAI, because of its combination of learning and logical reasoning, works better in such cases. Such a feature is critical for real-world applications such as autonomous transport and medicine, where systems need to perform well in uncontrolled environments. Apart from that, in an interdisciplinary engineering context such as 4D printing, which brings together materials science, additive manufacturing, and engineering, NSAI holds significant promise for improving both the interpretability and reliability of design decisions on the actuation and mechanical performance, and printability. Although these advantages seem promising, they remain

hypotheses requiring more extensive validation and industrial-scale testing. Ongoing research must demonstrate, through empirical studies and real-world implementations, how NSAI can reliably accelerate the discovery of smart materials and structures [11]. The second key benefit point of NSAI is that it has a reduced need for big data sets. Traditional AI systems usually require a tremendous amount of data to operate, which might be very time- and resource-consuming. NSAI, however, is able to do better with a much smaller set of data required, due to its symbolic reasoning ability. This makes it a more sustainable and viable option, especially for small organizations or new research areas with limited resources. Along with the aforementioned data efficiency, NSAI models also have the exceptional transferability, i.e., their capacity for using knowledge learned from one task and applying it in another with less need for retraining. Such a property is highly desirable in situations where there is little data related to a new task.

2.3 Neuro-Symbolic AI Architectures

This section provides an overview of various NSAI architectures, offering insights into their design principles, integration strategies, and unique capabilities. While Kautz's classification [138] serves as a foundational framework, we extend it by incorporating additional architectural perspectives—most notably the fibring architecture—and by introducing more granular subclassifications within the compiled architecture to capture the evolving landscape of NSAI systems. These proposed extensions should be considered as analytical refinements rather than as an established community consensus. These approaches range from symbolic systems augmented by neural modules for specialized tasks to deeply integrated models in which explicit reasoning engines operate within neural frameworks. This expanded categorization highlights the diversity of design strategies and the broad applicability of NSAI techniques, emphasizing their potential for more interpretable, robust, and data-efficient AI solutions.

2.3.1 Sequential: *Symbolic* $\rightarrow$ *Neuro* $\rightarrow$ *Symbolic*

As part of the sequential NSAI, the *Symbolic* $\rightarrow$ *Neuro* $\rightarrow$ *Symbolic* architecture involves systems where both input and output are symbolic, with a NN acting as a mediator for processing (Figure 2.1a). Symbolic input, such as logical expressions or structured data, is first mapped into a continuous vector space through an encoding process. The NN operates on this encoded representation, enabling it to learn complex transformations or patterns that are difficult to model symbolically. Once the processing is complete, the resulting vector is decoded back into symbolic form, ensuring that the final output aligns with the structure and semantics of the input domain. This framework is especially useful for tasks that require leveraging the generalization capabilities of NNs while preserving

symbolic interpretability [14, 140]. A formulation of this architecture is presented below:

$$y = f_{\text{neural}}(x) \tag{2.1}$$

where x is the symbolic input, $f_{\text{neural}}(x)$ represents the NN that processes the input, and y is the symbolic output.

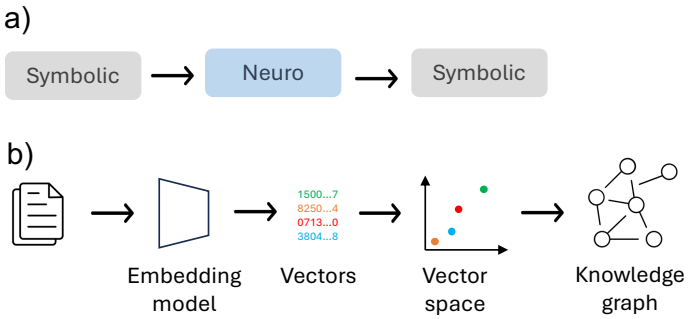

Figure 2.1: Sequential architecture: (a) Principle and (b) application to knowledge graph construction, reproduced from [208].

This architecture can be used in a semantic parsing task, where the input is a sequence of symbolic tokens (e.g., words). Here, each token is mapped to a continuous embedding via word2vec, GloVe, or a similar method [209, 210]. The NN then processes these embeddings to learn compositional patterns or transformations. From this, the network's output layer decodes the processed information back into a structured logical form (such as knowledge-graph triples), as illustrated in Figure 2.1b.

2.3.2 Nested: *Symbolic[Neuro]* and *Neuro[Symbolic]*

The nested NSAI category is composed of two different architectures. The first – *Symbolic[Neuro]* – places a NN as a subcomponent within a predominantly symbolic system (Figure 2.2a). Here, the NN is used to perform tasks that require statistical pattern recognition, such as extracting features from raw data or making probabilistic inferences, which are then utilized by the symbolic system. The symbolic framework orchestrates the overall reasoning process, incorporating the neural outputs as intermediate results [141]. This architecture can formally defined as follows:

$$y = g_{\text{symbolic}}(x, f_{\text{neural}}(z)) \tag{2.2}$$

where x represents the symbolic context, z is the input passed from the symbolic reasoner to the NN, $f_{\text{neural}}(z)$ expresses the neural model processing the input, and g_{symbolic} the symbolic reasoning engine that integrates neural outputs. A well-known instance of this architecture is AlphaGo [141], where a symbolic Monte-Carlo tree search orchestrates high-level decision-making, while a NN evaluates board states, providing a data-driven

heuristic to guide the symbolic search process [211] (Figure 2.2b). Similarly, in a medical diagnosis scenario, a rule-based engine oversees the core diagnostic process by applying expert guidelines to patient history, symptoms, and lab results. At the same time, a NN interprets unstructured radiological images, delivering key indicators such as tumor likelihood. The symbolic system then integrates these indicators into its final decision, combining transparent and rule-driven logic with robust pattern recognition.

The second architecture – *Neuro[Symbolic]* – integrates a symbolic reasoning engine as a component within a neural system, allowing the network to incorporate explicit symbolic rules or relationships during its operation (Figure 2.2c). The symbolic engine provides structured reasoning capabilities, such as rule-based inference or logic, which complement the NN's ability to generalize from data. By embedding symbolic reasoning within the neural framework, the system gains interpretability and structured decision-making while retaining the flexibility and scalability of neural computation. This integration is particularly effective for tasks that require reasoning under constraints or adherence to predefined logical frameworks [142, 143]. This configuration can be described as follows:

$$y = f_{\text{neural}}(x, g_{\text{symbolic}}(z)) \tag{2.3}$$

where x represents the input data to the neural system, z is the input passed from the NN to the symbolic reasoner, g_{symbolic} is the symbolic reasoning function, and f_{neural} denotes the NN processing the combined inputs.

This architecture is currently applied in automated warehouse, where a robot navigates dynamically changing aisles. During normal operation, it relies on a neural policy to select routes based on learned patterns. When it encounters an unexpected obstacle, it offloads route computation to a symbolic solver (e.g., a pathfinding or constraint-satisfaction algorithm), which returns an alternative path. The solver's output is then integrated back into the neural policy, and the robot resumes its usual pattern-based navigation. Over time, the robot also learns to identify which challenges call for the symbolic solver, effectively blending fast pattern recognition with precise combinatorial planning.

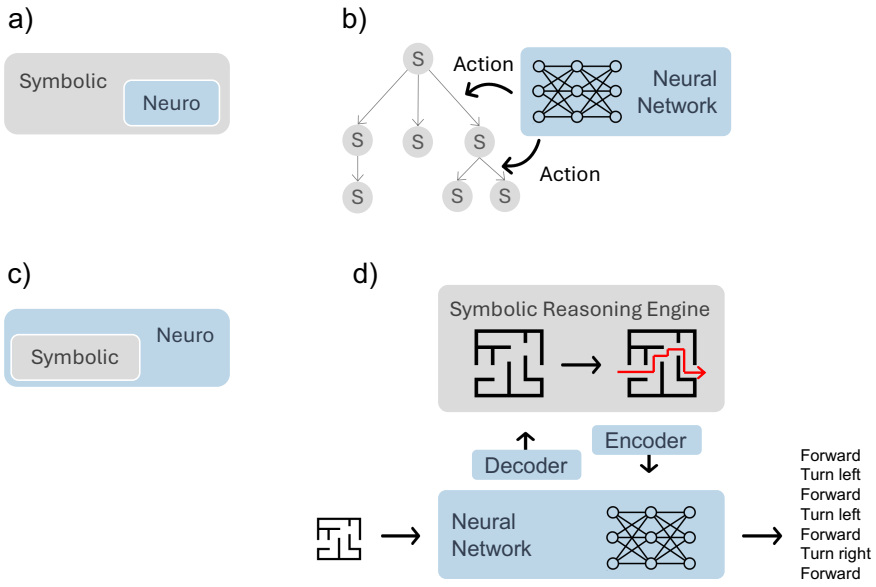

Figure 2.2: Nested architectures: (a) *Symbolic[Neuro]* principle and (b) its application to tree Search, (c) *Neuro[Symbolic]* principle and (d) its application to maze-solving, reproduced from [208].

Figure 2.2d illustrates this framework, a symbolic reasoning engine processes structured data, such as a maze, to generate a solution path. A NN encodes the problem into a latent representation and decodes it into a symbolic sequence of actions (e.g., forward, turn left, turn right).

2.3.3 Cooperative: *Neuro* | *Symbolic*

As a cooperative framework, *Neuro* | *Symbolic* uses neural and symbolic components as interconnected coroutines, collaborating iteratively to solve a task (Figure 2.3a). NNs process unstructured data, such as images or text, and convert it into symbolic representations that are easier to reason about. The symbolic reasoning component then evaluates and refines these representations, providing structured feedback to guide the NN's updates. This feedback loop continues over multiple iterations until the system converges on a solution that meets predefined symbolic constraints or criteria. By combining the strengths of NNs for generalization and symbolic reasoning for interpretability, this approach achieves robust and adaptive problem-solving [144]. This architecture can be described as follows:

$$z^{(t+1)} = f_{\text{neural}}(x, y^{(t)}), \qquad y^{(t+1)} = g_{\text{symbolic}}(z^{(t+1)}), \qquad \forall t \in \{0, 1, \ldots, n\}. \tag{2.4}$$

where x represents non-symbolic data input, $z^{(t)}$ is the intermediate symbolic representation at iteration t, $y^{(t)}$ is the symbolic reasoning output at iteration t, $f_{\text{neural}}(x, y^{(t)})$

expresses the NN that processes the input x and feedback from the symbolic output $y^{(t)}$, $g_{\text{symbolic}}(z^{(t+1)})$ is the symbolic reasoning engine that updates $y^{(t+1)}$ based on the neural output $z^{(t+1)}$, and n is the maximum number of iterations or a convergence threshold. The hybrid reasoning halts when the outputs $y^{(t)}$ converge (e.g., $|y^{(t+1)} - y^{(t)}| < \epsilon$)), where ϵ is a small threshold denoting minimal change between successive outputs, or when the maximum iterations n is reached.

For instance, this architecture can applied in autonomous driving systems, where a NN processes real-time images from vehicle cameras to detect and classify traffic signs. It identifies shapes, colors, and patterns to suggest potential signs, such as speed limits or stop signs. A symbolic reasoning engine then evaluates these detections based on contextual rules—like verifying if a detected speed limit sign matches the road type or if a stop sign appears in a logical position (e.g., near intersections). If inconsistencies are detected, such as a stop sign identified in the middle of a highway, the symbolic system flags the issue and prompts the neural network to re-evaluate the scene. This iterative feedback loop continues until the system reaches consistent, high-confidence decisions, ensuring robust and reliable traffic sign recognition, even in challenging conditions like poor lighting or partial occlusions (Figure 2.3b).

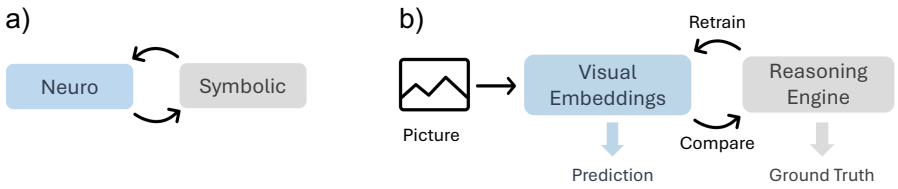

Figure 2.3: Cooperative architecture: (a) principle and (b) application to visual reasoning, reproduced from [208].

2.3.4 Compiled: $Neuro_{Symbolic_{Loss}}$, $Neuro_{Symbolic_{Neuro}}$ and *Neuro:Symbolic* $\rightarrow$ *Neuro*

As part of the compiled NSAI, $Neuro_{Symbolic_{Loss}}$ uses symbolic reasoning into the loss function of a NN (Figure 2.4a). The loss function is typically used to measure the discrepancy between the model's predictions and the true outputs. By incorporating symbolic rules or constraints, the network's training process not only minimizes prediction error but also ensures that the output aligns with symbolic logic or predefined relational structures. This allows the model to learn not just from data but also from symbolic reasoning, helping to guide its learning process toward solutions that are both accurate and consistent with symbolic principles.

$$\mathcal{L} = \mathcal{L}_{\text{task}}(y, y_{\text{target}}) + \lambda \cdot \mathcal{L}_{\text{symbolic}}(y) \tag{2.5}$$

where y is the model prediction,y_{target} represents the ground truth labels, $\mathcal{L}_{\text{task}}$ is the task-specific loss (e.g., cross-entropy), $\mathcal{L}_{\text{symbolic}}$ is the penalization for violating symbolic rules, λ the Weight balancing the two loss components, and $\mathcal{L}$ the final loss, combining both the task-specific loss and the symbolic constraint penalty to guide model optimization. This architecture is typically useful in the field of 4D printing, where structures need to be optimized at the material level to achieve a target shape. In such a case, a NN predicts the material distribution and geometric configuration that allows the structure to adapt under external stimuli. The training process incorporates a physics-informed loss function, where, in addition to minimizing the difference between predicted and desired mechanical behavior, the model is penalized whenever the predicted deformation violates symbolic mechanical constraints, such as equilibrium equations or the stress-strain relationship (Figure 2.4b). By embedding these symbolic equations directly into the loss function, the NN learns to generate designs that are not only data-driven but also physically consistent, ensuring that the final 4D-printed structure maintains the desired shape across different operational conditions.

A second compiled NSAI architecture, called $Neuro_{Symbolic_{Neuro}}$, uses symbolic reasoning at the neuron level by replacing traditional activation functions with mechanisms that incorporate symbolic reasoning (Figure 2.4c). Rather than using standard mathematical operations like ReLU or sigmoid, the neuron activation is governed by symbolic rules or logic. This allows the NN to reason symbolically at a more granular level, integrating explicit reasoning steps into the learning process. This fusion of symbolic and neural operations enables more interpretable and constrained decision-making within the network, enhancing its ability to reason in a structured and rule-based manner while retaining the flexibility of neural computations. This architecture can be described as follows:

$$y = g_{\text{symbolic}}(x) \tag{2.6}$$

where: x represents the pre-activation input, $g_{\text{symbolic}}(x)$ is the symbolic reasoning-based activation function, and y the final neuron. This architecture can find application in lean approval systems, where neural activations are driven by symbolic financial rules rather than traditional functions. One example is the collateral-based constraint neuron, which dynamically adjusts the risk score based on the value of the pledged collateral. When the collateral's value falls below a predefined threshold relative to the loan amount, the neuron applies a strict penalty that substantially increases the risk score, effectively preventing the system from underestimating the associated financial risk. This symbolic constraint ensures that, regardless of favorable patterns identified in other data, the model consistently accounts for the critical impact of insufficient collateral, leading to more reliable and regulation-compliant credit decisions (Figure 2.4d).

Finally, the last compiled architecture, *Neuro:Symbolic* $\rightarrow$ *Neuro*, uses a symbolic reasoner to generate labeled data pairs (x, y), where y is produced by applying symbolic rules or reasoning to the input x (Figure 2.4e). These pairs are then used to train a NN, which learns to map from the symbolic input x to the corresponding output y. The symbolic reasoner acts as a supervisor, providing high-quality, structured labels that guide the NN's learning process [148]. This architecture can be governed as follows:

$$\mathcal{D}_{\text{train}} = \{(x, g_{\text{symbolic}}(x)) \mid x \in \mathcal{X}\} \tag{2.7}$$

where $\mathcal{D}_{\text{train}}$ is the training dataset, x denotes the unlabeled data, $g_{\text{symbolic}}(x)$ represents symbolic rules generating labeled data, and $\mathcal{X}$ the set of all input data (Figure 2.4b).

Figure 2.4f illustrates this architecture, where a reasoning engine is used to label unlabeled training data, transforming raw inputs into structured (x, y) pairs, where symbolic rules enhance the data quality.

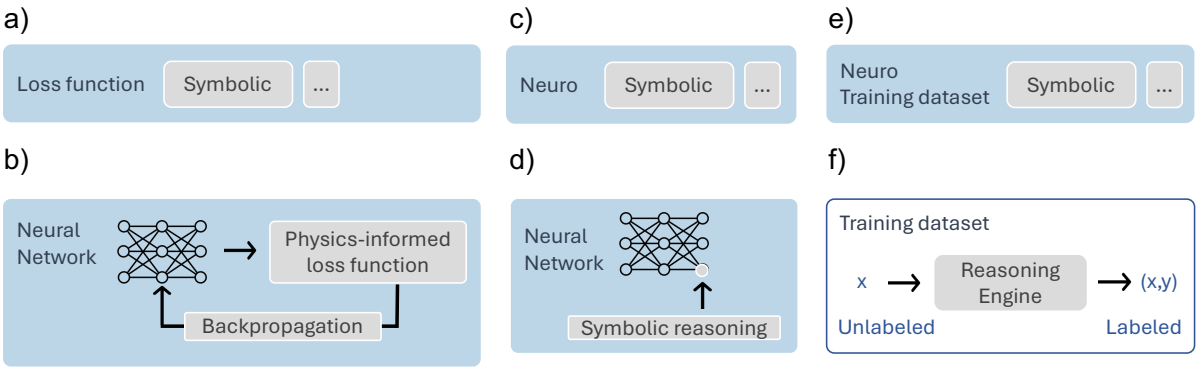

Figure 2.4: Compiled architectures: (a) $Neuro_{Symbolic_{Loss}}$ principle and (b) application to physics-informed learning; (c) $Neuro_{Symbolic_{Neuro}}$ principle and (d) application of symbolic reasoning in NNs; (e) *Neuro:Symbolic* $\rightarrow$ *Neuro* principle and (f) application to data labeling, reproduced from [208].

2.3.5 Fibring: *Neuro* $\rightarrow$ *Symbolic* $\leftarrow$ *Neuro*

Another architecture that we propose to distinguish taxonomically is *Neuro* $\rightarrow$ *Symbolic* $\leftarrow$ *Neuro*, in which multiple interconnected NNs are coordinated through a symbolic fibring function that enables them to collaborate and share information while remaining consistent with explicit symbolic constraints (Figure 2.5a). The symbolic function acts as an intermediary, facilitating communication between the networks by ensuring that their interactions respect predefined symbolic rules or structures. This enables the networks to exchange information in a structured manner, allowing them to jointly solve problems while benefiting from both the statistical learning power of NNs and the logical constraints imposed by the symbolic system [139]. We present fibring here as a proposed extension to Kautz's taxonomy rather than as an established consensus

category. Its motivation is to capture an integration principle that is not fully expressed by the existing cooperative or nested paradigms. In cooperative architectures, neural and symbolic components typically interact within a single iterative problem-solving loop. In nested architectures, one component is structurally embedded inside the other and remains subordinate to its host pipeline. By contrast, fibring is intended to describe settings in which multiple specialized neural components must remain mutually consistent through a shared symbolic coordination layer that performs aggregation, conflict resolution, and cross-model constraint enforcement. This distinction becomes especially relevant when no single neural-symbolic pair is sufficient to characterize the system, such as in multi-agent coordination, multi-scale design, or multi-view decision-making, where symbolic reasoning must mediate among several neural subsystems simultaneously. This architecture can formally defined as follows:

$$y = g_{\text{fibring}}(\{f_i\}_{i=1}^{n}) \tag{2.8}$$

where f_i represents the individual NN, g_{fibring} is the logic-aware aggregator that enforces symbolic constraints while unifying the outputs of multiple NNs, n the umber of NNs, and y is the combined output of interconnected NNs, produced through the symbolic fibring function g_{fibring}. For instance in smart cities and urban planning, multiple NNs can be employed, each handle a different urban data stream—such as real-time traffic flow, energy consumption, and air quality measurements. A symbolic fibring function then harmonizes these outputs, enforcing city-level constraints (e.g., ensuring pollution alerts match local environmental regulations, verifying that traffic predictions align with current road network rules). If one network forecasts a surge in vehicle congestion that would push pollution levels beyond acceptable thresholds, the symbolic aggregator identifies the conflict and directs all networks to converge on a coordinated strategy—such as adjusting traffic signals or advising public transport usage. By leveraging each network's specialized insight within logical urban-planning constraints, the system delivers efficient, consistent decisions across the city's complex infrastructure.

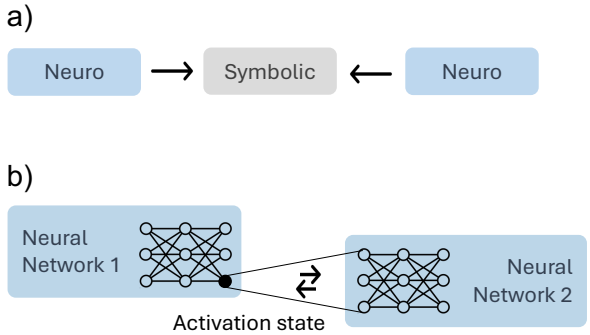

Figure 2.5: Fibring architecture: (a) principle and (b) application to NN collaboration, reproduced from [208].

Figure 2.5b illustrates this architecture, where two NNs (Neural Network 1 and Neural Network 2) communicate through activation states, which enables dynamic exchange of learned representations.

2.4 Leveraging NSAI in AI Technologies

Generative AI is advancing at a remarkable pace, addressing increasingly complex challenges through the integration of diverse methodologies. A key development is the combination of NNs with symbolic reasoning, resulting in hybrid systems that leverage both strengths. Recent studies have demonstrated the effectiveness of this approach in various applications, including design generation and enhancing instructability in generative models [36, 212]. This section examines how selected contemporary AI techniques can be discussed in relation to the NSAI framework. Rather than proposing definitive taxonomic assignments, it provides a structured perspective on where explicit symbolic elements are already present and where connections with NSAI remain indirect, conditional, or emerging.

Accordingly, the discussion distinguishes between (i) approaches that are already instantiated in the literature as genuinely neuro-symbolic systems and (ii) approaches whose relation to NSAI remains partial or prospective. This distinction is especially important for techniques which may support neuro-symbolic integration in some settings while relying on substantially different design principles.

In this way, the section helps clarify how these techniques relate to different forms of neural-symbolic integration and highlights where future neuro-symbolic extensions may plausibly emerge. It also supports a more structured discussion of their roles in addressing challenges such as interpretability, reasoning, and generalization.

2.4.1 Overview of Key AI Technologies

One of the most significant advancements is RAG, which integrates information retrieval with generative models to perform knowledge-intensive tasks. By combining a retrieval mechanism to extract relevant external data with Seq2Seq models for generation [213], RAG excels in applications such as question answering and knowledge-driven conversational AI [214]. Seq2Seq models themselves, built as encoder-decoder architectures, have been pivotal in machine translation, text summarization, and conversational modeling, providing the foundation for many generative AI systems. An extension of RAG is the GraphRAG approach [215], which incorporates graph-based reasoning into the retrieval and generation process. By leveraging knowledge graph (KGq) and ontologies structures to represent relationships between information elements, GraphRAG enhances query-focused summarization and reasoning tasks [216, 217]. This method has demon-

strated success in producing coherent and contextually rich summaries by integrating local and global reasoning.

GNNs [218] represent a breakthrough in extending neural architectures to graph-structured data, enabling advanced reasoning over interconnected entities. Their ability to model relationships between entities makes them indispensable for a range of tasks, including link prediction, node classification, and recommendation systems, with notable success in KG reasoning. GNNs have also proven highly effective in named entity recognition (NER) [219], where they can leverage graph representations to capture contextual dependencies and relationships between entities in text. This capability extends to relation extraction [220], where GNNs identify and classify semantic relationships between entities, crucial for building and enhancing KG.

Advances in agentic AI systems, which leverage Large Language Models (LLMs), have shown significant potential in enabling autonomous decision-making and task execution. These systems are designed to function independently, interacting with environments, coordinating with other agents, and adapting to dynamic situations without human intervention. A notable example is AutoGen [221], a framework that enables the creation of autonomous agents that can interact with each other to solve tasks and improve through continual interactions. Recent work has further enhanced these systems through MoE architectures, which integrate specialized sub-models ("experts") into multi-agent frameworks to optimize task-specific performance and computational efficiency. For instance, MoE-based coordination allows agents to dynamically activate subsets of experts based on context, enabling scalable specialization in complex environments [222, 223]. Xie et al. [224] explored the role of LLMs in these agentic systems, discussing their ability to facilitate autonomous cooperation and communication between agents in complex environments, and marking an important step toward scalable and self-sufficient AI. By combining MoE principles with multi-agent collaboration, systems can achieve hierarchical decision-making: LLMs act as meta-controllers, routing tasks to specialized agents (e.g., vision, planning, or language experts) while maintaining global coherence.

However, the growing autonomy of such systems underscores the importance of XAI [225] to ensure transparency and trust. XAI has gained prominence as a means to enhance accountability and support ethical AI adoption. By providing insights into model behavior, XAI ensures that even highly autonomous systems remain interpretable and accountable, addressing concerns about their decisions and actions in sensitive and dynamic environments.

Recent advancements in AI have demonstrated the potential of integrating fine-tuning, distillation, and in-context learning to enhance model performance. Huang et al. [226] introduced in-context learning distillation, a novel method that transfers few-shot learning capabilities from large pre-trained LLMs to smaller models. By combining in-context learning objectives with traditional language modeling, this approach allows

smaller models to perform effectively with limited data while maintaining computational efficiency.

Transfer learning [227] has similarly emerged as a foundational technique, enabling pre-trained models to adapt their extensive knowledge to new domains using minimal data. This capability is particularly advantageous in resource-constrained scenarios. Techniques such as feature extraction, where pre-trained model layers are repurposed for specific tasks, and fine-tuning, which involves adjusting the weights of the pre-trained model for new tasks, further illustrate its adaptability.

Complementing these methods, prompt engineering empowers LLMs to perform task-specific functions through carefully designed prompts. Techniques such as CoT prompting [206], zero-shot [228], and few-shot prompting enhance the ability of LLMs to reason and generalize across diverse tasks without extensive retraining [229]. Additionally, knowledge distillation plays a crucial role in optimizing AI models by transferring knowledge from larger, more complex models to smaller, efficient ones [230]. Variants of distillation, such as task-specific distillation, feature distillation, and response-based distillation, further streamline the process for edge computing and resource-limited environments.

Reinforcement learning and its variant RLHF [231], focus on training agents to make sequential decisions in dynamic environments. RLHF further aligns agent behavior with human preferences, fostering ethical and adaptive AI systems. Finally, continuous learning, or lifelong learning, addresses the challenge of adapting AI systems to new data while retaining previously learned knowledge, ensuring AI remains effective in changing environments [232].

These techniques represent prominent directions in contemporary generative AI, each contributing to the resolution of complex challenges across diverse applications. The discussion of these methods through an NSAI lens, developed in the following sections, provides a structured perspective on their synergies and practical relevance, without implying that all of them constitute neuro-symbolic systems by default or that their relationship to NSAI is equally established in the literature.

2.4.2 Classification of AI Technologies within NSAI Architectures

This section discusses how selected generative AI techniques may be interpreted in relation to the eight NSAI architectures, with attention to both their underlying integration principles and their practical use cases. The aim is not to impose definitive taxonomic assignments, but rather to clarify which techniques are already instantiated in the literature as neuro-symbolic systems and which are better understood as partial, approximate, or prospective alignments, as summarized in Figure 2.6.

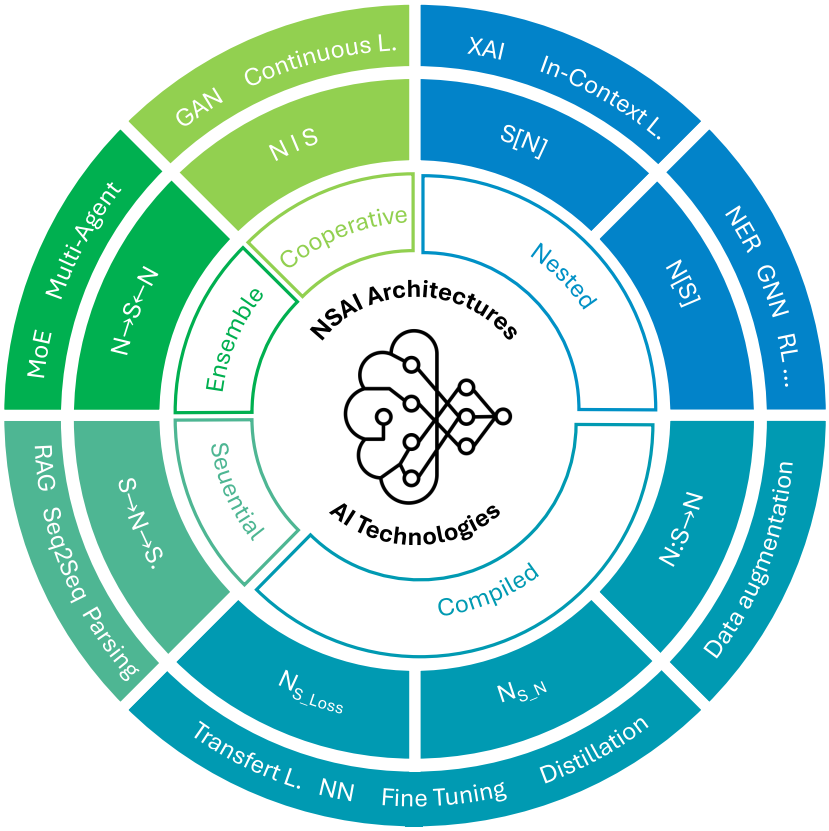

Figure 2.6: Classification of AI technologies into NSAI architectures, reproduced from [208].

The Sequential Paradigm: From Symbolic to Neural Reasoning

Techniques like RAG, GraphRAG, and Seq2Seq models (including LLMs, e.g., GPT [103]) often align with this paradigm because they encode symbolic inputs (e.g., text or structured information) into neural representations to perform complex transformations, and then produce outputs that are either discrete symbolic sequences (natural language) or, in some pipelines, explicitly structured symbolic forms. Similarly, semantic parsing benefits from this framework by leveraging NNs to uncover latent patterns in symbolic inputs and generating interpretable symbolic conclusions. For instance, RAG-Logic proposes a dynamic example-based framework using RAG to enhance logical reasoning capabilities by integrating relevant, contextually appropriate examples [233]. It first encodes symbolic input (e.g., logical premises) into neural representations using the RAG knowledge base search module. Neural processing occurs through the translation module, which transforms the input into formal logical formulas. Finally, the fix module ensures syntactic correctness, and the solver module evaluates the logical consistency of the formulas, decoding the results back into symbolic output. This process maintains the interpretability of symbolic reasoning while leveraging the power of NNs to improve flexibility and performance.

The Nested Paradigm: Embedding Symbolic Logic in Neural Systems

In-context learning mechanisms, such as few-shot learning and CoT reasoning, can instantiate different nested paradigms depending on how symbolic structure and neural generation are combined. In settings where an explicit symbolic framework (e.g., rules, knowledge bases, or logical templates) structures the reasoning task and the neural model operates within this framework, these mechanisms can be viewed through the lens of the *Symbolic[Neuro]* paradigm. XAI techniques can also be associated with this category, since a significant subset of post-hoc methods combine neural feature extraction with symbolic or rule-based structures to generate human-interpretable explanations [234–236].

Zhang et al. [237] presented a framework in which symbolic reasoning is enhanced by NNs. CoT is used as a method to generate prompts that combine symbolic rules with neural reasoning. For example, the task of reasoning about relationships between entities, such as "Joseph's sister is Katherine" is approached by generating a reasoning path through CoT. The reasoning path is structured using symbolic rules, such as $Sister(A, C) \leftarrow Brother(A, B) \wedge Sister(B, C)$, which define the relationships between entities. These rules are then used to form CoT prompts that guide the model through the reasoning steps. The NN processes these prompts, performing feature extraction and probabilistic inference, while the symbolic system (including the knowledge base and logic rules) orchestrates the overall reasoning process. In this approach, the symbolic framework is the primary system for structuring the reasoning task, and the NN acts as

a subcomponent that processes raw data and interprets the symbolic rules in the context of the query.

Methods like GNNs, NER, link prediction, and relation extraction fit into the *Neuro[Symbolic]* category. These methods often leverage symbolic relationships, such as ontologies or graphs, as integral components to enhance neural processing. In addition, they integrate symbolic reasoning subroutines to perform higher-order logical operations, enforce consistency, or derive insights from structured representations. RL and RLHF exemplify this approach, where symbolic reasoning is integrated into the reward shaping and policy optimization stages to enforce logical constraints, ensure decision-making consistency, and align neural outputs with human-like decision-making criteria. For instance, NeuSTIP [238] exemplifies this approach by combining GNN-based neural processing with symbolic reasoning to tackle link prediction and time interval prediction in temporal knowledge graphs (TKGs). NeuSTIP employs temporal logic rules, extracted via "all-walks" on TKGs, to enforce consistency and strengthen reasoning. By embedding symbolic reasoning subroutines into the neural framework, NeuSTIP demonstrates how such models can effectively derive structured insights and perform reasoning under constraints. Complementarily, Ma et al.'s Neural Symbolic Reinforcement Learning (NSRL) framework [239] encodes states and policies in first-order logic and uses a differentiable reasoning module to learn symbolic rules, illustrating a nested variant in which a predominantly symbolic RL agent incorporates a neural subcomponent for rule learning and multi-step reasoning.

The Cooperative Paradigm: Iterative Interaction Between Neural and Symbolic Modules

Rather than generic GANs, cooperative training schemes inspired by the GAN setting can instantiate the cooperative paradigm when at least one of the components is explicitly tied to symbolic rules or logic-based constraints. In such configurations, a neural generator proposes candidate solutions while a symbolic or rule-based critic evaluates them against formal constraints or domain knowledge, providing structured feedback that guides the generator. The resulting iterative feedback loop between neural generation and symbolic evaluation fits the *Neuro* | *Symbolic* framework, as it combines adaptive learning with constraint-driven refinement.

Continuous learning is inherently facilitated in this cooperative paradigm, in which both neural and symbolic modules undergo iterative refinement to enhance their performance over time. In this paradigm, NN continuously updates its internal representations and model parameters in response to feedback derived from the symbolic module's logical inferences and constraint evaluations. This adaptive process enables the NN to generalize more effectively across diverse and evolving data distributions. Simultaneously, the symbolic module is not static; it dynamically revises its rule-based

reasoning mechanisms and knowledge structures by integrating new information extracted from the NN's learned representations. An example of this approach in reinforcement learning is the detect-understand-act (DUA) framework [240], where neural and symbolic components collaborate iteratively to solve tasks in a structured manner. In DUA, the detect module uses a traditional computer vision object detector and tracker to process unstructured environmental data into symbolic representations. The understand component, which integrates symbolic reasoning, processes this data using answer set programming (ASP) and inductive logic programming (ILP), ensuring that decisions align with symbolic rules and constraints. The act component, composed of pre-trained reinforcement learning policies, acts as a feedback loop to refine the symbolic representations, allowing the system to converge on solutions that meet predefined criteria.

The Compiled Paradigm: Embedding Symbolic Reasoning Within Neural Computation

Approaches such as model distillation, fine-tuning, pre-training, and transfer learning are not neuro-symbolic methods by themselves. However, they can align with the $Neuro_{Symbolic}$ approach when symbolic constraints or objectives (e.g., logical consistency, relational structures) are explicitly integrated into the neural training process—either through the loss function or via neuron-level mechanisms (e.g., constrained activations). In such settings, the symbolic component is effectively compiled into the neural model during training, enabling the resulting network to adhere to predefined rules and support structured forms of reasoning. Parametric activation functions that implement fuzzy or logic-like operators provide one example of neuron-level compilation: as surveyed by Pusztaházi et al. [241], certain neuro-fuzzy activations can be interpreted as learnable AND/OR operators, embedding symbolic structure directly into the activation behaviour of neurons while remaining differentiable. Thus, these techniques become part of the compiled paradigm only when they explicitly transfer or enforce symbolic knowledge within the neural architecture. Mendez-Lucero and al. [242] exemplify this perspective by embedding logical constraints directly into the loss function. Their distribution-based method incorporates symbolic logic, such as propositional formulas and first-order logic, into the learning process. These constraints are encoded as a distribution and incorporated into the optimization procedure using measures such as the Fisher–Rao distance or Kullback–Leibler divergence, guiding the neural network to satisfy symbolic requirements. A related line of work is the Semantic Loss framework of Arrotta et al. [243], which introduces a logic-based loss term for context-aware human activity recognition; the loss penalizes predictions that violate symbolic constraints expressing activity patterns, thereby steering the network towards solutions that are both accurate and consistent with domain knowledge. Such integrations ensure that the model learns from data while simultaneously inheriting explicit logical structure, thereby strengthening the connection

between neural learning and symbolic reasoning and making the approach suitable for use within regimes such as knowledge distillation, fine-tuning, pre-training, and transfer learning.

Rather than generic data augmentation, we focus here on rule-constrained synthetic data generation, which leverages the *Neuro:Symbolic* $\rightarrow$ *Neuro* approach. In this setting, symbolic reasoning is used to generate logically valid synthetic examples that augment the training corpus of neural models. By producing high-quality labeled data through logical inference, symbolic solvers effectively compile structured knowledge into the data distribution, which is then absorbed by the neural network during training. This method seamlessly integrates the precision and structure of symbolic logic with the scalability and adaptability of NNs, resulting in more robust and efficient learning. Li et al. [244] proposed a methodological framework that exemplifies this approach. Their framework systematically generates labeled data pairs (x, y), where y is derived from x through symbolic transformations based on formal logical rules. The process begins with the formalization of mathematical problems in a symbolic space using mathematical solvers, ensuring the logical validity of the generated instances. Subsequently, mutation mechanisms are applied to diversify the examples, including simplification strategies (reducing the complexity of expressions) and complication strategies (adding constraints or variables to increase difficulty). Each transformation results in a new problem–solution instance with its corresponding solution, forming labeled pairs (x', y') that enrich the training corpus with controlled complexity levels and logically grounded synthetic data. In a complementary vein, Llugiqi et al. [245] use knowledge graphs as a symbolic backbone to perform semantic data augmentation for tabular prediction tasks: symbolic relations in the graph are first exploited to derive enriched semantic descriptors (e.g., graph-based embeddings and distance features), which are then used to train purely neural predictors.

The Fibring Paradigm: Connecting Neural Models Through Symbolic Constraints

Multi-agent AI align with this paradigm by leveraging symbolic functions to facilitate communication and coordination between agents (i.e., neural models). Symbolic reasoning mediates interactions, enforces constraints, and ensures alignment with predefined rules, while neural components adapt and learn from collective behaviors. This interplay enables robust and scalable problem-solving in complex, multi-agent environments. Belle et al. [246] explored how the combination of symbolic reasoning and agents can enable the development of advanced systems that are closer to human-like intelligence. They discusses how symbolic reasoning can mediate communication between agents, ensuring that they adhere to predefined rules while allowing the neural components to learn and adapt from collective behaviors. This directly aligns with the fibring paradigm, where multiple NNs are interconnected via a symbolic fibring function, enabling them

to collaborate and share information in a structured manner.

MoE models are neural architectures in which multiple specialized subnetworks ("experts") are trained jointly, and a learned gating/router network dynamically selects a small subset of experts for each input, enabling conditional computation and scalable capacity (i.e., large parameter counts with limited active compute). In modern LLMs such as Mixtral 8×7B, each Transformer layer contains several feed-forward experts, and a purely neural router chooses (e.g., top-2) experts per token and combines their outputs, without any explicit symbolic mediator or rule-based coordination layer [247]. DeepSeek-R1 builds on a large MoE base model and improves reasoning via reinforcement learning, but the interaction among experts remains governed by learned neural routing rather than formal symbolic constraints [248]. Therefore, standard MoE systems are not neuro-symbolic by default; they can only be interpreted as instantiations of a fibring-like neuro-symbolic paradigm in variants where expert routing or aggregation is explicitly controlled or constrained by symbolic rules, logical constraints, or knowledge-based controllers [249].

2.5 Evaluation of NSAI Architectures

Ensuring the reliability and practical applicability of NSAI architectures requires a systematic evaluation across multiple well-defined criteria. Such an evaluation not only identifies the strengths and limitations of the architectures but also fosters trust among stakeholders by emphasizing interpretability, transparency, and robustness—qualities essential in domains such as healthcare, finance, and autonomous systems. Moreover, a rigorous assessment provides benchmarks that can stimulate the development of next-generation models. The following sections delineate the key criteria for evaluating NSAI architectures.

2.5.1 Core Criteria

The evaluation framework for NSAI architectures is built upon several fundamental criteria: generalization, scalability, data efficiency, reasoning, robustness, transferability, and interpretability. Each criterion is elaborated below.

Generalization: Generalization is defined as the capability of a model to extend its learned representations beyond the training dataset to perform effectively in novel or unforeseen situations. This criterion is evaluated based on:

– *Out-of-distribution (OOD) performance* [250]: The ability to maintain performance on data that deviate from the training distribution.

- *Contextual flexibility* [251]: The capacity to adapt seamlessly to changes in context or domain with minimal retraining.
- *Relational accuracy* [252]: The capacity to identify and exploit relevant relationships in data while mitigating the influence of spurious correlations.

Scalability: Scalability assesses the performance of NSAI architecture under increasing data volumes or computational demands. A scalable system should remain efficient and effective as it scales. Key aspects include:

- *Large-scale adaptation* [253]: The ability to process and derive insights from massive datasets.
- *Hardware efficiency* [254]: Optimal utilization of available computational resources, enabling operation on both low-resource devices and high-performance infrastructures.
- *Complexity management* [255]: The ability to accommodate increased architectural complexity without compromising speed or deployment feasability.

Data Efficiency: Data efficiency measures how effectively an NSAI model learns from limited data, an important consideration in scenarios where labeled data are scarce or expensive to obtain. This criterion encompasses:

- *Data reduction* [256]: Achieving high performance with a reduced amount of training data.
- *Data optimization* [257]: Maximizing the utility of available data (both labeled and unlabeled), potentially through semi-supervised learning techniques.
- *Incremental adaptability* [258]: The capacity to incorporate new data progressively without undergoing complete retraining.

Reasoning: Reasoning reflects the model's ability to analyze data, extract insights, and draw logical conclusions. This criterion underscores the unique advantage of NSAI architectures, which combine neural learning with symbolic reasoning. This criterion evaluates:

- *Logical reasoning* [259]: The systematic application of explicit rules to derive precise and consistent inferences.
- *Relational understanding* [260]: The comprehension of complex relationships between entities within the data.
- *Cognitive versatility* [261]: The integration of various reasoning paradigms (e.g., deductive, inductive, and abductive reasoning) to tackle diverse challenges.

Robustness: Robustness measures the system's reliability and resilience to disruptions, including noisy data, adversarial inputs, or dynamic environments. The evaluation considers:

- *Resilience to perturbations/anomalies* [262]: The ability to sustain stable performance despite the presence of noise or adversarial data.
- *Adaptive resilience* [263]: The maintenance of functionality under changing or unpredictable conditions.
- *Bias resilience* [264]: The effectiveness in detecting and correcting biases to ensure fairness and accuracy in predictions.

Transferability: Transferability assesses the model's ability in applying learned knowledge to new contexts, domains, or tasks. This is essential for reducing the effort and time required for model adaptation. Its evaluation involves:

- *Multi-domain adaptation* [265]: The capacity to generalize across diverse domains with minimal modifications.
- *Multi-task learning* [266]: The capability to handle multiple tasks simultaneously through shared knowledge representations.
- *Personalization* [267]: The adaptability of the model to meet specific user or application requirements with limited additional effort.

Interpretability: Interpretability evaluates the model's ability to explain its decisions, ensuring transparency and trust in NSAI systems. This criterion assesses:

- *Transparency* [268]: The clarity with which the internal mechanisms and decision processes of the model are revealed.
- *Explanation* [269]: The ability to provide comprehensible justifications for predictions or decisions.
- *Traceability* [270]: The capability to reconstruct the sequence of operations and factors that contributed to a given outcome.

By systematically addressing these criteria, researchers and practitioners can ensure that NSAI architectures are not only scientifically rigorous but also practical, adaptable, and ready for real-world applications. This evaluation framework not only facilitates continuous improvement and innovation but also supports the broad adoption of NSAI systems across various industries and application domains.

2.5.2 Evaluation Methodology

The evaluation of NSAI architectures was conducted using a systematic approach to ensure a robust and transparent assessment of their performance across multiple criteria. This process relied on three key sources: scientific literature, empirical findings, and an analysis of the design principles underlying each architecture. **Table 2.1** summarizes the relevant research works associated with the identified NSAI architectures in Section 3. The scientific literature served as the primary source of qualitative insights, offering detailed analyses of the strengths and limitations of various architectures. Foundational research and state-of-the-art studies provided evidence of performance in areas such as scalability, reasoning, and interpretability, helping to guide the evaluation. Additionally, empirical results from experimental studies and benchmarks offered quantitative data, enabling objective comparisons across architectures. Metrics such as accuracy, adaptability, and efficiency were particularly valuable in validating the claims made in research papers. The design principles of each technology were also considered to understand how neural and symbolic components were integrated. This analysis provided insights into the inherent capabilities and constraints of each architecture, such as its suitability for handling complex reasoning tasks, scalability to large datasets, or adaptability to dynamic environments.

For each main criterion, architectures are rated on a four-point scale according to how many of its three sub-criteria using binary (*yes/no*) judgements grounded in the above methodology. If all three sub-criteria are met, the rating is *High*, reflecting consistently exceptional performance; if two are met, the rating is *Medium*, indicating generally satisfactory results with some limitations; if only one sub-criterion is met, the rating is *Low–Medium*, denoting limited strengths; and if none are met, the rating is *Low*, signifying significant weaknesses or inconsistent outcomes. In this way, each architecture receives a quantitative score ranging from 0 to 3, ensuring a balanced and evidence-based evaluation.

This evaluation framework should nevertheless be interpreted with caution. The proposed ratings do not constitute standardized benchmark scores obtained under a shared experimental protocol, but rather a qualitative synthesis of published evidence, architectural design principles, and representative empirical findings. As a result, the "yes/no" judgements and the resulting "High/Medium/Low" summaries should be read as structured comparative indicators rather than as definitive rankings. They are intended to make trade-offs explicit and to support discussion across paradigms, especially in a field where architectures are often evaluated on heterogeneous tasks, datasets, and objectives.

By combining insights from literature, empirical findings, and design analysis, this methodology ensures a balanced and evidence-based evaluation. It provides a clear understanding of the strengths and weaknesses of each architecture, enabling meaningful comparisons and guiding future advancements in NSAI research and applications.

Table 2.1: Set of relevant published NSAI architectures considered in the proposed study.

Architecture	**References**
Symbolic $\rightarrow$ *Neuro* $\rightarrow$ *Symbolic*	[271], [272], [273], [274], [275], [137], [276], [277], [278], [279], [280], [281], [282], [283], [284], [285], [286], [287], [288], [289], [290], [291], [292], [293], [294]
Neuro[Symbolic]	[143], [142]
Symbolic[Neuro]	[141], [295], [296], [297], [298], [299]
Neuro | *Symbolic*	[144], [300], [301], [302], [303], [304], [305]
Neuro $\rightarrow$ *Symbolic* $\leftarrow$ *Neuro*	[306], [139], [246], [248], [247], [197], [198], [199], [200]
Neuro:Symbolic $\rightarrow$ *Neuro*	[140], [307], [308], [309], [310], [311], [312], [313], [314], [315], [316], [147], [317], [318], [319], [320]
$Neuro_{Symbolic_{Loss}}$	[145], [25], [321], [322], [323], [324]
$Neuro_{Symbolic_{Neuro}}$	[325], [146]

2.5.3 Results and Discussion

Tables 2.2–2.3–2.4 provide the resulting qualitative evaluation of NSAI paradigms. Each table reports, for every main criterion, the satisfaction of its three sub-criteria (*yes*/*no*) for the corresponding architectures, together with a summary level (High/Medium/Low) derived directly from the number of satisfied sub-criteria. This organization allows Sequential, Fibring, Cooperative, Nested, and Compiled paradigms to be compared in a consistent and transparent manner with respect to their strengths and limitations. Taken together, the results reveal complementary profiles rather than a single universally optimal design. Sequential architectures perform strongly on reasoning, interpretability, and transferability, but typically remain more limited in scalability and robustness. Cooperative architectures provide rich iterative reasoning and good interpretability, yet their scalability and robustness profiles remain more constrained in large-scale settings. Within the Nested family, *Symbolic[Neuro]* models excel in data efficiency, reasoning, and interpretability but struggle with large-scale adaptation, whereas *Neuro[Symbolic]* variants often improve generalization and robustness by embedding symbolic modules into neural pipelines, while sacrificing some versatility and transferability. Compiled architectures show that embedding symbolic knowledge directly into losses or neuron-level mechanisms can strengthen reasoning and interpretability, although their generalization and multi-domain transfer capabilities remain more task-dependent.

Within this qualitative and literature-grounded comparison, the *Neuro* $\rightarrow$ *Symbolic* $\leftarrow$ *Neuro* architecture receives the broadest support across the selected criteria and appears particularly well balanced for scenarios requiring coordination, explicit constraints,

robustness, and interpretability. However, this observation should not be interpreted as a definitive empirical ranking. Rather, it indicates that this architecture offers a comparatively strong profile under the present evaluative lens, while alternative paradigms may remain preferable depending on the task, the integration mechanism, the data regime, and the deployment constraints. This interpretation is consistent with recent work showing that neuro-symbolic multi-agent settings can improve interpretability and probabilistic decision-making under uncertainty [326, 327]. Accordingly, the main contribution of these tables is not to identify a universally superior architecture, but to make explicit the trade-offs that characterize different NSAI integration strategies.

Table 2.2: Comparison of NSAI architectures (Part 1): Sequentiel, Fibring, and Cooperative

Main Criterion	Sub-Criterion	Symbolic Neuro Symbolic	Neuro → Symbolic ← Neuro	Neuro|Symbolic
Generalization	Out-of-dist.	yes [292]	yes [246]	yes [300]
	Continuous flex.	yes [271]	yes [248]	yes [304]
	Relative prec.	yes [286]	yes [139]	no
	Summary	High	High	Medium
Scalability	Large-scale adapt.	yes [137]	yes [200]	yes [302]
	Hardware efficiency	no	yes [247]	no
	Complexity	yes [283]	yes [198]	no
	Summary	Medium	High	Low–Medium
Data Efficiency	Reduction	no	yes [199]	yes [144]
	Optimization	yes [291]	yes [306]	yes [302]
	Incremental adapt.	yes [284]	yes [197]	no
	Summary	Medium	High	Medium
Reasoning	Logical reason.	yes [288]	yes [248]	yes [301]
	Comprehension	yes [289]	yes [198]	yes [305]
	Versatility	yes [279]	yes [139]	yes [303]
	Summary	High	High	High
Robustness	Perturbations	no	yes [248]	yes [302]
	Adaptability	yes [284]	yes [199]	yes [144]
	Bias handling	no	yes [197]	no
	Summary	Low–Medium	High	Medium
Transferability	Multi-domain	yes [272]	yes [198]	yes [304]
	Multi-task	yes [285]	yes [247]	no
	Personalization	yes [277]	yes [197]	no
	Summary	High	High	Low–Medium
Interpretability	Transparency	yes [278]	yes [246]	yes [304]
	Explanation	yes [282]	yes [139]	yes [303]
	Traceability	yes [294]	yes [306]	yes [301]
	Summary	High	High	High

Table 2.3: Comparison of NSAI architectures (Part 2): Nested

Main Criterion	Sub-Criterion	Symbolic[Neuro]	Neuro[Symbolic]
Generalization	Out-of-dist.	no	yes [142]
	Continuous flex.	yes [295]	no
	Relative prec.	yes [299]	no
	Summary	Medium	Low–Medium
Scalability	Large-scale adapt.	no	yes [143]
	Hardware efficiency	no	no
	Complexity	no	no
	Summary	Low	Low–Medium
Data Efficiency	Reduction	yes [296]	yes [142]
	Optimization	yes [297]	yes [142]
	Incremental adapt.	yes [141]	no
	Summary	High	Medium
Reasoning	Logical reason.	yes [295]	yes [143]
	Comprehension	yes [298]	no
	Versatility	yes [299]	no
	Summary	High	Low–Medium
Robustness	Perturbations	yes [141]	yes [142]
	Adaptability	yes [298]	no
	Bias handling	no	no
	Summary	Medium	Low–Medium
Transferability	Multi-domain	yes [298]	no
	Multi-task	no	no
	Personalization	no	no
	Summary	Low–Medium	Low
Interpretability	Transparency	yes [295]	yes [142]
	Explanation	yes [296]	yes [143]
	Traceability	yes [299]	yes [142]
	Summary	High	High

Table 2.4: Comparison of NSAI architectures (Part 3): Compiled

Main Criterion	Sub-Criterion	Neuro:Symbolic → Neuro	$\text{Neuro}_{\text{Symbolic}_{\text{Loss}}}$	$\text{Neuro}_{\text{Symbolic}_{\text{Neuro}}}$
Generalization	Out-of-dist.	yes [320]	no	no
	Continuous flex.	no	no	no
	Relative prec.	no	no	no
	Summary	Low–Medium	Low	Low
Scalability	Large-scale adapt.	yes [316]	yes [324]	yes [146]
	Hardware efficiency	yes [140]	yes [145]	yes [328]
	Complexity	no	no	no
	Summary	Medium	Medium	Medium
Data Efficiency	Reduction	no	yes [25]	yes [325]
	Optimization	yes [310]	yes [322]	yes [329]
	Incremental adapt.	no	no	no
	Summary	Low–Medium	Medium	Medium
Reasoning	Logical reason.	yes [313]	yes [145]	yes [325]
	Comprehension	yes [317]	yes [321]	yes [146]
	Versatility	yes [312]	no	no
	Summary	High	Medium	Medium
Robustness	Perturbations	yes [308]	no	yes [330]
	Adaptability	yes [315]	no	no
	Bias handling	yes [319]	no	yes [331]
	Summary	High	Low	Medium
Transferability	Multi-domain	no	no	no
	Multi-task	no	yes [324]	yes [325]
	Personalization	no	no	no
	Summary	Low	Low–Medium	Low–Medium
Interpretability	Transparency	yes [309]	yes [323]	yes [146]
	Explanation	yes [318]	yes [324]	yes [325]
	Traceability	yes [307]	yes [321]	yes [146]
	Summary	High	High	High

2.6 Neuro-Symbolic AI Architectures for 4D Printing: Proposed Applications

In recent years, artificial NNs have been explored in the 4D printing domain to tackle challenges such as predicting material response under stimuli, learning complex shape evolution dynamics, and supporting the inverse design of shape-changing structures [15, 15, 16, 19, 25, 61, 69, 332]. However, despite their effectiveness in pattern recognition and data-driven modeling, NN approaches face key limitations when applied to 4D printing. First, they often require large, high-quality datasets to achieve meaningful generalization, which is problematic in emerging fields where experimental data are sparse or costly to obtain [26]. Second, the black-box nature of NNs hinders the interpretability of results, limiting their adoption in safety-critical design contexts such as biomedical or aerospace applications. Third, NNs typically struggle with tasks that demand reasoning over structured knowledge, including the symbolic relationships between materials, geometries, and functional outcomes. These shortcomings impede the development of explainable and trustworthy design workflows, especially when targeting multifunctional or multi-material 4D-printed systems.

The integration of NSAI with 4D printing technology presents unprecedented opportunities to address the complex challenges inherent in designing, optimizing, and manufacturing stimuli-responsive structures. By combining the learning capabilities of NNs with the logical reasoning and interpretability of symbolic AI, researchers can develop robust frameworks that address the multidisciplinary demands of 4D printing [11]. This section proposes specific applications of various NSAI architectures throughout the 4D printing workflow, from material discovery to transformation control and structural optimization.

2.6.1 Sequential Architecture Applications

Sequential architecture (Symbolic $\rightarrow$ Neuro $\rightarrow$ Symbolic) offer promising applications in material property mapping for 4D printing. By first encoding material compositions and structural parameters into symbolic representations, then processing these through NNs, and finally decoding outputs back into interpretable symbolic form, researchers can establish explicit relationships between formulations and transformation behaviors. A sequential NSAI system for the design of shape memory polymers (SMPs), liquid crystal elastomers (LCEs), and hydrogels would begin with symbolic representations of polymer composition, crosslinking density, and thermal properties, encode these into continuous vectors for neural processing, utilize neural networks to predict complex transformation patterns, and conclude by decoding predictions into symbolic rules describing temperature-dependent folding angles and recovery rates. This approach maintains scientific traceability while leveraging NNs' capacity to model complex structure-property

relationships that are difficult to formulate using purely symbolic methods. The resulting framework would facilitate rapid design iteration with interpretable outputs that can guide experimental validation and material refinement.

2.6.2 Nested Architectures for Multi-scale Optimization

The nested Symbolic[Neuro] architecture presents significant advantages for rule-based design systems that incorporate neural material prediction. In this configuration, a symbolic reasoning framework establishes design constraints based on application requirements, while embedded NNs predict material distributions that achieve desired transformation properties. For instance, an expert system for designing 4D-printed medical stents could utilize symbolic reasoning to establish design constraints based on clinical requirements, employ NN subcomponents to predict material distribution patterns that achieve desired transformation properties, and evaluate neural predictions against clinical constraints to iteratively refine the design. This architecture maintains interpretable decision logic while leveraging NNs' pattern recognition capabilities for complex material behavior prediction.

Conversely, the Neuro[Symbolic] architecture enables generative design capabilities by embedding physical constraint solvers within neural frameworks. A design system for 4D-printed adaptive structures could leverage a generative NN to propose innovative material distributions, activate embedded symbolic solvers to verify physical feasibility when faced with complex transformation sequences, and integrate results back into the neural framework to guide subsequent design iterations. This approach would enable the exploration of novel design spaces while ensuring physical feasibility, effectively reducing the gap between computational design and manufacturability that currently hampers innovation in 4D printing applications.

2.6.3 Cooperative Architecture for Transformation Control

Cooperative NSAI architectures (Neuro | Symbolic) present compelling solutions for interactive transformation planning systems in 4D printing. Through iterative refinement of predictions based on physical models, these systems enable precise control of complex transformation sequences that are difficult to model using either purely data-driven or analytical approaches. A cooperative system for designing multi-stage transformation sequences would utilize NNs to process visual information of printed structures during initial transformation, employ symbolic reasoning to evaluate transformation against desired geometric goals and physical constraints, identify discrepancies between predicted and observed behavior, and continue this feedback loop until achieving the desired transformation sequence. This architecture is particularly suitable for applications requiring

precise spatiotemporal control of transformation, such as soft robotics and adaptive medical devices, where transformation pathways are as critical as final configurations.

2.6.4 Compiled Architectures for Physics-Informed Design

The incorporation of physical laws directly into neural network architectures through compiled approaches offers significant advantages for 4D printing applications. A physics-informed material design system utilizing $\text{Neuro}_{\text{Symbolic}_{\text{Loss}}}$ functions would enable more physically accurate prediction of material behavior by penalizing predictions that violate fundamental principles. For instance, a design system for 4D-printed hydrogel structures could employ NNs to predict swelling behavior based on material composition while incorporating physics-based loss terms that enforce conservation of mass, diffusion dynamics, and mechanical equilibrium constraints. This approach ensures that model predictions remain physically plausible even for novel material combinations or environmental conditions, substantially reducing the need for extensive experimental validation that currently limits exploration of new material formulations.

Similarly, constraint-aware material distribution optimization can be achieved through $\text{Neuro}_{\text{Symbolic}_{\text{Neuro}}}$ architectures, where traditional activation functions are replaced with logic-based elements that enforce manufacturing constraints. A voxel-based material distribution optimizer with symbolic neurons would enforce printability rules like minimum feature size, support requirements, and material interface limitations directly within the optimization process, eliminating post-processing steps and streamlining the transition from computational design to fabrication. This integration of manufacturing constraints at the neuron level represents a significant advancement over current approaches that typically separate design optimization from manufacturability verification.

The Neuro:Symbolic $\rightarrow$ Neuro approach addresses the critical challenge of limited experimental data in 4D printing through supervised training with synthetic data generation. A predictive model for LCEs behavior could utilize symbolic reasoning based on physical models to generate a comprehensive dataset of material compositions and corresponding transformation behaviors. The resulting neural model would combine the physical accuracy of analytical models with the computational efficiency of neural inference, enabling rapid exploration of material design spaces that would be prohibitively expensive to characterize experimentally.

2.6.5 Fibring Architecture for Multi-domain Optimization

The fibring architecture offers powerful capabilities for multi-scale material and structure optimization in 4D printing by connecting specialized neural networks through symbolic relationships. A comprehensive design system utilizing this approach would deploy specialized NNs for different aspects of the problem (molecular-scale material proper-

ties, meso-scale structural behavior, and macro-scale system performance), connect these networks through symbolic fibring functions that ensure consistency across scales, and resolve contradictions between predictions at different scales through the symbolic component. This architecture addresses the fundamental multi-scale challenge in 4D printing, where molecular-level interactions ultimately determine system-level performance, but direct modeling across all scales remains computationally intractable. By decomposing the problem into specialized networks while maintaining consistency through symbolic relationships, researchers can achieve simultaneous optimization of material composition and structural configuration.

The proposed applications of NSAI architectures in 4D printing demonstrate the potential for significant advancements throughout the material-design-manufacturing workflow. Each architecture addresses specific challenges inherent in 4D printing: sequential architectures provide interpretable material-property relationships; nested architectures balance creativity with physical constraints; cooperative architectures enable precise transformation control; compiled architectures integrate physical laws and manufacturing constraints; and fibring architectures connect predictions across multiple scales and domains. By systematically applying these approaches, researchers can accelerate innovation while maintaining scientific interpretability, potentially reducing the current gap between fundamental research and practical applications in 4D printing technology.

2.7 Conclusion

This study examined several NSAI architectures through a common conceptual framework and compared them across criteria including generalization, scalability, data efficiency, reasoning, robustness, transferability, and interpretability. The resulting analysis highlights the distinct strengths and limitations of each architecture and supports a more structured discussion of their potential roles in real-world applications. Among the architectures considered, *Neuro* $\rightarrow$ *Symbolic* $\leftarrow$ *Neuro* appears especially promising within the scope of our qualitative comparison, particularly for scenarios that require coordination across multiple components, explicit reasoning under constraints, and robust decision-making. However, this observation remains interpretive and literature-grounded rather than empirically definitive, and it should not be read as establishing universal superiority over other NSAI paradigms.

More broadly, this work argues that NSAI architectures are best understood as complementary integration strategies rather than as competitors in a single ranking. Their relevance depends on the form of neural-symbolic interaction required by the task, the level of interpretability sought, and the operational constraints of the target domain. Future work should therefore focus on comparative benchmarks under shared protocols,

clearer reporting standards for architectural claims, and more explicit distinctions between established neuro-symbolic instantiations and prospective integrations, particularly in the rapidly evolving context of generative AI.

This architectural analysis provides the conceptual basis for the computational contributions that follow. The next chapter begins with the knowledge-engineering dimension of the thesis by addressing how heterogeneous and rapidly evolving 4D-printing knowledge can be automatically extracted, structured, and continuously enriched through the combination of ontological representations, retrieval mechanisms, and multimodal language models.

Chapter 3

Structuring Scientific Knowledge for Smart Materials and 4D Printing: An Ontology Enrichment Framework Using Retrieval-Augmented Multimodal Large Language Models

3.1 Introduction

The scientific landscape of 4D printing is both rapidly evolving and inherently multidisciplinary, encompassing fields such as materials science, chemistry, mechanical engineering, process engineering, and biomimicry [10]. Since its inception in 2013, the field has experienced exponential growth, with more than 3,500 publications and an estimated annual growth rate of approximately 40%, according to the Web of Science database [34, 333, 334]. Key challenges in advancing 4D printing include improving the printability of smart materials, enhancing their mechanical and actuation performance, promoting safe and sustainable deployment, and ensuring reliability under cyclic stimuli and real-world conditions [10]. These challenges can be considered as interdependent, especially when designing and developing practical 4D-printed systems, where trade-offs between material properties, process parameters, and functional requirements must be carefully balanced [10]. While this diversity has accelerated innovation, it has also led to fragmented and rapidly evolving bodies of knowledge, therefore making it increasingly difficult to organize, reuse, and reason over existing research in a coherent manner.

In such a context, ontologies provide a formalized approach to capturing domain-specific knowledge. Several efforts have attempted to address knowledge organization in materials science through ontologies, including European Materials Modelling Ontology

(EMMO) [335], Materials Design Ontology (MDO) [336], Materials Ontology (MatOnto) [337], Materials Web Ontology Language ontology (MatOWL) [338], NanoParticle Ontology [339], eNanoMapper ontology [340], Metallic Materials Ontology (MMOY), derived from Yet Another Great Ontology (YAGO) [341], Materials and Molecules Basic Ontology (MAMBO) [342], Dislocation Ontology [343], and Platform Material Digital ontology (PMD) [344]. In particular, MDO enables semantic querying across diverse materials databases like the Materials Project [345], the Open Quantum Materials Database (OQMD) [346], the Novel Materials Discovery (NOMAD) [347], and the Automatic Flow Library for Materials Discovery (AFLOWLIB) [348], thus offering a unified framework for navigating the vast landscape of materials data. These ontologies provide structured, interpretable representations and support semantic querying across heterogeneous materials database. However, they remain largely static, require manual knowledge engineering, and struggle to adapt to fast-growing, interdisciplinary domains such as 4D printing. Among ongoing initiatives in material formalization [349] and AM knowledge capture [350, 351], taxonomic classifications [352] and design-oriented ontologies such as HERMES – an ontology integrates 3- and 4-dimensionalism theories to ensure semantic coherence across the involved disciplines and abstraction levels – [14, 353], offer valuable conceptual foundations but remain limited in scalability, automation, and continuous knowledge update. Consequently, no existing ontology adequately supports the dynamic and cross-disciplinary knowledge evolution required in 4D printing. This limitation is not only related to the conceptual coverage of existing ontologies, but also to the difficulty of continuously updating them from rapidly expanding scientific literature, heterogeneous datasets, and multimodal technical evidence.

To complement these structured but mostly static knowledge representations, artificial intelligence (AI) methods offer new opportunities for extracting, aligning, and updating domain knowledge from unstructured and multimodal sources. In particular, natural language processing (NLP) and machine learning (ML) techniques can support the automatic identification of concepts, relations, and evidence from scientific texts, figures, and datasets [354]. In this context, large language models (LLMs), such as GPT [355] and BERT [84], have transformed NLP by enabling advanced capabilities in translation, question answering, and text generation [356, 357]. Built on the Transformer architecture [85], these models leverage vast datasets to achieve high levels of contextual understanding. Scaling LLMs has led to emergent reasoning capabilities, including in-context learning (ICL) [358], chain-of-thought (CoT) [206], and retrieval-augmented generation (RAG) [359]. These advances mitigate some limitations of conventional AI models by enabling real-time knowledge retrieval and contextual inference. Additionally, the recent development of multimodal LLMs (MLLMs), such as GPT-4V [103] and Gemini [360], has further expanded AI's ability to integrate textual, visual, and symbolic information [361]. These models are particularly relevant for domains like 4D printing,

where information must be captured across modalities to support design synthesis.

Through these advancements, LLMs continue to face fundamental limitations in terms of interpretability and domain specificity. Their probabilistic generation process can result in hallucinations and unreliable outputs, which is particularly problematic in specialized and safety-critical domains such as 4D printing [362]. This contrast highlights a key research gap: ontologies offer rigor, interpretability, and logical consistency, while LLMs provide scalability and adaptability, but existing approaches rarely integrate both in a principled and domain-specific manner. Current ontology learning and enrichment methods either rely heavily on manual intervention or adopt fully automated pipelines that risk semantic drift, bias propagation, and error accumulation [363–365]. Traditional manual annotation remains the most reliable option, yet it is labor-intensive, subjective, and unscalable when dealing with large-scale or rapidly evolving datasets. To address this, semi-automatic approaches such as Phrase2Onto [366] have been proposed. This method applies phrase-level topic modeling to scientific corpora in order to identify recurring linguistic patterns that may correspond to new ontology concepts. Candidate terms are then suggested to experts for integration into the ontology. While effective in surfacing domain-specific terminology, Phrase2Onto still requires significant human input for validation and disambiguation, which can introduce subjectivity and limit scalability. In parallel, fully automated methods using NLP and ML (e.g., Text2Onto) expedite ontology population by extracting entities and relations directly from text [365]. Yet, these approaches remain highly dependent on the quality of training data, raising risks of bias propagation, semantic drift, or shallow representations when applied to domain-specific corpora [367]. More recent text-to-knowledge-graph (text2KG) approaches attempt to construct knowledge graphs (KGs) directly from unstructured text, integrating steps such as entity recognition, relation extraction, canonicalization, and ontology alignment [363, 368]. While these pipelines improve scalability, they often suffer from error accumulation across stages and lack mechanisms for continuous validation. Emerging solutions seek to reduce these limitations. For example, online clustering with LLM agents [369] allows for the incremental grouping of newly extracted knowledge units (entities, relations, or triplets) into coherent clusters without requiring extensive annotated datasets. These systems show potential for handling large-scale, continuously growing corpora, but they face challenges in maintaining logical consistency across clusters and preventing redundancy. Similarly, graph-enhanced LLM frameworks [370, 371] attempt to embed structural constraints from knowledge graphs into the generative process, thereby improving alignment between generated triplets and ontology schemas. Finally, RAG has emerged as a promising solution [189, 359, 372]. By retrieving external domain corpora during the generation process, RAG constrains LLM outputs to be grounded in verified evidence, thereby reducing hallucinations and improving semantic relevance, as also demonstrated in recent ontology-based industrial retrieval-

augmented generation systems [372]. Combined with post-generation filtering strategies (e.g., semantic similarity checks, predicate template matching, redundancy elimination), RAG represents a scalable pathway to bridge unstructured text and structured, domain-aligned ontologies in dynamic scientific fields.

In this chapter, we propose a retrieval-augmented MLLMs framework for active ontology enrichment in 4D printing. By integrating ontological structures with the generative and retrieval capabilities of MLLMs, the proposed framework enables scalable, evidence-grounded extraction of domain knowledge from both structured and unstructured sources. This hybrid approach preserves semantic rigor while allowing continuous adaptation to emerging research, thereby supporting knowledge discovery, reasoning, and reuse in the rapidly evolving field of 4D printing.

3.2 Description of Backbone Ontology For 4D Printing

The rapid advancements in 4D printing have introduced a need for a structured framework to manage and formalize the diverse knowledge involved in designing transformable systems. The HERMES ontology addresses this need by providing a semantic and logical foundation for representing the dynamic behavior of 4D-printed objects [14]. Built upon the Basic Formal Ontology [373] and mereotopology theory [374], this ontology is centered on key 4D printing views, namely AM, material, transformation process, and design and engineering. These views are formalized using interconnected classes, attributes, and relationships. **Figure 3.1** illustrates a part of the 4D printing ontology where views are highlighted.

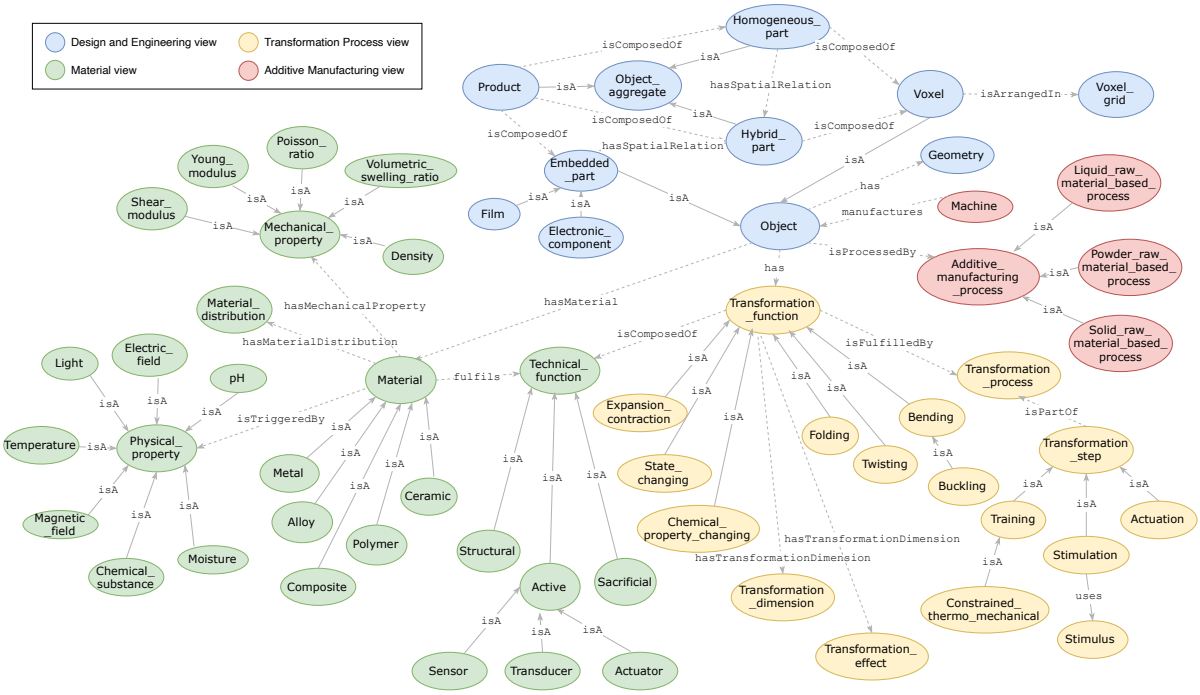

Figure 3.1: Extraction of the main classes of the backbone ontology for 4D printing, structured around four main views: design and engineering, material, AM, and transformation process [14] (Adapted with permission from Elsevier).

In this chapter, the following naming conventions are used as follows: the names of classes are written in capitalised/lower case Arial, the names of attributes and relationships are capitalised/lower case Courier New, while names of instances are in italics Arial. In the design and engineering view, key classes include Object, Homogeneous_Part, which captures single-material parts, and Hybrid_Part, which integrates different materials. These elements are linked by spatiotemporal relationships to enable comprehensive modeling of complex assemblies. The material view categorizes materials based on their Technical_Function – Active, Structural, or Sacrificial – and captures their mechanical and physical properties, as well as their spatial distribution. This view is critical for selecting and designing material combinations that respond predictably to stimuli. In addition, the AM view integrates manufacturing techniques, machines, and process parameters, linking these elements to specific material behaviors. This view provides a framework for planning fabrication steps that align with the requirements of 4D printing. Finally, the transformation process view formalizes the dynamic changes that define 4D printing. It includes Transformation_function such as Bending, Twisting, and Folding, along with property changes like color or stiffness. Transformation sequences are modeled as temporal entities, enabling the logical organization of Training, Stimulation, and Actuation steps required for shape and function changes in the structures. Beyond this broad set of concepts, the ontology also incorporates individuals which are tangible entities for representing classes. For instance, the Transformation_Effect class has the

following individuals, namely *One_way*, *Reversible*, and *Repeatable*. Object properties are then utilized to outline the relationships between individuals and classes, thus creating meaningful links within the ontology. Examples of these links are highlighted in dot-patterned arrows in **Figure 3.1**. The last component of the ontology concerns data properties, which specify attributes or characteristics of the classes, such as "Folding_angle", relevant to transformation function like Folding class.

The HERMES ontology extensively utilizes description logic (DL) rules to enhance its structural and functional capabilities. These DL rules enable precise definition and restriction of classes to support sophisticated inference mechanisms. By implementing these rules, HERMES facilitates key operations such as consistency checking, which verifies the logical coherence of the ontology, as well as subsumption to determine class hierarchies, and realization to identify class memberships. The application of DL rules not only strengthens the ontology's ability to manage and classify complex knowledge effectively but also ensures that it can be interacted with and interpreted by machines. Although structured around philosophical foundations and DL rules to ensure expressivity and reasoning across abstraction levels, this ontology – like most existing material ontologies – suffers from limited capabilities for automated and large-scale learning through enrichment and population. This limitation is particularly critical in emerging and rapidly evolving research domains like 4D printing, where knowledge consolidation is essential to enhance technological readiness levels and reach practical applications.

3.3 Ontology Enrichment Framework

Ontology enrichment enables the enhancement of an existing preliminary ontology by automatically adding new concepts (also considered as knowledge), relationships, and individuals (meaning information or data) to make it more comprehensive and practical for a specific domain or task. To ensure both the enrichment and population of the HERMES ontology, we employ an integrated framework combining information retrieval with advanced text generation capabilities (as illustrated in **Figure 3.2**).

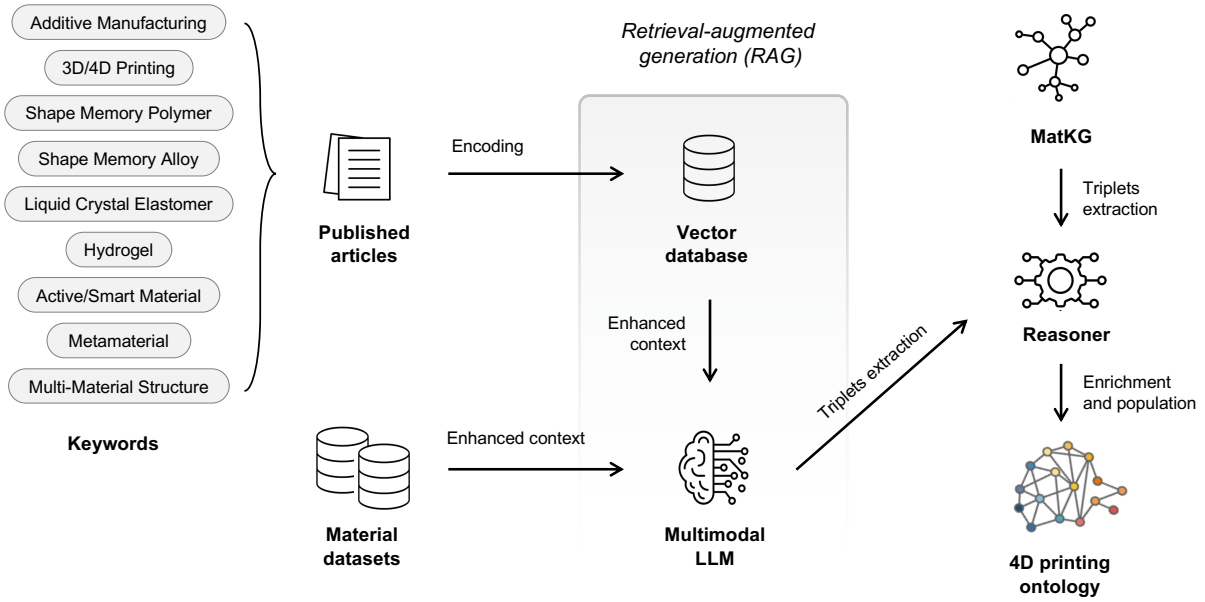

Figure 3.2: Retrieval-augmented ontology construction pipeline. Keywords related to 4D printing pillars are fed into ResearchRabbit to retrieve hundreds of published articles, which are then stored in a vector database. A RAG (retrieval-augmented generation) framework combines this vector database with a MLLM that is enriched by additional datasets to extract relational triplets. The resulting triplets are then combined with triplets from a large knowledge graph database on materials science (MatKG) to populate an existing 4D printing ontology, reproduced from [375].

Initially, we collected a curated corpus of published articles and domain-specific material datasets using targeted keywords. Each published article is split into discrete text segments and extracted figures, while each dataset table is parsed into individual records. All text and image snippets are then encoded using a fine-tuned MLLM into dense vectors and stored in a high-throughput vector index. At inference time, the LLM issues similarity queries against this index to retrieve the top-k relevant passages or images, which it incorporates as "context windows" into its prompts. From the generated and context-aware outputs, a downstream triplet-extraction module identifies candidate subject-predicate-object facts. These facts are merged with existing knowledge from MatKG [376] – one of the largest knowledge graph in materials science – and passed to a symbolic reasoner, which enforces ontology schema constraints, checks for logical consistency, and removes duplicates. Generated triplets are then translated into classes, properties, or instances, thereby populating and enriching the 4D printing ontology in a continuous loop that keeps our knowledge base both up to date and semantically rigorous.

3.4 Ontology Enrichment From Scientific Literature

To enrich the ontology, the process starts with the selection of key terms, ie., "Additive Manufacturing", "3D/4D Printing", "Shape Memory Polymer", "Shape Memory Alloy",

"Liquid Crystal Elastomer", "Hydrogel", "Active/Smart Material", "Metamaterial", and "Multi-Material Structure". By utilizing ResearchRabbit application tool [377], a scholarly publication discovery tool supported by AI, intersections among these keywords were explored to gather over 1,810 published articles. The key correlations extracted are illustrated through a Venn diagram in **Figure 3.3**.

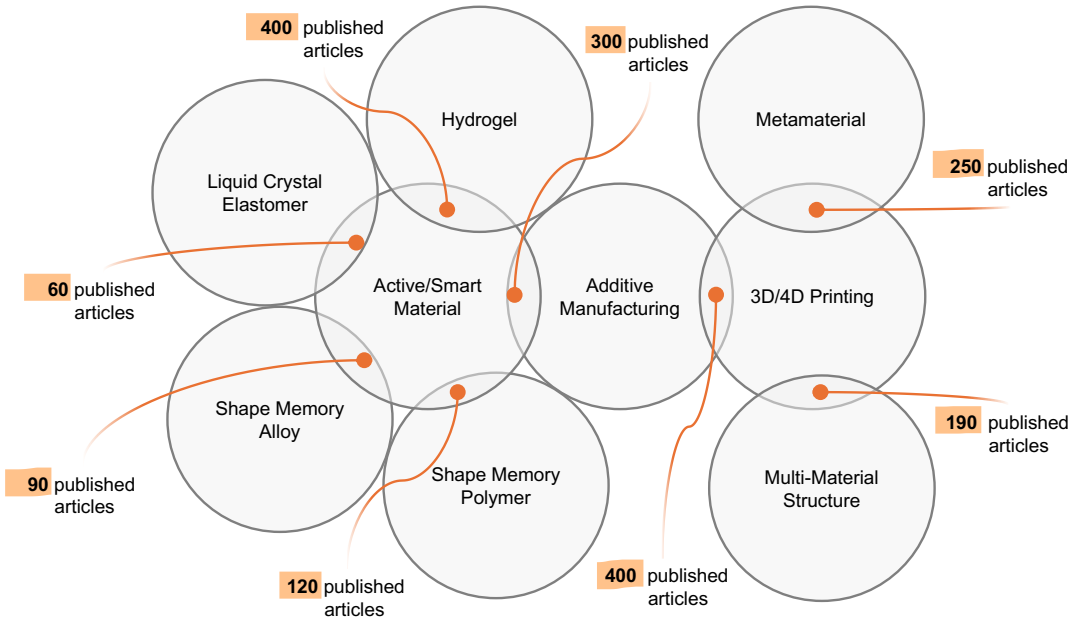

Figure 3.3: Venn diagram illustrating the selected cross-domain keywords used to retrieve relevant articles published in peer-reviewed journals, reproduced from [375].

By coupling the keywords "Additive Manufacturing" with both "3D/4D Printing" as the sole procedural node, the ontology captured the full spectrum of programmable, layer-by-layer processes, including studies involving smart materials that are not explicitly tagged "4D Printing". By linking each major material class ("Shape Memory Polymer", "Shape Memory Alloy", "Liquid Crystal Elastomer", and "Hydrogel") to the "Active/Smart Material" category, we broadened the scope to encompass the majority of experimental work that emphasizes stimuli-responsive functionality over processing terminology. Finally, by pairing "Metamaterial" and "Multi-Material Structure" with "4D Printing", we specifically targeted research on architectured and multi-material constructs whose geometries and properties evolve over time. The corpus of 1,810 published articles spans the period 2009–2024 and covers several peer-reviewed journals in the 4D printing domain. The full list of peer-reviewed journals is provided in Appendix B.

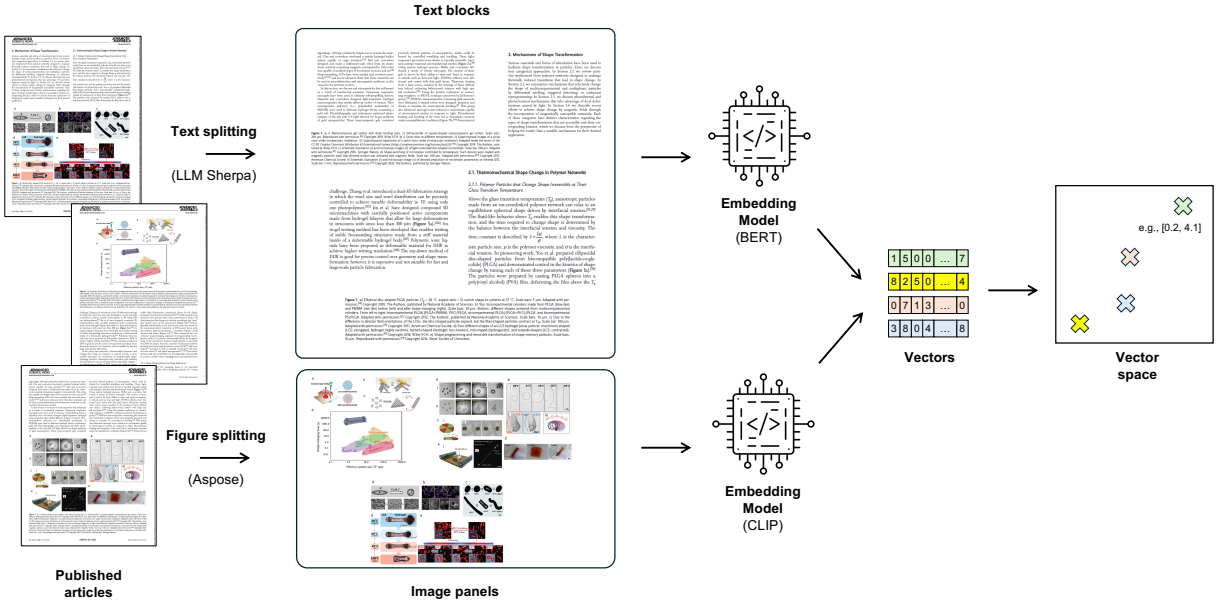

Figure 3.4: Document preprocessing and multimodal embedding pipeline. Multi-page articles were first divided into text blocks (using LLM Sherpa) and figure panels (using Aspose tool). Text segments were encoded with a BERT embedding model, while image panels were encoded with CLIP. The resulting text and image embeddings were then projected into a shared vector space for downstream retrieval and similarity tasks, reproduced from [375].

Then, we split these published articles using tools like LLM Sherpa [378] for robust text extraction and semantic chunking, which divided each paper into coherent chunks based on structural elements. This approach was designed to optimize both semantic completeness and computational efficiency, ensuring that each segment retained meaningful contextual information. Chunk boundaries followed the natural discourse flow (e.g., paragraphs or logical sections) rather than fixed lengths, thereby preserving local coherence throughout the segmentation process. The Aspose tool [379] was used for image extraction in order to isolate each figure into standalone image files. Each token was then embedded using BERT model and CLIP for images. This process converts the content into vector representations, allowing us to store them in a vector space, as shown in **Figure 3.4**.

Subsequently, detailed information concerning structures, material preparation processes, and methodologies was systematically extracted from articles. Leveraging BERT embeddings, queries were encoded within a text vector space to identify the 25 nearest vectors. As illustrated in **Figure 3.5**, selecting one text vector reveals the section highlighted in green as the most relevant to the query. This section is then utilized to identify and retrieve its corresponding relevant image through CLIP embedding model and image vector space.

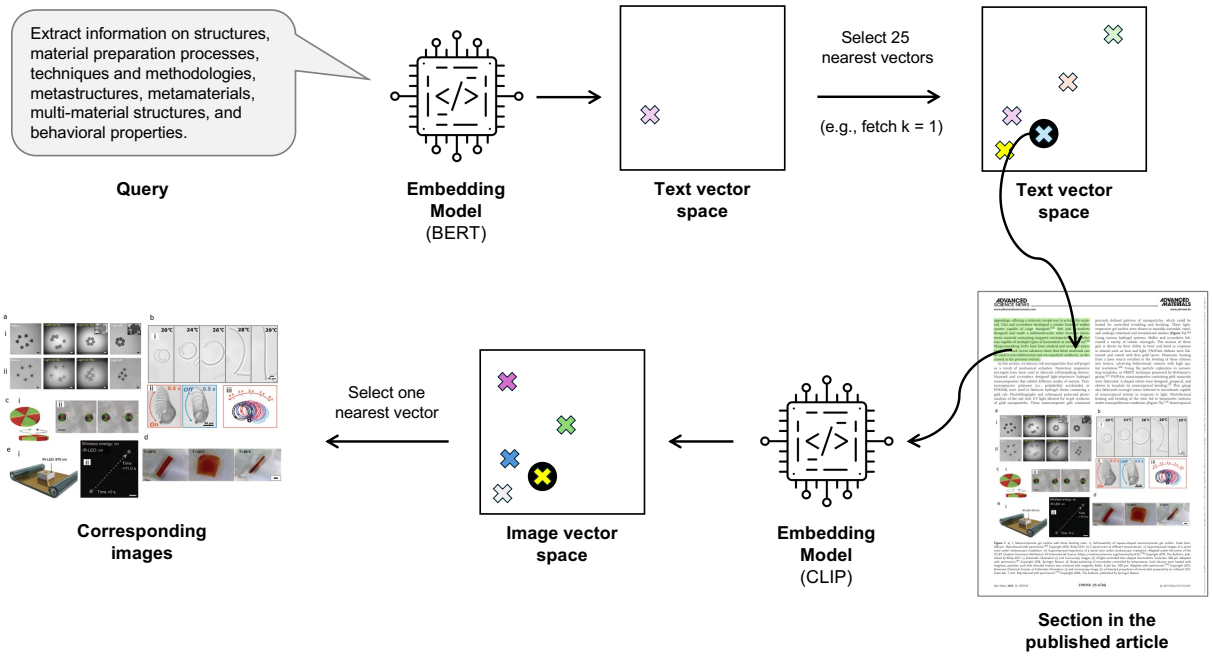

Figure 3.5: Cross-modal retrieval pipeline. A user's textual query was first encoded with BERT embedding model and used to retrieve the top-25 nearest text embeddings from a document corpus. The corresponding articles were then fetched, and all embedded figures within each article were encoded using CLIP embedding model. Finally, the CLIP embeddings are compared to the original query embedding to select the single most relevant image, reproduced from [375].

By pairing each textual section with its corresponding image, we enriched the LLaVA (Large Language-and-Vision Assistant) MLLM's input context [380–382]. The model-selection rationale and implementation details are provided in Appendix B. This RAG process combines information retrieval with text generation, so that it helps to address challenges such as hallucination, outdated knowledge, and opaque reasoning in language models. By incorporating data from external databases, RAG ensures more accurate and credible output, particularly in knowledge-intensive tasks. This integration facilitates ongoing updates and the inclusion of specialized information, therefore making RAG a dynamic solution that combines intrinsic model knowledge with extensive external data.

Initially, we employed the few-shot learning approach [383–385]; however, this method was inefficient due to its time and memory demands related to the excessive length of contexts and often yielding imprecise results. To enhance efficiency, we fine-tuned the LLaVA model specifically for triplet extraction [386–388]. This process involved embedding domain-specific knowledge within the LLaVA model, thus configuring the output format appropriately and ensuring consistent performance without the need for additional tokens. The fine-tuning dataset comprised prompts, relevant images, and targeted responses, enabling the model to align more closely with our extraction logic and generate outputs in the specified format. Moreover, during fine-tuning, we trained the LLaVA model to distinguish ontology classes, object properties, data properties,

and instances, while preserving the hierarchical relations among classes in accordance with ontology web language (OWL) formalism [389, 390]. This approach aims to refine the LLaVA model into a tool adept at identifying triplets specifically related to 4D printing. To construct the fine-tuned dataset, we opted for an approach to build a synthetic dataset by employing ChatGPT-4 [103] (see **Figure 3.6**). Relevant textual sections were extracted from the scientific corpus and paired with the most semantically aligned figure via CLIP-based cross-modal retrieval. Each image-text pair was then converted into a structured prompt comprising the raw section text, the retrieved figure, and explicit extraction constraints. In particular, ChatGPT-4 was instructed to rely exclusively on the provided sections and figures, to avoid introducing any information not supported by these inputs, and to output candidate triples strictly in a normalized subject-predicate-object format aligned with our ontology schema through controlled predicate templates and typing expectations. A one-shot demonstration example was included to stabilize the output structure and reduce format variability. Importantly, ChatGPT-4 outputs were treated as candidate annotations rather than authoritative truth: all 230 generated examples were subsequently reviewed by a 4D-printing domain expert with ontology-engineering expertise, who verified semantic correctness, removed unsupported claims, corrected entity typing and relation directionality, and resolved ambiguous cases by rewriting or discarding the corresponding triples. The resulting dataset therefore constitutes an expert-curated supervision set used to fine-tune the extractor toward ontology-aligned expression and typing, rather than learning to assert facts from the model itself. The resulting dataset follows a standardized format: [prompt (containing both the section and the image), triplets], and consisted of 230 ground truth examples.

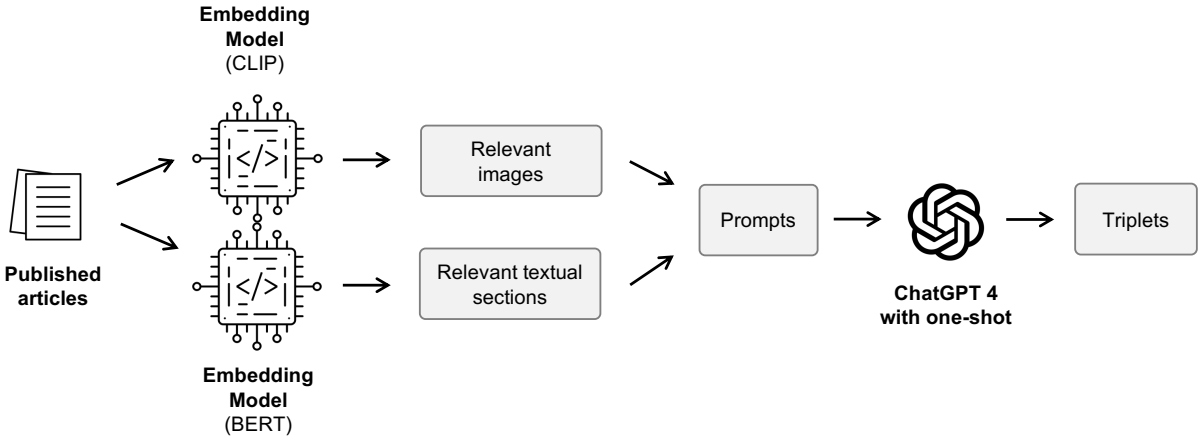

Figure 3.6: Pipeline for generating the fine-tuning dataset. (1) Section selection. Relevant textual sections are automatically extracted from a corpus of scientific articles using a BERT-based semantic similarity model. (2) Image association. for each section, the most semantically aligned image is selected using CLIP-based similarity matching. (3) Generation of textual prompts. For each selected image and its corresponding section, we insert them into the prompt template. (4) Generation of triplets. Using a one-shot prompt to ChatGPT-4, convert each prompt into triplets, reproduced from [375].

While specialized models may not always perform optimally on specific tasks, they require fine-tuning to enhance their effectiveness. Parameter efficient fine-tuning (PEFT) [391–395] is a technique for fine-tuning an LLM that requires fewer resources to train, making it a cost-effective option. It offers faster training times compared to traditional fine-tuning methods. By regularizing the training process, it helps maintain the model's performance on previously learned tasks. In addition, this allows for storing multiple specialized models using less disk space. For fine tuning the LLaVA model, we used low-rank adaptation (LoRA) [396], which is a method that is particularly relevant within the framework of parameter-efficient fine-tuning (PEFT). This technique allowed us to train only an adapter, which resulted in faster and more resource-efficient training. This adapter layer acts as a bridge between the pretrained LLaVA model and the specific task of triplet extraction from multimodal data that we aimed to optimize it for. By introducing this layer, we enabled the model to better align its image and text representations with the structural requirements of triplet extraction, such as identifying and extracting relevant information in the form of subject-predicate-object triplets.

During inference, a single multimodal prompt was constructed for each target section. This prompt included: (i) the raw section text, (ii) the associated figure or schematic, and (iii) a directive stating "Extract all domain-relevant triplets". This prompt was then processed by our MLLM, which jointly attended to textual tokens and image patches to generate a set of subject-predicate-object assertions. For figures, the model first employed an optical character recognition (OCR) module to detect and encode text regions, and to extract key graphical elements (i.e., shapes, connectors, symbols) as visual tokens. These

visual tokens interacted with text embeddings via cross-attention within the multimodal transformer. The text embeddings had been refined through our fine-tuning procedure, therefore allowing for better alignment with domain-specific semantics. This cross-modal mechanism enabled the model to infer high-level semantic relations that are not explicitly stated in the input but emerge from a combination of spatial configurations, textual cues, and prior knowledge encoded in the pretrained weights. For example, in a section from Peng et al.'s paper [163], illustrated in **Figure 3.7A**, describing liquid crystal elastomer (LCE) preparation, the spatial proximity and labels of "EDDET" and "PETA" enabled the model to infer an `isCrosslinkedWith` relation, which leverages both the visual structure and domain-specific patterns learned during pretraining. As shown in **Figure 3.7B**, relations grounded purely in the text (e.g., LCE_ink `isComposedOf` TEA) are rendered in blue, whereas those inferred from the figure's spatial layout and graphical elements (e.g., RM257 `isCombinedWith` RM82) appear in red. This integrated multimodal approach thus ensures a reliable extraction of triplets from both explicit textual descriptions and implicit visual patterns.

Our extraction pipeline is intentionally modular, and each component is selected for a distinct, non-overlapping functional role rather than as an interchangeable alternative. LLaVA serves as the multimodal extractor, translating paired textual evidence and figures into ontology-aligned subject-predicate-object triplets. CLIP is used upstream for cross-modal retrieval, aligning each target section with its most semantically relevant figure so that subsequent extraction is grounded in the appropriate visual evidence. A BERT-family encoder provides contextualized text embeddings for similarity-based retrieval and corpus navigation, supporting the evidence selection step that conditions extraction. GPT-4 is used only to construct the fine-tuning supervision set under strict evidence-use constraints; its outputs are treated as candidate annotations and are expert-curated prior to training, ensuring that fine-tuning does not operationalize model generations as ground-truth knowledge. Methodologically, RAG provides evidence grounding by conditioning extraction on retrieved passages and figures, thereby reducing unsupported assertions, while PEFT with LoRA enables resource-efficient adaptation of the large multimodal model under constrained compute by updating only a small subset of parameters. This functional separation clarifies why multiple models appear in the pipeline. They are different in kind – multimodal extraction, cross-modal retrieval, text embedding, and supervision construction – and each is chosen to address a specific bottleneck in the end-to-end knowledge engineering workflow, including grounding, multimodality, scalability, and reproducibility.

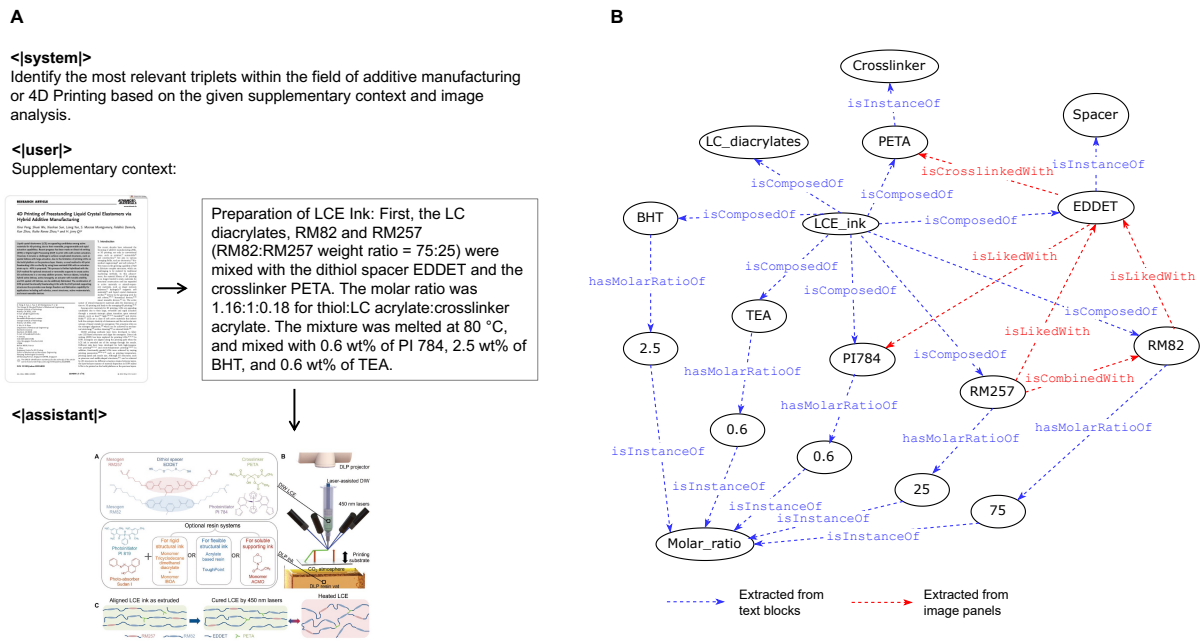

Figure 3.7: Example of the inference pipeline: (A) a section is selected from a published article in *Advanced Materials* Journal [163], its most relevant image was retrieved, and (B) a corresponding graph of triplets was produced, reproduced from [375].

3.5 Ontology Enrichment From Existing Datasets

To enhance our 4D printing ontology, we incorporated specific datasets that align with the domain's requirements. Our selection process involved choosing datasets based not only on their relevance but also on their compatibility with the format constraints. In this endeavor, we identified key resources including the Novel Materials Composition and Properties Database (Gnome) [397], sourced from Google DeepMind. This dataset catalogs the compositions and corresponding energies of over 2.2 million new crystalline materials, providing a comprehensive resource for new material discovery and validation. In addition, we utilized the polyOne dataset [398], which contains 100 million hypothetical polymers, each with 29 predicted properties computed using advanced ML models. The Hydrogel Design dataset [399–401] offers extensive insights into a variety of hydrogel properties. It includes key measurements such as swelling, diffusion, stiffness, and partitioning across different hydrogel formulations. The Smart Materials Database from NC State [402] aggregates extensive information on smart materials, including detailed shape memory alloy data such as constant-temperature stress-strain measurements, creep data, and minor loop data. Additionally, the Polymer Property Predictor and Database (PPPDB) from [403] offers a comprehensive collection of polymer data, enhanced with tools such as the Flory-Huggins Phase Diagram, the Random Phase Approximation Structure Factor, and a Binary Solution Cloud Point Estimator. Materials Project Data [345] offers an extensive collection of computed material properties. The dataset

is organized to present a comprehensive overview of each material's characteristics by integrating detailed information on composition, energy stability, electronic features, and structural configurations. Moreover, CoPolDB [404] focuses on copolymerization by offering two components: a repository of monomers, which serve as the building blocks for copolymers, and a catalog of copolymers generated through polymerization reactions. Although we initially planned to include data from NASA's Shape Memory repository [405], its lack of an API for automated data retrieval limited its integration into our framework.

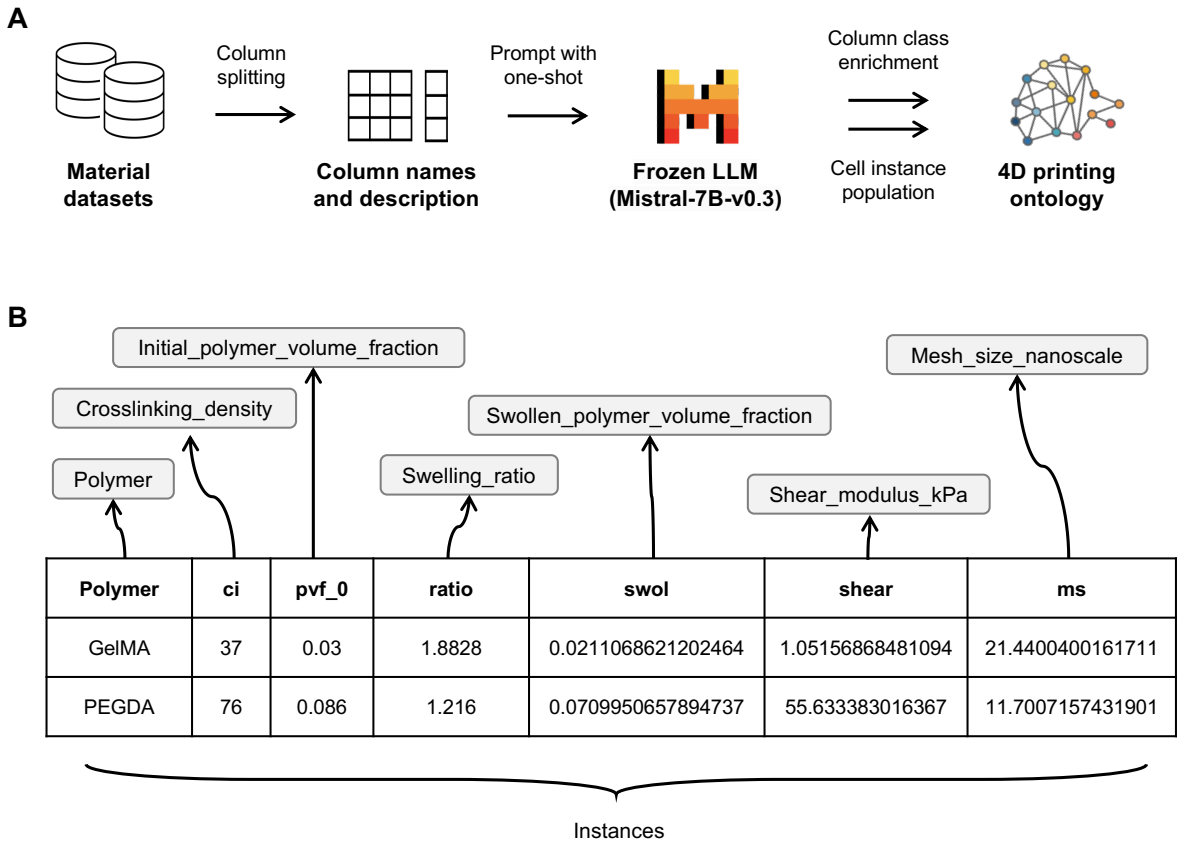

Figure 3.8: (A) Overview of the dataset pipeline, and (B) illustration of identified classes and intances related to a part of the Hydrogel Design dataset, reproduced from [375].

Our methodology for integrating these datasets into the ontology involves detailed data preparation and mapping. Each dataset was broken down into its constituent columns, which are then described and cataloged with examples provided for clarity. To categorize each column within the 4D printing ontology, we utilized a one-shot learning method [406, 407] with the Mistral-7B-v0.3 LLM model [408]. This choice reflects the fact that column-to-class mapping is a text-only task, for which a dedicated high-performing text LLM is sufficient and more appropriate than relying on a heavier multimodal model whose visual capacity would remain unused. The Supplementary Material provides more details about the justification underlying this model choice. We

then instantiate each cell in every row as an instance of its corresponding ontology class, as illustrated in **Figure 3.8** and extract the resource description framework (RDF) object properties that link these instances. For example, in the Hydrogel Design dataset we created instances for classes such as Polymer, SwellingRatio, and ShearModulus_kPa, and generate relations – e.g., Polymer linked via `hasShearModulus` to ShearModulus_kPa, Polymer linked via `hasSwellingRatio` to SwellingRatio, and SwellingRatio linked via `influencesShearModulus` to ShearModulus_kPa. This end-to-end pipeline yields a richly interconnected ontology graph that faithfully captures both the structural typology and the relational semantics of the original data.

In addition to relevance and format compatibility, dataset inclusion was expert-validated with respect to (i) source reliability (provenance, documentation quality, and methodological credibility), and (ii) utility for 4D-printing knowledge modeling (coverage of materials, AM processes, stimuli, and measurable properties). Datasets were excluded when they lacked a programmatic access route (e.g., no API or no feasible automated retrieval), had restrictive licensing terms, were not machine-parseable in practice (e.g., incompatible formats or unstable table structures), provided insufficient documentation, contained incomplete or low-coverage fields, or could not be traced to a reliable, citable source (e.g., no stable repository or peer-reviewed/referenceable record). To further increase the trustworthiness of the ingestion step, we conducted a targeted expert audit of the automatic mapping procedure: we randomly sampled 50 columns across 6 datasets and manually verified the column-to-class assignments proposed by the LLM, which showed high agreement with expert judgment. Finally, for instance enrichment, we treat the numerical and categorical values provided by these curated datasets as the primary data source, since they originate from established repositories or published resources; accordingly, our validation effort focused on semantic typing and alignment (i.e., whether values are assigned to the correct ontology classes and relations) rather than re-measuring or re-estimating the underlying experimental quantities.

3.5.1 Ontology Enrichment From Knowledge Graph

Furthermore, we harnessed the capabilities of MatKG – that is the most extensive knowledge graph in the field of materials science – to enrich our 4D printing ontology with passive materials. Derived from 5 million articles, MatKG contains over 1 million triplets, which were meticulously extracted using a named entity recognition (NER) model. Termed as MatBERT [409], this model is a BERT-based system specifically fine-tuned for materials science literature, enabling precise identification and categorization of entities such as materials, synthesis methods, and properties.

We designed a systematic transformation pipeline to feed our ontology with MatKG's triplets with the following subject-relation-object format. We implemented a targeted

relationship mapping strategy to enhance the semantic connections within our ontology. Specifically, we transformed MatKG's domain-specific relation codes into more semantically meaningful predicates: 'CHM-PRO' was converted to '`hasProperty`', where 'CHM' designates materials and 'PRO' refers to properties, while 'CHM-APL' was converted to '`isApplicatedTo`', where 'APL' denotes applications. Additionally, we systematically generated material instance declarations by identifying all chemical entities (CHM) involved in these relationships and explicitly defining them as instances of the Material class through an '`isInstanceOf`' predicate. This end-to-end pipeline thus enabled us to augment our ontology with richly typed data properties and new individual declarations. For example, the original MatKG triplets [PNIPAM, CHM-PRO, LCST] and [PNIPAM, CHM-APL, Drug Delivery] – where PNIPAM and LCST designate Poly(N-isopropylacrylamide) and Lower Critical Solution Temperature, respectively – were converted into the following semantically enriched relationships: [PNIPAM, `isInstanceOf`, Active_Material], [PNIPAM, `hasProperty`, LCST], and [PNIPAM, `isApplicatedTo`, Drug delivery].

3.6 Preprocessing and Construction of the Ontology

The results obtained from this framework were cleaned through a rigorous process that ensures only high-quality triplets are retained. Specifically, the framework evaluates the following aspects:

- **Domain relevance:** Each triplet's subject and object are transformed into contextualized embeddings using BERT and compared against embeddings derived from a curated list of domain-specific keywords. For each triplet element, the framework computes cosine similarity scores against all domain keywords and retains the maximum similarity value as the relevance indicator. The final domain relevance assessment combines both subject and object relevance scores in the overall evaluation function. This process ensures that the data is deeply aligned with the target field.

- **Semantic coherence:** The framework implements a comprehensive evaluation strategy to assess semantic meaningfulness. It computes direct BERT-based cosine similarity between subject and object embeddings to measure their semantic relatedness. The final coherence score integrates both the relation validity and subject-object similarity components. Additionally, predicate coherence is evaluated through template-based assessment, where the framework compares BERT embeddings of complete triplet phrases against baseline phrases and relationship templates to ensure predicate appropriateness within the semantic context.

- **Structural validity:** The framework checks the syntactic correctness of each triplet by verifying that all elements (subject, predicate, and object) are present, of

sufficient length, and follow expected formatting standards. This validation ensures data reliability for downstream applications.

- **Redundancy elimination:** Duplicate or highly similar triplets are identified through a two-stage process. First, exact duplicates are removed through string matching of subject-predicate-object combinations. Second, semantic duplicates are detected by computing BERT-based cosine similarity between triplet embeddings, where triplets exceeding a similarity threshold are flagged as redundant. This ensures that the final dataset is concise and free from both literal and semantic redundancy.

Together, these validation steps contribute to a robust and high-fidelity cleaning process that prepares the data for subsequent ontology construction and analysis. In addition to these quality control measures, the ontology construction phase integrated several advanced techniques to further enhance the ontology. First, entity names are normalized and cleaned to create valid uniform resource identifier fragments, thereby ensuring semantic consistency across the ontology. This preprocessing step effectively mitigates errors arising from formatting discrepancies or lexical variations. Furthermore, the framework incorporates a BERT-based similarity analysis that compares new class labels with those already present in the ontology. This mechanism dynamically identifies semantically similar classes and, when a sufficient similarity threshold is met, establishes subclass relationships. In doing so, the ontology consolidates redundant entities and organizes them hierarchically in a manner that mirrors the underlying domain structure. The pipeline also incorporates robust logging and error handling throughout each processing stage. Detailed logs capture any discrepancies or skipped entries, which facilitates iterative refinement and ensures full traceability of the data transformation process. Moreover, special attention has been given to maintaining the homogeneity of the complete ontology by enforcing uniform naming conventions and consistent semantic representations across all entities. This ensures that the entire knowledge base exhibits a high degree of internal consistency, which is critical for efficient reasoning and data integration.

3.7 Results

3.7.1 Final Enriched 4D Printing Ontology

Our framework initiates the ontology enrichment process with an initial 4D printing ontology, which comprises only 170 classes, 9 instances, 48 object properties, and 13 data properties. Through the successive integration of heterogeneous data sources and advanced validation techniques, the framework has dramatically enriched and populated

the ontology. In the first phase, the system processed a corpus of scientific articles by extracting triplets that describe various domain-specific relationships. In total, approximately 130,000 triplets were initially generated from articles, of which only about 28,000 were retained after applying the full consistency filtering pipeline, including semantic similarity checks, predicate template matching, and redundancy elimination. This stage resulted in the identification of 5,706 classes, 16,651 instances, 1,331 object properties, 4,390 data properties, and the establishment of 7,913 subclass relationships. In addition to this large-scale quantitative enrichment, we conducted a systematic evaluation of the triplet-extraction model itself, as detailed in Appendix B (See Figure S1). The consideration of MatKG further augmented the ontology by processing additional instance-of relationships. It was responsible for incorporating 6,629 new instances and two additional data properties with 445,370 relations. This considerable increase reflects the framework's ability to integrate detailed instance-level data from supplementary sources, thereby enhancing the granularity and applicability of the ontology. A further enrichment occurred through the automated ingestion of multiple datasets from an external directory. This step contributed 144 additional classes, 12,540,671 instances, 26 object properties, and 113 subclass relationships (see **Figure 3.9**). By parsing and merging these large-scale datasets, the framework ensured a comprehensive and diverse coverage of the domain knowledge, while maintaining structural validity and eliminating redundancy.

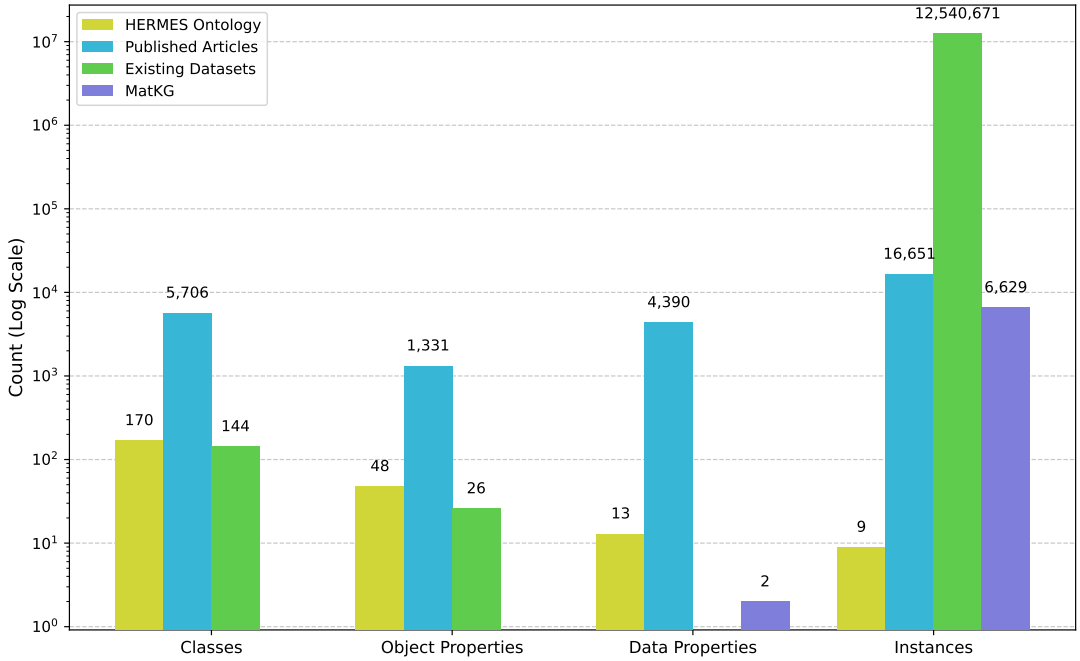

Figure 3.9: Comparison of ontology components between the baseline HERMES ontology and its extended counterparts derived from published articles processing, dataset parsing, and integration with MatKG triplets. The values demonstrate the significant scalability of automated multimodal extraction and knowledge graph integration, reproduced from [375].

After synthesizing the contributions from the scientific literature, the MatKG module, and additional datasets, the final ontology exhibits 5,849 classes, 12,563,951 instances, 1,357 object properties, 4,392 data properties, and 8,196 subclass relationships. This substantial ontology expansion demonstrates the efficacy of our multi-stage enrichment process, as illustrated in **Figure 3.10**, which highlights the material, design-and-engineering, and AM views, along with a subgraph centered on the stimuli-sensitive Hydrogel class (**Figure 3.11**).

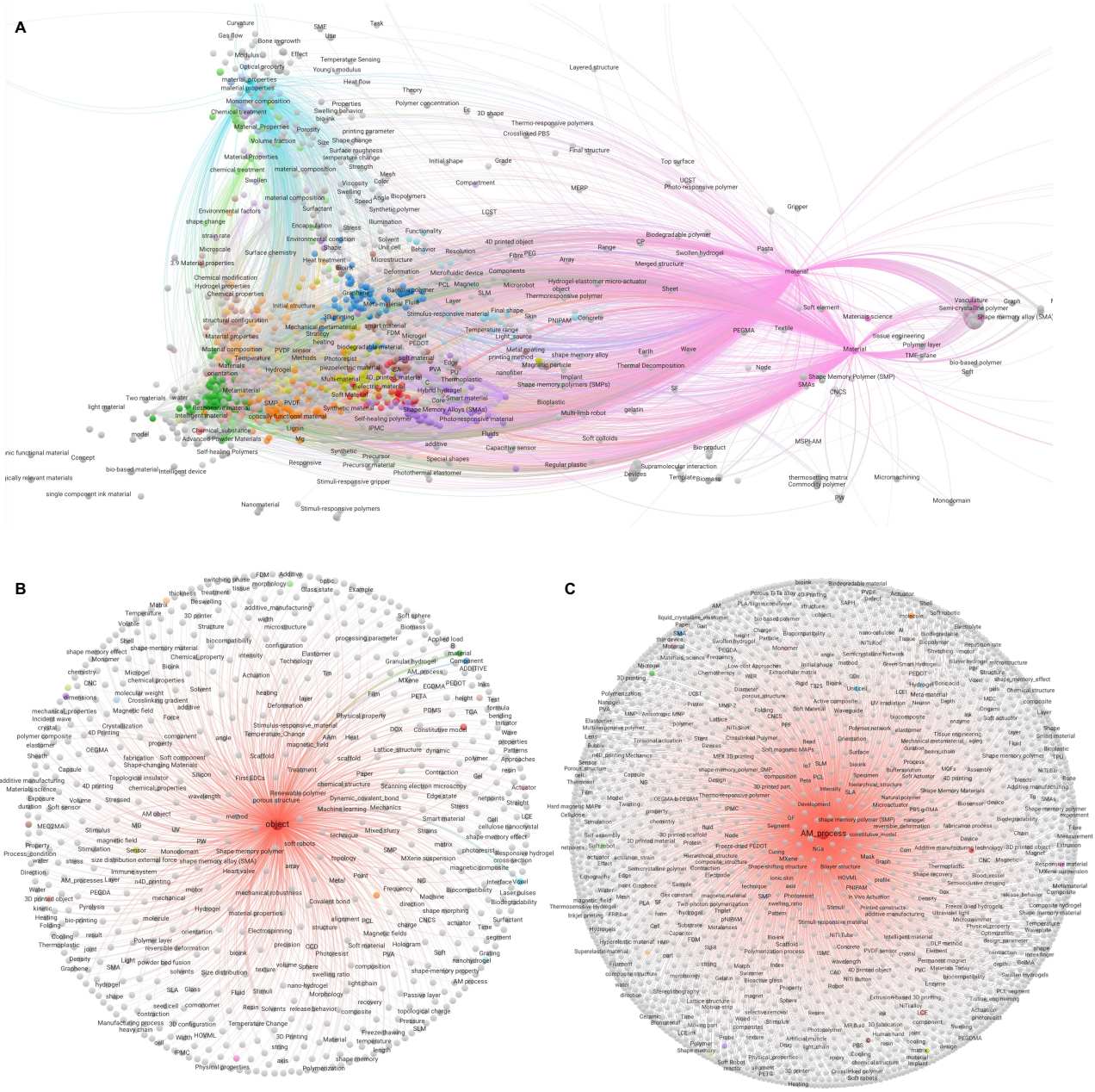

Figure 3.10: Final enriched 4D printing ontology illustrated through its (A) material view, (B) design and engineering view, and (C) AM view, reproduced from [375].

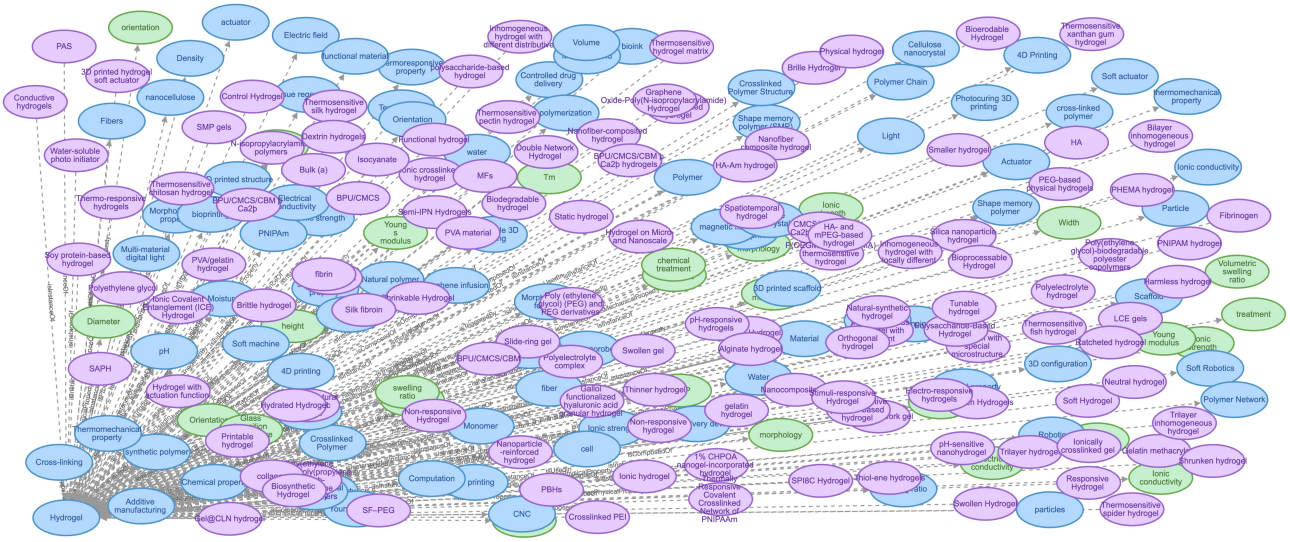

Figure 3.11: Final enriched 4D printing ontology through the lens of the stimuli-sensitive Hydrogel class. To ensure clarity, only a limited number of classes, object properties, and instances are displayed, reproduced from [375].

In summary, the integration of multiple data sources, coupled with advanced NLP and robust validation measures, has culminated in a high-fidelity, richly structured ontology. This framework is fully domain-agnostic—while it was demonstrated on 4D printing, it can just as easily be applied to any other field. The resulting ontology not only represents a substantial expansion in scale and detail compared to its initial state but also provides a solid foundation for downstream applications such as knowledge-based reasoning, data integration, and semantic information retrieval across complex scientific and technical domains.

3.7.2 Query–Answer Examples for Smart Materials and Structures

The enriched ontology enables the formulation and resolution of design-oriented queries in 4D printing, aligning with recent industrial question-answering systems that support technical knowledge access through natural language interfaces [410]. To demonstrate its relevance, we encoded a material selection need as a semantic query vector in the ontology embedding space and retrieved the most relevant materials, experiments, and architectures by ranking their embeddings using cosine similarity. In this illustrative case, the design intent was: "Identify a soft, biocompatible material that is chemically compatible with LCE and printable via direct ink writing (DIW)".

The highest-ranked neighborhood returned by cosine similarity was dominated by silicone-based elastomers. This ranking was supported by convergent evidence from the literature: (i) DIW processable silicones, including polydimethylsiloxane (PDMS) formulations, (ii) their established biocompatibility in biomedical and microdevice applications,

and (iii) explicit LCE-silicone co-fabrication reports indicating chemical/process compatibility at the interface.

As the top-ranked result, the ontology retrieved PDMS as a soft silicone elastomer that can be processed by DIW using embedded/freeform strategies, where uncured PDMS inks were extruded into a hydrophilic support bath and subsequently cured into stable 3D features [411, 412]. PDMS is also widely documented as biocompatible and chemically inert, which explains its routine use in soft implantable systems [413]. Crucially for LCE compatibility, PDMS has been used as the passive silicone layer in bilayer systems integrating cholesteric or nematic LCE actuators, demonstrating that silicone-LCE interfaces can be formed and operated without reported interfacial chemical inhibition. [414, 415] Representative extracted triples include [PDMS, isPrintedUsing, DIW], [PDMS, is, Biocompatible], and [LCE actuator, isIntegratedWith, PDMS layer], confirming PDMS as a DIW-printable, biocompatible material compatible with LCE systems.

Beyond PDMS, the ontology identified DIW-printable soft silicone elastomer inks formulated from platinum-cured or room-temperature-vulcanizing (RTV) systems. These materials are reported as DIW-processable into highly compliant parts for soft devices, and they share the same chemically inert polysiloxane backbone class as PDMS, which is the silicone family already demonstrated in LCE-integrated constructs [411]. Extracted triples such as [Soft silicone elastomer, isPrintedUsing, DIW], [Soft silicone elastomer, hasBackbone, Polysiloxane], and [Polysiloxane, isCompatibleWith, LCE layer] further support DIW-printable silicones as the closest validated compatibility class for co-printing with LCE actuators.

We then evaluated a second design need: "Determine the material distribution of LCE and PDMS to ensure actuation and biocompatibility of the structure". The ontology returned an embedded composite architecture, in which discrete LCE particles are dispersed within a continuous PDMS elastomer matrix. This configuration corresponds to reported polymer-dispersed LCE composites, where LCE particles provide localized anisotropic actuation and the PDMS matrix serves as a compliant host that transmits deformation while avoiding mechanical clamping of the LCE domains. When the PDMS matrix is sufficiently soft, the LCE inclusions can undergo reversible actuation, yielding macroscopic shape change of the composite. This architecture simultaneously benefits from the established biomedical compatibility of PDMS in soft device applications. Representative triples supporting this match include [LCE particles, isDispersedIn, PDMS matrix], [LCE composite, exhibits, Reversible actuation], and [PDMS, is, Biocompatible], all grounded in reported studies on LCE–PDMS composites and PDMS-based biomedical systems [413, 416].

The enriched ontology demonstrated strong potential as a design-support system for 4D printing by enabling high-level, intent-driven queries to be translated into relevant

recommendations. By embedding design intents into a semantic vector space and retrieving knowledge via similarity ranking, the ontology accurately identified DIW-printable, biocompatible silicone elastomers – particularly PDMS – as optimal matches for a active, LCE-compatible material, with results supported by literature-derived triples and experimental reports. It further inferred an appropriate LCE–PDMS composite architecture, retrieving evidence for embedded LCE domains within a compliant PDMS matrix that preserves actuation while ensuring biocompatibility. These examples illustrate the ontology's ability to integrate heterogeneous knowledge, infer mechanistically coherent solutions, and guide material and structural design decisions, highlighting its promise as a generalizable and evidence-backed tool for accelerating discovery in smart material systems.

To move beyond purely illustrative query–answer demonstrations, we compared the two competency questions against (i) the baseline HERMES ontology and (ii) our enriched ontology along three technically grounded dimensions: representational completeness (whether the ontology contains the required entities and relations to express the intent), instance-level operationality (whether the query can be resolved to concrete candidates rather than remaining at the schema level), and evidential traceability (whether returned assertions are supported by explicitly extracted triples anchored in the source corpus). For the first question, HERMES supports only a schema-level interpretation: it represents DIW as a manufacturing concept (e.g., `Direct_ink_writing` under `Material_extrusion`) and provides generic modeling patterns such as `Object` → `hasMaterial` → `Material`, but it neither populates candidate material instances (e.g., PDMS) nor represents the constraint vocabulary needed to formalize the design intent (e.g., biocompatibility and LCE–silicone chemical/process compatibility). As a result, the competency question cannot be instantiated into a design-resolvable query and yields, at best, high-level conceptual matches. For the second question, HERMES contains structural and functional primitives that could, in principle, encode transformable systems (e.g., voxel-based constructs such as `Voxel` arranged in a `Voxel_grid`, and patterns such as `Stimulus` → `enables` → `Transformation_function`); however, it does not represent compositional micro-architectures (e.g., "particles dispersed in a matrix") and again lacks the instance-level entities and typed relations required to connect materials, architectures, and performance-relevant properties, preventing resolution beyond generic statements about objects and transformations. In contrast, the enriched ontology resolves both questions at the instance and evidence levels by returning concrete candidates (e.g., PDMS and related DIW-processable silicone elastomers) and an embedded LCE–PDMS composite architecture, each supported by explicit, literature-derived triples. This enrichment strengthens the ontology's role as an AI-assisted design substrate in three complementary ways. From a reasoning perspective, this approach increases semantic controllability by encoding key design requirements as well-typed, cross-linked assertions (material-

process-property-architecture), enabling precise constraint-oriented querying (e.g., "DIW-printable" ∧ "soft/compliant" ∧ "biocompatible" ∧ "compatible with LCE interfaces") and reducing reliance on implicit narrative interpretation. In terms of decision support, it enables structured candidate narrowing by linking materials to processes, properties, and integration strategies, thereby supporting reproducible screening of material-process-architecture alternatives and providing evidence-backed rationales through the underlying extracted triples. From a knowledge discovery standpoint, it supports structural discovery from the aggregated corpus by (i) introducing new explicit relations between previously weakly connected entities, (ii) surfacing recurrent association patterns as graph motifs (e.g., bilayer and embedded-composite strategies coupling passive silicone matrices with active LCE phases), and (iii) revealing coverage gaps, i.e., plausible material–process-architecture links that are sparsely supported, thereby guiding follow-up experimentation or simulation and closing the loop between data-driven extraction, symbolic structuring, and design-relevant hypothesis generation.

3.8 Discussion

This substantial increase in conceptual scope provides a robust foundation for the precise representation of components and relationships inherent to transformable systems. Leveraging the breadth and depth of the modeled concepts, one can now intricately characterize both the functional and geometric configurations these systems may assume. Such enhanced granularity creates a fertile substrate for the development of more realistic simulation scenarios and the formulation of optimization criteria tailored to the specific constraints of each application. Ultimately, this enriched ontology could serve as a guiding framework to inform and streamline subsequent design phases. This originality aligns with recent perspectives [417], which highlight how novel frameworks integrating large-scale AI models with domain-specific tasks can expand the boundaries of knowledge representation.

The 4D printing ontology provides a formalized vocabulary and structure for the key 4D printing concepts, i.e, stimuli, smart materials, transformation functions, and AM techniques, into a coherent schema that supports both human understanding and machine processing [11]. By adhering to OWL [29] and Industry Ontology Foundry (IOF) guidelines, the ontology ensures that terms are precisely defined, relationships are explicitly modeled, and extensions remain compatible with related ontologies in materials science and manufacturing. This semantic rigor facilitates robust data exchange and tool integration across computer-aided design (CAD) systems, simulation packages, knowledge management platforms, and knowledge-driven digital twin environments [418].

Ontologies serve as the backbone for reasoning engines that can infer new knowledge, check consistency, and support design trade-offs. In the context of 4D printing, rules built on top of the ontology can evaluate which stimuli-material arrangements yield desired

transformations, recommend printing parameters, and flag incompatible design options. Such rule-based reasoning complements data-driven predictions from neural models, providing explainable decision paths that are critical for certification and compliance, and knowledge-driven decision support in engineering contexts [371, 418].

4D printing lies at the intersection of materials science, mechanical design, computational mechanics, chemistry, and computer science, each with its own lexicon and methodologies [353]. The ontology enables these diverse experts – product architects, material scientists, AM process planners – to collaborate within a unified semantic framework, reducing misinterpretations and ensuring that design intents are faithfully communicated, which is consistent with recent knowledge-augmented approaches for industrial human–machine collaboration [371]. By mapping domain-specific data (e.g., simulation outputs, experimental results) to shared ontology classes and properties, it ensures semantic alignment and coherent knowledge integration across multidisciplinary workflows.

When coupled with neuro-symbolic AI systems, the 4D printing ontology becomes a dynamic asset: NSAI can ingest unstructured text and visuals, map extracted entities and relations to ontology terms, and trigger reasoning processes for immediate insights. This tight integration supports continuous ontology evolution through new materials, stimuli, or manufacturing strategies discovered in the literature can be auto-added and validated against existing constraints. In practice, this accelerates the iteration cycle for 4D printed designs, enabling rapid prototyping informed by both data-driven predictions and symbolic rules, and ensuring the design pipeline remains synchronized with the cutting edge of materials research and data-driven design practices [419].

The proposed architecture enables a bidirectional enrichment process: symbolic knowledge guides neural extraction, while neural extraction discovers new candidate entities and relations that are then validated symbolically. This continuous *Symbolic* $\rightarrow$ *Neuro* $\rightarrow$ *Symbolic* loop, an instantiation of the sequential paradigm formalized in Chapter 2, ensures real-time knowledge updates and, through the LLM's generative and multimodal power, delivers enriched insights for the design and optimization of intelligent material systems.

Despite these encouraging outcomes, our approach is not without its critiques. The framework currently relies heavily on the quality and representativeness of the input data, and its scalability remains a challenge, particularly regarding processing time and resource demands. In addition, potential errors in the underlying datasets can propagate into the ontology. In this chapter, the datasets we used were validated by domain experts, which ensured a high level of reliability. However, as the framework scales to millions of instances, maintaining such expert validation becomes increasingly challenging, underlining the need for automated yet robust validation mechanisms in future work. Moreover, while this chapter focuses on 4D printing, the proposed methodology is designed to be transferable: the same combination of triplet extraction, consistency filtering, and ontology enrichment

could be applied to other domains, provided appropriate corpora and expert validation are available. However, we recognize that this claim of domain-agnosticism should be tempered until validated experimentally in additional fields. Our experiments were conducted under constrained hardware resources (e.g., CPU/GPU specifications). Within these constraints, we relied exclusively on open-source models to ensure transparency and reproducibility. Access to more powerful hardware would enable the use of more advanced LLMs, which we expect to improve both the precision and efficiency of the ontology-enrichment process. Such advancements would further enhance the system's ability to extract, reason, and update domain-specific knowledge in real time. In summary, while the contribution presented in this chapter constitutes a significant step towards more intelligent, adaptable, and scalable ontology-driven systems in the rapid evolving 4D printing, future work should focus on addressing the current limitations by incorporating broader datasets and leveraging next-generation language models, ultimately fostering more robust and comprehensive knowledge management solutions in emerging fields.

A central methodological consideration in applying probabilistic LLMs to OWL ontology enrichment is the epistemic risk introduced when stochastic generations are translated into symbolic commitments. This risk is compounded by corpus-level limitations: because the distribution of topics in the scientific literature is uneven, enrichment may over-represent well-studied domains while under-representing emerging or peripheral subfields. Moreover, there is a risk of overfitting the ontology to currently available literature, which may reduce its adaptability as new concepts and paradigms emerge. Addressing these limitations requires designing the ontology with mechanisms that preserve flexibility and scalability. This design tension echoes principles emphasized in organic architecture, namely maintaining structural rigor while preserving adaptability to changing contexts [420]. In this setting, four threats are particularly salient. First, semantic drift refers to progressive meaning shifts that can arise during iterative extraction, canonicalization, and re-alignment. Second, latent bias may be inherited from pretraining distributions and from uneven coverage in the retrieved corpus. Third, noisy co-occurrence patterns in scientific text can be reified into seemingly authoritative ontological axioms even when the underlying evidence is weak or highly context-dependent. Fourth, context loss can lead to over-generalization when conditional claims are expressed without the qualifiers needed to delimit their validity. Many statements in 4D printing are valid only under specific experimental conditions. Therefore, relations extracted from the literature, such as "DIW-printable" or "material compatibility", should be interpreted as context-dependent and should not be promoted to general ontological assertions unless the relevant qualifiers are captured, including the material formulation, the processing parameters, and the stimulus conditions. For this reason, the pipeline treats LLM outputs as candidate assertions rather than direct OWL commitments. Before any assertion is committed to the ontology, it is subjected to the quality gates described

earlier, which jointly enforce domain alignment, semantic plausibility of the relation, basic structural well-formedness, and redundancy control. This separation between probabilistic proposal and symbolic acceptance mitigates self-reinforcing feedback loops in which model-generated relations could otherwise be re-ingested as self-justifying axioms, and it helps maintain an auditable epistemic status suitable for OWL-based knowledge engineering. Nevertheless, residual model- and tool-induced errors may persist at scale, motivating future work on uncertainty-aware representations, stronger provenance formalization, and scalable expert-in-the-loop verification.

We acknowledge that, even with expert validation on curated subsets of articles and extracted outputs, human-in-the-loop verification cannot realistically scale to the full volume of documents, figures, and instances processed by our pipeline. Consequently, residual tool- and model-induced errors may persist and should be treated as threats to validity rather than assumed away. We further recognize that the software toolchain used for corpus construction and parsing can introduce systematic biases. ResearchRabbit may skew literature coverage toward highly connected regions of citation graphs seeded by the chosen keywords (topic-centrality bias), potentially under-representing emerging or peripheral subfields in smart materials and 4D printing. LLM Sherpa may introduce segmentation and chunking artifacts when parsing scientific PDFs (boundary bias), which can affect retrieval granularity and propagate to downstream extraction. Aspose-based conversion and figure extraction may also mis-handle complex layouts (e.g., multi-panel figures, caption alignment, or embedded tables), producing layout-dependent noise and occasional mis-associations between evidence and extracted claims. To mitigate these risks, we apply deduplication and controlled keyword expansion during corpus construction, perform sampling-based manual inspection of parsed outputs, and cross-check text–figure provenance during retrieval. In addition, we rely on downstream ontological validation (e.g., domain/range typing, redundancy filtering, and consistency-oriented reasoning checks) to reduce the likelihood that structurally inconsistent, weakly typed, or insufficiently grounded assertions are committed to the knowledge base. Nevertheless, these safeguards do not eliminate all errors, and future work should prioritize scalable verification strategies and uncertainty-aware representation of extracted knowledge.

Recent advancements in language model capabilities, including their use as digital intelligent assistants in manufacturing, suggest promising new directions [421]. One significant improvement could involve implementing a multi-agent LLM framework for automated, multimodal knowledge extraction and integration from unstructured scientific texts and visuals. A multi-agent LLM system could significantly enhance our ontology enrichment pipeline by deploying specialized agents that work in concert, each handling distinct aspects of knowledge ingestion—from text segmentation and entity recognition to figure interpretation, table parsing, relationship extraction, relation-type classification,

schema alignment, and conflict resolution [422, 423]. Such an architecture enables fine-grained processing of heterogeneous content, dramatically broadening the scope of automated ontology construction [424].

First, the division of labor among specialized agents enables more nuanced analysis of domain-specific terminology and modalities [425]. Dedicated text agents can be fine-tuned for extracting materials entities and technical nomenclature, while vision-language agents process figures (e.g., microstructure images, X-ray diffraction patterns) to recognize visual evidence of material phases and morphologies [426, 427], and table agents convert tabular data into structured triplets [428]. Relation classification agents specialize in categorizing extracted relations into classes, object properties, data properties, or instances, ensuring accurate semantic typing per OWL standards [429].

Second, the integration of cross-agent verification mechanisms would substantially increase the reliability of extracted knowledge [430]. By implementing a system where schema alignment agents validate candidate entities and relations against ontological constraints, vision agents perform cross-modal consistency checks between text descriptions and corresponding figures, and conflict resolution agents evaluate logical coherence among new and existing triplets, thus minimizing spurious connections in the ontology. Additionally, a cohesion verification agent continuously monitors the ontology's structural integrity—detecting hierarchy cycles, disjointness violations, and domain/range inconsistencies—while a triplet validation agent flags semantically incomplete or logically incoherent triples for further review [431].

Third, this multimodal approach could dramatically improve scalability. The current exponential growth in materials science literature – with thousands of new papers published monthly –demands automated systems capable of processing information at scale while maintaining precision [425]. A multi-agent framework could adaptively distribute computational resources based on document complexity and domain relevance [432]. Furthermore, such a system could implement domain-adaptive prompting strategies, allowing it to navigate the specialized terminology of different materials science subdomains while preserving semantic accuracy across the entire ontology [433].

Future work will focus on (i) designing specialized agent architectures that integrate vision encoders and domain-specific prompt templates for materials science modalities, (ii) implementing advanced verification heuristics that leverage both linguistic and visual ontological rules, and (iii) developing evaluation metrics for multimodal triplet extraction that reflect the unique challenges of materials knowledge representation, and (iv) creating dedicated relation classification agents for precise typing and specialized validation agents for ontology cohesion and triplet integrity. By embracing multimodal multi-agent systems, we can move toward adaptive ontologies that evolve seamlessly with the scientific literature, providing researchers with powerful tools for accelerated materials discovery and development.

3.9 Conclusion

In this chapter, we presented an innovative framework for ontology learning in the field of 4D printing, integrating MLLMs and RAG to overcome the limitations of traditional ontological systems. Our approach, successfully combines the formal rigor of structured knowledge representation with the adaptive and contextual capabilities of advanced language models, which systematically captures heterogeneous information from scientific literature, experimental databases, CSV files, and extensive knowledge graphs like MatKG. Experimental results demonstrate that our methodology significantly expanded an initial, rather limited ontology – starting from 170 classes and a few instances – to a comprehensive structure encompassing over 5,800 classes and more than 12.5 million instances.

The enriched and populated ontology promotes reuse and integration with other domain ontologies, therefore reducing silos and fostering cross-domain interoperability [434]. Importantly, this structured representation lays the groundwork for automated reasoning procedures, enabling qualitative decision support in material and process selection, design recommendations, and alignment with emerging standards. When embedded within a neuro-symbolic AI framework, the ontology can be updated in real-time and reasoned over alongside neural models, bridging symbolic and neural approaches for a context-aware design strategy [11].

While this enriched knowledge representation provides a structured foundation for reasoning and recommendation, knowledge organization alone cannot predict the physical response of candidate 4D-printed structures. The next chapter therefore moves from knowledge structuring to thermo-mechanical prediction, introducing automated dataset generation and graph-based surrogate modeling to accelerate numerical simulation and enable large-scale exploration of shape-morphing behaviors.

Chapter 4

Accelerating Thermo-Mechanical Simulation and Behavior Discovery in Voxelized 4D-Printed Active Structures through Automated Dataset Generation and Graph Neural Networks

4.1 Introduction

Among stimuli-responsive materials used in 4D printing, liquid crystal elastomers (LCEs) are a prominent class for thermally-induced shape morphing due to their ability to exhibit large, reversible, and strongly anisotropic deformation [163, 435–439]. Their behavior arises from the coupling between polymer network elasticity and liquid-crystal nematic ordering, which can be programmed through director alignment and processing conditions [440]. While recent advances in fabrication techniques – such as direct ink writing [163, 441], digital light processing [442–444], two-photon polymerization [436, 445], and electrospinning [446] – have significantly expanded the achievable design space of LCE-based actuators, their response remain highly nonlinear and strongly dependent on spatial arrangement and boundary conditions (BCs), therefore making predictive modeling and design non-trivial [437, 438, 447–449].

Consequently, numerical simulation remains essential for understanding, validating, and optimizing active 4D-printed structures. In particular, finite element (FE)-based computational design frameworks have become a standard route for predicting deformation and searching material layouts in multi-material active systems. Several works

proposed computational design frameworks – based on topology optimization or genetic algorithms – for multi-material active composites in 4D printing, demonstrating how voxelized material distributions can be optimized to achieve targeted actuation behaviors [13, 18, 66, 450–453]. These studies clearly demonstrate the importance of accurate simulation, but they also underline its computational cost when large combinatorial design spaces must be explored.

To mitigate this bottleneck, recent studies have increasingly investigated machine learning (ML)-based approaches for forward prediction and inverse design in 4D printing. Sun et al. [15] developed a ML-enabled framework for forward prediction and voxel-level inverse design of 4D-printed active plates, which demonstrates that learned surrogates can efficiently approximate complex shape transformations while accelerating 2D design space exploration. In a related contribution, Sun et al. [17] combined ML with sequential subdomain optimization to achieve ultrafast inverse design of 4D-printed active composite structures. Jin et al. [78] further extended this perspective to hierarchical 4D-printed architectures with arbitrary geometries, showing that data-driven models can move beyond simple plate-like benchmarks toward more general morphing structures. Recently, Poltue et al. [454] have developed an iterative data curation strategy combined with transfer learning to reduce the required training data in ML models for 4D printing active composites.

Although these studies clearly establish the promise of ML, several limitations remain. First, most existing frameworks focus primarily on forward prediction or inverse design, while the problem of automatically constructing diverse and physically admissible simulation datasets remains comparatively underexplored. This issue is particularly important in voxelized multi-material structures, where connectivity constraints, material combinatorics, and BCs jointly define the accessible design space. Second, current ML approaches in 4D printing are generally optimized for target matching rather than for identifying and organizing previously unseen actuation behaviors, even though multi-material active structures can exhibit a broad range of qualitatively distinct responses under subtle changes in material layout and constraints [15, 455]. Finally, and perhaps most fundamentally, the majority of existing approaches rely on regular tensor- or image-based representations that flatten the spatial structure of discretized systems into a fixed grid. Such representations do not explicitly encode adjacency relations between neighboring material regions, and therefore may fail to capture the local interaction structure governing thermo-mechanical deformation in heterogeneous active composites [456–458].

In this context, graph neural networks (GNNs) provide a particularly attractive alternative. Graph-based representations explicitly preserve adjacency relations and naturally encode discretized structures, irregular interfaces, and local interactions, which are central in computational mechanics [458]. More generally, graph-network simulators

have already demonstrated strong capabilities for learning complex physical dynamics and mesh-based simulation [456, 457]. Recent work on mesh-based GNN surrogates for time-independent partial differential equations has further shown that appropriately designed graph architectures can outperform strong baselines and generalize to unseen domains, BCs, and materials in solid mechanics settings [459]. Likewise, recent studies on GNN-driven surrogate modeling in computational mechanics highlight their relevance for accelerating repeated simulation tasks such as parameter identification, design optimization, and sensitivity analysis [460]. These properties make GNNs especially suitable for voxelized thermo-mechanical systems, where deformation emerges from structured interactions between neighboring material regions rather than from geometry alone.

Beyond prediction and inverse design, an additional challenge in 4D printing is the discovery of new shape morphing behaviors. The combinatorial freedom offered by multi-material voxelized structures creates a vast design space in which qualitatively different responses – such as bending, twisting, asymmetric folding, localized buckling, or mixed actuation modes – may emerge from subtle variations in material distribution and constraints. More broadly, data-driven discovery has become an important research direction in mechanics and materials science, with recent studies addressing the unsupervised discovery of interpretable constitutive laws, automated symbolic regression of material models, and structural identification of nonlinear assemblies [461–463]. In parallel, novelty-driven frameworks for autonomous experimentation have shown that explicitly optimizing for novelty rather than only for target performance can increase the diversity of discovered phenomena and improve the chances of uncovering unexpected behaviors [455]. These advances suggest that discovery-oriented pipelines can play a major role in moving from simple prediction toward genuine scientific exploration in active materials and 4D printing.

In this chapter, we propose a unified pipeline for accelerating thermo-mechanical simulation and discovering novel shape-morphing behaviors in voxelized 4D-printed structures based LCEs, where actuation is governed by specific mesogen alignments. The framework combines four complementary stages: (i) constrained generation of heterogeneous voxelized structures through random and logic-family generation modes, (ii) automated simulation for dataset construction and graph conversion, (iii) graph-based learning of nodal displacements through a GraphSAGE [3] surrogate model, and (iv) novelty-oriented exploration of shape-morphing behaviors through complementary discovery approaches. The discovery campaign explores approximately 8×10^5 candidate structures, approaching the scale of one million configurations, across four BC scenarios. A direct FE-based exploration of this design space would require several decades of sequential computation when both model setup and numerical solving are considered. By contrast, the proposed GNN surrogate enables direct prediction of deformation fields, achieving

a speedup of approximately achieving a speedup of approximately 6.69×10^4 compared with conventional FE simulations, corresponding to a reduction in prediction time of about 99.99925%–99.99975%. By integrating automated simulation, graph-based representation learning, and discovery-driven analysis into a unified workflow, the proposed approach not only reduces the computational burden of thermo-mechanical simulation but also enables structured exploration of large combinatorial design spaces in 4D printing.

4.2 Results and Discussion

Figure 4.1 introduces a complete pipeline for discovering novel shape-morphing behaviors in multi-material active structures. The process begins with the automated generation of a dataset that captures voxel-level material distributions and their corresponding thermo-mechanical responses, obtained through computer-aided design (CAD) and FE simulations. This dataset is used to train a GNN, which serves as a fast surrogate model for predicting shape change. Once trained, the framework generates new candidates designs using procedural design techniques that explore diverse voxel-level material distributions. These candidates are then evaluated by the GNN to efficiently predict their responses without requiring expensive simulations. Finally, a novelty reasoning module analyzes the predicted behaviors to identify unique and previously unseen shape transformations, enabling discovery of new functional material designs.

The study considers a 3D rectangular parallelepiped domain discretized into a regular $10 \times 3 \times 3$ voxel grid, corresponding to 90 voxels in total. This voxelization provides a controlled compromise between structural expressiveness and computational tractability, enabling large-scale automated generation while preserving sufficient spatial freedom for heterogeneous actuation patterns.

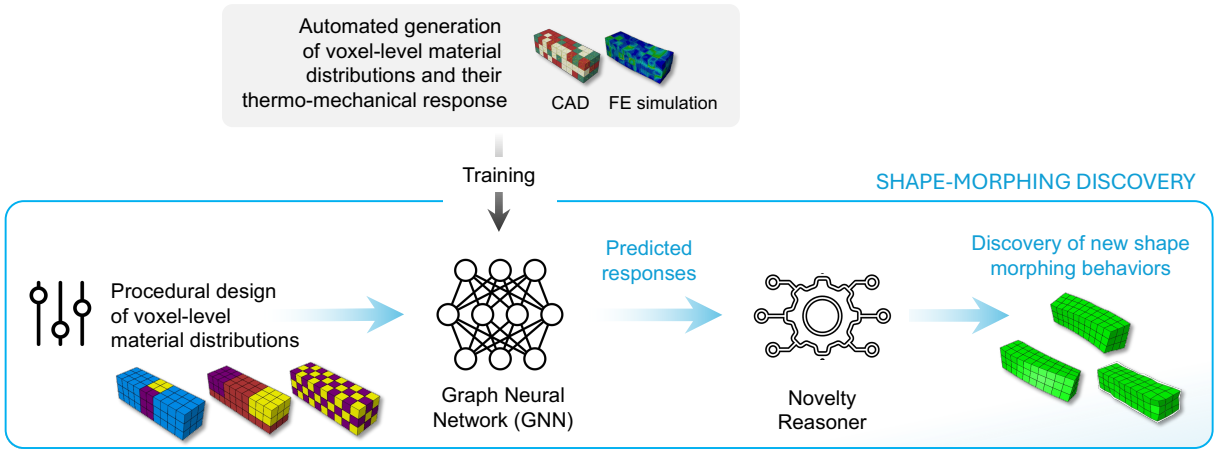

Figure 4.1: Overview of the proposed framework for discovering shape-morphing behaviors in thermo-responsive voxelized structures. A thermo-mechanical dataset is first automatically generated via computer-aide design and FE simulations, and used to train a GNN surrogate model. New voxel-level material distributions are then produced through random and procedural design strategies. The trained GNN rapidly predicts their deformation responses, while a novelty reasoning module identifies previously unseen shape-morphing behaviors.

The material system combines a thermo-responsive LCE phase, a passive isotropic LCE phase, and a void-like compliant phase introduced to represent empty regions. Rather than defining several different active materials, the framework considers a single LCE material with five distinct mesogen orientations of its principal actuation direction: three orientations aligned with the Cartesian axes and two diagonal in-plane (xy) orientations at $+45°$ and $-45°$ (**Figure 4.2**c). This representation enables the systematic study of anisotropic thermo-mechanical response while keeping the design space compatible with combinatorial generation.

To investigate the effect of constraint assignment on deformation behavior, four BCs were considered: (i) all translational degrees of freedom were fully constrained on the entire left-face boundary (Dirichlet BC applied over the complete face); (ii) two nodes located at the bottom-center of the structure were fully constrained in all translational degrees of freedom; (iii) four nodes situated at the geometric center of the left face were fully constrained; and (iv) four nodes located in the central region of the structure, corresponding to a localized interior cluster of nodes, were fully constrained (Figure 4.2d). These scenarios define distinct mechanical constraints for the same voxelized design domain and make it possible to evaluate the sensitivity of actuation to BCs.

4.2.1 Automated dataset generation under physical constraints

A first contribution of the proposed framework is the automated generation of a dataset comprising voxel-level material distributions and their corresponding thermo-mechanical responses (**Figure 4.2**). 4D-printed structures can be investigated across a range of

geometric design representations, including solids, lattices, surfaces, fibers, and lines (or skeletons). Among these, a solid-based representation of a 3D design domain provides a suitable starting point for subsequent decomposition, enabling systematic material assignment and compatibility with additive manufacturing capabilities (Figure 4.2a). Beginning with a voxelized 3D structure, the workflow assigns materials and local actuation orientations, defines voxel-voxel interactions, applies thermal loading and mechanical BCs, generates the FE mesh, and computes the resulting displacement field, which is subsequently used for downstream learning tasks (Figure 4.2b).

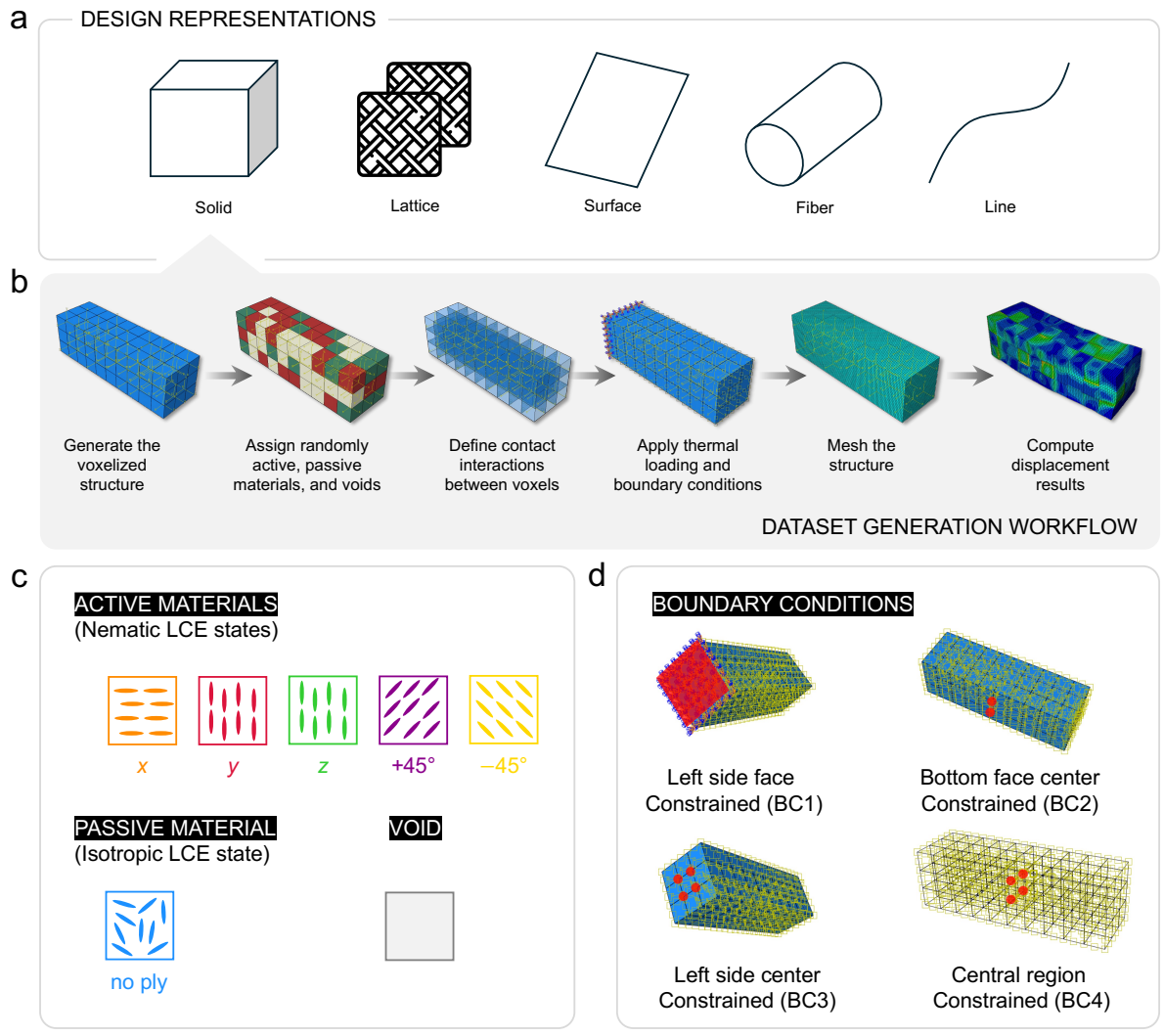

Figure 4.2: Automated dataset generation based on the generation of voxel-level material distributions and their thermo-mechanical response. (a) Available geometric design representations for material assignment. (b) Workflow for voxel-based decomposition, material assignment, thermal loading, BCs, meshing, and displacement computation. (c) Considered materials, including active LCE in the nematic state via distinct mesogen orientations, passive LCE in the isotropic state, and void. (d) BCs designed to promote generalization and facilitate the discovery of new deformation behaviors.

The training dataset construction is organized as a set of randomized material

distribution campaigns carried out under different BC scenarios. More specifically, for each BC, three groups of voxelized structures are generated. The first group consists of 500 random structures combining diagonally oriented active LCE regions at $+45°$ and $-45°$, together with passive and void regions. The second group consists of 500 random structures combining active LCE orientations aligned with the Cartesian axes, namely x, y, and z, together with passive material (LCE material in its isotropic state) and void regions (Figure 4.2c). The third group consists of 1,000 random structures combining all available active LCE orientations, together with passive and void regions. This strategy makes it possible to balance targeted exploration of specific orientation subsets with broader coverage of the full material-orientation design space.

The generation process is explicitly constrained to preserve physical admissibility. In particular, the material assignment remains embedded in a connected, voxelized domain, and the resulting designed structures are simulated under prescribed thermal and mechanical conditions to produce consistent displacement fields. This point is essential because the objective is not merely to enumerate symbolic voxel patterns, but to build a mechanically meaningful dataset suitable for surrogate modeling. This automated generation strategy provides two major advantages. First, it enables the large-scale production of labeled thermo-mechanical data related to voxel-level material distribution without manual intervention. Second, it promotes diversity in material layout and orientation patterns while preserving the physical consistency required for the learning stage.

4.2.2 From voxelized structure to graph representation

To enables direct use of the simulated structures in ML pipelines, each voxelized structure is converted into a graph-based representations, as illustrated in **Figure 4.3**. For every simulated instance, two graph states are constructed: a graph representing the "as designed" structure and another corresponding to "as actuated" structure. The "as designed" graph is generated procedurally from the underlying voxelized geometry. Each voxel is mapped to a set of characteristic nodes located at geometrically meaningful positions, including corners, edge midpoints, and face centers. In the present implementation, this results in 27 nodes per voxel, allowing the representation to encode both the local geometric structure and the spatial organization of the material distribution. This approach provides a richer description than a single voxel labelling, as it explicitly captures the internal geometric support needed to model deformation and interfacial interactions.

Each node is assigned both geometric and material attributes. Geometric attributes correspond to spatial coordinates, while material attributes encode whether the surrounding region belongs to the active LCE phase, the passive phase, or the void. A key feature

of this construction is the treatment of interface nodes. Because nodes may lie at the boundary between adjacent voxels, they can be influenced by multiple materials. Rather than simplifying or discarding this information, the graph retains a primary material label along with secondary material attributes when applicable, thereby preserving the local heterogeneity that governs thermo-mechanical coupling.

Edges are defined based on spatial adjacency inherited from the voxel grid, ensuring that local neighborhood relations are explicitly maintained. This results in a sparse yet graph that remains computationally efficient. Compared to regular tensor-based encodings, this formulation is better aligned with the mechanics of heterogeneous active structures, as it preserves relational and topological information instead of flattening the system to an image-like representation. Consequently, the graph serves as a physically meaningful intermediate representation, bridging voxelized geometry and learned deformation prediction.

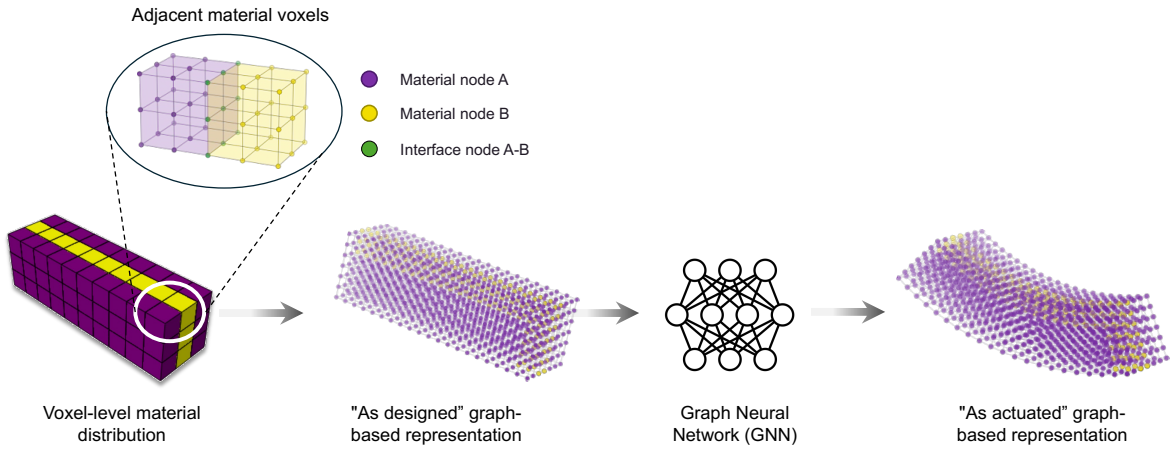

Figure 4.3: Graph construction and graph-based learning pipeline. The voxelized multi-material structure is converted into a graph in which nodes encode geometric position and material attributes, while edges preserve local neighborhood relations. The resulting "as designed" graph is then processed by the proposed GNN surrogate for nodal displacement prediction through a "as actuated" graph.

4.2.3 Graph neural network

The surrogate learning stage aims to predict nodal displacements directly from the graph representation of the "as designed" structure. The GNN serves as a surrogate for repeated thermo-mechanical simulation, enabling rapid inference once trained on physics-generated graph pairs. Its role is therefore not only to reduce computational cost, but also to support the efficient screening of large numbers of candidate structures prior to downstream discovery analysis.

More formally, each "as designed" structure is represented by an attributed graph $\mathcal{G}^0 = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, where $\mathcal{V}$ denotes the set of nodes, $\mathcal{E}$ the set of edges encoding spatial

neighborhood relations, and $\mathbf{X}$ the node-feature matrix containing geometric and material information. The surrogate model learns a mapping

$$f_\theta : \mathcal{G}^0 \longrightarrow \hat{\mathbf{U}}, \tag{4.1}$$

where $\hat{\mathbf{U}} = \{\hat{\mathbf{u}}_i\}_{i=1}^{|\mathcal{V}|}$ is the predicted displacement field, and $\hat{\mathbf{u}}_i = (\hat{dx}_i, \hat{dy}_i, \hat{dz}_i)$ is the predicted displacement vector of node i. The learning target is obtained from the difference between the deformed ($\mathbf{p}_i^{\text{def}}$) and initial nodal coordinates ($\mathbf{p}_i^{ini}$),

$$\mathbf{u}_i = \mathbf{p}_i^{\text{def}} - \mathbf{p}_i^{ini}, \tag{4.2}$$

or, in normalized coordinate space, from the corresponding normalized positions. This formulation makes the GNN an explicit data-driven approximation of the FE solution operator. Instead of solving the thermo-mechanical equilibrium equations for each new candidate, the trained model directly infers the nodal displacement field from the initial graph. This is consistent with recent graph-based learning approaches for physical simulation and mesh-based surrogate modeling, where local interactions are learned through message passing on graph-structured representations [3, 456, 457].

The adopted model combines enriched spatial encoding with graph-based neighborhood aggregation. Node coordinates are first normalized and mapped through Fourier positional encoding, which augments the raw three-dimensional coordinates with multi-frequency sinusoidal features. For a normalized coordinate vector $\bar{\mathbf{p}}_i = (\bar{x}_i, \bar{y}_i, \bar{z}_i)$, the positional encoding can be written as

$$\gamma(\bar{\mathbf{p}}_i) = \left[\bar{\mathbf{p}}_i, \sin(2^0\pi\bar{\mathbf{p}}_i), \cos(2^0\pi\bar{\mathbf{p}}_i), \ldots, \sin(2^{L-1}\pi\bar{\mathbf{p}}_i), \cos(2^{L-1}\pi\bar{\mathbf{p}}_i)\right], \tag{4.3}$$

where L is the number of frequency bands. This transformation increases the expressive capacity of the model by allowing it to capture both smooth global displacement trends and more localized spatial variations in the deformation field [464]. This positional encoding is concatenated with material descriptors before graph processing, such that the input feature vector of node i becomes

$$\mathbf{x}_i = \left[\gamma(\bar{\mathbf{p}}_i) \,\|\, \mathbf{m}_i\right], \tag{4.4}$$

where $\mathbf{m}_i$ encodes the local material state and possible material-interface information.

The graph backbone is composed of six GraphSAGE message-passing layers. This architecture relies on stable neighborhood aggregation to propagate local geometric and material information across the structure. At each layer (k), the hidden representation of a node is updated by combining its current embedding with an aggregated representation

of its neighboring nodes:

$$\mathbf{h}_i^{(k+1)} = \sigma\left(\mathbf{W}^{(k)}\left[\mathbf{h}_i^{(k)} \parallel \mathrm{AGG}\big(\{\mathbf{h}_j^{(k)} : j \in \mathcal{N}(i)\}\big)\right]\right). \tag{4.5}$$

where $\mathcal{N}(i)$ denotes the neighborhood of node i, $\mathrm{AGG}(\cdot)$ is a mean aggregation operator, $\mathbf{W}^{(k)}$ is a trainable weight matrix at layer k, and $\sigma(\cdot)$ is a nonlinear activation function. Here, this neighborhood aggregation is mechanically meaningful because the deformation of a node depends not only on its own material and position, but also on the surrounding material arrangement, interfaces, and local connectivity. Residual connections, layer normalization, Gaussian Error Linear Unit (GELU) nonlinear activations, and dropout regularization are used throughout the network to improve optimization stability and reduce overfitting. The residual connections help preserve information across multiple message-passing layers, while layer normalization stabilizes the distribution of hidden features during training. Dropout further reduces the risk of over-specialization to the training set, which is important because the trained surrogate is later applied to large numbers of novel candidate structures.

The final node embeddings are mapped to the three nodal displacement components (dx, dy, dz), and the decoding stage is anisotropy-aware. Axial displacement is predicted through a relatively compact branch, whereas the two transverse displacement components are handled by reinforced decoding branches with deeper transformations and skip connections. This asymmetry is mechanically motivated, since the transverse components are more directly associated with bending- and twisting-dominated responses. In thermo-responsive voxelized LCE structures, small differences in material orientation and spatial arrangement can produce significant transverse deformation, including bending, twisting, mixed shear-bending, or localized shape changes. For this reason, the transverse displacement components are not treated as secondary outputs, but as key quantities for preserving the deformation signature used in the downstream behavior discovery stage.

In addition, the training pipeline incorporates lightweight data augmentation through perturbations of node features and target displacements, together with a weighted Huber-based loss that places stronger emphasis on the transverse directions. The training objective can be expressed as

$$\mathcal{L} = w_x\mathcal{L}_x^{\mathrm{Huber}} + w_y\mathcal{L}_y^{\mathrm{Huber}} + w_z\mathcal{L}_z^{\mathrm{Huber}} + \lambda\mathcal{L}_{\mathrm{top}}^{yz}, \tag{4.6}$$

where $\mathcal{L}_x^{\mathrm{Huber}}$, $\mathcal{L}_y^{\mathrm{Huber}}$, and $\mathcal{L}_z^{\mathrm{Huber}}$ are Huber losses computed independently for the three displacement components, and $\mathcal{L}_{\mathrm{top}}^{yz}$ penalizes the largest transverse prediction errors [465]. The Huber loss is used because it combines the stability of a quadratic loss for small residuals with improved robustness to larger errors. The additional transverse top-error term encourages the model to reduce large local errors in the y and z directions, which is particularly important for avoiding the loss of localized bending or twisting features.

This design reflects the central objective of the surrogate, namely accurate prediction of deformation patterns that remain informative for subsequent behavior analysis.

The proposed GNN model was systematically evaluated and compared with state-of-the-art methods, with the complete evaluation protocol, architectural comparison, statistical analysis, and computational-performance results reported in Appendix C. Results show that the GraphSAGE-GNN demonstrates superior performance for large-scale screening and exploration of candidate structures.

4.2.4 Multi-level discovery of shape morphing behaviors

To assess novelty in a mechanically meaningful manner, the proposed pipeline integrates a novelty reasoner that evaluates each generated structure using six complementary discovery approaches (DAs). This novelty reasoner is applied after the trained GNN has predicted the deformation responses of the candidate structures generated by the random and procedural design techniques. Rather than relying on a single novelty score, the novelty reasoner analyzes the predicted responses through multiple views of deformation, each emphasizing a different aspect of the thermo-mechanical response. These approaches are applied independently to both generation modes and to each of the four BC scenarios. This comparison strategy is particularly important in 4D printing, where distinct actuation mechanisms may lead to similar global amplitudes while differing substantially in deformation profile, symmetry, localization, or node-level displacement exceedance [455, 466].

Generation modes used for behavior discovery

The discovery stage is applied to two complementary sets of candidate structures. The first set is produced by a random generation mode, while the second set is produced procedural design techniques. Both generation schemes provide a wide spectrum of material distributions. They operate on the same $10 \times 3 \times 3$ voxelized domain and use the same materials: five oriented active LCE states, one passive material, and one void state.

Random generation mode. In the random generation mode, the algorithm enumerates all non-empty subsets of the five active LCE orientations. Since the available orientations are aligned with the Cartesian axes and the two diagonal directions, namely x, y, z, $+45°$, and $-45°$, this produces: $2^5 - 1 = 31$ active-orientation combinations. Each active subset is then combined with the passive and void states to form a material pool. For each material pool, random voxel distributions are sampled over the 90-voxel domain. The generated structure is admissible only if it satisfies two constraints: the fraction of

void voxels must remain below the prescribed maximum value, and all non-void voxels must form a single connected component under 6-neighbor connectivity.

Procedural design mode. The procedural design approach adopts a more structured exploration strategy. Rather than sampling voxel labels independently, the algorithm initializes candidate design from predefined topology families. Each family encodes a specific spatial arrangement of passive regions, void regions, and one or more active regions. The active regions correspond to independent connected domains, each of which can be assigned a distinct LCE mesogen orientation. For a topology family containing N independent active regions, the design space is constructed by enumerating all ordered combinations of the five active admissible mesogen , yielding up to 5^N configurations. To main computational tractability, a maximum sampling threshold is imposed per family, thereby limiting the total number of generated candidates. This approach enables systematic exploration of structured and physically interpretable regions of the design space. In particular, it captures a diverse set of of topology classes, including layered architectures, axial gradients, torsional configurations, checkerboard-like patterns, localized void distributions, wave-like motifs, and serpentine layouts, among others. **Figure 4.4** presents representative examples of these structured patterns, incorporating active and passive materials as well as void regions, as introduced in Figure 4.2c.

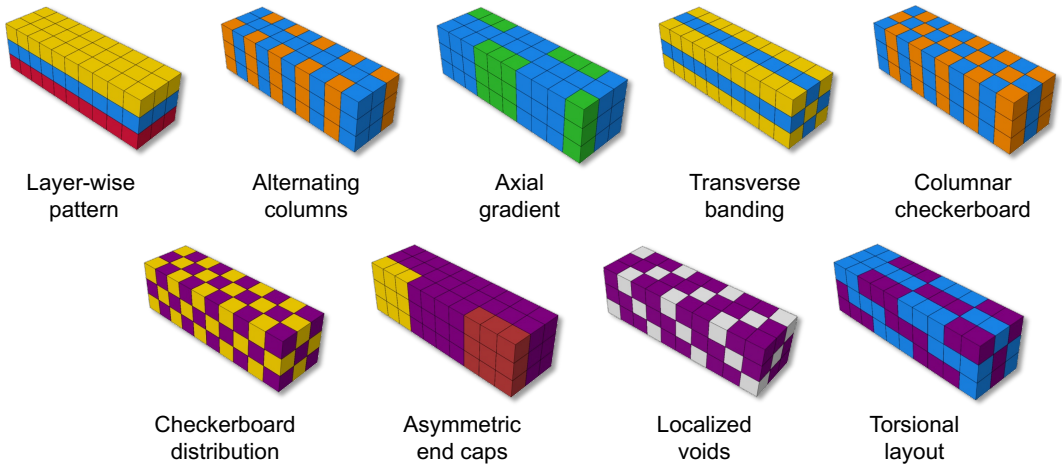

Figure 4.4: Representative topology families obtained using the procedural design approach. Each voxel-based design encodes a structured arrangement of active material, passive material and voids, following predefined spatial templates rather than independent voxel sampling. The shown families include layer-wise patterns, alternating columns, axial gradients, transverse banding, columnar checkerboard structures, checkerboard distributions, asymmetric end caps, localized voids, and torsional layouts. These parameterized templates define the connectivity and spatial organization of active regions, which are subsequently assigned distinct LCE orientations to generate diverse and physically interpretable design candidates.

Both generation modes were evaluated under the four BC scenarios considered in

this study. Consequently, the complete discovery campaign contains 400,000 random candidates and 400,000 parameterized candidates, corresponding to a total of 800,000 candidate structures.

Discovery approaches

The discovery stage is organized around six complementary criteria: two descriptor-based comparisons, namely a FE simulation-referenced comparison and a self-referential comparison, descriptor cosine similarity, maximum displacement amplitude, asymmetry-based response, and per-node displacement exceedance. A reference behavior library is also used for the approaches that require comparison with known deformation responses. This library was built from FE-simulated structures representing deformation modes reported in the literature and reproduced under the considered BC settings. It includes representative behaviors such as twisting, axial expansion combined with twisting, shrinking, bending, localized twisting, non-uniform twisting, double bending or hinge-like deformation, and coupled shrinking/expansion along different axes. The objective of this library is not to exhaustively cover all possible behaviors in 4D-printed active structures, but to provide a physically meaningful baseline of known responses against which newly generated candidates can be compared.

The descriptor-based discovery approaches, DA1 and DA2, rely on physically interpretable descriptors extracted along representative fibers of the voxelized structure. Fibers are constructed from the graph nodes of the voxelized structure. Each fiber is obtained by selecting nodes that share the same transverse position within the cross-section and by ordering them along the longitudinal axis of the structure. Each fiber therefore defines a one-dimensional path through the graph, extending from one end of the structure to the other. The same ordered node sequence is then tracked between the "as designed" and "as actuated" states, allowing mechanical descriptors to be computed from the resulting changes in fiber geometry. These descriptors include axial strain ε, curvature κ, torsion τ, shear-related deformation γ, growth ratio Λ, localized curvature extent, critical-strain-related quantities ε_{cr}, and cross-fiber strain contrasts $\Delta\varepsilon_t$ and $\Delta\varepsilon_w$. These quantities are used to interpret stretching, bending, twisting, shearing, folding, and buckling-like responses in slender active structures [467–470]. This descriptor-based family is divided into two distinct approaches: a FE simulation-referenced descriptor comparison and a self-referential descriptor comparison. To support these comparisons, each known motion type is associated with physics-informed descriptor ranges. These ranges are not used as rigid class definitions, but rather as reference signatures for interpreting deformation regimes and guiding novelty assessment. The corresponding descriptor thresholds are summarized in **Table 4.1**.

Table 4.1: Physics-informed descriptor ranges associated with known deformation modes in the descriptor-based comparison framework. The listed ranges are used as reference signatures for novelty assessment and should not be interpreted as strict class boundaries.

Primitive	Parameter	Symbol	Threshold range	Notes	Reference
Stretching	Axial strain	ε	0.01–0.30	Larger values may indicate non-recoverable deformation	[467, 468]
	Curvature	κ	< 0.05 m^{-1}	Near-zero; larger values suggest bending onset	[468, 469]
	Torsion	τ	< 0.01 rad/m	Near-zero by definition	[467, 469]
Bending	Strain across thickness	$\Delta\varepsilon_t$	0.01–0.15	Through-thickness gradient must be present	[468, 470]
	Curvature	κ	0.1–10 m^{-1}	Below this range: stretching-like; above it: folding-like	[468, 469]
Twisting	Strain across width	$\Delta\varepsilon_w$	0.01–0.15	Must be width-directed rather than thickness-directed	[467, 468]
	Torsion	τ	0.1–2.0 rad/m	Main distinguishing feature relative to bending	[467, 469]
	Curvature	κ	< 0.05 m^{-1}	Near-zero; twist without significant bend	[468, 469]
Folding	Localized curvature	κ	> 10 m^{-1}	Strongly concentrated at a crease	[468, 470]
	Spatial extent	–	$< 10\%$ of total length	Distinguishes folding from distributed bending	[468]
Buckling	Curvature	κ	> 5 m^{-1}	Post-critical, mode-shape dependent	[467, 470]
	Critical strain	ε_{cr}	Material-dependent	Often estimated from Euler-type instability relations	[470]
Shearing	Curvature	κ	< 0.05 m^{-1}	Approximately zero; no dominant bending	[467, 470]
	Torsion	τ	< 0.01 rad/m	Approximately zero; no dominant twist	[467, 469]
	Shear strain	γ	0.01–0.20	Primary identifying quantity	[467, 470]

An important feature of the discovery framework is that each approach is evaluated at multiple threshold levels rather than at a single fixed operating point. This multi-level strategy makes it possible to analyze how sensitive each discovery criterion is to the strictness of the novelty definition. Conservative levels retain only the most extreme candidates, whereas more permissive levels provide a broader view of potentially interesting structures. In the implementation, the descriptor-based comparisons are evaluated using fiber-agreement thresholds ranging from 1% to 80%. The cosine-similarity approach is evaluated using low-percentile thresholds, namely 1%, 0.5%, 0.3%, and 0.1%, where lower similarity indicates stronger novelty. The maximum-displacement approach is evaluated using increasingly selective upper-tail thresholds, which retain only the candidates with the largest predicted amplitude responses, from approximately the top 1% of the population down to the top 0.1%. The asymmetry-based response is evaluated using normalized thresholds from 0.5 to 0.99. Finally, the per-node displacement exceedance approach is evaluated using node-exceedance fractions ranging from 1% to 80%. The fiber-agreement percentage represents the fraction of representative fibers that must satisfy the descriptor-based agreement or deviation criterion, thereby controlling whether novelty is detected as a localized deviation or as a more global deformation signature.

FE simulation-referenced descriptor comparison (DA1). Here, novelty is assessed relative to the reference behavior library described above, which contains FE-simulated structures representing known deformation responses reported in the literature. The method compares each GNN-predicted candidate against descriptor ranges extracted from the FE simulation reference set. These ranges are computed from the 5th and 95th percentiles of the reference fiber descriptors, thereby providing data-driven reference

intervals rather than fixed class boundaries. For comparability between FE simulation references and GNN-predicted candidates, this mode uses a restricted set of descriptors. Stretching is evaluated using axial strain and growth ratio, bending using cross-thickness strain contrast and axial strain, shearing using shear-related deformation and axial strain, and twisting using cross-width and cross-thickness strain contrasts. A candidate structure is considered consistent with a known response when a sufficient fraction of its fibers falls within the corresponding reference descriptor ranges. This fraction is varied from 1% to 80% to control the strictness of the comparison. Structures that do not satisfy the required fiber agreement for the reference response families are identified as potentially novel.

Self-referential descriptor comparison (DA2). In this approach, novelty is assessed relative to the behavior distribution generated by the candidate population itself. Unlike the FE simulation-referenced descriptor comparison, this self-referential mode can use the full descriptor framework, because all candidates are compared within the same predictive setting and are therefore affected by similar model and discretization biases. Percentile-based descriptor ranges are computed from the generated structures themselves, using the 5th and 95th percentiles. Structures that fall outside these internal ranges are interpreted as outliers with respect to the generated population. As in the FE simulation-referenced descriptor comparison, this approach is evaluated at multiple fiber agreement levels ranging from 1% to 80%. This enables the detection of increasingly strict outliers associated with stretching, bending, twisting, folding, buckling, and shearing-like descriptor signatures.

Descriptor cosine similarity (DA3). This approach evaluates novelty through similarity in descriptor space. For each candidate structure, a behavior signature is built from fiber-level and cross-fiber descriptors. The candidate signature is then compared with the reference signatures using cosine similarity. Since cosine similarity emphasizes profile direction rather than absolute magnitude, low similarity indicates a deformation pattern that differs from previously observed reference behaviors. In the implementation, the distribution of maximum cosine similarities is analyzed, and candidates below the 1%, 0.5%, 0.3%, and 0.1% percentiles are classified as novel at increasingly selective levels.

Maximum displacement (DA4). This approach focuses on the amplitude of the predicted deformation response. It computes a displacement-related score from the maximum combined contribution of axial strain and shear-like deformation descriptors across the fibers of each candidate structure. This criterion is not intended to classify the underlying deformation mechanism by itself. Instead, it identifies structures whose global actuation amplitude is unusually large compared with the rest of the explored population.

In practice, candidates are ranked according to this amplitude score, and only the structures located in the extreme upper tail of the distribution are retained. The different threshold levels correspond to selecting increasingly small fractions of the population with the highest amplitude responses, ranging from approximately the top 1% of candidates to the top 0.1%. Therefore, this approach progressively filters the candidate set from broadly high-displacement structures to only the most extreme predicted actuation cases.

Asymmetry-based response (DA5). This approach evaluates the degree to which the deformation field departs from a symmetric or homogeneous response. In the implementation, asymmetry is estimated from the normalized inter-fiber variability of key descriptors, including axial strain, shear-related deformation, and curvature. High asymmetry indicates that the structure exhibits spatially differentiated deformation, which is especially relevant in heterogeneous active systems where nonuniform material placement can trigger unconventional shape changes. The normalized asymmetry score is evaluated using thresholds from 0.5 to 0.99, allowing the analysis to range from broadly asymmetric structures to only the most extreme asymmetry cases.

Per-node displacement exceedance (DA6). This approach operates at the graph-node level. For each reference structure, the maximum observed transverse displacement is stored at each node. A candidate structure is then evaluated by measuring the fraction of nodes whose transverse displacement exceeds the reference maximum at the corresponding node. This criterion complements the fiber-level and global criteria by detecting localized deformation anomalies that may not strongly affect averaged descriptors. The node-exceedance fraction is evaluated using thresholds ranging from 1% to 80%, making it possible to distinguish between localized exceedance events and more spatially widespread deviations from known thermo-mechanical responses.

These discovery approaches are applied independently to the two generation modes and to the four BC scenarios (see Figure 4.2d). This organization is important because novelty in 4D-printed active structures is not an intrinsic property of the material distribution alone. It also depends on the generation strategy and on the mechanical boundaries, both of which can strongly modify the accessible deformation response. The resulting framework therefore compares candidate behaviors under a common multi-approach discovery protocol rather than through a single scalar novelty score. It is worth noting that although the growth ratio is used as an auxiliary descriptor in the implementation, volumetric expansion or shrinkage is not treated as an independent deformation class in the present discovery framework.

Acceleration of shape-morphing discovery

The random generation mode produced 100,000 candidate structures per BC, whereas the procedural design mode produced 100,000 candidate structures per BC. The resulting discovery results are summarized in **Table 4.2**. For readability, threshold levels for which no structure was identified as novel in any generation mode or BC are omitted from the table.

Table 4.2: Number of newly identified structures through discovery approaches, threshold levels, generation modes, and BCs. Threshold rows for which all values are zero across all generation modes and BCs are omitted for readability.

		Random generation				Procedural design			
Discovery approach	Level	BC1	BC2	BC3	BC4	BC1	BC2	BC3	BC4
DA1 – FE simulation-referenced descriptor comparison	40%	0	1	0	0	0	0	0	0
	45%	0	29	0	0	0	17	0	0
	50%	0	189	0	0	0	90	0	0
	55%	0	839	0	6	0	1,411	0	0
	60%	0	4,528	0	112	0	10,375	0	5
	65%	0	11,771	0	439	0	18,282	0	50
	70%	0	33,075	0	2,442	0	54,023	0	598
	75%	0	52,061	0	6,044	0	74,267	0	2,144
	80%	118	79,161	1,525	18,180	4	92,061	1,135	11,041
DA2 – Self-referential descriptor comparison	70%	0	11	0	2	0	2	1	0
	75%	26	139	32	46	2	82	16	0
	80%	1,364	2,716	1,447	1,724	637	2,334	1,269	1,346
DA3 – Descriptor cosine similarity	1%	930	930	930	930	1,001	1,001	1,001	1,001
	0.5%	279	465	279	465	301	501	301	501
	0.3%	93	279	93	279	101	301	101	301
	0.1%	93	93	93	93	101	101	101	101
DA4 – Maximum displacement	99%	930	930	930	930	1,001	1,000	1,001	1,001
	99.3%	651	651	651	651	701	701	701	443
	99.5%	465	465	465	465	501	501	501	443
	99.7%	279	279	279	279	301	301	301	282
	99.9%	93	93	93	93	101	100	101	91
DA5 – Asymmetry-based response	0.5	93,000	92,723	92,993	92,990	100,010	99,960	99,994	100,010
	0.6	91,797	81,017	89,867	89,736	99,786	95,632	93,969	99,496
	0.7	66,070	38,471	50,824	54,210	82,898	41,033	39,911	86,545
	0.8	14,457	5,896	6,867	9,846	19,729	2,872	2,307	33,127
	0.9	519	193	142	295	494	65	25	984
	0.93	138	51	36	68	99	30	10	213
	0.95	39	23	11	24	31	17	9	46
	0.97	10	6	3	7	9	5	4	9
	0.99	3	3	2	1	1	3	2	1
DA6 – Per-node displacement exceedance	1%	0	0	1,035	0	0	0	1,869	0

For a large-scale discovery campaign of approximately 8×10^5 candidate structures, direct FE simulation would be computationally prohibitive. Considering an end-to-end FE workflow time of approximately 3600–10,800 s per structure, including model setup and numerical solution, evaluating the complete population would require between 2.88×10^9 and 8.64×10^9 s. This corresponds to approximately 800,000–2,400,000 hours,

33,333–100,000 days, or 91.3–274.0 years of sequential computation. By contrast, the retained GraphSAGE surrogate requires only 0.0269 s per structure for pure inference. Predicting the deformation of 8×10^5 structures therefore requires approximately 21,520 s, corresponding to 358.7 min, or 5.98 h, excluding structure generation, data conversion, file I/O, graph construction, and behavior post-processing. Relative to the complete FE workflow, this corresponds to a prediction speedup of approximately 1.34×10^5–4.01×10^5, equivalent to a reduction in pure prediction time of approximately 99.99925%–99.99975%.

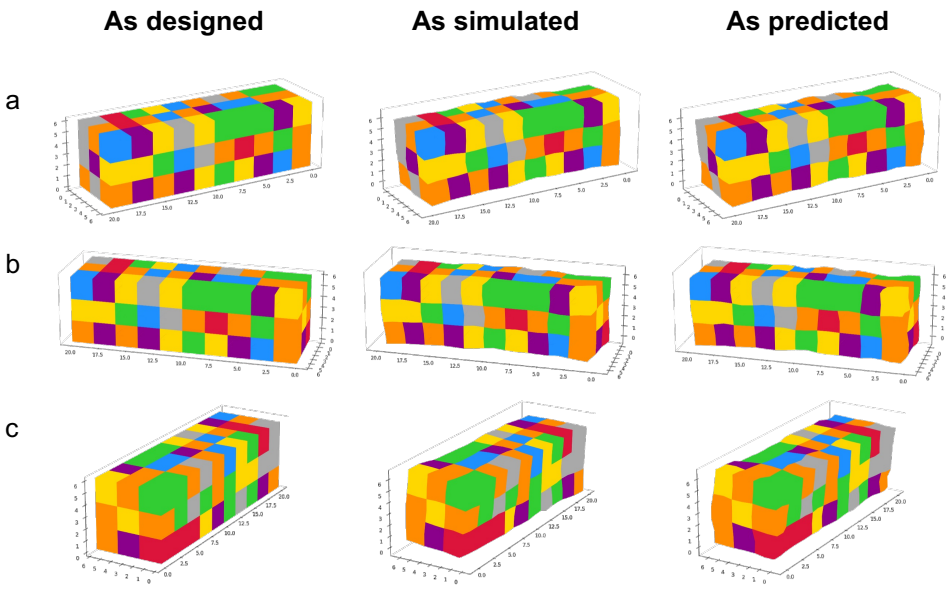

Figure 4.5: Visualization of a voxelized multi-material structure showing its "as designed" configuration, "as simulated" deformation obtained via FE simulation, and "as predicted" deformation using a GraphSAGE-based GNN. (a) Front view, (b) back view, and (c) side view. The retained surrogate model predicts the deformed configuration with an inference time of 0.0269 s per structure. Material colors are defined as follows: LCE_X in orange, LCE_Y in red, LCE_Z in green, LCE_{45} in violet, LCE_{-45} in yellow, passive material in blue, and void regions in light gray, as defined in Figure 4.2c.

Quantitative analysis. In Table 4.2, the descriptor-based criteria show a strong dependence on both the selected threshold level and the BC. The FE simulation-referenced descriptor comparison is highly selective at low fiber agreement levels and only begins to identify substantial numbers of structures at high agreement thresholds. In the random generation mode, BC2 is the most sensitive to this criterion: the number of detected structures increases from 189 at the 50% level to 79,161 at the 80% level, corresponding to 85.1% of the 93,000 generated random structures. By contrast, BC4 reaches 18,180 structures at the 80% level, corresponding to 19.5% of the generated random population, while BC3 and BC1 remain much less affected, with 118 and 1,525 detections, respectively. A similar trend is observed in the procedural design mode, where BC2 again dominates, reaching 92,061 detections at the 80% level, corresponding to 92.1% of the 100,010

procedurally generated structures. BC4 reaches 11,041 detections, whereas BC1 and BC3 remain limited to 4 and 1,135 detections, respectively.

The self-referential descriptor comparison is substantially more conservative. No structure is detected below the 70% level, and the highest counts appear only at the 80% threshold. At this level, the random generation mode identifies 2,716, 1,724, 1,364, and 1,447 structures for BC2, BC4, BC1, and BC3, respectively. These values correspond to approximately 1.5–2.9% of the random population, depending on the BC. In the procedural design mode, the same criterion identifies 2,334, 1,346, 637, and 1,269 structures, corresponding to approximately 0.6–2.3% of the generated population. This confirms that the self-referential descriptor comparison acts as a stricter outlier detector than the FE simulation-referenced descriptor comparison, because it evaluates novelty relative to the internal distribution of the generated population.

The descriptor cosine similarity criterion behaves as expected from its percentile-based definition. In the random generation mode, the 1%, 0.5%, 0.3%, and 0.1% levels identify 930, 465, 279, and 93 structures, respectively, for each BC. These values exactly correspond to the selected proportions of the 93,000 generated structures. In the procedural design mode, the same levels identify 1,001, 501, 301, and 101 structures, respectively, which similarly correspond to the selected proportions of the 100,010 generated structures. This criterion therefore provides a controlled way to select the most directionally dissimilar deformation signatures relative to the reference set.

The maximum displacement criterion also follows a percentile-based behavior, but with a notable exception for BC1 in the procedural design mode. In the random generation mode, the successive upper-tail selection levels consistently identify 930, 651, 465, 279, and 93 structures for each BC, corresponding approximately to the top 1%, 0.7%, 0.5%, 0.3%, and 0.1% of the generated population. For the procedural design mode, BC2, BC4, and BC3 show the expected behavior, with approximately 1%, 0.7%, 0.5%, 0.3%, and 0.1% of structures retained across the percentile levels. However, under BC1, the procedural design identifies only 443 structures for the first three upper-tail levels, 282 structures for the next stricter level, and 91 structures for the strictest level. This indicates a less continuous or more clustered displacement distribution under this BC, where several percentile thresholds map to the same displacement value.

The asymmetry-based response criterion is the most permissive at low threshold levels and the most discriminative at high threshold levels. At the 0.5 level, nearly all structures are identified as asymmetric in both generation modes. In the random generation mode, the number of detected structures ranges from 92,723 to 93,000, corresponding to 99.7–100% of the generated population. In the procedural design mode, the same level identifies between 99,960 and 100,010 structures, again corresponding to almost the entire population. As the threshold increases, the number of detected structures decreases sharply. At the 0.9 level, only 142–519 structures are retained in the random mode and 25–

984 structures in the logic-family mode. At the strictest 0.99 level, the criterion isolates only one to three structures depending on the generation mode and BC. This strong decrease shows that asymmetry is broadly present in the explored design space, but that extreme asymmetry remains rare.

The per-node displacement exceedance criterion is the most localized and selective of the six discovery approaches. Non-zero detections occur only at the 1% node-exceedance level and only for BC3. In the random generation mode, this criterion identifies 1,035 structures, corresponding to 1.1% of the random Left-voxel population. In the procedural design mode, it identifies 1,869 structures, corresponding to 1.9% of the procedurally generated population while considering BC3. No structures are detected for the other BCs or for higher node-exceedance thresholds. This suggests that node-level exceedance is primarily triggered by localized BCs, where small changes in material distribution can generate localized displacement peaks relative to the reference maximum.

Overall, the quantitative results show that the six discovery approaches capture complementary notions of novelty. The descriptor cosine similarity and maximum displacement criteria provide controlled percentile-based selection of unusual response profiles or high amplitude responses. The descriptor-based comparisons reveal sensitivity to the physical interpretation of deformation regimes, especially under BC2. The asymmetry-based response criterion shows that most generated heterogeneous structures produce nonuniform deformation fields, while its high threshold levels isolate only the most extreme asymmetric responses. Finally, the per-node displacement exceedance criterion identifies localized deviations that are not necessarily captured by global or fiber-averaged criteria. These trends confirm the relevance of considering a multi-approach discovery framework: no single criterion captures all forms of novelty, and the interpretation of newly discovered behaviors depends jointly on the generation mode, the BC, and the selected novelty threshold.

Qualitative analysis. A qualitative inspection of the retained candidates reveals representative deformation responses identified across the six discovery criteria, and four BC scenarios, as shown in **Figure 4.6**. These examples are drawn from both generation modes. Each panel shows the deformed configuration colored by material type (see Figure 4.2c), with the initial geometry superimposed as a reference outline, allowing the relationship between material distribution and deformation response to be directly observed.

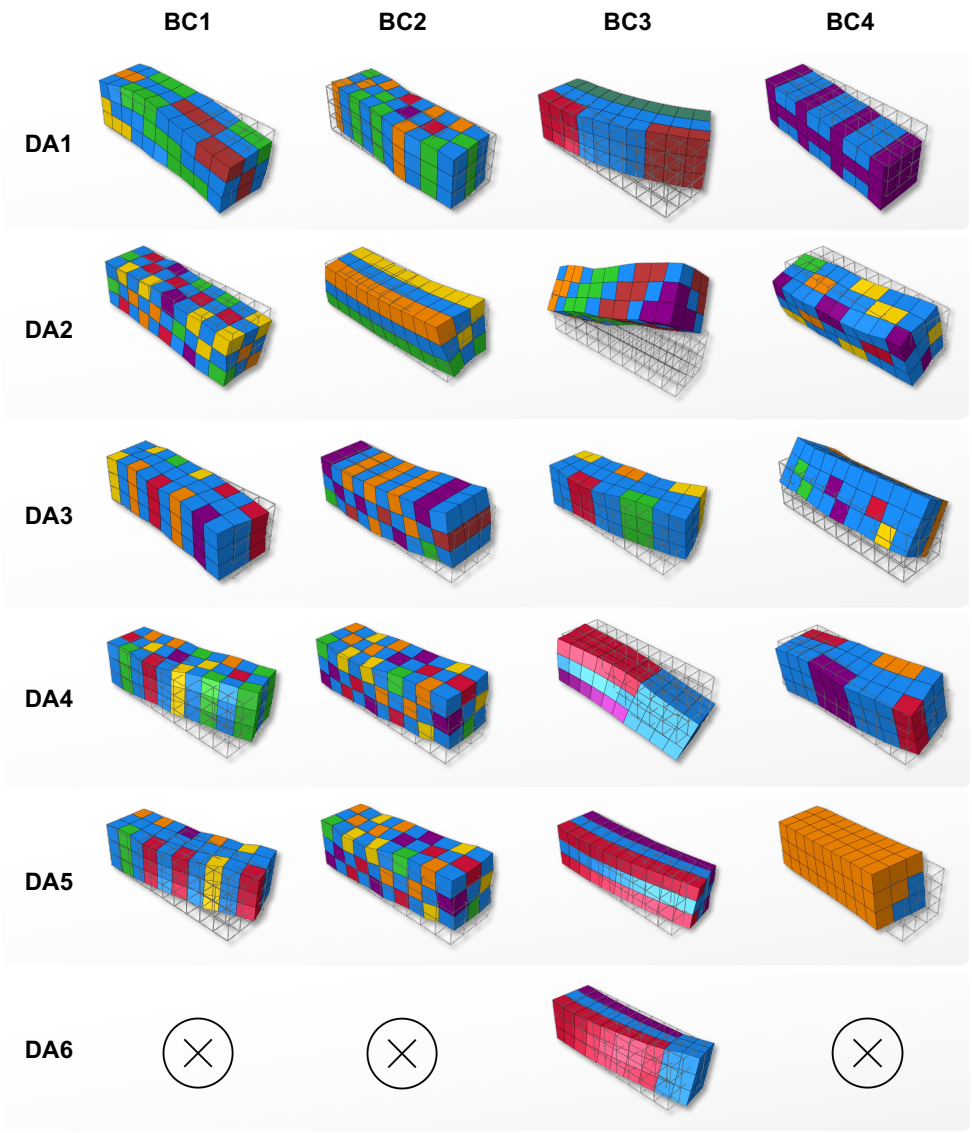

Figure 4.6: Qualitative examples of candidate deformation behaviors discovered within the explored voxelized multi-material design space. From left to right, candidates correspond to boundary condition scenarios (BC1 to BC4). From top to bottom, rows correspond to the different discovery approaches (DA1 to DA4). Cross symbols indicate that no valid candidates were generated for the corresponding combination of discovery approach and BC.

The panels reveal qualitatively distinct deformation responses across both criteria and BCs. Within each row, the character of the deformation reflects the specific notion of novelty associated with the corresponding discovery criterion. Descriptor-based comparisons tend to identify responses in which multiple deformation components are active simultaneously, while maximum displacement retains structures with unusually large actuation amplitudes. Asymmetry-based criteria select responses with strongly non-uniform deformation fields, and per-node displacement exceedance identifies cases where deformation is concentrated at specific nodes rather than distributed globally. Within each column, the influence of the BC is visible in the spatial distribution and orientation

of the deformation field, consistent with the quantitative results showing strong sensitivity of novelty detection to the support configuration.

Among the four BCs, BC2 tends to produce the richest set of qualitatively interpretable candidate behaviors. This trend is consistent with the quantitative results, where BC2 leads to the largest number of structures identified by the FE simulation-referenced descriptor comparison. Mechanically, this can be explained by the fact that a localized BC on the bottom region constrains the structure while still allowing substantial global freedom for bending, twisting, shear-like deformation, and curvature redistribution along the beam. Therefore, BC2 appears to promote structured and mechanically interpretable morphing responses more effectively than the other BCs.

A qualitative difference is observed between the two generation modes. The procedural design mode produces the most interpretable behavior-level discoveries because it relies on predefined spatial rules, such as gradients, alternating regions, checkerboards, and sandwich-like arrangements. These organized layouts promote mechanically structured responses and make it easier to relate a given material distribution to a visible morphing mechanism. In contrast, the random generation mode also produces candidates identified as novel, but these cases more often correspond to unusual displacement amplitudes, localized displacement exceedance, or irregular deformation fields. Therefore, random generation is particularly useful for detecting outlier responses, whereas logic-family generation is more effective for identifying structured and explainable deformation behaviors.

Among the six discovery criteria, the FE simulation-referenced descriptor comparison provides the most direct qualitative support for behavior-level novelty, since it compares each candidate against a reference library of known deformation responses. Therefore, it directly highlights structures whose descriptor signatures deviate from already documented behaviors. The other approaches remain complementary, each capturing the aspects of novelty not reflected by descriptor deviation alone. Across all panels in Figure 4.6, the material distributions further indicate that the identified responses arise from voxel organisations that simultaneously engage multiple known deformation modes, rather than from material layouts associated with a single known behavior.

4.2.5 Digital light processing 3D printing OR Direct Ink Write 3D printing

To validate predicted and discovered shape-morhing behaviors.ELEONORE PART OR AARJU PART

4.2.6 Neuro-Symbolic Positioning: The *Symbolic[Neuro]* Paradigm

The complete simulation-and-discovery workflow can be interpreted as a workflow-level instantiation of the nested *Symbolic[Neuro]* paradigm defined in Chapter 2. The symbolic control component comprises the graph schema, the discrete material vocabulary, the connectivity and material-validity rules, and the physics-based novelty reasoner. Within this structured workflow, GraphSAGE is invoked as a neural subroutine to map a material-aware structural graph to a nodal displacement field. The predicted field is then returned to the symbolic component, which reconstructs the actuated graph and evaluates its behavior using explicit deformation descriptors and threshold-based rules.

4.2.7 Limitations and remaining challenges

Several challenges and limitations remain before the proposed framework can be considered a fully validated discovery tool for 4D-printed active structures. The first limitation concerns the computational nature of the study. The discovered behaviors are predicted from a GraphSAGE surrogate trained on FE simulation-generated data. Although this enables large-scale screening, the surrogate inherits the assumptions, approximations, and possible biases of the FE simulations used for training. Therefore, the identified behaviors should be considered as computational candidates that require further confirmation by high-fidelity modeling and FE simulations.

A second limitation is related to the size and resolution of the voxelized domain. The present study uses a controlled $10 \times 3 \times 3$ beam-like grid, which is useful for systematic generation and large-scale exploration. However, this discretization restricts the geometric complexity of the structures and limits the range of possible deformation mechanisms. Some fine-scale phenomena, localized instabilities, or complex three-dimensional shape changes may not be fully captured at this resolution. Future work should therefore investigate finer voxel grids, different aspect ratios, and more complex geometries to assess whether the discovered behaviors remain stable across scales and shapes.

A third challenge concerns the definition of novelty. In this chapter, novelty is evaluated through descriptor-based comparison, cosine similarity, maximum displacement, asymmetry, and node-level exceedance. These criteria capture complementary aspects of the deformation response, but none of them alone provides a definitive definition of a new behavior. In particular, some candidates may appear novel because they produce large or localized displacements, while their underlying deformation mechanism remains close to known bending, twisting, or folding responses. Conversely, some mechanically interesting behaviors may not appear as extreme outliers in terms of displacement amplitude. This highlights the need to combine quantitative criteria with qualitative mechanical interpretation.

A geometric perspective on the descriptor space may further strengthen such validation rules. If each known behavior in the reference library is characterised by a bounded region in the descriptor space, a candidate whose descriptor signature falls outside all such regions, or spans multiple regions simultaneously, satisfies a stricter novelty criterion that is independent of displacement amplitude and complementary to the qualitative mechanical interpretation. Finally, a formal characterisation of the descriptor space, mapping the regions occupied by known behaviors and the zones that separate them, could provide a geometric basis for novelty definition and guide the development of more principled classification criteria for discovered behaviors [471].

The descriptor framework also presents limitations. The current descriptors are sufficient to describe major deformation mechanisms such as bending, twisting, shearing, folding, and buckling-like responses. However, some of the observed candidate behaviors require more specific descriptors. For example, S-shaped bending requires a descriptor able to count curvature sign reversals along the beam. Segmented or accordion-like bending requires a descriptor related to the spatial frequency of curvature oscillations. Localized kink deformation requires a more precise localization index, while oblique shear–bending coupling would benefit from descriptors coupling longitudinal curvature, transverse curvature, and shear strain. Enriching the descriptor set is therefore necessary to make the classification of new behaviors more rigorous.

Future work should address these limitations in several directions. First, selected candidate behaviors should be re-simulated using high-fidelity FE simulation to verify the surrogate predictions and quantify the error associated with each morphology. Second, explicit validation rules should be introduced to determine when a deformation response can be considered a new behavior rather than a variant of an already known mode. These rules should combine quantitative descriptors, qualitative deformation signatures, recurrence across structures, and agreement with FE simulation or experimental results. Third, the discovery campaign should be extended to a larger number of structures and to different voxelized resolutions. Exploring finer grids, longer beams, thicker cross-sections, and alternative aspect ratios would make it possible to test whether the discovered behaviors are robust or specific to the current $10 \times 3 \times 3$ discretization. Fourth, the framework should be applied to other active material-stimuli systems beyond the LCE configuration considered here, such as shape-memory polymers, thermo-responsive hydrogels, magneto-active elastomers, or multi-material combinations with different stiffness and actuation contrasts. This extension would help evaluate whether the proposed discovery workflow is material-specific or can generalize to broader classes of 4D-printed active structures.

Finally, uncertainty-aware GNN models and active learning strategies could be introduced to identify regions of the design space where predictions are less reliable or where behavior novelty is expected to be higher. Active learning could automatically select

the most informative candidates for additional FE simulation, improving both surrogate accuracy and behavior discovery efficiency while reducing unnecessary computations.

4.3 Conclusion

In this chapter, we presented a GNN-assisted framework for accelerating thermo-mechanical simulation and supporting shape-morphing behavior discovery in voxelized 4D-printed structures. The proposed pipeline combines automated generation of physically admissible voxelized structures, graph-based representation, GNN surrogate prediction, and multi-criteria novelty assessment. This makes it possible to explore a large combinatorial design space that would be impractical to investigate using direct FE simulation alone.

The discovery analysis shows that different criteria capture different forms of novelty. Descriptor-based comparison identifies deviations from known deformation signatures, cosine similarity detects unusual response profiles, maximum displacement highlights high-amplitude actuation, asymmetry captures nonuniform responses, and per-node exceedance reveals localized deviations. This confirms that behavior discovery in active multi-material structures requires a multi-criteria analysis rather than a single novelty measure. The qualitative analysis identified representative candidate deformation behaviors within the explored voxelized LCE design space. These behaviors are interpreted as candidate discoveries within the considered design space, not as universally new physical classes. The results also suggest that procedural design technique provides more structured and interpretable behavior-level discoveries, whereas random generation is more effective for revealing displacement outliers and irregular responses.

Overall, the proposed framework demonstrates that graph-based learning extends beyond efficient deformation prediction to enable a more principled and structured exploration of the design space. By embedding spatial relationships and material connectivity directly into the learning process, the approach facilitates not only rapid forward evaluation but also improved interpretability of complex shape-morphing behaviors in 4D-printed active structures. This capability is particularly significant in uncovering non-intuitive deformation mechanisms and in linking local material arrangements to global functional responses. Moreover, the integration of structured procedural strategies with predictive modeling establishes a foundation for informed design iteration, where candidate architectures can be systematically generated, evaluated, and refined. Such a paradigm shifts the design process from heuristic trial-and-error toward a more data-driven and physically grounded methodology. Therefore, these advances open promising pathways toward the deployment of sophisticated, programmable materials in real-world applications. In soft robotics, they enable the design of adaptive actuators with tailored motion profiles; in biomedical engineering, they support the development of responsive

implants and minimally invasive devices; and in architecture, defense, and space systems, they offer new opportunities for deployable, lightweight, and adaptive structures capable of operating under complex environmental constraints.

The fast forward-prediction capability established in this chapter provides the computational basis for addressing the complementary inverse problem. Building on the graph representation and surrogate thermo-mechanical model developed here, the next chapter investigates how prescribed target shapes can be translated into suitable geometries and material distributions through graph-based representation learning and hybrid search strategies.

Chapter 5

Multidimensional Inverse Design of Multi-Material 4D-Printed Structures via Graph-Based Learning

5.1 Introduction

By combining additive manufacturing with stimuli-responsive materials, such as shape-memory polymers, hydrogels, liquid crystal elastomers, and active composites, 4D printing enables the fabrication of adaptive systems with potential applications in soft robotics, biomedical devices, aerospace components, energy systems, and deployable structures [1, 163, 472–474]. In these applications, functionality is not only determined by the material response itself, but also by the ability of the printed structure to transform into a prescribed target shape with sufficient accuracy, robustness, and repeatability.

The need for accurate target-shape control is driven by a wide range of applications. In biomedical engineering, deployable scaffolds, stents, and drug delivery devices require controlled deformation under physiological stimuli to ensure minimally invasive deployment and site-specific functionality [22, 475, 476]. In soft robotics, 4D-printed actuators and grippers exploit programmable bending, twisting, and gripping motions to interact with complex environments [474, 477, 478]. In aerospace and civil engineering, morphing components, deployable structures, adaptive trusses, and vibration-damping systems require reliable shape transformation under operational constraints [1, 22]. These examples show that target-shape matching is not only a geometric objective, but also a functional requirement for many 4D-printed systems.

Despite significant progress in materials development, process control, and stimulus-responsive behavior, the design of 4D-printed structures that achieve a desired shape

transformation remains a major challenge [10, 22]. The final activated shape emerges from a complex and highly coupled interplay between geometry, boundary conditions and spatially distributed materials properties. This complexity is further amplified in multi-material structures, where heterogeneous mechanical, thermal, and chemical properties must be spatially arranged while preserving interfacial compatibility and functional coherence under actuation [10, 13, 70]. Consequently, the design of structures capable of morphing into a desired target shape constitutes an inherently inverse problem. The high dimensionality and nonlinearity of this design space make such inverse design tasks particularly challenging for conventional modeling and optimization approaches.

Several computational design strategies have been proposed to address this problem. Voxel-based modeling discretizes the design domain into material-bearing voxels, making it possible to explicitly represent heterogeneous material distributions in 4D-printed structures [13, 450]. Topology optimization provides a physics-based framework for distributing materials according to mechanical or morphing objectives [18, 479]. Finite element (FE) simulations remain essential for high-fidelity evaluation of nonlinear thermo-mechanical responses, but they become expensive when repeated over large design spaces. Moreover, direct optimization is difficult when the design variables include both continuous geometric parameters and discrete material assignments. The high dimensionality of voxel-level material distributions, the nonlinear nature of smart material actuation, and the sensitivity of morphing behavior to local material arrangements limit the scalability of conventional approaches.

Machine learning (ML) has recently been explored as a promising alternative to accelerate both forward prediction and inverse design in 4D printing. Early studies demonstrated the potential of data-driven models for designing active composite structures and predicting shape-morphing behavior [16, 66]. More recently, ML-enabled inverse design strategies have been proposed for 4D-printed active plates, combining learned forward models with gradient-based or evolutionary optimization to recover material distributions that match prescribed three-dimensional target shapes [15]. Sequential subdomain optimization has further improved the efficiency of inverse design for large-scale 4D-printed active composite structures [17]. In parallel, ML-driven frameworks have been developed for forward prediction and inverse design of 4D-printed hierarchical architectures with arbitrary shapes [78]. These studies highlight the potential of learned surrogate models, but also reveal remaining limitations in representing irregular structural connectivity, handling discrete material choices, and exploring high-dimensional design spaces efficiently.

Graph neural networks (GNNs) provide a natural framework for addressing part of these limitations because 4D-printed structures can be represented as graphs. Unlike grid-based neural networks, GNNs can operate directly on irregular geometries, lattice-like architectures, and mesh-based representations. Message-passing mechanisms allow local

interactions between connected nodes to be learned and propagated across the structure, making GNNs suitable for approximating mechanical responses and deformation fields [195, 456, 460, 480]. In the context of inverse design, a GNN-based surrogate model can replace repeated high-fidelity simulations by rapidly predicting the deformed configuration of candidate designs while preserving structural connectivity.

However, using a surrogate model alone does not fully solve the inverse design problem, since the search space of possible geometries and material distributions remains large, non-convex, and partly discrete. Generative models, particularly variational autoencoders (VAEs), offer a complementary solution by learning a compact latent representation of valid designs [481–483]. By mapping complex structures into a continuous latent space, VAEs enable smoother exploration of geometric variations and reduce the dimensionality of the optimization problem. Nevertheless, in multi-material 4D printing, material assignment remains a categorical decision, which is not naturally handled by purely continuous latent-space optimization. Evolutionary and black-box strategies, such as the covariance matrix adaptation evolution strategy (CMA-ES), are therefore attractive because they can operate on nonlinear, non-convex, and non-differentiable objectives without requiring explicit gradients [484].

In this chapter, we present a scalable inverse design framework for multi-material 4D-printed structures targeting prescribed shape transformations. The proposed framework combines a GNN surrogate model for fast deformation prediction, a graph-based VAE for learning a compact latent representation of candidate geometries, and optimization strategies for material distribution refinement. These strategies include retrieval-based screening, stochastic latent-neighborhood sampling, CMA-ES, Bayesian optimization, local material mutation, and a combined top-k search with local mutation strategy. Given a target shape, the method explores candidate geometries and material configurations, predicts their activated shapes using the learned GNN surrogate, and minimizes the geometric discrepancy between the predicted deformation and the prescribed target. This formulation enables efficient target-shape-driven design while accounting for graph connectivity, discrete material assignment, and the multi-material nature of 4D-printed structures.

5.2 Results and Discussion

Figure 5.1 presents a scalable inverse design framework for multi-material 4D-printed structures, which enables prescribed shape morphing across varying dimensionalities and geometries. The framework couples a deformation-aware graph-based VAE for latent space representation learning, a GNN surrogate model for forward prediction of deformation responses, and an optimization layer for efficient refinement of material distribution. This unified pipeline establishes a consistent mapping between shapes, graph-

based structural representations, voxel-level material distributions, and the resulting deformation behavior, therefore enabling efficient inverse design under diverse boundary conditions.

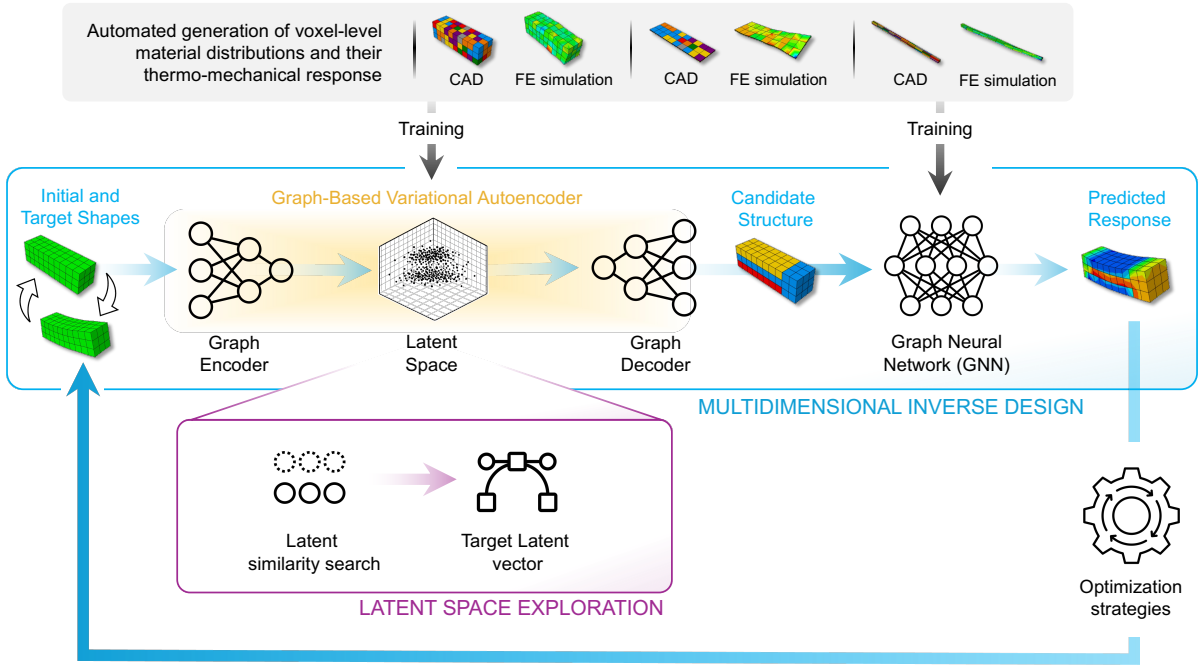

Figure 5.1: Overview of the scalable inverse design framework for multi-material 4D-printed structures across 1D, 2D, and 3D design domains. Automated generation of voxel-level material distributions and corresponding deformation response is first performed via computer-aided design and finite element simulations for training. The inverse design workflow integrates a graph-based VAE learning to encode voxel-level initial and target shapes into a continuous, deformation-aware latent space, followed by decoding to generate candidate material distributions. These candidates can be retrieved through the latent space exploration, including latent similarity search and target latent vector. A GNN surrogate model predicts the deformation response of candidate designs, while a subsequent optimization step refines material distributions.

Within this framework, three families of voxel-based structures are considered to systematically evaluate performance across increasing dimensional complexity. These include one-dimensional beams, two-dimensional plates, and three-dimensional solids. These structure families are selected to represent canonical shape-morphing archetypes commonly encountered in 4D-printed systems, ranging from bending-dominated beam actuation to surface morphing in plates and fully volumetric deformation in 3D architectures. **Figure 5.2a** illustrates representative examples of these geometries across multiple dimensions. For each structure, two boundary conditions are considered in combination with heterogeneous material distributions to ensure broad generalization of deformations (Figure 5.2c). Boundary condition BC1 imposes full kinematic constraints, fixing all translational degrees of freedom along the entire left-face boundary (Dirichlet condition applied to the full surface). Boundary condition BC2 constrains two nodes at the bottom center

of the structure, where all translational degrees of freedom are fixed. These mechanical conditions are combined with spatially heterogeneous material distributions, including thermally-responsive liquid crystal elastomer (LCE), passive elastomer, and void.

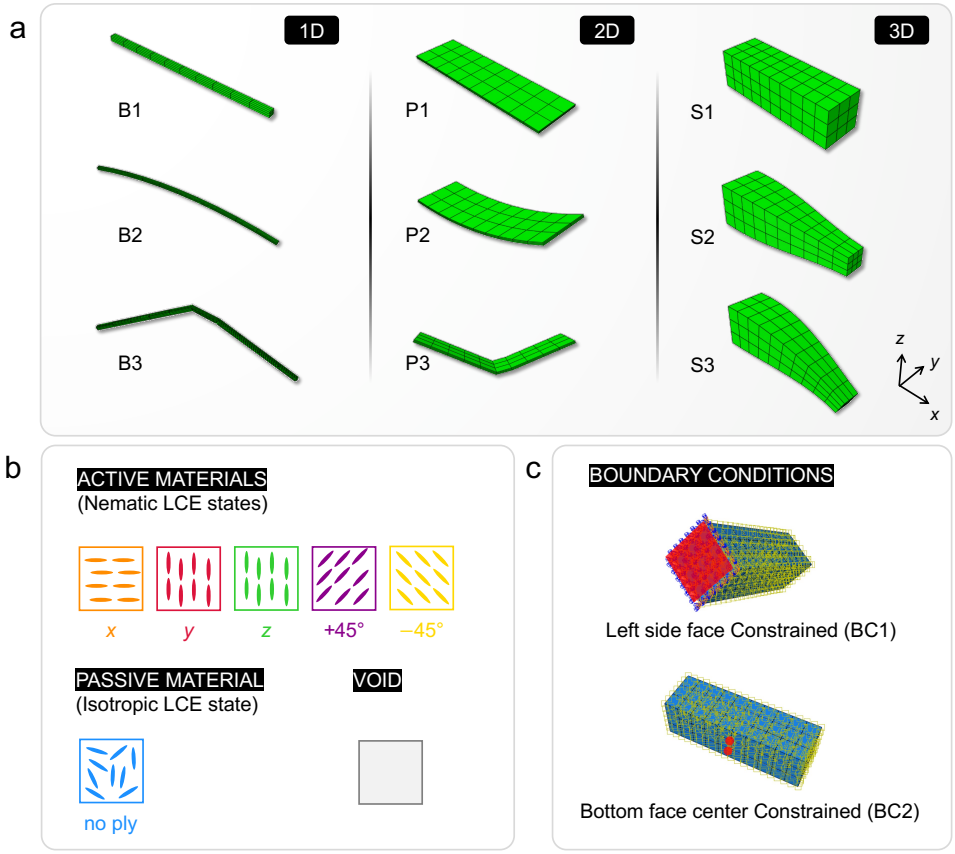

Figure 5.2: Considered geometries, materials and boundary conditions for dataset construction. (a) Representative voxel-based structures: 1D beams (B1 to B3), 2D plates (P1 to P3), and 3D solids (S1 to S3). These multidimensional structures are versioned according to typical deformation modes observed in 4D-printed systems, to better accommodate future complex forms that need to be designed. (b) Considered materials, including active LCE in the nematic state via distinct mesogen orientations, passive LCE in the isotropic state, and void. (c) Boundary conditions designed to ensure generalization.

LCEs are stimuli-responsive polymer networks that couple rubber-like elasticity with liquid crystalline order, which enable large, reversible, and anisotropic deformation under thermal actuation. Upon heating, the active LCE domains contracts along the programmed director orientation (nematic state) while slightly expanding in the transverse directions. Here, multiple director configurations are considered, including Cartesian-aligned orientations and diagonal in-plane (xy) orientations at $+45°$ and $-45°$, enabling spatial programming of anisotropic strain fields (Figure 5.2b). Passive regions – in which LCE material is in its isotropic state – do not actively respond to thermal stimulation but provide mechanical coupling between active domains, while void-like regions are modeled with significantly reduced stiffness to locally tune compliance and

deformation pathways. The combination of boundary conditions, material heterogeneity, and director programming enables rich deformation behaviors including bending, twisting, and out-of-plane morphing. The complete dataset-generation procedure, including sample counts, material distributions, finite element settings, and graph-pair construction, is provided in Appendix E.

5.2.1 Graph-based VAE for geometric latent design

The first component of the proposed framework aims to learn a compact latent representation of shape-morphing structural geometries. VAEs are well suited for this purpose because they provide a probabilistic latent space in which complex high-dimensional configurations can be encoded, sampled, and interpolated [481]. In inverse design, such a latent representation is particularly relevant because it transforms the search from a high-dimensional geometric space into a lower-dimensional continuous space where candidate designs can be explored more efficiently.

The Graph-VAE is used to learn a deformation-aware latent representation from pairs of original ("as designed") and deformed ("as actuated") structural graphs. Both graphs share the same connectivity, while their nodal coordinates encode the geometric evolution of the structure (**Figure 5.3a**). This formulation allows the latent space to capture feasible shape transformation patterns and to provide a compact continuous representation for subsequent inverse design exploration. Coordinates are read from the graph attributes associated with each node, while edges define local adjacency relationships between nodes. This graph formulation is more suitable than a purely vectorial representation because it preserves the spatial organization of the structure and the local relationships through which deformation patterns propagate. Graph-based latent representations have been widely explored to extend autoencoding models from Euclidean data to relational and non-Euclidean domains [480, 482, 483]. At this stage, only geometric information is considered. Material descriptors are intentionally excluded from the graph-based VAE input so that the latent space captures deformation-aware geometric patterns independently of the specific material assignments.

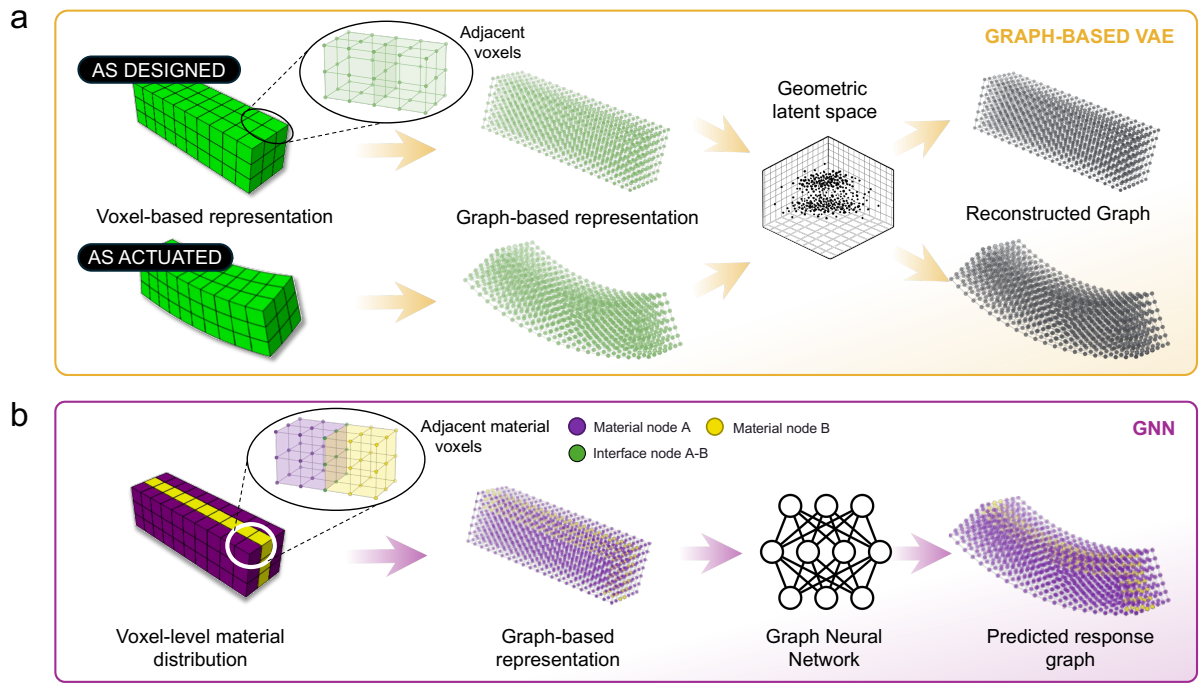

Figure 5.3: Graph construction and graph-based learning pipeline. (a) Workflow of the graph-based VAE: a voxel-based decomposition of the "as designed" and "as actuated" structures is encoded into a deformation-aware latent space through a graph-based representation. This latent space is then used to retrieve candidate structures for target-shape inverse design. For each optimization iteration, the retrieved candidate structure is associated with its voxel-level material distribution before being evaluated by the GNN surrogate. (b) The voxelized multi-material structure is converted into a graph in which nodes encode geometric position and material attributes, while edges preserve local neighborhood relations. The resulting graph is processed by the GNN surrogate to predict nodal displacements in a predicted response graph.

A key preprocessing step is the use of a balanced coordinate normalization. Instead of mapping coordinates to a strictly positive interval, the coordinates are normalized into a centered range, which preserves negative coordinate values and avoids suppressing deformation patterns involving negative z values. This is particularly important for the considered shape-morphing structures, where the sign and magnitude of the out-of-plane coordinate can be physically meaningful. In the complete dataset used for the VAE analysis, the nodal coordinates span $x \in [-0.639, 10.575]$, $y \in [-5.719, 5.426]$, and $z \in [-5.007, 5.946]$, corresponding to global ranges of 11.214, 11.144, and 10.953, respectively. The dataset contains 58,089,108 nodal samples across all analyzed graph configurations, confirming the large scale of the geometric reconstruction task. The presence of both positive and negative values along the z-axis motivates a dedicated evaluation of negative z reconstruction, since these regions correspond to downward out-of-plane deformation patterns. After reconstruction, the inverse transformation is used to recover the coordinates in the original physical space.

The VAE is trained with a composite objective designed to balance geometric

reconstruction, latent space regularization, local structural consistency, and preservation of physically meaningful out-of-plane deformation patterns. The latent regularization is enforced through the Kullback–Leibler divergence term ($\mathcal{L}_{\mathrm{KL}}$), which encourages the learned latent posterior to remain close to a Gaussian prior. The loss function is defined as:

$$\mathcal{L} = \mathcal{L}_{\mathrm{coord}} + 1.5\,\mathcal{L}_{z^-} + \mathcal{L}_{\mathrm{dist}} + \mathcal{L}_{\mathrm{nodes}} + \beta(e)\,\mathcal{L}_{\mathrm{KL}} + \mathcal{L}_{\mathrm{spatial}} \tag{5.1}$$

where e denotes the training epoch, and $\beta(e)$ is an epoch-dependent weighting coefficient applied to the KL divergence term. This coefficient is gradually increased during training following a KL annealing strategy, allowing the model to first prioritize coordinate reconstruction before progressively enforcing a smoother and more regular latent distribution. The coordinate reconstruction term $\mathcal{L}_{\mathrm{coord}}$ measures the reconstruction error for both the original and target graph coordinates. In practice, the deformed state is assigned a higher weight than the original one, as it carries the deformation information used later for inverse design. The z coordinate is also weighted more strongly to better preserve out-of-plane deformation patterns. The term $\mathcal{L}_{z^-}$ penalizes errors associated with negative z regions. It is used to preserve the sign and magnitude of physically meaningful negative vertical coordinates, rather than forcing all points toward negative values. During training, this behavior is monitored using precision, recall, F1-score, accuracy, over-prediction rate, sign accuracy, and negative z mean squared error (MSE). The local distance term $\mathcal{L}_{\mathrm{dist}}$ promotes geometric consistency by penalizing deviations in inter-node distances along graph edges. The node-count term $\mathcal{L}_{\mathrm{nodes}}$ encourages consistency in the reconstructed number of nodes, while $\mathcal{L}_{\mathrm{spatial}}$ regularizes coordinates outside the normalized admissible range. The KL divergence term is computed as

$$\mathcal{L}_{\mathrm{KL}} = -\frac{1}{2}\sum_{j=1}^{d_z}\left(1 + \mathrm{logvar}_j - \mu_j^2 - \exp(\mathrm{logvar}_j)\right), \tag{5.2}$$

where d_z denotes the latent space dimension, μ_j is the mean of the j-th latent variable, and $\mathrm{logvar}_j = \log\sigma_j^2$ is the log-variance predicted by the encoder. The variance is therefore recovered as $\sigma_j^2 = \exp(\mathrm{logvar}_j)$. In the implementation, the encoder predicts $\boldsymbol{\mu}$ and `logvar`, from which the latent variable is sampled using the reparameterization trick. The prior distribution is assumed to be a standard Gaussian distribution ($p(\boldsymbol{z}) = \mathcal{N}(0, \mathbf{I})$). This KL divergence term penalizes deviations of the learned posterior from the Gaussian prior and encourages a smooth, continuous latent space suitable for interpolation and sampling [481]. This objective provides a compact geometric latent representation that supports the subsequent target-shape inverse design stage.

To select the best suitable VAE, we performed a comparative assessment of VAE encoder backbones is reported in Appendix E. Built on this, EdgeConv was retained as the graph-based VAE encoder because it achieved the best balance in reconstruction

performance. The selected graph-based VAE follows an encoder-decoder formulation. The encoder maps the original and deformed graph configurations into a latent distribution, while the decoder reconstructs both configurations from a sampled latent vector. Let $V = \{v_i\}_{i=1}^{N}$ denote the set of graph nodes, and let $X_o \in \mathbb{R}^{N\times d}$ and $X_d \in \mathbb{R}^{N\times d}$ denote the nodal coordinate matrices of the original and deformed configurations, respectively, where N is the number of nodes and d is the spatial dimension. Given an original graph $G_o = (V, E, X_o)$ and a deformed graph $G_d = (V, E, X_d)$, where both graphs share the same node set V and edge set E, the encoder estimates the parameters of a variational posterior:

$$q_\phi(\mathbf{z}|G_o, G_d) = \mathcal{N}\left(\boldsymbol{\mu}_\phi(G_o, G_d), \mathrm{diag}(\boldsymbol{\sigma}_\phi^2(G_o, G_d))\right), \tag{5.3}$$

where $\boldsymbol{\mu}_\phi(G_o, G_d)$ and $\boldsymbol{\sigma}_\phi(G_o, G_d)$ denote the mean and standard deviation vectors predicted by the encoder, respectively, and ϕ represents the learnable parameters of the encoder. A latent vector $\mathbf{z} \in \mathbb{R}^{d_z}$ is then sampled from the variational posterior using the reparameterization trick:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}), \tag{5.4}$$

where $\mathbf{z}$ is the sampled latent vector, $\boldsymbol{\mu}$ is the latent mean vector, $\boldsymbol{\sigma}$ is the latent standard deviation vector, $\boldsymbol{\epsilon}$ is a random noise vector sampled from the standard normal distribution $\mathcal{N}(0, \mathbf{I})$, and $\odot$ denotes the Hadamard product. This reparameterization enables gradient-based optimization of the variational objective [481].

In practice, the encoder first projects nodal coordinates into a hidden representation, then applies graph-based message passing to capture spatial dependencies between neighboring nodes. The original and deformed graphs are encoded separately and then pooled into graph-level embeddings. These two embeddings are combined to produce the latent mean and log-variance. This formulation allows the latent space to encode not only the initial geometry, but also the geometric transformation between the original and deformed states. The decoder maps the sampled latent vector $\mathbf{z}$ back to the graph coordinate space. Formally, it predicts the reconstructed original coordinates $\hat{X}_o$, the reconstructed deformed coordinates $\hat{X}_d$, and the node-count logits $\boldsymbol{\ell}_n$ as $(\hat{X}_o, \hat{X}_d, \boldsymbol{\ell}_n) = D_\theta(\mathbf{z})$, where D_θ denotes the decoder network parameterized by the learnable weights θ.

The coordinate outputs are constrained within a normalized range using a bounded activation function, after which the inverse normalization is applied to recover the physical coordinate space. This decoder design is shared by all encoder comparisons, ensuring that the reported differences are due to the encoder backbone rather than the reconstruction head. Overall, the encoder learns a deformation-aware latent distribution, while the decoder reconstructs both the initial and displaced geometries from this compact representation.

5.2.2 GNN surrogate for forward prediction

After the graph-based VAE has organized deformation-aware candidate geometries in the latent space, the forward surrogate evaluates whether these candidates can produce the desired material-dependent deformation response. This step explicitly connects latent-space exploration with physical response prediction. The VAE provides a compact geometric search space from which candidate structures are retrieved, while the GNN surrogate acts as the material-aware forward model that predicts the activated configuration of each candidate.

The forward simulation problem is formulated as a graph-to-displacement regression task. For each candidate geometry retrieved from the VAE latent space, the corresponding material-aware graph is defined as $G_M = (V, E, X_o, M)$, where V is the set of nodes, E is the edge set, $X_o \in \mathbb{R}^{N\times 3}$ denotes the initial nodal coordinates, and M represents the voxel-level material descriptors attached to the graph nodes. The GNN surrogate then predicts a nodal displacement field (Figure 5.3b). Importantly, the model does not generate a new graph topology; instead, it preserves the initial connectivity and updates the nodal positions according to the learned deformation field.

The retained surrogate model is based on a GraphSAGE backbone introduced and validated in Chapter 4. The surrogate architecture, input representation, training strategy, automated dataset generation, predictive performance, and computational acceleration relative to finite element simulation are presented in Chapter 4. The complete implementation details, comparative model assessment, evaluation protocol, runtime analysis, and statistical robustness results are reported in Appendix C, Section C.2.

5.2.3 Inverse design

After learning a compact geometric latent representation with the graph-based VAE and validating the forward GNN surrogate, the final stage of the proposed framework addresses the target-shape inverse design problem. The objective is to identify a candidate structure with material distribution which matches a prescribed target geometry as closely as possible. This stage therefore connects the two previous components of the framework: the VAE organizes deformation-aware geometric configurations into a compact latent space, while the GNN surrogate provides a fast approximation of the forward deformation response for each candidate material distribution.

The inverse design problem is formulated as an optimization problem in which candidate structures are evaluated according to their predicted deformed shape. Given an initial candidate geometry and a material distribution, the GNN surrogate predicts the nodal displacement field. The resulting displaced coordinates are then compared with the target shape using a face-based geometric error. In the implemented objective, each predicted face is compared with the corresponding target face using a symmetric Chamfer

distance [485], which is robust to differences in point ordering and face discretization. An additional centroid-alignment penalty is added by comparing the centroids of the predicted and target faces, so that geometrically close but globally shifted or incorrectly aligned deformations are penalized. The optimization objective is therefore defined as:

$$\min_{G,M} \mathcal{J}(G,M) = \frac{1}{|\mathcal{F}|} \sum_{f \in \mathcal{F}} \left[d_{\mathrm{CD}} \left(\widehat{X}_d^f, X_{\mathrm{target}}^f \right) + \lambda_c \left\| \widehat{\mathbf{c}}_d^f - \mathbf{c}_{\mathrm{target}}^f \right\|_2^2 \right], \tag{5.5}$$

where G denotes the structural graph, M is the material distribution, $\mathcal{F}$ is the set of evaluated structural faces, typically including the top, bottom, left, right, front, and back faces, d_{CD} is the symmetric Chamfer distance, $\widehat{X}_d^f$ represents the surrogate-predicted displaced coordinates of face f, X_{target}^f denotes the corresponding target-face coordinates, and $\widehat{\mathbf{c}}_d^f$ and $\mathbf{c}_{\mathrm{target}}^f$ are the centroids of the predicted and target faces, respectively. The coefficient $\lambda_c \geq 0$ weights the centroid-alignment penalty. In the present implementation, $\lambda_c = 1$, which corresponds to the unweighted penalty used in the objective function.

The candidate material distributions are defined over the same discrete material library used during dataset generation (see Figure 5.2b). This library enables the optimizer to explore both actuation directionality and passive structural support. The active LCE orientations provide different deformation tendencies, while the passive and void regions allow the algorithm to modulate stiffness, continuity, and local compliance. As a result, the inverse design problem becomes a mixed geometric and discrete material search problem.

At this stage, the inverse design problem is treated as a surrogate-assisted search problem over candidate geometries and discrete material distributions. For each prescribed target shape, candidate structures are selected from the available design pool and evaluated using the same GNN surrogate and the same face-based objective function. This provides a common evaluation criterion for all optimization approaches and ensures that the comparison focuses on the search strategy rather than on differences in the forward model or error metric.

Optimization strategies

To evaluate the effectiveness of the inverse design framework, seven optimization strategies were applied and compared. For each target shape, the best result obtained by each strategy was selected according to the final target shape error, and the best overall method was defined as the one reaching the lowest objective value. All strategies are evaluated using the same GNN surrogate model and the same face-based objective, ensuring that the comparison reflects the search strategy rather than differences in the evaluation function. The optimization strategies are described hereafter:

1. **Initial retrieval without mutation.** This baseline selects the best candidate

based on target-shape matching and serves as a reference for retrieval-only performance.

2. **Top-k candidate search.** Evaluates the top-k candidates and retains the one with the lowest target-shape error, improving diversity over the baseline without iterative refinement.

3. **Stochastic latent-neighborhood sampling.** Performs stochastic neighborhood sampling around promising candidates, generating nearby variants within a perturbation radius to increase local diversity and find better solutions at lower cost than full population-based optimization.

4. **CMA-ES-based optimization.** Uses covariance matrix adaptation evolution strategy (CMA-ES), a black-box evolutionary optimizer for nonlinear, non-convex spaces [484], to minimize target shape error by adaptively focusing the search on promising regions of the design space.

5. **Bayesian optimization.** Builds a probabilistic model of the objective function to balance exploration and exploitation when selecting new candidates [486].

6. **Local material mutation.** Defines candidates via discrete mutations on voxel segments, where each segment is jointly reassigned to a material state with a controllable granularity from local to global edits. Mutations are retained only if they reduce shape error and satisfy material constraints. It enables direct exploration of the discrete space but is sensitive to initialization and local minima.

7. **Top-k search followed by local mutation.** Selects a promising candidate from a top-k pool and refines it using local discrete material mutations. It assesses whether retrieval-based initialization improve subsequent local refinement.

The optimization workflow starts from a prescribed target shape and evaluates candidate designs using the GNN surrogate. Depending on the strategy, candidates are selected by direct retrieval, top-k screening, stochastic latent-neighborhood sampling, Bayesian optimization, CMA-ES, local material mutation, or a combination of top-k search and local mutation. For each candidate, the surrogate predicts the deformed geometry, and the face-based target-shape error is used as the optimization criterion. This unified evaluation loop makes it possible to compare all strategies under the same objective while avoiding computationally expensive FE simulations during optimization.

Average performance over the benchmark target shapes

Table 5.1 reports the average performance of the seven inverse design strategies over a benchmark set of $T = 153$ target shapes. This benchmark target set was constructed

from the generated shape-morphing dataset to evaluate the optimization strategies under the same target-shape matching conditions. Each target shape defines an independent inverse design task, in which the objective is to identify a candidate geometry and material distribution whose predicted deformed configuration matches the prescribed target. The comparison is based on three quantitative metrics: the mean final target-shape error, the mean improvement relative to the initial retrieval baseline, and the mean computational time. These metrics quantify both target-shape matching accuracy and computational efficiency in the surrogate-assisted inverse design process.

For a given target shape j and optimization strategy os, the final target-shape error $\mathcal{J}_j^{(os)}$ is defined as the final value of the face-based objective in Equation (5.5) after optimization. It therefore measures the discrepancy between the surrogate-predicted deformed configuration and the prescribed target shape using the same Chamfer-based face error and centroid-alignment penalty for all methods. A lower value of $\mathcal{J}_j^{(os)}$ indicates better target-shape matching.

The mean final target-shape error $\overline{\mathcal{J}}^{(os)}$ reported in Table 5.1 is obtained by averaging the final objective values over all target shapes

$$\overline{\mathcal{J}}^{(os)} = \frac{1}{T}\sum_{j=1}^{T} \mathcal{J}_j^{(os)}. \tag{5.6}$$

where $T = 153$ is the total number of target shapes in the benchmark set.

The improvement metric quantifies the relative reduction in target-shape error with respect to the initial retrieval baseline. It is computed from the average final objective values as

$$\text{Improvement}^{(os)} = \frac{\overline{\mathcal{J}}^{(\text{baseline})} - \overline{\mathcal{J}}^{(os)}}{\overline{\mathcal{J}}^{(\text{baseline})}} \times 100. \tag{5.7}$$

where $\text{Improvement}^{(os)}$ is the mean relative improvement, expressed as a percentage, obtained by optimization strategy os, and $\overline{\mathcal{J}}^{(\text{baseline})}$ is the mean final target-shape error of the initial retrieval without mutation baseline.

A positive improvement value indicates that the optimization strategy reduces the average target-shape error compared with the initial retrieval without mutation. The mean computational time measures the average elapsed time required by each approach over the full set of evaluated targets:

$$\bar{t}^{(os)} = \frac{1}{T}\sum_{j=1}^{T} t_j^{(os)}, \tag{5.8}$$

where $\bar{t}^{(os)}$ is the mean computational time of optimization strategy os, and $t_j^{(os)}$ is the computational time required by strategy os for target shape j.

Table 5.1: Average performance of inverse design approaches over the benchmark set of 153 target shapes. Lower final target-shape error indicates better target-shape matching, higher improvement indicates stronger optimization gain, and lower mean time indicates higher computational efficiency. A win corresponds to the lowest final target-shape error for a given target shape.

Optimization strategy	Mean final target-shape error	Mean improvement (%)	Mean time (s)	No. of wins
Initial retrieval without mutation	43.82	0.00	0.49	0
Top-k candidate search	42.60	2.78	29.06	1
Stochastic latent-neighborhood sampling	41.85	4.50	83.76	4
Bayesian optimization	41.30	5.75	149.29	2
Local material mutation	40.85	6.78	10.43	10
CMA-ES-based optimization	39.95	8.83	441.36	24
Top-k search + local mutation	39.20	10.54	71.76	112

The average results show a progressive improvement from simple retrieval-based strategies to more advanced hybrid optimization schemes. The initial retrieval baseline obtains a mean final target-shape error of 43.82, while the top-k candidate search reduces this value to 42.60 by evaluating a broader set of candidate structures. Stochastic latent-neighborhood sampling further improves the mean final target-shape error to 41.85, indicating that local exploration around promising latent candidates can provide additional shape-matching gains.

Bayesian optimization and local material mutation provide stronger improvements, with mean final target-shape error values of 41.30 and 40.85, respectively. These results suggest that both probabilistic search and direct discrete material refinement can improve target-shape matching when the candidate design space is sufficiently constrained. Local material mutation is particularly efficient, reaching a mean runtime of only 10.43 s, which makes it the fastest optimization strategy among the non-baseline methods.

CMA-ES-based optimization achieves a strong mean final target-shape error of 39.95 and a mean improvement of 8.83%, confirming the effectiveness of population-based black-box optimization for exploring non-convex inverse design spaces [487]. However, this accuracy comes with a substantial computational cost, with an average runtime of 441.36 s. In contrast, the combined top-k search with local mutation strategy achieves the lowest mean final target-shape error, with a value of 39.20, and the highest mean improvement, reaching 10.54%. Compared with CMA-ES, this strategy provides a lower final target-shape error while requiring a much lower average computational time.

These results indicate that the main advantage of the proposed hybrid strategy lies in its accuracy–efficiency trade-off. By combining retrieval-based candidate selection with local material refinement, the method benefits from a strong initialization while avoiding the high computational cost of population-based global optimization. Compared with the initial retrieval baseline, the proposed strategy reduces the mean final target-shape error from 43.82 to 39.20, corresponding to an absolute reduction of 4.62 objective-error units and a relative improvement of 10.54%. In addition to averaged metrics, a target-level win-

count analysis is used to evaluate the robustness of each approach across the benchmark. For each target shape j, the winning method is the one that achieves the lowest final target-shape error:

$$os_j^\star = \arg\min_{os} \mathcal{J}_j^{(os)}, \tag{5.9}$$

where $os_j^\star$ denotes the winning optimization strategy for target shape j, i.e., the strategy that achieves the lowest final target-shape error among all compared optimization strategies.

The number of wins for optimization strategy os is then computed as

$$\mathrm{Wins}^{(os)} = \sum_{j=1}^{T} \mathbb{I}\left(os_j^\star = os\right), \tag{5.10}$$

where $\mathrm{Wins}^{(os)}$ is the number of target shapes for which optimization strategy (os) achieves the lowest final target-shape error, and $\mathbb{I}(\cdot)$ is the indicator function, equal to 1 if the condition is true and 0 otherwise. When two approaches achieve numerically equivalent final target-shape error values, the corresponding win is shared between them. The win-count analysis in Table 5.1 confirms the same trend. The proposed top-k search combined with local mutation obtains the largest number of wins, achieving the best final target-shape error on 112 out of 153 targets. This win rate indicates that the proposed two-stage strategy is competitive on average and more robust across individual target-shape matching problems. CMA-ES obtains 24 wins, showing that global evolutionary search remains useful for difficult targets where the material distribution landscape is highly non-convex. Local material mutation obtains 10 wins, confirming that purely local refinement can be effective when the initial retrieved candidate is already close to the target. However, the proposed combined strategy clearly outperforms local mutation alone, demonstrating the benefit of first selecting a stronger candidate through top-k search before applying local material refinement.

Here, the global benchmark shows that the proposed top-k search combined with local mutation provides the most favorable balance between accuracy, robustness, and computational cost. These results support a hierarchical inverse design policy, as illustrated in **Figure 5.4**, where fast retrieval and local refinement can be used as the default strategy for efficient target-shape matching, while CMA-ES can be reserved for particularly difficult targets requiring deeper global exploration.

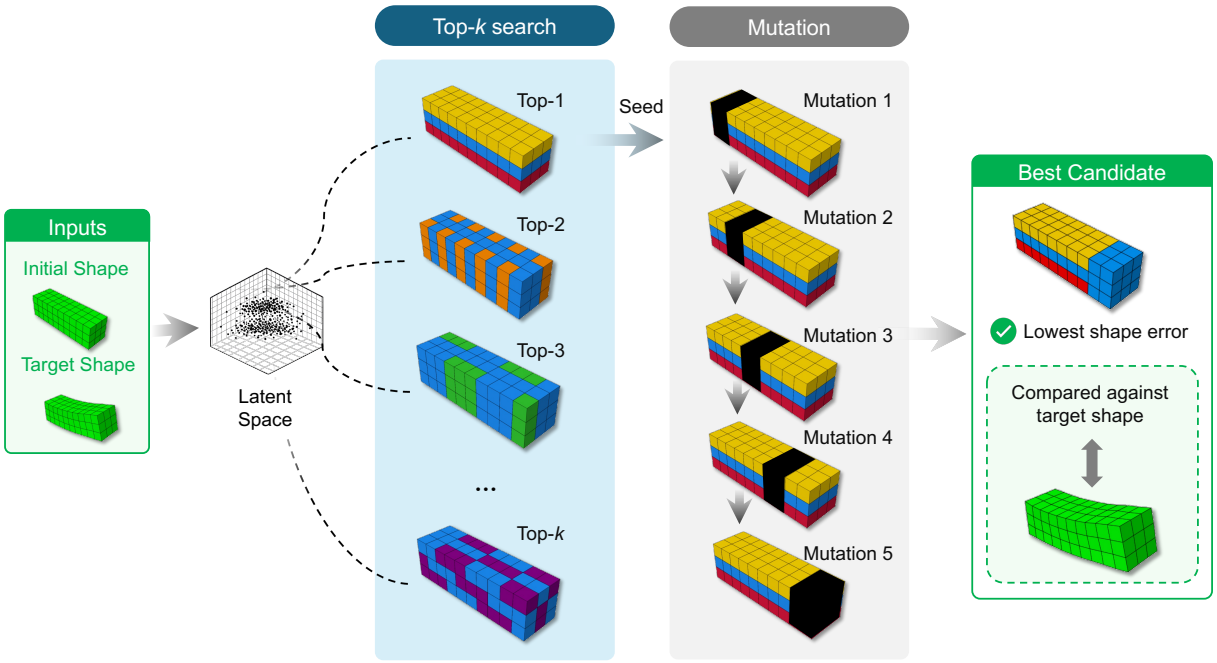

Figure 5.4: Overview of the retained two-stage optimization pipeline combining top-k and local mutation. Given initial and target shapes, candidates are first encoded in a latent space a top-k set of material distributions is retrieved based on similarity to the target. The best candidate (Top-1) is then used as a seed for iterative local mutations, where discrete voxel segments are modified while enforcing a material assignment among materials from Figure 5.2b. Mutations are accepted only if they improve the target-shape error. The process yields a final candidate with the lowest shape error relative to the target. For clarity, the encoder, decoder and GNN are not shown.

4D printing applications

Beyond surrogate-based optimization, the framework is evaluated at the application scale to assess its ability to generate complex 4D-printed structures through the assembly of ,optimized voxel-based components. Rather than optimizing full structures directly, the approach reuses material distributions derived at the representative, multidimensional structure level (Figure 5.2a).

The objective is to demonstrate that these components can be transferred, repeated, and assembled into larger structures while preserving coherent global morphing behavior, therefore highlighting the scalability of the approach. To this end, optimized material layouts are mapped onto FE simulations of seven representative applications: a soft robotic hand, a deployable morphing structure, an adaptive enclosure, a cross-shaped facade panel, a programmable gripper, a stent-like expandable lattice, and a shape-morphing tubular structure. Each application is designed by assembling one or more voxel-based structures with consistent material assignments (Figure 5.2b). FE simulations are performed under isothermal loading using application-specific boundary conditions. The resulting application-level responses are shown in **Figure 5.5**.

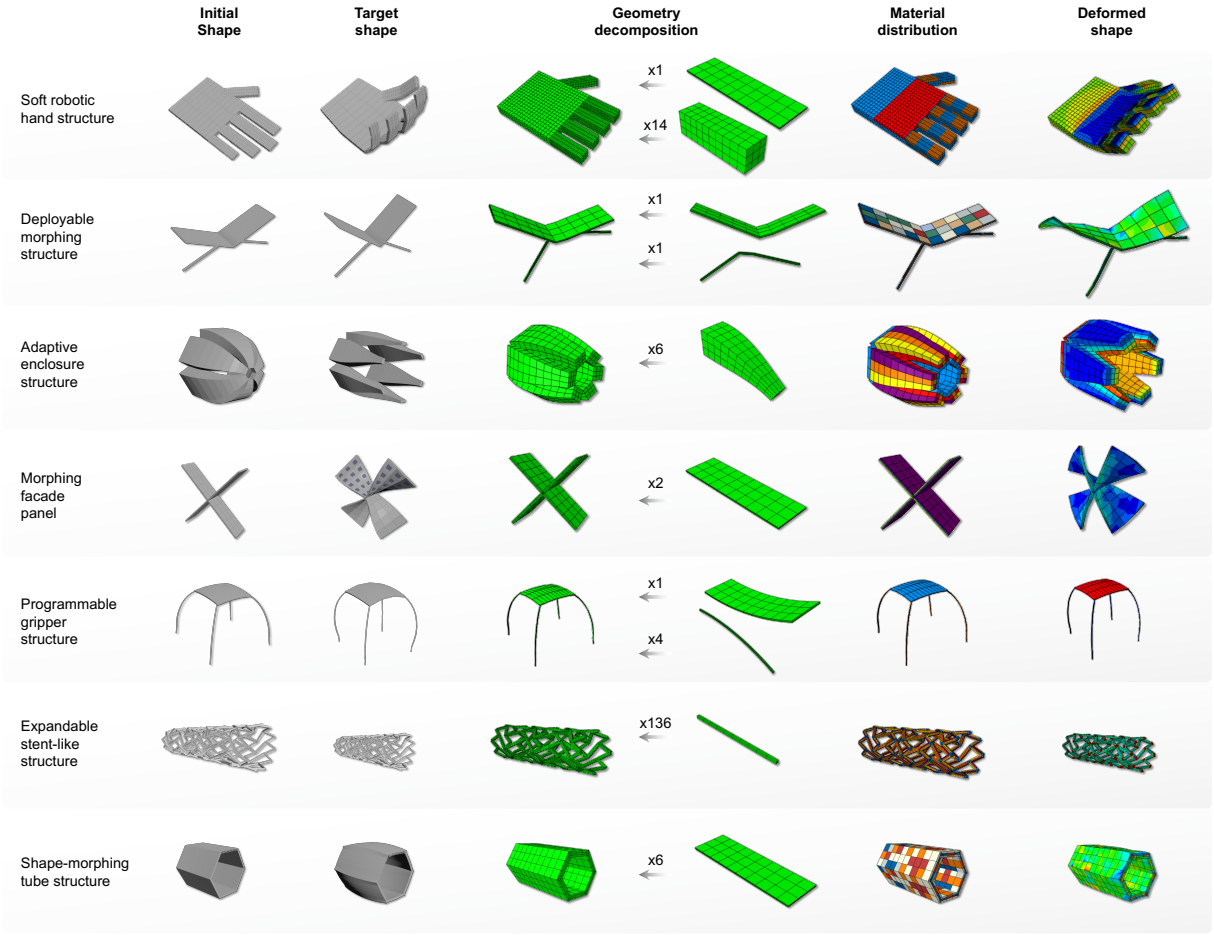

Figure 5.5: Exploitation of optimized material distributions through targeted 4D printing applications. For each application, the initial geometry and prescribed target shape are first defined, then decomposed into elementary voxel-based families. The optimized material distributions obtained at the component level are transferred, repeated, and assembled to generate the complete application geometry. The final column shows the thermally activated deformed configuration after assigning the optimized material patterns. This illustrates that a limited number of optimized voxel-based components can be reused across soft robotic, deployable, enclosure, facade, gripper, stent-like, and tubular structures while preserving coherent global morphing responses. The corresponding thermo-mechanical properties, programmed orientations, and finite element material formulation are detailed in Appendix D.

The soft robotic hand is assembled from a P1-based dome and fourteen S2 finger-like members. The dome forms the palm-like support, while the S2 components act as distributed fingers. Each part receives its optimized material distribution before assembly, and a BC1 is applied to enable thermally induced deformation of the complete structure. This case demonstrates that separately optimized dome and finger components can be integrated into a coordinated hand-like morphing system. The deployable morphing structure combines a P3 top panel with a B3 inclined support. The B3 member is placed below the panel and connected through a local voxel contact region, while BC2 is applied to its lowest nodes. This configuration evaluates whether a deployable panel

can be actuated through a supporting member while maintaining mechanical transfer at the interface without overconstraining the assembly. The adaptive enclosure structure is obtained by repeating the same optimized S3 volumetric part six times around a circular layout. Each member is oriented to form a closed enclosure-like geometry and is subjected to BC1. This case shows that a single optimized 3D component can be reused to produce a coherent global enclosure deformation. The morphing facade uses two P1 panels arranged perpendicularly to create a cross-shaped adaptive structure. The same optimized P1 material pattern is mapped onto both panels, with boundary conditions applied at the bottom-center region of the intersecting structure. This validates the reuse of a panel-level design in a crossed configuration while preserving global bending and morphing behavior. The programmable gripper combines one P2 surface with four B2 branches attached to its corners. The branches are oriented downward and mechanically connected to the upper surface through joint couplings, with BC1 applied to each member. This example shows that repeated optimized branch components can generate a coordinated gripper-like deformation after assembly. The stent-like structure is built from 136 repeated B1 parts arranged into connected rings and linking sections. The same optimized B1 material distribution is assigned to all members, while joint couplings preserve continuity at the intersections. This case illustrates how a simple one-dimensional optimized component can be scaled up into a three-dimensional expandable lattice structure. Finally, the tubular structure is generated by repeating one optimized B1 voxel-based face six times around a hexagonal cross-section. Each face receives the same material distribution and is subjected to BC1. The resulting response confirms that a local panel-like pattern can be transferred to a closed tubular geometry while maintaining coherent shape morphing.

A complementary quantitative metric was introduced to evaluate the physical agreement between each thermally-activated structure and its target shape. This metric complements the target-shape error reported in Table 5.1, as the assembled structures may influence the ordering of nodes between the deformed and target shapes. Therefore, a face-based physical shape error is used to compare the deformed and target surfaces without requiring a strict one-to-one node correspondence. For each extracted face f, the predicted deformed face is represented by a point cloud $P_f = \{\mathbf{p}_i\}$, where each point $\mathbf{p}_i \in \mathbb{R}^3$ denotes a sampled 3D position on the predicted surface. Similarly, the corresponding target face is represented by a point cloud $T_f = \{\mathbf{t}_i\}$, where each point $\mathbf{t}_i \in \mathbb{R}^3$ denotes a sampled 3D position on the target surface. The geometric discrepancy between faces is measured using a symmetric Chamfer distance [485, 488]:

$$d_{\text{Chamfer}}(P_f, T_f) = \frac{1}{|P_f|} \sum_{\mathbf{p} \in P_f} \min_{\mathbf{t} \in T_f} \|\mathbf{p} - \mathbf{t}\|_2^2 + \frac{1}{|T_f|} \sum_{\mathbf{t} \in T_f} \min_{\mathbf{p} \in P_f} \|\mathbf{t} - \mathbf{p}\|_2^2 . \tag{5.11}$$

For each point $\mathbf{p}_i$, the operator $\min_{\mathbf{t} \in T_f}$ selects the nearest point $\mathbf{t}_i$ in the target face, while for each point $\mathbf{t}_i$, the operator $\min_{\mathbf{p}_i \in P_f}$ selects the nearest point $\mathbf{p}_i$ in the predicted

face. In addition to local surface proximity, a centroid-based penalty is introduced to enforce global geometric consistency and to penalize physically inconsistent deformation directions [489]. The resulting physical shape error is defined as

$$\varepsilon_{\text{shape}} = \frac{1}{|\mathcal{F}|} \sum_{f \in \mathcal{F}} \left[d_{\text{Chamfer}}(P_f, T_f) + \lambda_c \left\| \mathbf{c}(P_f) - \mathbf{c}(T_f) \right\|_2^2 \right], \tag{5.12}$$

where $\mathcal{F}$ is the set of extracted faces, and $\mathbf{c}(\cdot)$ denotes the centroid operator, which returns the geometric center of a face point cloud. For comparison across applications with different geometric scales, the error is normalized by the squared diagonal length of the target bounding box (D_{bbox}) as follows

$$\widetilde{\varepsilon}_{\text{shape}} = \frac{\varepsilon_{\text{shape}}}{D_{\text{bbox}}^2}. \tag{5.13}$$

Lower values of $\widetilde{\varepsilon}_{\text{shape}}$ indicate better agreement between the deformed application geometry and the target shape. The application-level normalized physical shape error, based on the face-wise Chamfer distance and centroid-alignment penalty, was evaluated for seven validation cases. The cross-shaped facade panel achieved the lowest normalized physical shape error, with a value of 0.069, followed by the deployable morphing structure with 0.074. The shape-morphing tube and adaptive enclosure also showed low errors, with values of 0.083 and 0.086, respectively. Higher errors were observed for the programmable gripper, soft robotic hand, and stent-like expandable lattice, with normalized physical shape errors of 0.102, 0.118, and 0.131, respectively. Across all seven application-level cases, the average normalized physical shape error was 0.095, indicating that the optimized material distributions remain physically consistent after being transferred from elementary voxel-based families to complete application-level assemblies.

These results show that the inverse design target-shape error and the face-based physical shape error provide complementary information. The benchmark objective evaluates target-shape matching during inverse design using the surrogate-predicted candidate geometries, whereas $\widetilde{\varepsilon}_{\text{shape}}$ evaluates application-level geometric consistency after assembly. The residual physical shape error suggests that the optimized candidates provide satisfactory agreement with the target deformations, while further refinement could still improve surface-level matching. A natural extension is therefore to include $\varepsilon_{\text{shape}}$ directly as an additional objective during mutation-based refinement, Bayesian optimization, or CMA-ES search.

An important advantage of this validation strategy is its computational efficiency. Instead of optimizing the full application-level model directly, only a reduced number of elementary voxel-based components is optimized and then reused during assembly. This reduces the number of distinct optimization problems required to generate complex application-level structures. For example, the enclosure requires only one optimized S3

member to generate six assembled members, corresponding to a six-fold reduction in distinct optimized components. Similarly, the tube uses one optimized P1 face repeated over six faces, and the stent reuses a single optimized B1 member to construct a 136-member lattice. The gripper requires only two optimized components, one P2 surface and one B2 branch, to generate a five-member assembly. The facade uses one P1 optimized panel pattern to generate two perpendicular panels. The hand combines one P1 dome and fourteen S2 finger components to form a larger soft robotic hand without optimizing the complete assembled geometry directly.

Overall, the physical validation demonstrates that the optimized material distributions are not limited to isolated benchmark targets. They can be transferred to larger 4D-printing applications through geometric repetition, orientation, scaling, and mechanical connection of voxel-based structure families. The resulting assemblies exhibit coherent global deformation responses, confirming that the proposed decomposition strategy preserves the essential morphing behavior at the application scale. This supports the use of voxel-based family optimization as a scalable design strategy for complex 4D-printed systems, where full direct optimization of the complete application would be computationally more expensive.

5.2.4 Discussion

The results presented in this work demonstrate the potential of combining graph-based representation learning, surrogate simulation, and hybrid optimization for inverse design of multi-material 4D-printed structures. The framework addresses a central difficulty in 4D-printed morphing systems: identifying a material distribution and structural configuration capable of producing a prescribed deformation under thermal activation. This problem is challenging because the final shape depends simultaneously on geometry, boundary conditions, local material states, programmed LCE orientations, passive regions, void-like compliant zones, and mechanical interactions between neighboring voxels.

A first important outcome of this study is the effectiveness of graph-based learning for representing voxelized 4D-printing structures. The graph-based VAE provides a compact deformation-aware latent space by encoding original and displaced structural graphs while preserving their connectivity. The encoder comparison shows that graph message passing improves the reconstruction of physically relevant deformation patterns compared with a purely node-wise MLP baseline. In particular, EdgeConv achieves the best average negative-z reconstruction performance, confirming that local geometric relationships are important for capturing downward out-of-plane deformation. This is consistent with the physical nature of the considered structures, where local voxel interactions and neighborhood connectivity strongly influence the global morphing response.

The forward surrogate results further confirm the relevance of graph neural networks

for thermo-mechanical deformation prediction. The retained GraphSAGE surrogate achieves the best predictive performance among the evaluated learning-based models, with the lowest MAE, transverse displacement errors, and RMSE, as well as the highest R^2 scores along the y- and z-directions. This is particularly important because the transverse components are the most informative for bending, twisting, and out-of-plane shape transformation. The comparison with FE simulation highlights the main computational advantage of the surrogate: while a high-fidelity FE simulation may require between 1800 and 3600 s per structure, the GraphSAGE surrogate performs inference in 0.0269 s per structure. This acceleration makes it possible to evaluate a very large number of candidate material distributions, which would be impractical using direct FE-based optimization.

The hybrid inverse design results show that no single optimization strategy is universally optimal for all target shapes. Retrieval alone provides a fast baseline, but its accuracy remains limited because it only selects existing candidates without adapting their material layouts. Top-k candidate search improves the baseline by increasing the diversity of evaluated candidates, while stochastic latent-neighborhood sampling provides additional exploration around promising regions of the design space. Bayesian optimization and CMA-ES offer stronger search capabilities, but their computational cost is higher. Local material mutation is computationally efficient and can significantly improve retrieved candidates when the initial geometry is already close to the target. The best overall compromise is obtained by combining top-k candidate search with local material mutation, which achieves the lowest mean final target-shape error and the highest number of target-level wins.

This result is significant because it shows that the inverse design problem benefits from a two-stage strategy. The first stage identifies a promising candidate structure through retrieval or candidate screening, while the second stage refines the discrete material distribution through local mutations. This avoids the high cost of fully global black-box optimization while still improving the final target-shape match. Compared with CMA-ES, the proposed top-k search combined with local mutation achieves better average accuracy while requiring a much lower average computational time. Therefore, the main contribution of the proposed strategy is not only improved accuracy, but also a favorable accuracy–efficiency trade-off.

The win-count analysis reinforces this conclusion. The proposed two-stage strategy obtains the largest number of wins across the evaluated target shapes, indicating that its performance is not only good on average but also robust across different morphing configurations. This is important for practical 4D-printing design, where a method should not only perform well on a few favorable cases but should also remain reliable across diverse geometries, boundary conditions, and deformation modes. The relatively smaller number of wins obtained by CMA-ES suggests that global evolutionary search remains useful for difficult cases, but it is not always necessary when a strong retrieved candidate can be

locally refined.

The physical validation at the application scale provides an additional level of evidence. Instead of evaluating only isolated benchmark structures, the optimized material layouts are transferred to full application-level assemblies, including a soft robotic hand, a deployable morphing structure, an adaptive enclosure, a cross-shaped facade panel, a programmable gripper, a stent-like expandable lattice, and a shape-morphing tube. These examples show that optimized voxel-based structure families can be transformed, repeated, connected, and assembled into larger systems while preserving coherent global deformation behavior. This is a critical step because practical 4D-printing applications rarely consist of a single isolated voxelized member; they are generally composed of multiple interacting components.

The application-level validation also demonstrates the scalability of the proposed decomposition strategy. By optimizing elementary voxel-based families and then reusing them in larger assemblies, the number of distinct optimization problems is reduced. This component-level reuse provides a practical route toward scalable 4D-printing inverse design, where full direct optimization of an entire application-level structure would be computationally expensive.

The additional face-based physical shape error further complements the inverse design target-shape error used in the benchmark. While the inverse design target-shape error evaluates candidate matching during the surrogate-based benchmark, the physical shape error compares deformed and target surfaces using a Chamfer-based face metric and a centroid-alignment penalty. This is more appropriate for assembled applications, where geometric transformations and repeated components can alter node ordering. The observed application-level errors indicate that the optimized assemblies remain physically close to their target deformations, while also revealing residual discrepancies for more complex systems such as the soft robotic hand, gripper, and stent-like lattice. These results suggest that latent-space similarity and surrogate-based target-shape error are useful for candidate selection, but surface-level metrics should also be considered when validating full application geometries.

Despite these promising results, several limitations remain. First, the inverse design framework relies on the quality and diversity of the simulated training dataset. If certain deformation modes, boundary conditions, or material combinations are underrepresented, the surrogate and optimizer may have limited extrapolation capability. Second, the current optimization strategies operate primarily on discrete material assignments, while geometry and topology are mostly selected from existing candidate structures. Extending the framework to jointly optimize geometry, topology, and material distribution would increase design freedom. Third, the physical validation is performed numerically using FE simulations; experimental fabrication and actuation tests are still required to confirm the printability and real-world performance of the optimized designs. Finally, although

the surrogate provides strong acceleration compared with FE simulation, uncertainty quantification should be further developed to identify predictions that may be unreliable and require high-fidelity verification.

Future work should therefore focus on several directions. A first perspective is to integrate the face-based physical shape error directly into the optimization objective, rather than using it only as a post-validation metric. This would allow the optimizer to minimize not only nodal discrepancies but also surface-level geometric mismatch and deformation-direction inconsistency. A second direction is to develop multi-objective optimization strategies that simultaneously consider target-shape accuracy, actuation amplitude, stiffness, printability, material usage, and robustness to uncertainty. A third direction is to incorporate active learning, where uncertain or high-error candidate regions are automatically selected for additional FE simulations and added to the training dataset. This would progressively improve the surrogate in the most relevant regions of the design space.

Another promising direction is the extension toward experimental validation. The optimized material layouts could be fabricated using multi-material additive manufacturing or programmable LCE-based printing, then tested under thermal activation to compare the measured deformation with the predicted shape. Such experiments would make it possible to assess the influence of fabrication defects, material anisotropy variations, interfacial bonding, and actuation hysteresis. Finally, the framework could be extended to more complex stimuli and multi-physics settings, including humidity-responsive hydrogels, magnetic composites, light-responsive polymers, or coupled thermo-mechanical and fluidic actuation. This would broaden the applicability of the proposed graph-based hybrid inverse design strategy to a wider class of adaptive 4D-printed systems.

The inverse design workflow can be interpreted as a workflow-level instantiation of the nested *Neuro[Symbolic]* paradigm described in Chapter 2. Its neural core comprises the graph-based VAE for deformation-aware latent representation and candidate retrieval, and the GraphSAGE surrogate for material-dependent forward prediction. Seven numerical search and optimization strategies operate around these neural models, including top-k screening, latent-neighborhood sampling, Bayesian optimization, CMA-ES, local mutation, and their hybrid combinations.

The symbolic subcomponent is implemented as an explicit constraint module invoked within the predominantly neural inverse design loop. It represents the admissible material states, connectivity requirements, voxel-level material-validity rules, segmentation rules, and candidate-acceptance conditions. A candidate generated or modified by the search process is passed through this constraint module before surrogate evaluation. Invalid candidates are rejected, whereas admissible candidates are converted into material-aware graphs and evaluated by GraphSAGE. This organization is consistent with $y = f_{\text{neural}}(x, g_{\text{symbolic}}(z))$, because the output of the symbolic constraint module becomes an

input to the neural forward predictor.

5.3 Conclusion

In this chapter, we presented a graph-based hybrid inverse design framework for target-shape matching in multi-material 4D printing. By combining deformation-aware graph representation learning, a fast GraphSAGE thermo-mechanical surrogate, and hybrid material-search strategies, the proposed approach enables efficient exploration of discrete material distributions while preserving structural connectivity. The results show that the combined top-k search with local material mutation provides the most favorable accuracy–efficiency compromise among the evaluated strategies, while application-level validation demonstrates that optimized voxel-based families can be transferred, repeated, and assembled into larger 4D-printing structures with coherent global deformation behavior. Overall, the framework supports a multi-dimensional component-based design strategy for complex morphing structures, where elementary optimized voxel families can be reused to reduce the cost of full application-level inverse design. Future work will focus on incorporating surface-level physical shape error directly into the optimization loop, extending the method toward joint geometry–material optimization, and validating the optimized designs experimentally under fabrication and actuation conditions.

At this stage, the thesis has established complementary capabilities for structured knowledge representation, accelerated forward prediction, behavior discovery, and target-driven inverse design. However, these capabilities remain individual computational components. The next chapter therefore addresses their system-level integration by introducing an orchestrator-centered multi-agent framework in which these specialized capabilities cooperate within request-specific workflows for scientific reasoning, design, simulation, and discovery in 4D printing.

Chapter 6

A Multi-Agent System for Scientific Reasoning, Design, and Discovery in 4D Printing

6.1 Introduction

Four-dimensional (4D) printing extends additive manufacturing by integrating stimuli-responsive materials into structures that undergo programmed changes in shape, properties, or functionality after fabrication [10]. Designing such systems requires coordinating interdependent decisions across materials, geometry, manufacturing, actuation, and structural response, which jointly determine the programmed behavior of the printed structure [9].

Recent computational methods have advanced several stages of this process. Physics-based simulation remains central to predicting stimulus-induced deformation, while machine-learning surrogates can accelerate forward prediction and support inverse design across large material-distribution spaces [15, 78]. Knowledge representations can additionally organize heterogeneous information on materials, manufacturing processes, stimuli, and actuation mechanisms [14]. These capabilities, however, are often developed and evaluated as separate computational components, with limited exchange of intermediate representations and results across the design cycle. Developing an integrated workflow that coordinates knowledge, design, simulation, and exploration therefore remains a system-level challenge.

Agents based on large language models (LLMs) offer a computational mechanism for addressing this fragmentation. ReAct interleaves reasoning with actions and observations [490], Toolformer enables language models to interact with external functions [491], and retrieval-augmented generation conditions model outputs on retrieved evidence [189]. Multi-agent systems extend these capabilities by assigning distinct roles and tools to

interacting agents and organizing their collaboration through centralized, distributed, or peer-to-peer structures [197, 492]. Frameworks such as AutoGen further support configurable interactions among language models, external tools, and human users [221]. Together, these developments provide the planning, specialization, and tool-integration capabilities required to coordinate interdependent scientific and engineering operations.

Tool-augmented agents have already been demonstrated in scientific and engineering applications. ChemCrow combines an LLM with chemistry tools for molecular design, synthesis planning, and experimental execution [493], while Coscientist coordinates information retrieval, code execution, experimental planning, and laboratory automation [494]. Conversely, representative computational methods for 4D printing focus on individual capabilities such as forward deformation prediction or inverse material design [15, 78]. Consequently, coordinating heterogeneous knowledge sources, design operations, predictive models, and discovery procedures within a unified workflow remains an open system-level challenge in 4D printing.

To address this challenge, we present an orchestrator-centered multi-agent system for scientific reasoning, design, and discovery in 4D printing. The system relies on a request-driven coordination mechanism based on explicit agent capability specifications and typed intermediate artifacts. Four specialized agents support ontology enrichment and querying, material assignment and inverse design, surrogate thermo-mechanical simulation and behavior discovery. The architecture is then evaluated through five coordinated workflows covering deployable-structure inverse design, literature-based material selection, question-to-design-to-simulation, controlled hypothesis verification, and large-scale behavior discovery. Together, these components integrate complementary scientific and engineering capabilities within an end-to-end computational workflow for 4D printing.

6.2 Proposed Multi-Agent Framework

The proposed framework is organized around a central orchestrator, powered by the open-weight gpt-oss-120b reasoning model, and connected to a set of specialized agents, as illustrated in Figure 6.1. Given a natural-language request, the orchestrator determines the scientific and computational operations required to address it, constructs an ordered execution plan, selects the appropriate agents and tools, and coordinates the intermediate results produced during execution. The reasoning model is used for request interpretation, task decomposition, operation ordering, and final synthesis, whereas domain-specific calculations are delegated to the corresponding agents and computational tools. This separation prevents the language model from directly generating numerical or physical results that should instead be obtained from scientific models.

The available agents are described through capability specifications defining the

operations they can perform, the tools they expose, and the representations they consume and produce. Using these specifications, the reasoning model constructs a request-specific execution plan rather than selecting a fixed workflow from a predefined catalog. Each planned operation identifies the agent and tool to invoke, the required input artifacts, the expected output representation, and its dependencies on preceding operations. Before invocation, the orchestrator checks that the required inputs are available and compatible with the selected capability. This process determines both the execution order and how the output of one agent becomes the input of another.

Information is transferred between agents through a shared execution state rather than through unconstrained natural-language messages. Each operation registers its output as a typed artifact that can be explicitly referenced by subsequent operations. Depending on the task, these artifacts may include ontology entities, extracted triples, supporting scientific evidence, voxelized structures, material maps, graph representations, displacement fields, deformed geometries, physical descriptors, candidate rankings, and scientific summaries. For example, a material distribution produced by one agent can be converted into a graph and passed to the simulation agent, while the resulting displacement field can subsequently be transferred to the discovery agent for behavior analysis. This artifact-mediated communication preserves the numerical and structural content of intermediate results and allows heterogeneous agents to operate on compatible representations.

During execution, the orchestrator updates the shared state after each completed operation and invokes dependent operations only when their required inputs become available. Once all operations required by the plan have been completed, the synthesis stage combines the retrieved evidence, generated configurations, numerical predictions, and behavior analyses into a structured scientific response. The resulting answer is therefore derived from the artifacts produced by the specialized agents rather than from the parametric knowledge of the reasoning model alone.

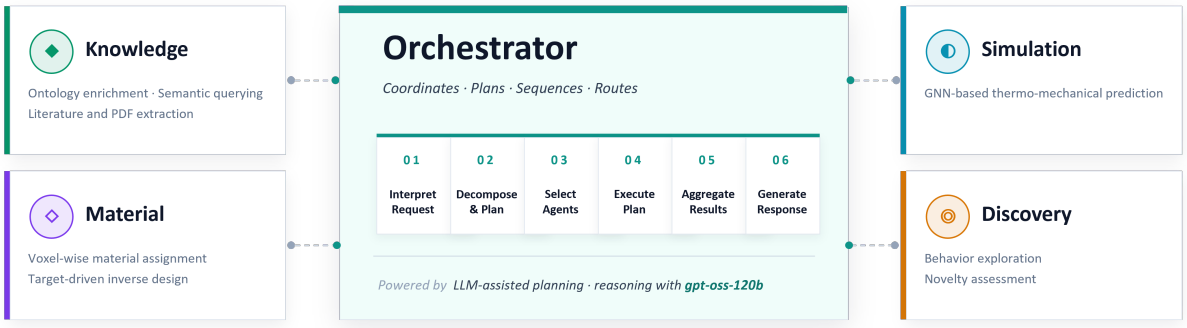

Figure 6.1: Overall Architecture of the Proposed Multi-Agent System.

6.3 Specialized Agent Roles and Capabilities

The framework relies on specialized agents for scientific knowledge processing, thermo-mechanical simulation, material inverse design, and behavior discovery. Each agent performs a dedicated operation and produces structured outputs that can be transferred to other agents through the orchestrator. The architecture therefore comprises one central orchestrator and four specialized agents:

- **Scientific knowledge agent.** This agent exposes the ontology-enrichment, knowledge-graph querying, semantic retrieval, multimodal information extraction, and symbolic validation capabilities developed in Chapter 3. It receives scientific questions, documents, or material-selection requirements and produces retrieved evidence, ontology entities, validated triples, and knowledge-grounded recommendations. The complete architecture and enrichment procedure are presented in Chapter 3, while the component-level evaluation and supplementary results are reported in Appendix B.

- **Material inverse design agent.** This agent wraps the graph-based inverse design framework developed in Chapter 5. Given a prescribed target deformation, it uses deformation-aware latent-space retrieval and material-distribution refinement to identify candidate structures whose predicted responses match the requested target. It produces candidate material maps, predicted target-shape errors, and ranked design solutions. The main inverse design methodology and results are presented in Chapter 5, while the VAE implementation, optimization-strategy definitions, encoder comparison, and supplementary evaluation are reported in Appendix E.

- **Thermo-mechanical simulation agent.** This agent exposes the GraphSAGE surrogate developed and validated in Chapter 4. It receives a material-aware structural graph and predicts the nodal displacement field and deformed geometry under thermal activation. It therefore provides the rapid forward-evaluation capability required by the inverse design, hypothesis-verification, and behavior-discovery workflows. The automated dataset-generation framework, surrogate methodology, and principal results are presented in Chapter 4, while the implementation details, comparative model assessment, evaluation protocol, and runtime analysis are reported in Appendix C.

- **Behavior discovery agent.** This agent wraps the candidate-generation and novelty-driven exploration methods introduced in Chapter 4. It analyzes surrogate-predicted deformation fields using physically interpretable descriptors, reference-based comparisons, displacement criteria, and deformation-asymmetry measures. Its outputs include behavior descriptors, novelty scores, ranked candidates, and

potentially novel coupled deformation responses. The discovery methodology and principal results are presented in Chapter 4, while the complete descriptor definitions, boundary condition specifications, thresholds, runtime calculations, and additional qualitative results are reported in Appendix C.

6.4 System-Level Workflow Evaluation

The proposed system can support a range of 4D-printing tasks and is evaluated through five representative workflows spanning inverse design, literature-based material selection, question-to-simulation, hypothesis verification, and behavior discovery. For each request, the orchestrator constructs an execution plan and coordinates the required agents and intermediate representations.

Workflow 1 — Inverse design of a deployable structure. This workflow addresses a user request for a deployable stent-like structure capable of expanding from an outer radius of 4.0 mm at 25°C to 8.0 mm at 38°C. A parametric serpentine geometry comprising six longitudinal rings, eight peaks per ring, 96 nodes, and 136 interconnected struts was generated (See Figure 5.5). Rather than optimizing the entire structure simultaneously, the framework decomposed it into repeated voxelized components. For a representative component, the target deformation was encoded in the learned latent space, and candidate material distributions were retrieved through cosine-similarity search. The material inverse design agent identified candidate voxel-wise material distributions, and the simulation agent predicted their thermo-mechanical responses. The best distribution achieved a cosine similarity of 0.98 with the target representation and a physical shape error of $E_{\text{shape}} = 20.43$. This material distribution was then assigned to the 136 repeated components before reassembling the complete expandable structure. The experiment demonstrates how component-level inverse design and surrogate simulation can be coordinated and transferred to a larger application-scale geometry.

Workflow 2 — Knowledge extraction from the literature. The enriched ontology was evaluated through a design-oriented material-selection query: "Identify a soft, biocompatible material that is compatible with liquid crystal elastomers (LCEs) and printable via direct ink writing (DIW)." The request was encoded as a semantic query vector, and the ontology entities were ranked by cosine similarity in the embedding space. Polydimethylsiloxane (PDMS) emerged as the highest-ranked candidate, supported by literature evidence showing that PDMS-based inks can be processed by DIW using embedded or freeform printing strategies [411, 412], that PDMS is widely used as a biocompatible and chemically inert elastomer [413], and that silicone layers have been integrated with cholesteric or nematic LCE actuators in bilayer structures

[414, 415]. Representative supporting triples included [PDMS, isPrintedUsing, DIW], [PDMS, hasProperty, Biocompatible], and [LCE actuator, isIntegratedWith, PDMS layer]. The retrieved neighborhood also contained platinum-cured and room-temperature-vulcanizing silicone elastomers, but PDMS provided the strongest combined evidence across printability, biocompatibility, and compatibility with LCE-based structures.

Workflow 3 — Question → design → simulation. Starting from a scientific design question on how to convert the predominantly tensile actuation of LCEs into compressive or pushing forces, the system was asked to select an appropriate material or structural strategy, generate a minimal actuator design, and validate its behavior through physics-based simulation. The knowledge agent queried the ontology and retrieved evidence indicating that LCEs primarily contract along their molecular alignment direction. Based on this mechanism, the system proposed a bilayer configuration coupling an active LCE layer with a passive layer, such that the differential strain between the two materials converts axial contraction into a bending response capable of producing the desired pushing motion. The resulting concept was generated as a $10 \times 3 \times 3$ voxelized structure, with active and passive materials assigned to the corresponding regions. The simulation agent predicted a mean displacement of 0.668 and a maximum displacement of 1.959. The resulting bending response was consistent with the mechanism retrieved from the ontology, demonstrating the system's ability to transform a scientific question into a knowledge-informed design and evaluate its predicted thermo-mechanical behavior.

Workflow 4 — Hypothesis → verification. The workflow evaluates the hypothesis that a heterogeneous anisotropic LCE orientation pattern produces a coupled, non-single-mode deformation response. Three configurations with identical geometry and BCs were generated: Case A combined active LCE orientations along x, y, z, $+45°$, and $-45°$ with passive and void regions; Case B preserved the same voxel layout but assigned a uniform LCE-X orientation to all active voxels; and Case C replaced all voxels with passive material to provide a non-actuating baseline. The simulation agent evaluated the final thermo-mechanical response of each case. Case A produced a mean displacement of 0.586 and a maximum displacement of 2.314, with coupled bending, localized shear, and asymmetric out-of-plane warping. Case B reached a higher mean displacement of 0.804 but a lower maximum displacement of 1.924, corresponding to a more spatially uniform response dominated by the common actuation direction. Case C produced a mean displacement of 0.231, representing the passive baseline. These results support the hypothesis that heterogeneous LCE orientations enrich the deformation response by introducing coupled and spatially non-uniform modes, although they do not necessarily maximize the mean displacement.

Workflow 5 — Behavior discovery. This workflow addresses a user request to discover non-single-mode 4D behaviors by generating 100 random material configurations on a $10 \times 3 \times 3$ voxel grid. Each voxel is assigned one of six material states: active LCE oriented along x, y, z, $+45°$, or $-45°$, or passive LCE. The simulation agent predicts the deformation induced by heating using the GNN surrogate. For each predicted response, the discovery agent extracts physics-based descriptors and compares them with a reference library of known deformation modes. Four candidates whose responses could not be explained solely by a known single-mode behavior were retained as potentially novel coupled deformation structures.

6.5 Discussion, Limitations, and Future Work

Given a natural-language request, the orchestrator constructs a request-specific execution plan, selects the required agents and tools, and manages the intermediate representations exchanged during execution. The specialized agents support scientific knowledge retrieval from an enriched ontology, voxel-wise material inverse design, accelerated thermo-mechanical prediction using a GNN surrogate, and descriptor-based behavior discovery. The five evaluated workflows demonstrate how these capabilities can be combined to address deployable-structure design, literature-based material selection, knowledge-guided simulation, controlled hypothesis verification, and the discovery of coupled deformation behaviors.

The proposed multi-agent system can be interpreted as a system-level instantiation of the Neuro $\rightarrow$ Symbolic $\leftarrow$ Neuro fibring paradigm introduced in Chapter 2. In this architecture, the central orchestrator implements the fibring operator $g_{\text{orchestrator}}$, which coordinates the heterogeneous neural, symbolic, and computational components exposed by the specialized agents.

The neural components include the multimodal extraction and retrieval models, the graph-based VAE used for inverse design, the GraphSAGE surrogate used for thermo-mechanical prediction, and the reasoning model used for request interpretation and plan construction. The symbolic resources include the domain ontology, the knowledge graph, the ontology-aligned entities and relations, and the symbolic validation rules used by the scientific knowledge agent.

The orchestrator connects these components through explicit capability specifications, typed intermediate artifacts, operation dependencies, and input–output compatibility constraints. The orchestration controller checks whether the required inputs are available, verifies their compatibility with the selected agent capabilities, enforces the execution dependencies, and routes validated outputs to subsequent operations. The orchestrator therefore performs the aggregation and coordination function assigned to g_{fibring}.

Several limitations remain. The orchestrator relies on LLM-based interpretation and

planning, meaning that ambiguous or underspecified requests may result in incomplete or inappropriate execution plans. Errors may also propagate across agents, as incomplete retrieved evidence, suboptimal inverse design candidates, or inaccurate surrogate predictions can affect subsequent operations and the final response. The current framework does not propagate calibrated uncertainty estimates across its agents or automatically revise the execution plan when an intermediate result is unreliable. Moreover, the GNN surrogate may be less accurate for geometries, material distributions, stimuli, or BCs outside its training distribution. The system-level evaluation is limited to five representative workflows and does not yet constitute a standardized assessment across a broad range of scientific requests. Finally, the current design space is restricted to voxelized structures, LCE-based material states, thermal actuation, and four boundary condition configurations.

Future work will develop a standardized benchmark containing diverse scientific questions, design requests, hypotheses, and discovery tasks. Uncertainty estimates will be integrated into knowledge retrieval, inverse design, surrogate simulation, and novelty assessment, enabling the orchestrator to revise its execution plan when intermediate outputs are incomplete, inconsistent, or unreliable. Out-of-distribution detection will also be introduced to identify uncertain GNN predictions and selectively trigger high-fidelity FEM verification. The framework will be extended to additional responsive materials, non-thermal stimuli, higher-resolution geometries, BCs, and manufacturing constraints. Finally, fabrication and physical testing will be integrated to support a closed-loop scientific workflow in which simulation and experimental results continuously update the ontology, datasets, predictive models, and subsequent design decisions.

6.6 Conclusion

In this chapter, we presented an orchestrator-centered multi-agent system that converts natural-language requests into request-specific execution plans coordinating scientific knowledge retrieval, material inverse design, accelerated thermo-mechanical simulation, and behavior discovery for 4D printing. The framework was evaluated through five workflows covering deployable-structure inverse design, literature-based material selection, knowledge-guided design and simulation, hypothesis verification, and behavior discovery. The results demonstrate that heterogeneous computational capabilities can be combined within a unified workflow to transform high-level scientific requests into structured designs, numerical predictions, and candidate discoveries. Although further work is required to improve uncertainty handling, out-of-distribution detection, physical validation, and generalization to broader material and manufacturing domains.

General Conclusion and Perspectives

4D printing combines AM with stimuli-responsive materials to produce structures whose shape, properties, or functionality evolve after fabrication. Designing these systems requires the coordinated consideration of material composition, stimulus conditions, manufacturing processes, spatial material distribution, structural geometry, boundary conditions, and expected deformation. These variables are strongly coupled, while the scientific knowledge required to configure them remains distributed across publications, experimental datasets, numerical models, and domain-specific expertise. Moreover, high-fidelity numerical simulations remain computationally expensive, and the inverse relationship between a prescribed behavior and the material distribution capable of producing it is generally non-unique.

This thesis addressed the following central research question: *How can knowledge be recommended when designing smart materials and transformable structures via 4D printing to achieve a desired property, behavior, and performance?* To address this question, the thesis developed conceptual, knowledge-engineering, predictive-modeling, inverse design, and system-integration contributions. These contributions connect scientific knowledge acquisition, thermo-mechanical modeling, computational design, and coordinated scientific workflows within a common framework for accelerated 4D-printing design.

The thesis produced six complementary scientific contributions. First, it established a critical analysis of NSAI for accelerated 4D-printing design. Second, it formalized and qualitatively evaluated alternative neuro-symbolic architectures according to generalization, scalability, data efficiency, reasoning, robustness, transferability, and interpretability. Third, it developed a retrieval-augmented multimodal framework that expanded an initial 4D-printing ontology from approximately 170 classes and a limited number of instances to more than 5,800 classes and over 12.5 million instances. Fourth, it introduced an automated numerical workflow and a GraphSAGE surrogate capable of predicting thermo-mechanical displacement fields in approximately 0.0269 s per structure. On the complete finite element workflow runtime basis adopted in this thesis, this corresponds to an acceleration factor of approximately 1.34×10^5–4.01×10^5. This reduction in computational time supported the screening of 800,000 candidate structures under four boundary condition scenarios. Fifth, the graph-based inverse design framework evaluated seven

search and optimization strategies on 153 target shapes, with the combination of top-k retrieval and local material mutation achieving the lowest final target-shape error for 112 targets. Finally, four specialized agents were integrated through an orchestrator and applied to workflows involving scientific knowledge retrieval, material inverse design, thermo-mechanical simulation, hypothesis verification, and deformation-behavior discovery.

Across the four computational contributions, NSAI functions as a system-level design principle rather than as a single architecture reproduced identically in every chapter. In the ontology-enrichment framework, neural multimodal models extract and map information from heterogeneous scientific sources, whereas the ontology, knowledge graph, and validation rules provide the symbolic representation against which candidate relations are evaluated. In the thermo-mechanical contribution, GraphSAGE constitutes the neural predictor, while the typed material graph, explicit spatial adjacency, boundary condition representation, and physics-informed deformation descriptors provide structured domain information. This contribution therefore implements a structured–neural interface, although graph representation alone should not be interpreted as symbolic reasoning. In the inverse design framework, learned graph representations and surrogate predictions are combined with explicit target-shape metrics, material-assignment constraints, and defined search and mutation operators. In the agent-based contribution, the orchestrator coordinates neural models, the knowledge graph, numerical tools, and rule-governed operations through explicit execution plans and dependencies. The neuro-symbolic character of the thesis consequently arises from the coordinated interaction between learned prediction and structured scientific knowledge across the complete design workflow, without requiring every individual component to constitute a stand-alone neuro-symbolic architecture.

Taken together, these contributions provide a direct response to the central research question. Knowledge recommendation for designing smart materials and transformable 4D-printed structures can be supported by combining structured scientific knowledge, data-driven physical prediction, constrained inverse exploration, and computational orchestration. The ontology and knowledge graph identify relations among materials, stimuli, processes, properties, behaviors, and applications. The surrogate model estimates the physical response of candidate material distributions at a computational cost compatible with large-scale exploration. The inverse design framework translates a prescribed deformation into ranked candidate configurations. The orchestrator connects these capabilities and manages the dependencies between the scientific objective, the retrieved knowledge, the generated candidate, and its predicted evaluation.

In this framework, recommendation is formulated as a sequence of knowledge retrieval, candidate construction, response prediction, quantitative evaluation, and iterative refinement. Its output is not limited to a material name or a statistically

similar existing design. It consists of a structured candidate characterized by a material distribution, activation conditions, a predicted deformation field, and a quantitative measure of agreement with the prescribed objective. The results obtained in this thesis indicate that these operations can be connected through explicit representations and interoperable computational components. They do not establish that a single computational model can solve every stage of 4D-printing design or that the resulting recommendations are valid outside the investigated domain.

Several limitations define the scope of these conclusions. The evaluation of neuro-symbolic architectures is qualitative and literature-grounded. It provides a systematic framework for comparing architectural properties but does not constitute a controlled empirical ranking performed on identical datasets, tasks, and computational resources. The suitability of an architecture remains dependent on the available data, required reasoning operations, knowledge representation, application scale, and evaluation objectives. Standardized engineering benchmarks will therefore be required to compare neuro-symbolic architectures quantitatively.

The ontology-enrichment framework processes a volume of scientific knowledge that cannot be exhaustively verified through manual inspection. Its content may include incomplete relations, terminology ambiguities, extraction errors, and biases inherited from the retrieved sources. Furthermore, the quality of the resulting knowledge graph depends on the coverage of the scientific literature, the structure of the ontology, and the performance of the multimodal extraction and mapping models. Confidence-aware relation management, systematic provenance recording, contradiction detection, incremental ontology updating, and targeted expert validation should consequently accompany future extensions of the knowledge base.

The numerical and learning-based contributions were evaluated within a defined thermo-mechanical domain. The finite element simulations rely on constitutive assumptions, discretization choices, prescribed material properties, thermal activation conditions, and selected boundary conditions. The GraphSAGE surrogate inherits these assumptions and is expected to provide reliable predictions only within or near the domain represented by its training data. Predictions for substantially different geometries, materials, stimuli, or deformation regimes may be affected by distribution shifts. Because the inverse design and behavior-discovery workflows use the surrogate as a forward evaluator, their outputs also inherit its prediction errors and domain limitations.

The voxel representation facilitates material assignment, automated graph construction, and large-scale dataset generation, but it restricts the accessible geometric resolution and does not represent all fabrication constraints or material-interface effects. In addition, the current inverse design evaluation mainly measures agreement with a prescribed target deformation. It does not yet incorporate all requirements related to printability, material compatibility, actuation stability, reversibility, fatigue, energy consumption, pro-

cessing time, or fabrication cost. The computationally discovered deformation behaviors and inverse designed structures should therefore be interpreted as candidates requiring high-fidelity numerical confirmation and physical validation.

The orchestrated workflows demonstrate the functional integration of heterogeneous scientific capabilities, but they do not constitute a complete autonomous engineering system. An error introduced during retrieval, prediction, inverse design, or geometric processing may propagate between agents. The language-based orchestrator also requires further evaluation concerning planning reliability, failure recovery, dependency management, reproducibility, and uncertainty communication. Systematic benchmarks should compare complete orchestrated workflows with the independent use of the underlying tools and determine whether orchestration improves task success, computational efficiency, and scientific consistency.

The immediate continuation of this work should focus on experimental validation. Selected inverse designed and behavior-discovery candidates should be fabricated and characterized under controlled activation conditions. Their measured deformation fields should be compared with both finite element and surrogate predictions. This comparison would quantify modeling and manufacturing errors, evaluate the physical reproducibility of the discovered deformation modes, and provide additional observations for calibrating the constitutive models and retraining the surrogate.

The computational domain should also be extended to additional smart-material families, stimulus mechanisms, geometric resolutions, and manufacturing processes. Uncertainty quantification and out-of-distribution detection should be incorporated into the forward predictor so that each recommendation is accompanied by an estimate of its reliability. Candidates associated with high predictive uncertainty could then be directed toward finite element analysis or physical experimentation, whereas candidates within the validated domain could be evaluated using the surrogate. Manufacturability, reversibility, structural stability, and lifecycle performance should also become explicit objectives or constraints within a multi-objective inverse design formulation.

The principal long-term perspective is the development of a self-driving laboratory for smart materials and 4D printing organized around a scientific world model. A world model learns an internal representation of an environment and predicts how its state may evolve in response to candidate actions, thereby supporting decision-making through simulated trajectories before those actions are executed physically [4, 5]. In the context of 4D printing, the state represented by this model would include material composition, spatial material distribution, structural geometry, manufacturing parameters, stimulus history, boundary conditions, measured deformation, and predictive uncertainty. Its actions would correspond to selecting or modifying a material formulation, voxel distribution, geometry, printing condition, activation protocol, simulation request, or physical experiment.

The objective of this scientific world model would extend beyond predicting a single

deformation field. It would represent the coupled evolution of the material–process–structure–behavior system and estimate how alternative design and experimental actions could modify that system. The world model should not be understood as a single neural network. It would constitute a multi-fidelity computational representation combining learned dynamics, finite element simulations, experimental observations, structured scientific knowledge, and explicit physical and manufacturing constraints.

The contributions of this thesis provide several components from which such a world model could be developed. The ontology and knowledge graph would constitute its structured scientific memory by representing known materials, stimuli, processes, properties, prior experiments, and compatibility relations. The GraphSAGE surrogate would provide rapid estimates of thermo-mechanical response, whereas the finite element model would provide selective high-fidelity verification. The inverse design framework would propose candidate actions intended to reach a prescribed deformation, and the behavior-discovery methods would identify unexplored response regions. The orchestrator would coordinate knowledge retrieval, candidate generation, simulated evaluation, fabrication requests, characterization operations, and model updates.

When a target property, behavior, or performance is specified, the world model could first retrieve relevant materials, activation conditions, manufacturing processes, and prior observations from the structured knowledge base. The inverse design framework would then produce candidate material distributions. Instead of evaluating only their final predicted response, the world model could simulate alternative design and experimental trajectories and estimate their expected consequences. Candidate structures could be ranked according to their expected performance, manufacturability, uncertainty, and potential contribution to improving the current model. This procedure would allow the system to select experiments that are useful both for achieving the design objective and for reducing uncertainty in insufficiently explored regions.

The selected structures would subsequently be fabricated and characterized automatically. Their measured responses would be compared with the world-model predictions, and the resulting prediction errors, successful responses, and experimental failures would be converted into structured observations. These observations would update the training datasets, material models, ontology relations, surrogate parameters, and candidate-selection strategy before the next experimental cycle. The resulting closed loop would follow the sequence of scientific-objective definition, knowledge retrieval, candidate generation, imagined evaluation, experiment selection, fabrication, characterization, and model revision.

Existing self-driving laboratories have demonstrated that iterative experiment selection, automated execution, and model updating can accelerate thin-film optimization and inorganic materials discovery [6, 7]. Reinforcement-learning-guided platforms have also demonstrated the autonomous exploration of sequential, high-dimensional experimental

processes [8]. Extending these principles to 4D printing would require automated multi-material fabrication, controlled stimulus application, geometric and thermo-mechanical characterization, and reliable conversion of experimental measurements into representations compatible with the world model.

The proposed platform would require uncertainty-aware decision making, out-of-distribution detection, calibrated sensors, standardized metadata, experimental provenance, and explicit safety and manufacturability constraints. Human supervision should remain part of the decision loop, particularly when proposed experiments involve new material formulations, equipment operating limits, or unexpected physical responses. The role of the world model would be to predict, prioritize, and learn from experimental actions rather than to replace scientific judgment.

A practical implementation could initially be restricted to a single thermo-responsive material family, a controlled set of voxelized geometries, and a defined collection of target deformation modes. The first stage could automate candidate selection, printing, thermal activation, deformation measurement, and comparison with predicted responses. The world model could then be updated after each experimental campaign while a human expert validates proposed experiments and examines anomalous results. Once the reliability of the individual components has been established, the system could progressively expand toward additional materials, stimuli, geometries, and multi-objective design requirements.

The development of this platform would transform the contributions of the thesis from a sequence of connected computational capabilities into an experimentally closed learning system. The knowledge graph would provide its evolving semantic memory, the neural models would learn predictive relations from simulation and experimental data, the finite element model would supply selective high-fidelity verification, and the physical laboratory would provide observations of the real system. The interaction between these components would result in a neuro-symbolic and multi-fidelity scientific world model that learns progressively from literature, numerical analysis, and physical experimentation.

This thesis provides conceptual and computational foundations for knowledge-guided recommendation in 4D-printing design. It structures scientific information, accelerates thermo-mechanical prediction, explores large design spaces, produces candidate material distributions for prescribed deformations, and coordinates these operations through specialized computational capabilities. These results remain bounded by the investigated materials, numerical assumptions, datasets, and validation conditions. Closing the loop through a world-model-based self-driving laboratory represents the principal perspective for transforming the proposed framework into a system that iteratively designs, fabricates, observes, and improves smart materials and transformable structures.

Scientific and Academic Achievements

6.7 Peer-Reviewed Journal Papers

During this doctoral work, several research articles were published, submitted, or prepared in international journals and preprint platforms. These contributions address different aspects of artificial intelligence for 4D printing, including neuro-symbolic artificial intelligence, ontology enrichment, reinforcement learning, generative artificial intelligence, and graph-based learning.

- **Oualid Bougzime, Christophe Cruz, Jean-Claude André, Kun Zhou, H. Jerry Qi, and Frédéric Demoly.** *Neuro-symbolic artificial intelligence in accelerated design for 4D printing: Status, challenges, and perspectives. Materials & Design*, vol. 252, article 113737, 2025. DOI: 10.1016/j.matdes.2025.113737 [11].
- **Hichem Seriket, Oualid Bougzime, Yuyang Song, H. Jerry Qi, and Frédéric Demoly.** *Reinforcement learning-enabled design of topological interlocking materials for sustainable multi-material additive manufacturing. Additive Manufacturing*, vol. 111, article 104992, 2025. DOI: 10.1016/j.addma.2025.104992 [495].
- **Oualid Bougzime, Samir Jabbar, Christophe Cruz, and Frédéric Demoly.** *Unlocking the Potential of Generative AI through Neuro-Symbolic Architectures: Benefits and Limitations. arXiv preprint*, arXiv:2502.11269, 2025. DOI: 10.48550/arXiv.2502.11269 [496].
- **Oualid Bougzime, Christophe Cruz, Kun Zhou, H. Jerry Qi, and Frédéric Demoly.** *Structuring scientific knowledge for smart materials and 4D printing: An ontology enrichment framework using retrieval-augmented multimodal large language models. Computers in Industry*, article 104522, 2026. DOI: 10.1016/j.compind.2026.104522 [375].

- **Oualid Bougzime, Samir Jabbar, Christophe Cruz, and Frédéric Demoly.** *Neuro-Symbolic Architectures: Formalizing, Extending, Evaluating, and Empowering Generative Artificial Intelligence. Neurosymbolic Artificial Intelligence*, vol. 2, 2026. DOI: 10.1177/29498732261469485 [208].

6.8 National and International Conferences

Research conducted during the doctoral period was presented or accepted for presentation at several national and international scientific events. These contributions supported the dissemination of the research results, exchanges with the scientific community, and the development of collaborations in 4D printing, artificial intelligence, neuro-symbolic reasoning, knowledge engineering, and computational mechanics.

- **Oualid Bougzime, Hichem Seriket, Christophe Cruz, Jean-Claude André, H. Jerry Qi, and Frédéric Demoly.** *Construction of a 4D Printing Ontology Supporting Knowledge Graphs Using a Multimodal Large Language Model.* Presented at the *4D Materials Design and Additive Manufacturing (4DMDA 2024)*, Ajaccio, France, 2024.

- **Oualid Bougzime, Christophe Cruz, and Frédéric Demoly.** *L'Intelligence Artificielle Neuro-symbolique dans la Conception Accélérée pour l'Impression 4D : État, Défis et Perspectives.* Presented at the *19th Colloque national S.Mart 2025*, Vogüé, France, 2025. HAL: hal-05244947. [497].

- **Oualid Bougzime, Hichem Seriket, Christophe Cruz, H. Jerry Qi and Frédéric Demoly** *Accelerating Structure Design in 4D Printing through AI.* Presented at the *4D Materials Design and Additive Manufacturing Conference (4DMDA 2025)*, Singapore, 2025.

- **Oualid Bougzime, Samir Jabbar, Christophe Cruz, and Frédéric Demoly.** *Evaluating Neuro-Symbolic AI Architectures: Design Principles, Qualitative Benchmark, Comparative Analysis and Results.* Poster presentation at the *19th International Conference on Neurosymbolic Learning and Reasoning (NeSy 2025)*, Santa Cruz, USA, 2025. Published in *Proceedings of Machine Learning Research (PMLR)*, vol. 284, pp. 1119–1143, 2025 [498].

- **Oualid Bougzime, Christophe Cruz, and Frédéric Demoly.** *Accélérer la conception de structures en impression 4D grâce à l'IA.* Presented at the *Journées GDR TRANS4M*, Belfort, France, 2025. [499].

- **Oualid Bougzime, Samir Jabbar, Christophe Cruz, and Frédéric Demoly.** *Agent-Based and Knowledge-Graph-Grounded Tree-of-Thought Reasoning for Expert*

Question Answering. Presented at the *World Conference on eXplainable Artificial Intelligence (xAI 2026)*, Fortaleza, Brazil, 2026.

- **Oualid Bougzime, Sameera Bommisetty, Christophe Cruz, and Frédéric Demoly.** *Neuro-Symbolic Automated Discovery of Novel Deformation Behaviors in Multi-Material 4D-Printed Structures Using Graph Neural Surrogates and Novelty-Driven Search.* Presented at the *World Congress on Computational Mechanics and ECCOMAS Congress (WCCM–ECCOMAS 2026)*, Munich, Germany, 2026.
- **Oualid Bougzime, Sameera Bommisetty, Christophe Cruz, and Frédéric Demoly.** *An Explainable Multi-Agent Orchestrator for End-to-End 4D Printing Design.* Presented at the *5th International Conference on 4D Materials Design and Additive Manufacturing (4DMDA 2026)*, Dublin, Ireland, 2026. **Best Presentation Award**.
- **Samir Jabbar, Oualid Bougzime, Christophe Cruz, and Maria Alice Bertolim.** *DocAgent-XAI: An Entropy-Guided Agentic RAG System for Adaptive and Explainable Clinical History-Taking.* In *Artificial Intelligence in Healthcare: Third International Conference, AIiH 2026, Proceedings, Part I*, Lecture Notes in Computer Science, vol. 16875, pp. 501–515, Springer, 2027. DOI: 10.1007/978-3-032-35387-0_36 [500].
- **Oualid Bougzime, Hichem Seriket, Christophe Cruz and Frédéric Demoly** *Applying Neuro-Symbolic AI Architectures to Engineering Design in 4D Printing.* Presented at the *ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE 2026)*, Houston, Texas, USA, 2026.
- **Participation in World Summit AI, Amsterdam, Netherlands, 2026.**

6.9 Intellectual Property

The doctoral period also contributed to intellectual property activities, including seven computer software patents related to the research developments conducted during the thesis.

6.10 Technology Transfer and Entrepreneurial Development

Building on the research outcomes and software developments produced during the thesis, several activities were undertaken to explore their potential transfer toward industrial

applications and entrepreneurial development. Discussions were held with industrial stakeholders, institutional actors, and local innovation partners, including representatives from the City of Belfort, to assess the potential for developing a deeptech project based on the technologies and methods developed during this doctoral work.

These exchanges aimed to explore how the artificial intelligence methods developed during the thesis could be transferred toward industrial applications. The discussions focused on the valorization of research outcomes, the identification of potential use cases, the analysis of market needs, and the positioning of an future deeptech project at the intersection of artificial intelligence, engineering design, smart materials, additive manufacturing, and 4D printing. They also helped clarify the possible industrial relevance of explainable recommendation systems for supporting material selection, design automation, simulation acceleration, and decision-making in advanced manufacturing.

In 2026, this entrepreneurial perspective was further strengthened through participation in the *DeepTech in BFC* training program, organized within the Bourgogne-Franche-Comté innovation ecosystem. This program supports researchers and project holders working on disruptive technological innovations by providing collective workshops, expert guidance, and individual coaching for the maturation of deeptech projects [501].

These activities contributed to broadening the impact of the thesis beyond academic research. They provided an opportunity to confront the scientific developments with industrial expectations, refine the value proposition of the proposed technologies, and evaluate possible pathways for transferring research results toward real-world applications. This valorization process reinforces the applied dimension of the thesis and highlights the potential of neuro-symbolic artificial intelligence and 4D printing as a foundation for future deeptech innovation.

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Appendix A

Hardware and Runtime Environment

All experiments were executed locally on a mobile workstation equipped with a 13th Gen Intel® Core™ i9-13900H CPU (20 logical cores, base frequency ∼2.6 GHz), 64 GB of RAM, and an NVIDIA RTX 3000 Ada Generation Laptop GPU with 8 GB of dedicated VRAM (about 40 GB total GPU memory reported by the driver). The operating system is Windows 11 Education 64-bit.

Appendix B

Ontology Enrichment: Model Assessment and Experimental Details

B.1 AI Model Assessment

To determine the optimal temperature – a key parameter that regulates the level of randomness in the model's output during inference – we evaluated the model's performance across a range of temperature settings. Specifically, we tested two configurations: the fine-tuned model on its own, and the fine-tuned model combined with one-shot learning. Temperature plays an important role in balancing determinism and creativity in language model outputs. Lower temperatures make the model's responses more focused and predictable, while higher temperatures increase variability and originality. This trade-off impacts the accuracy and relevance of extracted triplets [502]. As illustrated in **Figure B.1**, the standard fine-tuning approach without any in-context learning demonstrated higher stability and improved performance when compared to the fine-tuning approach with one-shot across metrics which represent n-gram-based metrics encompassing precision (Bilingual Evaluation Understudy, termed as BLEU), Recall-Oriented Understudy for Gisting Evaluation (termed as ROUGE), and F1-score (combining BLEU and ROUGE metrics) [386]. These n-gram-based metrics rely on the comparison of overlapping word sequences (called n-grams) between the generated and reference texts. For instance, an e-gram refers to a contiguous sequence of e words, 1-grams are unigrams (single words), 2-grams are bigrams, and so on, thus providing nuanced evaluation of fluency and relevance in generated text [503]. Details of the metric computation are provided in the next section.

In addition to large-scale quantitative enrichment, we performed a qualitative assessment by manually reviewing a representative subset of extracted triples (top-k highest

confidence predictions and a random sample of 200 instances) compared to expert knowledge. The evaluation considered four dimensions: (i) relevance of the extracted relation to the 4D printing domain, (ii) factual correctness with respect to the source text, (iii) clarity of entity boundaries and relation semantics, and (iv) actionability in terms of whether the triple can be meaningfully integrated into downstream ontology reasoning. For instance, the extracted triple [Hydrogel, `isInstanceOf`, Biocompatible_Material] was judged as fully correct and relevant, whereas [3D_printing, `converts`, Robust_manufacturing_process], although syntactically valid, was marked as semantically unclear and thus of limited actionability. Overall, 84% of the reviewed triples were rated as both factually correct and relevant, 10% as partially correct but ambiguous (e.g., inconsistent entity typing), and 6% as incorrect due to model hallucinations or misalignment. This analysis highlights that while large-scale automatic enrichment can produce "shallow" or noisy entries, systematic sampling and expert review confirm that the majority of extracted triples are suitable for ontology integration. Qualitatively, these proportions indicate that high-confidence extractions are immediately usable, whereas most residual errors stem from entity typing and boundary delineation; this highlights schema-aware normalization and predicate templating as the primary levers for improvement. Error forensics further show that low-evidence contexts and predicate drift account for the majority of the remaining 6% failures, motivating a quarantine queue and tighter domain/range constraints prior to graph insertion. Taken together, these observations align with our objectives – precision under ontological constraints and provenance-backed integration – and directly inform corrective actions in the pipeline (stricter typing, evidence thresholds, and conflict-aware deduplication). Beyond this qualitative view, we performed a quantitative analysis showing that the stand-alone fine-tuned model consistently outperforms the fine-tuned-plus-one-shot configuration, which exhibits pronounced variability and uniformly lower scores. Triplet-matching F1 peaks at $T \approx 0.55$, while G-BLEU and G-ROUGE remain optimal over $T \in [0.55, 0.70]$; G-BERTScore precision is maximal near $T \approx 0.55$, indicating fine-grained semantic alignment between predicted and reference graphs. Taken together, these results show that $T \approx 0.55$ offers the best precision–recall trade-off, combining robust F1 with sensitivity captured by both n-gram overlap and contextualized embeddings. Consequently, plain fine-tuning delivers superior extractive accuracy and stability for relational triple extraction while avoiding the complexity and instability introduced by one-shot in-context learning. Analytically, this justifies fixing $T = 0.55$ as the default decoding regime to prioritize precision without collapsing recall; operationally, it implies fewer low-confidence triples entering pre-insertion quarantine and, downstream, fewer ontology-coherence violations – i.e., a larger yet cleaner graph.

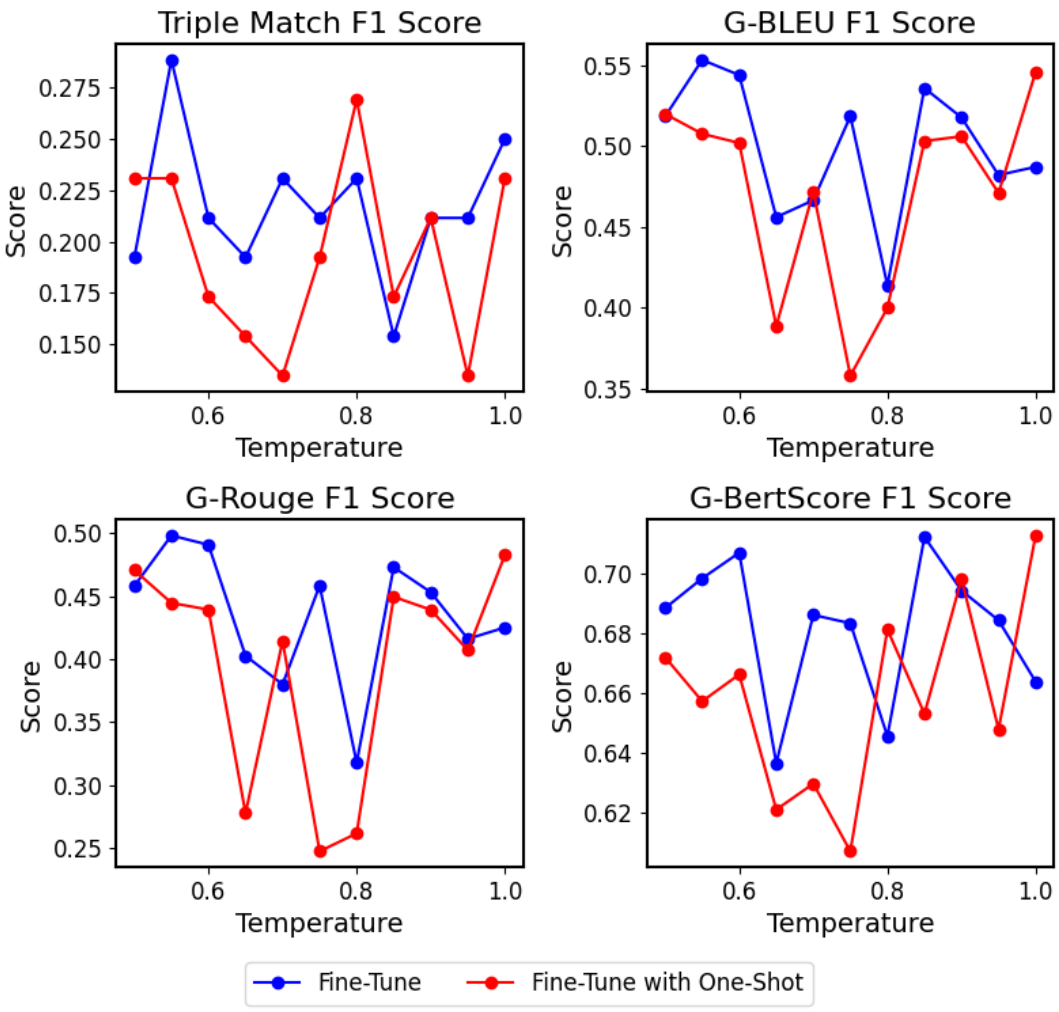

Figure B.1: Comparison metrics between prompt with the fine-tuned model vs. prompt with the fine-tuned model using one-shot technique performance across various temperatures, reproduced from [375].

B.2 Metric Computation

BLEU-F1 Score [504]

Let C_{gen} be the number of 4-grams in the generated graph, C_{ref} the number of 4-grams in the reference graph, and C_{match} the number of matching 4-grams. Then:

$$P_{\text{Bleu}} = \frac{C_{\text{match}}}{C_{\text{gen}}} \tag{B.1}$$

$$R_{\text{Bleu}} = \frac{C_{\text{match}}}{C_{\text{ref}}} \tag{B.2}$$

$$F1^{\text{Bleu}} = \frac{2\,P_{\text{Bleu}}\,R_{\text{Bleu}}}{P_{\text{Bleu}} + R_{\text{Bleu}}} \tag{B.3}$$

ROUGE-F1 Score [505]

For ROUGE-2 (bigrams), let $bigram_{\text{cand}}$ and $bigram_{\text{ref}}$ be the sets of bigrams in the candidate and reference, respectively. Then

$$P_{\text{ROUGE}} = \frac{|bigram_{\text{cand}} \cap bigram_{\text{ref}}|}{|bigram_{\text{cand}}|} \tag{B.4}$$

$$R_{\text{ROUGE}} = \frac{|bigram_{\text{cand}} \cap bigram_{\text{ref}}|}{|bigram_{\text{ref}}|} \tag{B.5}$$

$$F1^{\text{ROUGE}} = \frac{2\, P_{\text{ROUGE}}\, R_{\text{ROUGE}}}{P_{\text{ROUGE}} + R_{\text{ROUGE}}} \tag{B.6}$$

Graph BERTScore (G-BS) [506]

G-BS takes graphs as a set of edges and solves a matching problem which finds the best alignment between the edges in the predicted graph and those in the ground-truth graph. Each edge is considered as a "sentence" and BERTScore is used to calculate the similarity between a pair of predicted and ground-truth edges. Based on the optimal alignment and the overall matching score, the computed F1 score is used as the final G-BERTScore.

Considering x_i as a reference token (entity or relation) and $\hat{x}_j$ as a generated token (entity or relation), the complete score matches each token in x to a generated token in $\hat{x}$ to compute recall, and each token in $\hat{x}$ to a token in x to compute precision. A greedy matching was used to maximize the total similarity score, where each token was matched to the most similar token in the other graph. Then precision and recall were combined to compute an F1 measure.

For a reference sequence $x = \{x_1, \ldots, x_{|x|}\}$ and a candidate sequence $\hat{x} = \{\hat{x}_1, \ldots, \hat{x}_{|\hat{x}|}\}$, the recall R, precision P, and F1 scores are defined as:

$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} x_i^\top \hat{x}_j \tag{B.7}$$

$$P_{\text{BERT}} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} x_i^\top \hat{x}_j \tag{B.8}$$

$$F_1^{\text{BERT}} = \frac{2\, P_{\text{BERT}}\, R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}} \tag{B.9}$$

Triples Matching (T-F1)

The F1 score for triple matches T-F1 was calculated as follows:

$$T\text{-}F1 \;=\; \frac{2\,\text{TP}}{2\,\text{TP} + \text{FP} + \text{FN}} \tag{B.10}$$

where TP is the number of true positive triple matches, FP is the number of false positive triple matches, and FN is the number of false negative triple matches. These metrics could potentially yield even better results if synonyms of entities or relations are considered as exact matches. This evaluation serves as a benchmark analysis for assessing the efficacy

of our ontology enrichment methodology.

B.3 Peer-Reviewed Journals Covered for Ontology Enrichment

Concerning metadata, the 1,810 scientific articles span the period 2009–2024 and cover peer-reviewed journals. The full list of journals is below: *Additive Manufacturing*, *Advanced Materials*, *Materials & Design*, *Polymers*, *Materials*, *Journal of Inorganic and Organometallic Polymers and Materials*, *International Journal of Advanced Manufacturing Technology*, *Scientific Reports*, *Micromachines*, *Engineering Structures*, *Nanomaterials*, *Virtual and Physical Prototyping*, *Nature*, *Soft Robotics*, *Small*, *Sensors and Actuators*, *Composite Structures*, *Smart Materials and Structures*, *Composite Structures*, *Frontiers in Materials*, *Advanced Functional Materials*, *Journal of Materials Chemistry B*, *Journal of Materials Chemistry C*, *Journal of Materials Research*, *Journal of Polymer Science*, *Nature Communications*, *Journal of King Saud University*, *Journal of King Saud University - Engineering Sciences*, *International Journal of Nanomedicine*, *International Journal of Bioprinting*, *Composite Part B*, *Journal of Colloid and Interface Science*, *Journal of Materials Chemistry A*, *International Journal of Biological Macromolecules*.

B.4 Experimental Setup

LLaVA (Large Language and Vision Assistant) [99] demonstrates significant effectiveness in visual and language understanding by connecting a vision encoder with a language model, offering an end-to-end trained large multimodal model for general-purpose tasks. It notably excels in visual instruction tuning, the first attempt to extend instruction tuning to the language-image multimodal space, showing impressive capabilities in multimodal chat and yielding a relative score of 85.1% compared with GPT-4 on a synthetic multimodal instruction-following dataset. Further, when fine-tuned on Science QA [507], LLaVA and GPT-4 together achieve a state-of-the-art accuracy of 92.53%. Additionally, LLaVA achieves remarkable scores across various benchmarks, including a score of 63.3% on GQA [508] and 61.3% on TextVQA [509], when scaled up to a 13B model with a 336px resolution [510]. The LLaVA model has been particularly effective when fine-tuned with academic task-oriented data, incorporating data such as VQA-v2 [511], which enhances its multimodal understanding capabilities. Simple modifications, like using an multilayer perceptron (MLP) cross-modal connector [512, 513], have proven instrumental in boosting its performance across 11 benchmarks [510]. The results underscore LLaVA's robust performance across diverse, application-oriented benchmarks, reinforcing its utility in enhancing instruction-following models with a visual component. Beyond performance,

LLaVA provides open-source code and publicly available checkpoints, which facilitates reproducibility and integration in domain-specific pipelines.

LLaVA was fine tuned using LoRA to improve triplet extraction. This technique allowed us to efficiently adjust the model's parameters while minimizing resource consumption. The fine-tuning process utilized a synthetic dataset specifically designed to include textual and visual contexts aligned with the needs of 4D printing. In this process, LoRA was enabled with a rank of 128 and an alpha value of 256, and the multimodal projector learning rate was set to 2×10^{-5}. The base model used was `liuhaotian/llava-v1.5-7b` (version v1), and the training data comprised a synthetic JSON dataset alongside an image folder containing the extracted images. The vision tower was configured to use OpenAI's CLIP ViT-Large-Patch14-336, while the multimodal projector was set to an MLP with GELU activation (mlp2x_gelu), selecting the penultimate layer for vision features. Notably, both image start-end tokens and image patch tokens were disabled, and images were preprocessed with a padded aspect ratio. The training procedure grouped samples by modality length and leveraged bfloat16 precision. Training was conducted for 5 epochs with a per-device batch size of 8 for training (and 4 for evaluation), accumulating gradients over 2 steps. The evaluation strategy was disabled during training, and model checkpoints were saved every 50,000 steps with a total checkpoint limit of one. The learning rate was fixed at 2×10^{-4} with no weight decay and a warmup ratio of 3%, using a cosine learning rate scheduler. Additionally, TF32 was enabled, the maximum model length was set to 2,048 tokens, gradient checkpointing was employed to conserve memory, and 4 dataloader workers were used in conjunction with lazy preprocessing. **Figure B.2** shows a rapid early loss drop followed by a low-variance plateau ($\sim$0.05), indicating stable convergence under LoRA (rank= 128, α = 256) and supporting this fine-tuned LLaVA configuration as an appropriate and reliable model for triplet extraction.

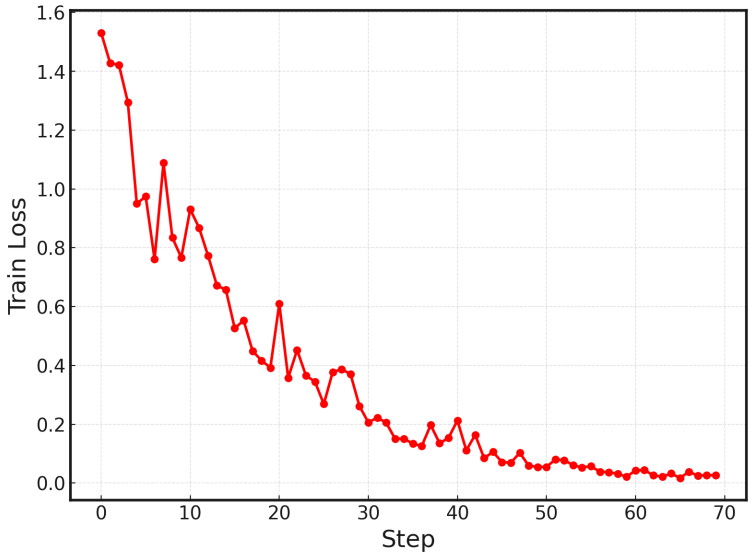

Figure B.2: Training loss during LoRA fine-tuning of LLaVA for triplet extraction, reproduced from [375].

BERT is particularly effective for text embedding due to its deep bidirectional nature, which allows it to capture context from both the left and the right sides of a token within a sentence. This approach contrasts with traditional unidirectional models that only consider context from one direction, either left-to-right or right-to-left, which can limit their understanding of language context. BERT is trained on a large corpus with a wide variety of language modeling tasks, which allows it to develop a nuanced understanding of language syntax and semantics. This makes it highly effective for generating dense and informative embeddings that represent the semantic meaning of the text, making these embeddings very useful for subsequent machine learning tasks.

CLIP excels in mapping textual descriptions to relevant visual content, making it an ideal tool for cross-modal retrieval and text-to-image linking where text is used to query and retrieve corresponding images. By encoding both text and images into a shared vector space, CLIP enables the effective comparison of textual input to a vast database of pre-encoded images. This capability is particularly useful for retrieving the most relevant image based on a textual description, utilizing cosine similarity to determine the best match. The flexibility of CLIP, trained on a diverse set of image-text pairs, allows it to perform robustly across various domains without the need for task-specific training. This makes CLIP a powerful component in systems where dynamic linkage between descriptive text and applicable images is necessary, enhancing content accessibility and user interaction in digital archives, educational platforms, and content management systems.

LLM Sherpa was employed to extract relevant information from scientific articles and efficiently partition textual and visual data. LLM Sherpa enables article analysis by parsing PDF documents into a structured format (sections, paragraphs, tables,

and figures) and producing clean, semantically coherent chunks that are subsequently embedded and directly integrated into our vector database for retrieval and evidence-grounded triplet extraction, which is then used to enrich the ontology.

GPT-4 played a pivotal role in constructing the training dataset. It was employed to generate synthetic data, including prompts, contexts, and expected responses, specifically designed to fine-tune the LLaVA model for triplet extraction. Importantly, GPT-4 outputs were treated as candidate annotations rather than ground truth: they were generated under explicit constraints to rely exclusively on the provided section text and aligned figure, and all examples were subsequently reviewed by a 4D-printing domain expert with ontology-engineering expertise to remove unsupported claims and to correct entity typing and relation directionality. This process ensured that the dataset aligned with the specific requirements of 4D printing applications.

The Mistral-7B-v0.3 model was used to process and classify datasets for ontology enrichment. Leveraging a one-shot learning approach, Mistral efficiently mapped dataset attributes – such as material properties, printing parameters, and experimental results – to the relevant ontology classes, enabling seamless integration of structured data into the ontology. This model was chosen for its strong balance between performance and computational efficiency: it outperforms larger models such as LLaMA 2–13B and even LLaMA 1–34B on reasoning, mathematics, and code benchmarks, despite having fewer parameters. Its architecture integrates Grouped Query Attention (GQA) and Sliding Window Attention (SWA), which accelerate inference while enabling effective handling of long sequences [408]. These properties make Mistral particularly well-suited for processing large scientific corpora, where long contextual spans are common and where attribute descriptions can be technical and context-heavy, requiring precise alignment to ontology classes. Furthermore, its open-source availability under the Apache 2.0 license facilitates reproducibility and integration in domain-specific pipelines.

Appendix C

Supplementary Experimental Details for GraphSAGE-Based Thermo-Mechanical Simulation and Shape-Morphing Discovery

C.1 Experimental Setup

C.1.1 Training Objective and Loss Function

The GraphSAGE surrogate was trained using an axis-weighted Huber loss augmented with a transverse top-error penalty. This loss was designed to account for the anisotropic nature of the predicted displacement field and to emphasize the displacement components most relevant to bending- and twisting-dominated responses. Since the transverse displacement components are more strongly associated with these deformation modes, larger weights are assigned to the y and z axes than to the axial x component.

The training objective is defined as:

$$\mathcal{L} = w_x \mathcal{L}_x^{\text{Huber}} + w_y \mathcal{L}_y^{\text{Huber}} + w_z \mathcal{L}_z^{\text{Huber}} + \lambda \mathcal{L}_{\text{top}}^{yz}$$

where $\mathcal{L}_x^{\text{Huber}}$, $\mathcal{L}_y^{\text{Huber}}$, and $\mathcal{L}_z^{\text{Huber}}$ are smooth-ℓ_1 losses computed independently on the predicted displacement components along the x, y, and z axes, respectively. The weights used in the implementation are $w_x = 0.5$, $w_y = 5.0$, and $w_z = 6.0$, with a Huber transition parameter $\beta = 0.02$.

For a residual r, the Huber loss is defined as:

$$\ell_\beta(r) = \begin{cases} \dfrac{r^2}{2\beta}, & |r| \le \beta, \\ |r| - \dfrac{\beta}{2}, & |r| > \beta. \end{cases}$$

The additional term $\mathcal{L}_{\text{top}}^{yz}$ focuses on the largest transverse prediction errors. It is computed from the mean absolute error of the top 10% largest errors along the y and z directions:

$$\mathcal{L}_{\text{top}}^{yz} = \frac{1}{2}\left(\frac{1}{k_y}\sum_{i\in\mathcal{T}_y}|\hat{u}_i^y - u_i^y| + \frac{1}{k_z}\sum_{i\in\mathcal{T}_z}|\hat{u}_i^z - u_i^z|\right),$$

where $\mathcal{T}_y$ and $\mathcal{T}_z$ denote the sets of nodes corresponding to the largest 10% absolute prediction errors in the y and z directions, respectively. The coefficient of this penalty is set to $\lambda = 2.0$.

Overall, this loss formulation combines robustness to outliers through the Huber terms with a targeted emphasis on high-error transverse displacements. This is particularly important for the proposed discovery pipeline, because localized or high-amplitude transverse deformations are strongly associated with bending, twisting, and other behaviorally relevant actuation modes.

C.1.2 Training-Loss Evolution

The evolution of the training loss as a function of the number of epochs is shown in **Figure C.1**. The curve exhibits a rapid decrease during the first epochs, with the loss dropping from a high initial value to below 0.5 after only a few training iterations. This sharp reduction indicates that the model quickly learns the dominant relationships between the graph-based structural representation and the corresponding thermo-mechanical displacement response.

After this initial phase, the loss continues to decrease more progressively. Between approximately 20 and 100 epochs, the model refines its internal representation and gradually reduces the prediction error. A small temporary increase can be observed around 120–140 epochs, which may be associated with optimization fluctuations or learning-rate scheduling effects. However, this perturbation remains limited and does not prevent the model from continuing its convergence.

Beyond approximately 150 epochs, the training loss enters a slower convergence regime. The curve decreases smoothly and progressively approaches a stable low-loss region toward the end of training. This behavior suggests that the model does not exhibit strong training instability and that the optimization process remains well controlled. Overall, the loss evolution confirms that the retained GraphSAGE surrogate effectively learns the deformation mapping from the voxel-based graph representation while maintaining a

stable convergence trend throughout training.

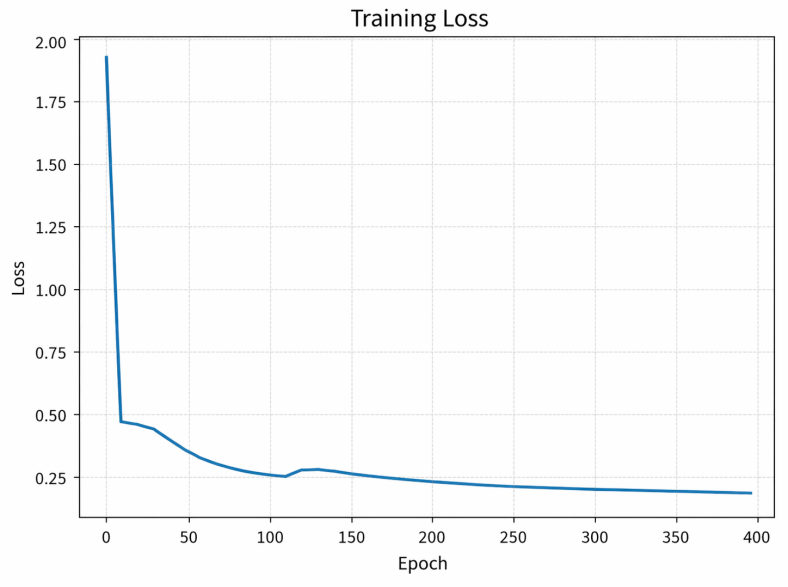

Figure C.1: Evolution of the training loss as a function of the number of epochs for the retained GraphSAGE surrogate model. The curve shows a rapid initial decrease followed by a slower and stable convergence regime, indicating progressive learning of the thermo-mechanical displacement response.

C.2 Evaluation protocol and comparative assessment of the GraphSAGE-GNN surrogate model

The surrogate is evaluated using both standard regression metrics and task-oriented indicators. The training and test pipeline computes global and axis-wise mean absolute error (MAE), root mean squared error (RMSE), coefficient of determination (R^2), and percentile-based error measures. Particular attention is paid to the transverse displacement components, which are more challenging to predict and more directly linked to bending- and twisting-related deformation modes. In addition to standard metrics, the evaluation also measures target-satisfaction and violation rates for the transverse axes, thereby quantifying how often the predictor remains within predefined error tolerances relevant to downstream interpretation.

To assess the architectural relevance of the adopted surrogate, a dedicated comparison pipeline evaluates several competing models under aligned optimization settings. The benchmark includes the retained GraphSAGE surrogate together with a hybrid SAGE–GAT model, a graph transformer model [514], a graph attention network (GAT), a graph convolutional network (GCN) [515], and a multilayer perceptron (MLP) baseline. All models are assessed under harmonized preprocessing, optimization strategy, loss design, and y/z-oriented evaluation criteria, so that the comparison primarily reflects differences

in model family rather than differences in tuning.

The comparison reported in **Table E.1** corresponds to the BC scenario in which two nodes located at the bottom-center of the structure were fully constrained (see BC2 in Figure 4.2d). Under this condition, GraphSAGE achieved the best overall predictive performance among the evaluated models, combining the lowest global error, the best axis-wise errors on the transverse directions, and the highest coefficients of determination. This result motivated the selection of GraphSAGE as the main surrogate architecture retained in the present study.

Table C.1: Comparison of predictive performance across numerical and ML-based models for the BC2 scenario. The FE model is included as the ground truth reference. Best results are highlighted in bold (Inf. denotes inference).

Model	MAE	MAE_y	MAE_z	R_y^2	R_z^2	RMSE	Train(s)	Inf./structure (s)
FE Model	–	–	–	–	–	–	–	1800 - 3600
Multilayer Perceptron (MLP)	0.02417	0.03554	0.02985	0.0387	0.0730	0.03873	32617.7	**0.0102**
Graph Convolutional Network (GCN)	0.01420	0.01929	0.01700	0.5860	0.5560	0.02623	283917.7	0.0660
Graph Transformer	0.01295	0.01741	0.01555	0.6173	0.5869	0.02523	**105747.2**	0.0637
Hybrid SAGE–GAT	0.01283	0.01717	0.01529	0.6211	0.5855	0.02520	351582.0	0.0787
GraphSAGE (retained model)	**0.01237**	**0.01641**	**0.01459**	**0.6293**	**0.6015**	**0.02485**	130176.9	0.0269

The results highlight that GraphSAGE provides the best overall compromise between predictive accuracy and computational efficiency for the considered BC. Among all evaluated learning-based surrogates, it achieved the lowest global MAE (0.01237), the lowest transverse errors ($MAE_y = 0.01641$ and $MAE_z = 0.01459$), the highest coefficients of determination ($R_y^2 = 0.6293$ and $R_z^2 = 0.6015$), and the lowest RMSE (0.02485). Although the Graph Transformer achieved a lower training time than GraphSAGE, its predictive performance remained slightly inferior. The hybrid SAGE–GAT model also remained competitive, but it was both less accurate and slower at inference than GraphSAGE for this benchmark. By contrast, the MLP baseline achieved the fastest inference among the learning models, but at the cost of a major degradation in prediction quality, which confirms the relevance of graph-based relational modeling for thermo-mechanical displacement prediction in voxelized active structures.

To quantify statistical robustness, the retained GraphSAGE surrogate was evaluated on an independent test set of $n_{\text{test}} = 300$ structures, corresponding to 295,323 nodes. The dataset was split into training, validation, and test subsets using a 70/15/15 split. In addition to reporting point estimates, 95% confidence intervals were estimated using bootstrap resampling with $B = 1000$ replicates [516, 517]. In this procedure, the test set is repeatedly resampled with replacement, and the evaluation metrics are recomputed for each resampled set. This makes it possible to estimate the variability of the reported metrics without requiring additional FE simulations.

The retained GraphSAGE surrogate achieved a global mean absolute error (MAE) of $0.01237 \pm 5 \times 10^{-5}$, with a 95% confidence interval of $[0.01227, 0.01247]$. The global

root mean squared error (RMSE) was 0.02485 ± 10^{-4}, with a 95% confidence interval of [0.02465,0.02505] [518, 519]. Axis-wise errors were also computed to assess directional prediction quality. The surrogate achieved $0.01641 \pm 6 \times 10^{-5}$ for MAE_y, with a 95% confidence interval of [0.01629,0.01653], and $0.01459 \pm 5 \times 10^{-5}$ for MAE_z, with a 95% confidence interval of [0.01449,0.01469]. These narrow confidence intervals indicate that the predictive performance is statistically stable over the test set.

A particularly important outcome concerns computational efficiency relative to the complete FE workflow. Including model setup and numerical solution, the reference FE workflow requires approximately 3600–10,800 s per structure, whereas the retained GraphSAGE surrogate requires only 0.0269 s for pure inference. This corresponds to an acceleration factor of approximately 1.34×10^5–4.01×10^5, equivalent to a reduction in prediction time of approximately 99.99925%–99.99975%. This computational gain enables the large-scale screening of candidate structures that would otherwise require decades of sequential FE computation.

Appendix D

Materials, Fabrication, and Thermo-Mechanical Modeling of LCEs

D.1 Materials

1, 4-Bis[4-(3-acryloyloxypropyloxy)benzoyloxy]-2- methylbenzene (≥ 95%, RM257) was purchased from Smolecule (San Antonio, USA). 2,2-(Ethylenedioxy)diethanethiol (≥95%, EDDET), Toluene (≥99, 5%), 2, 6-di-tert-butyl-4-methylphenol (≥99%, BHT), phenyl-bis(2,4,6- trimethylbenzoyl)phosphine oxide (≥97%, BAPO), Iron(II, III) oxide (50–100 nm, 97% purity, magnetite), and Bisphenol A ethoxylate diacrylate (BEDA) were purchased from Sigma–Aldrich (France). 4-Cyano-4'-pentylbiphenyl (≥98%, E7) was purchased from TCI (via https://zentek.it/). All chemicals were used as-received without further purification. This formulation was adopted from Aïdonidis et al. [443].

D.2 Material Preparation

RM257 and the nematic liquid crystal E7 were combined in an amber flask at a weight ratio of 1:1.5. BAPO (2 wt.%) was then added. The mixture was heated to 90 °C for 5 min and subsequently cooled to room temperature, at which point the mixture remained in the nematic phase due to the stabilizing effect of E7. EDDET was added dropwise (acrylate-to-thiol molar ratio of 1:1), and the formulation was homogenized by vortex mixing.

D.3 DLP 3D Printing

All geometries were designed using Rhinoceros3D/Grasshopper and exported as STL files.

Three-dimensional structures were fabricated using a DLP printer (ASIGA Max X27) equipped with a 385 nm UV light source operating at an intensity of 59 mW cm^{-2}. The layer thickness was fixed at 50 μm for all prints. The objects were printed with an illumination time of 4.6 s for the first two layers and 1.0 s for the subsequent layers when using the isotropic formulation.For the nematic formulation, the exposure time was reduced to 0.6 s for the initial two layers and 0.1 s for the following ones. For the nematic LCE formulation, mesogen alignment was achieved by exposing the resin to a 100 mT magnetic field for 15 min. Following the alignment phase,the printing process was initiated without disturbing the resin. After printing, the structures were rinsed in ethanol to remove uncured resin. Control samples were fabricated in the absence of any magnetic field, under otherwise identical printing conditions, to serve as unaligned references.

D.4 Finite Element Modeling of Thermo-Responsive LCEs

Stimuli-responsive materials are central to 4D printing because they enable printed structures to undergo programmable shape transformations under external activation. In this work, the active material is modeled as a LCE, a thermo-responsive polymer network combining the entropic elasticity of elastomers with the molecular anisotropy of liquid-crystalline mesogens. This combination allows LCEs to generate large, reversible, and direction-dependent deformations under thermal stimulation, making them suitable for soft robotic, morphing, and adaptive structural applications.

The actuation mechanism originates from the temperature-dependent change in mesogen order. When heated toward the nematic–isotropic transition, the LCE contracts along its programmed director direction and expands in the transverse directions. This anisotropic thermal response is exploited in the present voxel-based structures by assigning different programmed orientations to active voxels. The active material library includes LCE domains aligned with the Cartesian directions, denoted as LCE_x, LCE_y, and LCE_z, as well as two diagonal in-plane orientations (xy), LCE_{45} and LCE_{-45}. These active voxels are combined with passive material regions and void-like highly compliant regions to create heterogeneous thermo-mechanical responses.

In the FE simulations, LCEs are modeled as mechanically isotropic but thermally orthotropic materials. This assumption reflects the fact that the dominant programmed response is introduced through the anisotropic thermal expansion tensor, while the small-strain elastic response is approximated using an isotropic constitutive law. The elastic properties used for the LCE are

$$E = 1.4 \text{ MPa}, \qquad \nu = 0.49,$$

where E is the Young's modulus and ν is the Poisson's ratio. The near-incompressible value of ν is consistent with elastomeric behavior.

The temperature-dependent orthotropic thermal expansion coefficients used in the simulations are reported in Table D.1. The coefficient α_{11} corresponds to the programmed director direction and is negative, representing contraction upon heating. The transverse coefficients α_{22} and α_{33} are positive, representing expansion in the directions perpendicular to the director.

Table D.1: Temperature-dependent orthotropic thermal expansion coefficients used for the LCE material model.

Temperature (°C)	α_{11}	α_{22}	α_{33}
0	-3.30×10^{-4}	1.65×10^{-4}	1.65×10^{-4}
20	-5.075×10^{-4}	2.5375×10^{-4}	2.5375×10^{-4}
40	-8.31667×10^{-4}	4.15833×10^{-4}	4.15833×10^{-4}
60	-1.14875×10^{-3}	5.74375×10^{-4}	5.74375×10^{-4}
80	-1.469×10^{-3}	7.345×10^{-4}	7.345×10^{-4}
100	-1.640833×10^{-3}	8.20417×10^{-4}	8.20417×10^{-4}

The thermoelastic constitutive response is expressed as

$$\boldsymbol{\sigma} = \mathbf{C} : \left[\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{\mathrm{th}}\right],$$

where $\mathbf{C}$ is the elastic stiffness tensor, $\boldsymbol{\varepsilon}$ is the total strain, and $\boldsymbol{\varepsilon}^{\mathrm{th}}$ is the thermal strain. The thermal strain is computed from the temperature-dependent expansion tensor as

$$\boldsymbol{\varepsilon}^{\mathrm{th}} = \int_{T_0}^{T} \boldsymbol{\alpha}(T')\, dT',$$

with

$$\boldsymbol{\alpha}(T) = [\alpha_{11}(T), \alpha_{22}(T), \alpha_{33}(T), 0, 0, 0]^T.$$

Here, T_0 denotes the reference temperature of the simulation, set to 20°C, and thermal activation is applied up to 80°C unless otherwise specified.

Different programmed LCE orientations are implemented by assigning local material coordinate systems to the active voxels. For LCE_X, LCE_Y, and LCE_Z, the contraction direction is aligned with the corresponding Cartesian axis. For LCE_{45} and LCE_{-45}, the local director is rotated by $+45°$ or $-45°$ in the plane. Passive regions are modeled as mechanically inactive support domains with a lower thermal response, while void-like regions are assigned a very low stiffness to approximate highly compliant or removed material.

This material model enables voxel-level programming of anisotropic actuation and

provides a consistent finite element representation of the thermo-mechanical behavior of active, passive, and void-like regions under thermal loading.

Appendix E

Supplementary Modeling and Comparative Evaluation for Graph-Based Multidimensional Inverse Design

E.1 Experimental Setup

E.1.1 Finite Element Simulations

All finite element simulations were performed using Abaqus/Standard 2023 (Dassault Systèmes, Simulia). The simulation models were generated automatically through dedicated Python scripts executed with Python 3.11.5. These scripts created the Abaqus input files from the voxel geometries and optimized material-distribution CSV files by defining the nodal coordinates, element connectivity, material sections, local material orientations, boundary conditions, thermal loading steps, and output requests before submitting the corresponding Abaqus jobs. The voxelized structures were discretized using eight-node three-dimensional hybrid brick elements, C3D8H, which are suitable for nearly incompressible elastomeric materials. This automated Python–Abaqus workflow ensured a consistent generation of all voxel-based structure families and application-level validation models.

E.1.2 Training-Dataset Generation

The experimental study is organized around three families of voxel-based 4D-printed structures, corresponding to one-dimensional, two-dimensional, and three-dimensional design configurations. These families were selected because they represent common structural abstractions encountered in shape-morphing applications: beam-like structures for

bending-dominated actuation, plate-like structures for surface morphing, and volumetric structures for three-dimensional deformation. Representative examples of these families and their deformation responses are shown in Figure 5.2.

The dataset is generated for nine structural configurations: three one-dimensional structures ($R1, R2, R6$), three two-dimensional structures ($S1, S2, S3$), and three three-dimensional structures ($C1, C2, C3$). The material library is based on liquid crystal elastomers (LCEs), which are anisotropic active materials whose deformation depends on the programmed director orientation. In the simulations, LCE_X, LCE_Y, and LCE_Z correspond to LCE domains aligned with the Cartesian axes, while LCE_{45} and LCE_{-45} correspond to diagonal in-plane orientations at $+45°$ and $-45°$, respectively. These orientations are implemented through orthotropic thermal expansion definitions. The passive material is used to provide mechanically inactive support, while the void material represents removed or nearly inactive voxels. The Abaqus generation scripts define the active LCE materials with temperature-dependent orthotropic expansion, together with passive and void material models.

For each structure type and each boundary condition, 1500 simulated samples are generated. The generation process is organized into three material-distribution subsets of 500 structures each. The first subset uses diagonal LCE orientations together with passive and void states:

$$\{\mathrm{LCE}_{45}, \mathrm{LCE}_{-45}, \mathrm{PASS}, \mathrm{VOID}\}.$$

The second subset uses the complete material library:

$$\{\mathrm{LCE}_{45}, \mathrm{LCE}_{-45}, \mathrm{LCE}_X, \mathrm{LCE}_Y, \mathrm{LCE}_Z, \mathrm{PASS}, \mathrm{VOID}\}.$$

The third subset focuses on axis-aligned LCE orientations with passive and void states:

$$\{\mathrm{LCE}_X, \mathrm{LCE}_Y, \mathrm{LCE}_Z, \mathrm{PASS}, \mathrm{VOID}\}.$$

This organization ensures that the training set contains both simple orientation-controlled cases and more diverse multi-orientation material configurations.

For each generated sample, a voxel-wise material distribution is assigned, an Abaqus input file is created, and a thermo-mechanical simulation is executed. The generation script uses $N = 500$ samples per structure type and iterates over the nine configurations, producing material and coordinate files for each simulated structure. The automatic pipeline generates the input file, launches the Abaqus job, extracts the displacement field (U_1, U_2, U_3), creates the corresponding CAE file when available, and stores the resulting material and coordinate outputs.

Two boundary conditions are considered in the dataset. The first corresponds to a support applied on the bottom region of the structure, representative of base-constrained

or cantilever-like actuation. The second corresponds to a support applied on the left face, allowing the same structural family to be studied under a lateral constraint. These boundary conditions provide complementary deformation scenarios and allow the learning models to observe how support placement affects the resulting shape transformation.

The simulation outputs are converted into graph-based training pairs. Each sample contains an original graph and a displaced graph. The original graph stores the initial nodal coordinates, material information, and connectivity, while the displaced graph stores the corresponding deformed coordinates obtained from the thermo-mechanical simulation. These graph pairs are then used in the subsequent stages of the framework: the VAE learns a compact geometric latent representation from original–displaced graph pairs, the GNN surrogate learns the mapping from an initial material graph to nodal displacement, and the hybrid optimizer searches for material distributions that minimize the target-shape error.

The complete pipeline therefore combines physics-generated data, graph representation, latent geometric learning, surrogate deformation prediction, and material-aware inverse design. The following subsections analyze each component: first the graph-based VAE representation, then the forward GNN surrogate, and finally the hybrid optimization strategy for target-shape matching.

Training convergence of the forward simulation surrogate To further assess the stability of the forward simulation surrogate, we analyzed the evolution of the training and validation curves across the different structural configurations. These curves correspond to the GNN-based forward simulator, not to the VAE stage. The objective of this analysis is to verify whether the surrogate learns stable deformation mappings for the different 1D, 2D, and 3D structure families and for the considered boundary conditions.

Figure E.1 reports the validation loss evolution across the simulated configurations. Overall, the loss decreases rapidly during the first training epochs, followed by a slower convergence regime. This behavior indicates that the model quickly captures the dominant deformation trends before progressively refining the prediction of more localized displacement patterns. The same trend is observed across the 1D beam-like structures, the 2D plate-like structures, and the 3D volumetric configurations, suggesting that the graph-based surrogate provides a stable learning behavior across different structural dimensionalities.

In addition to the global loss, the axis-wise MAE and R^2 curves show that the y and z displacement components are progressively improved during training. This is particularly important for the present shape-morphing problem, since these transverse components are more closely related to bending and out-of-plane deformation modes. The increase of the R^2 values over training confirms that the surrogate progressively explains a larger fraction of the displacement variance, while the decrease of the MAE indicates improved

local prediction accuracy.

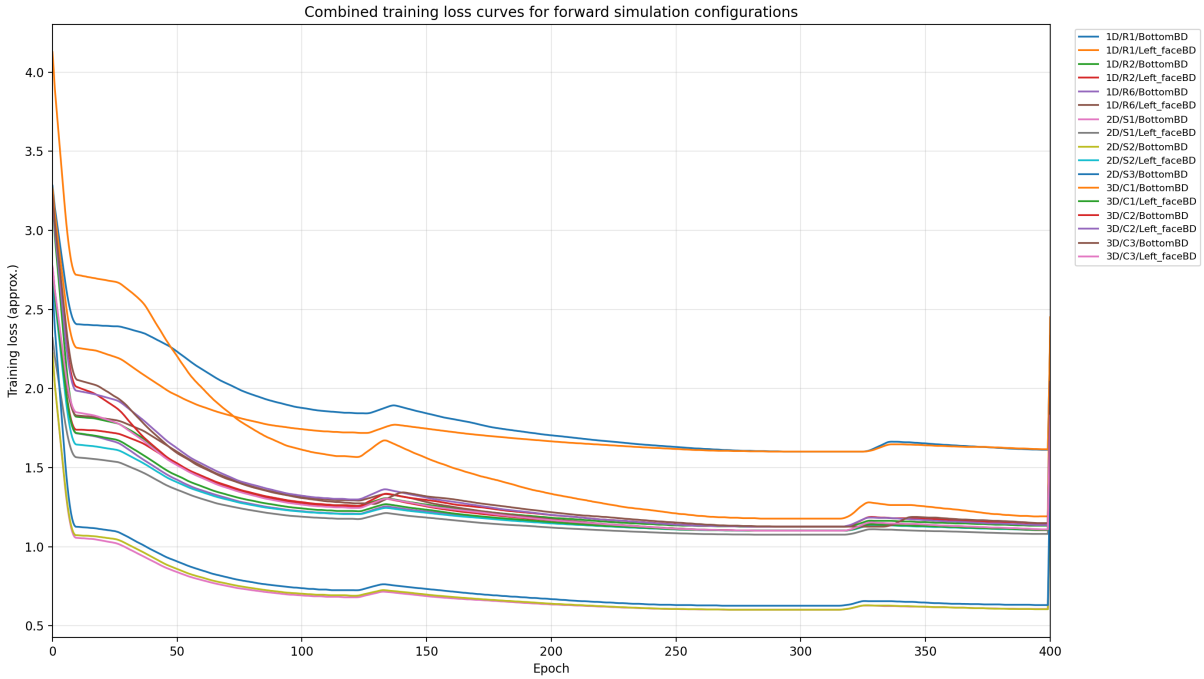

Figure E.1: Validation loss evolution of the forward GNN surrogate across the different 1D, 2D, and 3D structural configurations and boundary conditions. The curves correspond to the simulation surrogate training stage.

The convergence behavior supports the use of the trained GNN surrogate in the subsequent hybrid inverse design stage. Since the optimizer evaluates many candidate material distributions, the surrogate must provide both fast inference and stable deformation prediction. The observed loss reduction and the improvement of the transverse-axis metrics indicate that the model learns meaningful graph-to-displacement mappings, making it suitable for rapid target-shape evaluation.

E.2 Evaluation protocol and comparative assessment of the GraphSAGE-GNN surrogate model

The surrogate is evaluated using both standard regression metrics and task-oriented indicators. The training and test pipeline computes global and axis-wise mean absolute error (MAE), root mean squared error (RMSE), coefficient of determination (R^2), and percentile-based error measures. Particular attention is paid to the transverse displacement components, which are more challenging to predict and more directly linked to bending- and twisting-related deformation modes. In addition to standard metrics, the evaluation also measures target-satisfaction and violation rates for the transverse axes, thereby quantifying how often the predictor remains within predefined error tolerances relevant to downstream interpretation.

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The comparison reported in **Table E.1** corresponds to the BC scenario in which two nodes located at the bottom-center of the structure were fully constrained (see BC2 in Figure 4.2d). Under this condition, GraphSAGE achieved the best overall predictive performance among the evaluated models, combining the lowest global error, the best axis-wise errors on the transverse directions, and the highest coefficients of determination. This result motivated the selection of GraphSAGE as the main surrogate architecture retained in the present study.

Table E.1: Comparison of predictive performance across numerical and ML-based models for the BC2 scenario. The FE model is included as the ground truth reference. Best results are highlighted in bold (Inf. denotes inference).

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GraphSAGE (retained model)	**0.01237**	**0.01641**	**0.01459**	**0.6293**	**0.6015**	**0.02485**	130176.9	0.0269

The results highlight that GraphSAGE provides the best overall compromise between predictive accuracy and computational efficiency for the considered BC. Among all evaluated learning-based surrogates, it achieved the lowest global MAE (0.01237), the lowest transverse errors ($\text{MAE}_y = 0.01641$ and $\text{MAE}_z = 0.01459$), the highest coefficients of determination ($R_y^2 = 0.6293$ and $R_z^2 = 0.6015$), and the lowest RMSE (0.02485). Although the Graph Transformer achieved a lower training time than GraphSAGE, its predictive performance remained slightly inferior. The hybrid SAGE–GAT model also remained competitive, but it was both less accurate and slower at inference than GraphSAGE for this benchmark. By contrast, the MLP baseline achieved the fastest inference among the learning models, but at the cost of a major degradation in prediction quality, which confirms the relevance of graph-based relational modeling for thermo-mechanical displacement prediction in voxelized active structures.

To quantify statistical robustness, the retained GraphSAGE surrogate was evaluated on an independent test set of $n_{\text{test}} = 300$ structures, corresponding to 295,323 nodes. The dataset was split into training, validation, and test subsets using a 70/15/15 split.

In addition to reporting point estimates, 95% confidence intervals were estimated using bootstrap resampling with $B = 1000$ replicates [516, 517]. In this procedure, the test set is repeatedly resampled with replacement, and the evaluation metrics are recomputed for each resampled set. This makes it possible to estimate the variability of the reported metrics without requiring additional FE simulations.

The retained GraphSAGE surrogate achieved a global mean absolute error (MAE) of $0.01237 \pm 5 \times 10^{-5}$, with a 95% confidence interval of $[0.01227, 0.01247]$. The global root mean squared error (RMSE) was 0.02485 ± 10^{-4}, with a 95% confidence interval of [0.02465,0.02505] [518, 519]. Axis-wise errors were also computed to assess directional prediction quality. The surrogate achieved $0.01641 \pm 6 \times 10^{-5}$ for MAE_y, with a 95% confidence interval of [0.01629,0.01653], and $0.01459 \pm 5 \times 10^{-5}$ for MAE_z, with a 95% confidence interval of [0.01449,0.01469]. These narrow confidence intervals indicate that the predictive performance is statistically stable over the test set.

A particularly important outcome concerns computational efficiency relative to the complete FE workflow. Including model setup and numerical solution, the reference FE workflow requires approximately 3600–10,800 s per structure, whereas the retained GraphSAGE surrogate requires only 0.0269 s for pure inference. This corresponds to an acceleration factor of approximately 1.34×10^5–4.01×10^5, equivalent to a reduction in prediction time of approximately 99.99925%–99.99975%. This computational gain enables the large-scale screening of candidate structures that would otherwise require decades of sequential FE computation.

E.2.1 Evaluation protocol and comparative assessment of the Forward GraphSAGE Surrogate

To assess the relevance of the retained surrogate, a dedicated comparison was performed on the C1 structure left-face support condition, in which all nodes located on the left face of the structure are fixed. The retained GraphSAGE surrogate is subsequently used as a forward evaluator to predict the material-dependent deformation of each candidate structure. Its architecture selection, comparative evaluation, regression metrics, bootstrap confidence intervals, and computational speedup are reported in Appendix C, Section C.2.

E.2.2 Evaluation protocol and comparative assessment of the Graph-based VAE

To identify the most suitable graph encoder for the geometric VAE, we performed a systematic comparison across all structure families and boundary conditions defined above. This results in 18 evaluated structure–boundary condition configurations. For a fair comparison, all models use the same decoder, loss function, optimizer, hidden

dimension, latent dimension, and training protocol; only the graph encoder backbone is changed. In this comparison, all encoders receive three-dimensional normalized nodal coordinates as input. The shared architecture uses a hidden dimension of 128, a latent dimension of 64, two encoder message-passing layers, ReLU nonlinear activations, layer normalization, dropout regularization with a rate of 0.2, and residual connections in the encoder. The decoder is kept identical across all models and reconstructs both the original and displaced coordinates from the sampled latent vector. This setup ensures that the comparison primarily reflects the effect of the encoder backbone rather than differences in representation size, decoder capacity, or training protocol.

The compared encoders are a graph convolutional network (GCN) [515], a graph attention network (GAT) [520], GraphSAGE [3], TransformerConv [514], EdgeConv [521], and an multilayer perceptron (MLP) baseline without graph message passing [522]. These backbones cover complementary graph aggregation mechanisms, including spectral-like convolution, attention-based aggregation, neighborhood sampling, transformer-style message passing, and edge-based local feature extraction. The MLP baseline is included to assess whether graph connectivity provides an advantage over a purely node-wise feed-forward representation.

The evaluation is focused on the reconstruction of negative-z regions, since these regions correspond to downward out-of-plane deformation patterns that are important for shape-morphing behavior. This evaluation is formulated as a binary detection problem at the node level: a reconstructed node is considered negative when its recovered z-coordinate lies below zero. Precision measures the fraction of predicted negative-z nodes that are truly negative, and therefore quantifies the reliability of the predicted negative regions. Recall measures the fraction of true negative-z nodes that are successfully recovered, quantifying the ability of the model to avoid missing physically meaningful downward deformations. The F1-score, defined as the harmonic mean of precision and recall, is used as the primary selection criterion because it balances false positives and false negatives. Accuracy is also reported, but it is interpreted with caution because it can be overly optimistic when negative-z nodes represent a minority of the geometry [523, 524]. Finally, training time and inference time are included to quantify the trade-off between reconstruction quality and computational cost.

Table E.2: Average VAE encoder comparison over all evaluated dimensions and boundary conditions (values in bold indicate the best performance).

Encoder	Precision (%)	F1 (%)	Recall (%)	Accuracy (%)	Train time (s)	Inference time (s)
EdgeConv	**55.99**	**55.55**	**66.76**	94.49	14,085.56	11.83
GAT	55.82	55.54	66.74	**94.82**	10,934.54	9.44
GraphSAGE	55.74	55.21	66.01	94.52	9,561.72	6.06
GCN	55.72	54.62	64.62	**94.82**	8,422.54	6.61
Transformer	55.51	53.96	63.34	94.65	15,945.99	10.66
MLP	54.06	52.96	62.31	94.71	**6,813.64**	**2.88**

The results in **Table E.2** show that EdgeConv achieves the best average F1-score, precision, and recall among all evaluated encoders. This indicates that EdgeConv provides the most balanced reconstruction of negative-z deformation patterns. However, the performance difference with GAT is very small: GAT reaches an almost identical F1-score while requiring less training and inference time. GraphSAGE also provides a competitive compromise, with slightly lower reconstruction metrics but lower inference time than both EdgeConv and GAT. The MLP baseline is the fastest model, but it obtains the lowest F1-score and recall, confirming the benefit of graph-based message passing for learning deformation-aware latent representations.

Appendix F

Implementation Details of the Orchestrator-Centered Multi-Agent Framework

F.1 Orchestrator Reasoning Model

The orchestrator is implemented using `gpt-oss-120b`, an open-weight mixture-of-experts reasoning model with 117 billion total parameters, of which 5.1 billion are activated per token [525]. The model was selected because it supports function calling, structured outputs, and configurable reasoning effort, which are required by the proposed orchestration workflow. Specifically, it is used to interpret scientific requests, construct ordered execution plans, select the appropriate agents and tools, manage dependencies between operations, and synthesize the intermediate results produced by the specialized agents.

The reasoning model is not used to generate numerical or physical predictions directly. Material optimization, displacement prediction, geometric processing, and behavior evaluation are delegated to the corresponding scientific models and computational tools. This separation maintains a clear distinction between language-based orchestration and domain-specific computation, while ensuring that the final response is grounded in the structured artifacts produced during execution.